Using the Chapter Problems

Every core teaching chapter ends with a two part problem program. The research synthesis on niche consumption instead supplies a reading guide connecting existing exercises. Part I contains Core benchmarks to work without software: show the derivation, check feasibility or the defining identity, and state the economic point. Part II uses Proof and Applied problems as bridges to a named topic case, which appears last within its topic group. The unified general equilibrium chapter contains three such groups and preserves all three cases. Technical appendices supply proofs rather than a separate weekly assignment. Problem identifiers are stable across the book, course outline, and released handouts. Problem identifiers are permanent catalog labels. They can differ from the printed chapter number after topics are combined, and they remain unchanged in weekly assignments, case studies and the solutions handbook.

Before using Codex, Python, or Lean, write four lines: Seed identifies the hand result being extended; Fixed records the assumptions that do not change; Scaled names the single enlarged dimension or strengthened claim; Risk states the new failure mode. A verified artifact must reproduce the seed, include a boundary or deliberately failing test, report the validation command, and separate the checked mathematical statement from the economic interpretation.

Weekly handouts are release views of this architecture. The chapter problem section is the question-only source of record; solutions and instructor notes are maintained separately.

Microeconomic Theory I

Victor H. Aguiar

Fall 2026 edition

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Preface to the Fall 2026 Edition

These notes provide a rigorous first course in graduate microeconomic theory. The Fall 2026 edition preserves the substantive coverage of the Fall 2024 course while improving the exposition, notation, organization, and portability of the source files. The central discipline throughout is to distinguish primitives, behavioral assumptions, optimization problems, equilibrium conditions, and empirical implications.

The material is cumulative. Part I begins with utility maximization and consumer demand, then develops duality, comparative statics, welfare, revealed preference and abstract choice. Part II studies expected utility, stochastic choice and attention. Part III turns to aggregation. Part IV develops production, competitive markets and one unified general equilibrium chapter, including existence, welfare, equilibrium behavior and empirical restrictions. Part V begins with niche consumption and product space, connecting consumer theory to the preceding equilibrium analysis, then treats sequential trade, matching and computable general equilibrium. Appendix A collects order theory, Appendix B develops fixed points and equilibrium existence, Appendix C records the exact versions of starred consumer theorems, and the final glossary provides linked notation.

Notation and Conventions

The following conventions apply unless a chapter explicitly states otherwise. They are collected here so that the same symbol does not silently change meaning across topics. In the PDF, a symbol shown in dark blue at its first substantive use links to its entry in the Glossary of Symbols; the symbol in the glossary links back to that first use. The LyX manuscript is the authoritative source for both the text and these links.

- There are L commodities, indexed by $\ell = 1, \ldots, L$. Commodity bundles are column vectors $x \in X \subseteq \mathbb{R}^L_+$.
- Consumers are indexed by $i = 1, \ldots, I$; firms are indexed by $j = 1, \ldots, J$. Consumption is denoted by x, production by y, and endowments by ω.
- Prices are column vectors $p \in \mathbb{R}^L_{++}$, unless a result explicitly permits zero boundary prices. The scalar $w > 0$ denotes wealth. The budget set is $B(p, w) = \{x \in X : p \cdot x \leq w\}$.
- Set difference is denoted by $A \setminus B$. We write $\max, \min, \arg\max, \arg\min$ for the value and optimizer-set operators; each is typeset as a mathematical operator.
- Direct utility is u. Marshallian demand is $x(p, w)$, Hicksian demand is $h(p, \overline{u})$, indirect utility is $v(p, w)$, and the expenditure function is $e(p, \overline{u})$.
- The expression $a \cdot b$ denotes the Euclidean inner product. The Euclidean norm is $\|a\|_2$, and $\mathbf{1}_n$ denotes the all-ones column vector in $\mathbb{R}^n$. The notation $a^\top$ denotes transpose. A prime marks an alternative object of the same type, as in a'; it is not used for transpose.
- Weak preference, strict preference, and indifference are denoted by $\succeq, \succ, \sim$, respectively. A choice correspondence is written $c : \mathcal{A} \rightrightarrows X$; the surrounding section defines the menu domain and states when the rule is single-valued.
- For a differentiable vector-valued function, $D_z f$ is the Jacobian of a vector-valued map

with respect to z, while $\nabla_z f$ is the column gradient of a scalar-valued function. Hessians are written $D_z^2 f$.

- For vectors, $a \leq b$ means coordinatewise inequality and $a \gg 0$ means every coordinate is strictly positive. The power set is 2^X.
- Probability and expectation are written $\Pr, \mathbb{P}, \mathbb{E}$ after the relevant probability space has been declared. The indicator is $\mathbb{I}\{\cdot\}$, and $\log$ denotes the natural logarithm.

Part I

The Classical Rational Consumer

Chapter 1

Utility Maximization and Consumer Demand

Question: What is the benchmark model, and what behavior does it generate?

The classical rational consumer chooses a feasible bundle to maximize a utility function. This chapter states that model, proves existence of demand, and develops its first-order characterization. Duality and welfare are treated in the next chapter. Throughout, $X = \mathbb{R}^L_+$, $L \geq 1$ is the fixed number of commodities, $\mathsf{P} = \mathbb{R}^L_{++}$, $\mathsf{W} = \mathbb{R}_{++}$.

1.1 The Benchmark Model

The benchmark model has three primitives: a consumption set X, a utility function $u : X \to \mathbb{R}$, and a feasible-set correspondence $(p, w) \mapsto B(p, w)$. At prices and wealth (p, w), the consumer solves

$$x^* \in x(p, w) := \arg \max_{y \in B(p,w)} u(y).$$

Everything in the chapter follows from this optimization problem plus assumptions that are displayed where they are used.

Utility is ordinal in the deterministic benchmark: if ϕ is strictly increasing on $u(X)$, then u and $\phi \circ u$ generate exactly the same demand. We therefore take a utility representation as the primitive here without attaching cardinal or interpersonal meaning to its levels. Chapter 4 returns to the logically prior question of when an observed preference relation admits such a representation.

1.2 Commodity Space and Budget Sets

An economic commodity is a good or service whose quantity matters to the consumer. There are $L \geq 1$ commodities. A consumption bundle is a column vector $x = (x_1, \ldots, x_L)^\top \in \mathbb{R}^L$, and x_ℓ denotes the quantity of commodity $\ell \in \{1, \ldots, L\}$.

The consumption set $X \subseteq \mathbb{R}^L$ records the bundles that are feasible for the consumer. Unless a section states otherwise, we take $X = \mathbb{R}^L_+$, so negative consumption is ruled out. Other applications may impose upper bounds or interpret coordinates as net trades.

The consumer cannot generally choose every bundle in X. In the benchmark model, prices and wealth are the only constraints. Let $w > 0$ denote wealth and let $p \in \mathbb{R}^L_{++}$ denote the price vector. The budget set is $B(p, w) = \{x \in X : p \cdot x \leq w\}$, a subset of the consumption set.

1.3 Utility Maximization

We assume that u is continuous and $X = \mathbb{R}^L_+$.

The utility maximization problem (UMP) is:

$$\max_{x \geq 0} u(x)$$

$$\text{subject to} \quad p \cdot x \leq w.$$

Equivalently, define the budget set as $x \in B(p, w) = \{x \in \mathbb{R}^L_+ : p \cdot x \leq w\}$.

Proposition 1.1. *If $p \gg 0$ and u is continuous, then the UMP has a solution.*

Proof. Because $B(p, w)$ contains the zero bundle, it is nonempty. For each $x \in B(p, w)$ and each commodity ℓ, strict positivity of prices gives $0 \leq x_\ell \leq w/p_\ell$; hence the budget set is bounded.

The map $x \mapsto p \cdot x$ is continuous, so $B(p, w) = \mathbb{R}^L_+ \cap \{x : p \cdot x \leq w\}$ is closed. The Heine--Borel theorem therefore implies that $B(p, w)$ is compact.

The Weierstrass maximum theorem therefore gives a solution to the UMP.

□

We now prove further properties of the demand correspondence.

Proposition 1.2. *The Marshallian demand correspondence is*

$$x(p, w) := \arg \max_{y \in B(p,w)} u(y).$$

It is non-empty, compact-valued and upper hemicontinuous. The indirect utility function is

$$v(p, w) = \max_{y \in B(p,w)} u(y),$$

and it is continuous.

Proof. The result follows from Berge's maximum theorem once we verify that the budget correspondence $B(\cdot)$ is continuous. We already know that u is continuous and that $B(p, w)$ is compact, so $B(\cdot)$ is compact-valued.

Let $(p^n, w^n) \to (p, w)$. If $x^n \in B(p^n, w^n)$ and $x^n \to x$, then $p^n \cdot x^n \leq w^n$; taking limits gives $p \cdot x \leq w$, so the graph is closed. Moreover, near any (p, w), prices are bounded away from zero and wealth is bounded above; hence the nearby budget sets are uniformly bounded. Closed graph plus this local boundedness gives upper hemicontinuity.

For the reverse inclusion, let $x \in B(p, w)$. If $p \cdot x < w$, then $x \in B(p^n, w^n)$ for all sufficiently large n. If $p \cdot x = w$, set

$$x^n = \frac{x}{\max\{1, (p^n \cdot x)/w^n\}}.$$

Then $x^n \in B(p^n, w^n)$ and $x^n \to x$. Hence the budget correspondence is continuous on $\mathbb{R}^L_{++} \times \mathbb{R}_{++}$, and Berge's theorem gives the stated properties.

□

The next definitions separate the assumptions used in the comparative-statics results.

Definition 1.3. Local nonsatiation. A preference relation $\succeq$ is locally nonsatiated if, for every $x \in X$ and every $\varepsilon > 0$, there is $y \in X$ with $\|y - x\|_2 < \varepsilon$ and $y \succ x$.

Proposition 1.4. *If a utility function representing a locally nonsatiated preference relation generates the Marshallian demand correspondence, then every demanded bundle exhausts the budget: $p \cdot x = w$ for every $x \in x(p, w)$.*

Proof. If $p \cdot x < w$, local nonsatiation supplies a strictly preferred bundle arbitrarily close to x. For a sufficiently small neighborhood the bundle remains affordable, contradicting utility maximization. Therefore the budget constraint binds.

Proposition 1.5. *The same demand correspondence is homogeneous of degree zero. If preferences are convex, demand is convex-valued; if preferences are strictly convex, demand is single-valued.*

Proof. Homogeneity follows because $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$. If $x, y \in x(p, w)$ and preferences are convex, every mixture $\alpha x + (1 - \alpha)y$ is feasible and weakly preferred to either

maximizer, so it is also demanded. Under strict convexity, two distinct maximizers would have a feasible mixture strictly preferred to both, a contradiction. □

1.4 Demand Correspondences

Definition 1.6. A Walrasian demand correspondence is a map

$$x : \mathsf{P} \times \mathsf{W} \rightrightarrows X$$

with nonempty values such that every $q \in x(p, w)$ satisfies $p \cdot q \leq w$. If demand is generated by a choice correspondence c, then $x = c \circ B$, where

$$B(p, w) = \{q \in X : p \cdot q \leq w\}.$$

Example 1.7. With two perfect-substitute goods and utility $u(q) = q_1 + q_2$,

$$x(p, w) = \begin{cases} \{(w/p_1, 0)\}, & p_1 < p_2, \\ \{(0, w/p_2)\}, & p_2 < p_1, \\ \{q \in \mathbb{R}^2_+ : p_1 q_1 + p_2 q_2 = w\}, & p_1 = p_2. \end{cases}$$

The equal-price value is the entire budget line, not a pair of independent intervals.

For later use, distinguish the following properties.

- A correspondence has a closed graph if $(p^n, w^n) \to (p, w)$,$q^n \in x(p^n, w^n)$, and $q^n \to q$ imply $q \in x(p, w)$. Closed graph and upper hemicontinuity are not equivalent without additional compactness or local-boundedness conditions.
- It is compact-valued if $x(p, w)$ is compact for every (p, w). In Euclidean space, compactness is equivalent to being closed and bounded.
- It is convex valued if $\alpha q + (1 - \alpha) r \in x(p, w)$ whenever $q, r \in x(p, w)$ and $\alpha \in [0, 1]$.

1.5 First-Order and KKT Conditions

Assume that u is continuously differentiable and that a constraint qualification holds at the candidate optimum. The KKT conditions are then necessary. If u is concave, they are also sufficient for a global optimum; differentiability alone does not provide sufficiency.

The KKT conditions are as follows.

$\mathcal{L} = u(x) + \lambda(w - p \cdot x)$

Primal and dual feasibility require $x^* \geq 0$, $w - p \cdot x^* \geq 0$, and $\lambda^* \geq 0$.

$\partial_{x_\ell} u(x^*) \leq \lambda^* p_\ell$ and $x^*_\ell[\partial_{x_\ell} u(x^*) - \lambda^* p_\ell] = 0$ for all $\ell \in \{1, \ldots, L\}$.

Complementary slackness is $\lambda^*(w - p \cdot x^*) = 0$.

This is a system with $L + 1$ unknowns.

Local nonsatiation implies that the consumer is on the budget hyperplane (the frontier). This means that

$$w - p \cdot x^* = 0.$$

For an interior solution with a positive multiplier and nonzero marginal utilities, the KKT conditions reduce to

$$\frac{\partial_{x_\ell} u(x^*)}{\partial_{x_k} u(x^*)} = \frac{p_\ell}{p_k} \quad \forall \ell, k \in \{1, \ldots, L\}.$$

Note that for $L = 2$,$-\frac{p_1}{p_2}$ is the slope of the budget line and $-\frac{\partial_{x_1} u(x^*)}{\partial_{x_2} u(x^*)}$ is the slope of the indifference curve. Their equality is the familiar tangency condition expressed in terms of the marginal rate of substitution (MRS).

Problem 1.6 applies these conditions to the fertility--education household block in Unified Growth Theory [Galor, 2005, Galor and Weil, 2000]. The problem uses a closed-form child-human-capital technology for teaching; it does not reproduce the full dynamic growth model.

1.5.1 Expenditure-Augmented Utility

The Marshallian problem treats x as the complete consumption bundle and wealth w as observed. In many applications, however, the data record only the goods in one market. Spending $p \cdot x$ on those goods reduces what remains for outside consumption, saving, or other financial decisions. Expenditure-augmented utility represents this opportunity cost directly [Deb et al., 2023].

Definition 1.8. Expenditure-augmented utility. Fix a consumption set $X \subseteq \mathbb{R}^L_+$. An expenditure-augmented utility function is $U^A : X \times \mathbb{R}_- \to \mathbb{R}$, where $U^A(x, m)$ is strictly increasing in $m = -e$; equivalently, holding the bundle fixed, utility strictly decreases as spending rises. At $p \gg 0$, define augmented demand and the value of the price environment by

$$D^A(p) := \arg\max_{x \in X} U^A(x, -p \cdot x), \qquad V^A(p) := \sup_{x \in X} U^A(x, -p \cdot x).$$

Unlike Marshallian demand $x(p, w)$, choice is over all of X and expenditure is endogenous. A maximizer must therefore be assumed or established.

Quasilinear utility is the special case, not another name for the general model:

$$U^A(x, m) = \widetilde{u}(x) + m \quad \Longrightarrow \quad D^A(p) = \arg\max_{x \in X} \{\widetilde{u}(x) - p \cdot x\}.$$

This objective follows from overall utility $\widetilde{u}(x) + z$ with an outside numeraire z priced at one, provided wealth is high enough that the optimizer leaves $z = M - p \cdot x \geq 0$. A general U^A instead allows both the marginal value of outside resources and substitution among the observed goods to vary with total spending.

For $X = \mathbb{R}^L_+$, suppose U^A is differentiable and concave and that $x^* \in D^A(p)$ exists. The KKT conditions are

$$\begin{aligned} U^A_{x_\ell}(x^*, -p \cdot x^*) - p_\ell U^A_m(x^*, -p \cdot x^*) &\leq 0, \\ x^*_\ell \left[U^A_{x_\ell}(x^*, -p \cdot x^*) - p_\ell U^A_m(x^*, -p \cdot x^*)\right] &= 0, \qquad \ell = 1, \ldots, L. \end{aligned}$$

Concavity makes these conditions sufficient. If $x^*_\ell > 0$ and $U^A_m > 0$, then $U^A_{x_\ell}/U^A_m = p_\ell$. In the quasilinear case $U^A_m = 1$, recovering $\partial_{x_\ell} \widetilde{u}(x^*) = p_\ell$ for every consumed good. For finite data $\{(p^t, x^t)\}_{t=1}^T$, the corresponding revealed-price test is GAPP: p^s is directly revealed preferred to p^t when $p^s \cdot x^t \leq p^t \cdot x^t$, and the transitive closure may contain no cycle with a strict comparison. Deb, Kitamura, Quah, and Stoye prove that GAPP is equivalent to finite-data rationalization by an augmented utility function. This price-ranking condition is distinct from GARP and WGARP, which compare bundles on observed budget sets.

1.6 Workhorse Utility Specifications

A useful specification gives more than a closed-form answer: it identifies the assumptions behind expenditure shares, substitution, and corner choices. We derive several families from the same budget problem. Throughout, prices are strictly positive and wealth is positive; additional consumption-domain restrictions are stated for each model.

1.6.1 Homotheticity and Cobb-Douglas Demand

Definition 1.9. Homothetic preferences. A preference relation is homothetic if $x \succeq y$ implies $\alpha x \succeq \alpha y$ for every $\alpha > 0$ whenever the scaled bundles are feasible. Under the usual continuity and monotonicity assumptions, it admits a positively homogeneous utility representation; an arbitrary representation need not itself be homogeneous.

The Cobb--Douglas representation $u(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}$ is a standard example.

For L goods, fix positive taste weights a_ℓ with $\sum_{\ell=1}^{L} a_\ell = 1$. Cobb-Douglas utility is $u(x) = \prod_{\ell=1}^{L} x_\ell^{a_\ell}$. Every boundary bundle has utility zero and a strictly positive affordable bundle has positive utility, so the optimum is interior. Maximizing its logarithm gives $a_\ell / x_\ell = \lambda p_\ell$. Multiplying by x_ℓ and summing gives $\lambda = 1/w$, hence

$$x_\ell(p, w) = a_\ell \frac{w}{p_\ell}, \qquad b_\ell(p, w) := \frac{p_\ell x_\ell(p, w)}{w} = a_\ell.$$

The expenditure share b_ℓ is the fraction of the budget allocated to good ℓ; the share vector is $\boldsymbol{b} = (b_1, \ldots, b_L)^\top$. Cobb-Douglas shares are constant even though quantities respond to prices. Each wealth elasticity is one, each own-price elasticity is minus one, and each uncompensated cross-price elasticity is zero.

Example 1.10. A fixed budget share. Let $a = (1/4, 3/4)$, $p = (2, 1)$, and $w = 80$. Demand is $(10, 60)$ and expenditure is $(20, 60)$. Doubling only the first price halves x_1 while leaving x_2 unchanged. Doubling wealth doubles both quantities. A constant share is a restriction on behavior, not a property of all homothetic preferences.

1.6.2 CES: A Family of Substitution Patterns

The constant-elasticity-of-substitution (CES) family varies substitution while retaining explicit demand. Keep the positive normalized weights above. Let $\rho < 1$ and define the substitution elasticity $\sigma = 1/(1 - \rho) > 0$. For $\rho \neq 0$, use

$$u_\rho(x) = \left(\sum_{\ell=1}^{L} a_\ell x_\ell^{\rho} \right)^{1/\rho}.$$

For $0 < \rho < 1$ this formula is defined on $\mathbb{R}^L_+$. For $\rho < 0$, define it first on $\mathbb{R}^L_{++}$ and extend it continuously by assigning zero utility to bundles with any zero coordinate. At $\rho = 0$, take the continuous limit $u_0(x) = \prod_\ell x_\ell^{a_\ell}$. Each representation is homogeneous of degree one. CES preferences are therefore homothetic, including cases in which expenditure shares vary with

relative prices.

The optimum is interior. For $\rho \leq 0$, a positive bundle dominates every boundary bundle in utility. For $0 < \rho < 1$, a small transfer of expenditure toward a missing good is beneficial because its marginal utility becomes arbitrarily large near zero. For $\rho \neq 0$, an equivalent objective on the positive orthant, including the sign needed when $\rho < 0$, is

$$\widetilde{u}_\rho(x) = \frac{\sum_\ell a_\ell x_\ell^\rho - 1}{\rho}, \qquad \partial^2_{x_\ell x_\ell} \widetilde{u}_\rho = a_\ell(\rho - 1)x_\ell^{\rho-2} < 0.$$

At $\rho = 0$, instead use $\widetilde{u}_0(x) = \sum_\ell a_\ell \log x_\ell$, with own second derivatives $-a_\ell/x_\ell^2 < 0$. In both cases all cross derivatives vanish, so the objective is strictly concave. Its first-order conditions $a_\ell x_\ell^{\rho-1} = \lambda p_\ell$, together with budget exhaustion, identify the unique maximizer. For negative ρ, maximizing the untransformed sum of powers would reverse the intended ranking. Taking ratios and then imposing the budget gives

$$\frac{x_\ell}{x_k} = \left(\frac{a_\ell/p_\ell}{a_k/p_k}\right)^\sigma, \qquad x_\ell(p, w) = \frac{w a_\ell^\sigma p_\ell^{-\sigma}}{\sum_{k=1}^L a_k^\sigma p_k^{1-\sigma}}.$$

The same demand formula applies at $\sigma = 1$ and gives Cobb-Douglas. It satisfies Walras' law, has degree zero in (p, w), and scales linearly with wealth. Its shares are

$$b_\ell(p) = \frac{a_\ell^\sigma p_\ell^{1-\sigma}}{\sum_k a_k^\sigma p_k^{1-\sigma}}.$$

Except in the Cobb-Douglas case, a taste weight is generally not an expenditure share. Raising a good's price lowers its share if $\sigma > 1$ and raises its share if $0 < \sigma < 1$, even though its purchased quantity falls in both cases. The name CES refers to the response of a quantity ratio to a relative price:

$$-\frac{d\log(x_\ell/x_k)}{d\log(p_\ell/p_k)} = \sigma.$$

This relative-demand elasticity differs from an individual good's own-price elasticity. Differentiating the demand formula at fixed wealth gives, for $k \neq \ell$,

$$\begin{aligned}
\frac{\partial \log x_\ell}{\partial \log w} &= 1, \\
\frac{\partial \log x_\ell}{\partial \log p_\ell} &= -\sigma(1 - b_\ell) - b_\ell, \\
\frac{\partial \log x_\ell}{\partial \log p_k} &= (\sigma - 1)b_k.
\end{aligned}$$

Thus every good is normal. Distinct goods are gross substitutes when $\sigma > 1$ and gross complements when $0 < \sigma < 1$: these terms describe the cross-price response at fixed wealth. Chapter 2 separates this response into substitution and purchasing-power effects. In particular, a negative uncompensated cross effect need not imply a negative compensated cross effect.

Example 1.11. Substitution and relative expenditure. Let $a_1 = a_2 = 1/2$, $\rho = 1/2$, $p = (1, 2)$, and $w = 120$. Then $\sigma = 2$ and the denominator is $1/4 + 1/8 = 3/8$, giving $x = (80, 20)$. The quantity ratio is four but the expenditure ratio is two. The budget check is $80 + 2(20) = 120$. The own-price elasticity of good 1 is $-4/3$, rather than -2.

Example 1.12. Gross complementarity. Let $a_1 = a_2 = 1/2$, $\rho = -1$, $p = (1, 4)$, and $w = 90$. Here $\sigma = 1/2$, demand is $(30, 15)$, and shares are $(1/3, 2/3)$. Raising only p_2 to 9 gives $(22.5, 7.5)$. Both quantities fall: the greater purchasing-power burden outweighs the relative-price incentive to consume more of good 1.

Example 1.13. Weights versus shares in three goods. Let $a = (0.5, 0.3, 0.2)$, $\sigma = 2$, $p = (1, 1, 1)$, and $w = 38$. The denominator is $0.25 + 0.09 + 0.04 = 0.38$, so $x = (25, 9, 4)$. Shares are $(25/38, 9/38, 4/38)$, not the taste weights. The same weights under Cobb-Douglas give $(19, 11.4, 7.6)$. A change in curvature with weights fixed generally changes the reference allocation as well as price responses.

Calibration makes the distinction operational. Given positive reference shares $b^0_\ell = p^0_\ell x^0_\ell / w^0$ and a specified σ, weights reproducing that observation are

$$a_\ell = \frac{(b^0_\ell)^{1/\sigma}(p^0_\ell)^{(\sigma-1)/\sigma}}{\sum_k (b^0_k)^{1/\sigma}(p^0_k)^{(\sigma-1)/\sigma}}.$$

One positive observation therefore cannot identify the substitution elasticity: each specified σ admits calibrated weights. Different elasticities can fit the same initial allocation and predict different responses to a price change.

1.6.3 Limiting Cases: Substitutes and Complements

Holding weights fixed, $\rho \to 0$ gives Cobb-Douglas, $\rho \uparrow 1$ gives $u(x) = \sum_\ell a_\ell x_\ell$, and $\rho \to -\infty$ gives $u(x) = \min_\ell x_\ell$. Fixed weights disappear in the last limit. Unequal required proportions instead arise by applying the CES aggregator to x_ℓ / d_ℓ, with $d_\ell > 0$, before taking that limit. The endpoint models must be solved directly; their solutions need not satisfy the interior CES conditions.

Example 1.14. Perfect substitutes and a demand correspondence. For $u(x) = 3x_1 + x_2$, $p = (2, 1)$, and $w = 12$, utility per dollar is $3/2$ for good 1 and 1 for good 2, so the unique choice is $(6, 0)$. At $p = (3, 1)$ the two ratios tie: every nonnegative bundle satisfying $3x_1 + x_2 = 12$ maximizes utility. Reporting a single demand formula would omit valid choices.

Example 1.15. Perfect complements and a kink. For $u(x) = \min\{x_1/2, x_2\}$, the consumer buys $x_1 = 2x_2$. At $p = (1, 3)$ and $w = 20$, the chosen bundle is $(8, 4)$ because $2(4) + 3(4) = 20$. In general $x_2 = w/(2p_1 + p_2)$ and $x_1 = 2w/(2p_1 + p_2)$. Tangency of differentiable indifference curves is unavailable at the optimum.

1.6.4 Subsistence and Stone-Geary Demand

To allow expenditure shares to vary with wealth, introduce a nonnegative subsistence bundle $\gamma \in \mathbb{R}^L_+$. On $X = \gamma + \mathbb{R}^L_+$, let $u(x) = \prod_\ell (x_\ell - \gamma_\ell)^{a_\ell}$ with positive normalized weights. Suppose $w > p \cdot \gamma$. Setting $z = x - \gamma$ leaves discretionary wealth $w - p \cdot \gamma$, so Cobb-Douglas demand in z gives

$$x_\ell(p, w) = \gamma_\ell + a_\ell \frac{w - p \cdot \gamma}{p_\ell}.$$

At $w = p \cdot \gamma$, the only feasible bundle is γ; below that level this subsistence-constrained model has no feasible bundle. It is not a model of behavior below subsistence. Except in special cases, these preferences are not homothetic about the origin and their budget shares depend on wealth.

Example 1.16. A necessity and discretionary spending. Let $\gamma = (2, 1)$, $a = (1/2, 1/2)$, $p = (2, 1)$, and $w = 15$. Subsistence costs 5, leaving 10 to allocate; demand is $(4.5, 6)$ and shares are $(0.6, 0.4)$. At wealth 25, demand is $(7, 11)$ and shares are $(0.56, 0.44)$. The first good's share falls even though its quantity increases.

1.6.5 Quasilinearity and Wealth-Dependent Corners

Definition 1.17. Quasilinear preferences. A numeraire good L is quasilinear when preferences admit a representation

$$u(x) = g(x_1, \ldots, x_{L-1}) + x_L.$$

Such preferences are invariant to a common translation of both bundles in the numeraire whenever the translated bundles remain feasible. The converse is not implied by translation invariance alone. On a product domain $Z \times \mathbb{R}$, one sufficient formulation adds strict monotonicity in the

numeraire and numeraire solvability: for a fixed $z_0 \in Z$ and every $z \in Z$, there is a unique $g(z)$ such that $(z, 0) \sim (z_0, g(z))$. Translation invariance then gives $(z, m) \sim (z_0, g(z) + m)$, which yields the displayed representation.

A nonnegative numeraire creates a boundary that matters for demand. Let $u(x_1, x_2) = 2\sqrt{x_1} + x_2$ on $\mathbb{R}^2_+$, normalize $p_2 = 1$, and write $r = p_1 > 0$. Substituting the budget gives the strictly concave problem $\max_{0 \le x_1 \le w/r}\{2\sqrt{x_1} - rx_1 + w\}$. Its unique solution is

$$x_1(r, w) = \min\{r^{-2}, w/r\}, \qquad x_2(r, w) = \max\{w - r^{-1}, 0\}.$$

Example 1.18. When zero wealth effects stop applying. At $r = 2$ and $w = 3$, demand is $(1/4, 5/2)$: extra wealth is spent entirely on good 2. At $w = 1/4$, demand is $(1/8, 0)$ and all wealth is spent on good 1. The claim that quasilinear demand has no wealth effect for the non-numeraire good requires an interior numeraire choice, here $w > 1/r$.

1.6.6 From CES to Flexible Translog Demand

CES imposes one common, constant substitution elasticity. Applied demand analysis often needs substitution patterns that vary with relative prices. A translog specification uses a quadratic polynomial in logarithms, allowing such variation. Direct-utility, indirect-utility, and expenditure translog specifications are different parametric restrictions: the dual of a direct translog utility need not be translog. Chapter 2 develops a homothetic translog expenditure model, derives its demand shares, and checks the price region on which it represents a classical rational consumer. This order lets us explain the model through expenditure minimization rather than present an unexplained demand formula.

For further practice, Problems 1.2-1.6 compare these specifications. The artists case study, Problem 10.4, uses CES fan demand as the first stage of a model of streaming revenue, concert production, and ticket pricing; its consumer part can be completed now.

1.7 Problems

Part I: By hand

Problem 1.1 [Core]. Budget scaling and Cobb-Douglas demand.

Let $X = \mathbb{R}^2_+$ and $B(p, w) = \{x \in X : p \cdot x \le w\}$, where $p \gg 0$ and $w > 0$.

(a) Prove that $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$. Explain why every demand correspondence generated by preference maximization is homogeneous of degree zero in (p, w).

(b) Suppose preferences are locally nonsatiated. Prove that every chosen bundle satisfies Walras' law, $p \cdot x = w$.

(c) For $u(x) = x_1^{\alpha} x_2^{1-\alpha}$ with $0 < \alpha < 1$, derive Marshallian demand and verify both properties directly.

Problem 1.2 [Core]. CES curvature, demand, and expenditure shares.

For $\rho < 1$ and $\rho \neq 0$, let $u_\rho(x) = (\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho)^{1/\rho}$, with the Cobb-Douglas limit used at $\rho = 0$ and the boundary extension stated in the text. Write $\sigma = 1/(1-\rho)$.

(a) Derive demand from the first-order conditions. Explain why the optimum is interior for $\rho < 1$ and why maximizing the sum of powers is incorrect when $\rho < 0$.

(b) At $p = (1, 4)$ and $w = 120$, calculate quantities and expenditure shares for $\sigma \in \{1/2, 1, 2\}$. Check budget exhaustion.

(c) Raise only the second price to 9. Recalculate the three demands and explain why a smaller quantity can have a larger expenditure share.

(d) Derive the own-price, cross-price, and wealth elasticities. Distinguish the elasticity of a quantity ratio from an individual demand elasticity.

Problem 1.4 [Core]. Substitutes, complements, and boundary choices.

Consider separately $u^A(x) = 2x_1 + x_2$ and $u^B(x) = \min\{x_1/3, x_2\}$ on $\mathbb{R}^2_+$, at strictly positive prices and wealth.

(a) Derive the full demand correspondence for u^A, including the price ratio at which it is set-valued.

(b) Derive demand for u^B. At $p = (2, 3)$ and $w = 36$, compute both models' choices.

(c) Explain why interior differentiable tangency conditions do not characterize these optima.

(d) Obtain the ratio $x_1 = 3x_2$ as a CES limit. Explain why holding the usual taste weights fixed does not preserve unequal Leontief proportions.

Part II: Household demand across subsistence and discretionary regimes — chapter use case and verified scale-up

Problem 1.3 [Applied]. CES calibration and observational equivalence.

At prices $p^0 = (1, 2, 4)$, a household with wealth $w^0 = 120$ chooses $x^0 = (30, 25, 10)$. Consider Chapter 1's CES representation with positive normalized weights.

(a) For a specified $\sigma > 0$, derive the weights reproducing this observation.

(b) Calibrate the weights for $\sigma = 1/2$ and $\sigma = 2$. For each calibration predict demand at $p^1 = (2, 2, 4)$ and wealth 120.

(c) Explain why fitting one observation does not identify substitution. Describe additional observations that could distinguish the models.

Problem 1.6 [Applied]. KKT regimes and the quantity--quality tradeoff.

Unified Growth Theory links the demographic transition to a shift from child quantity toward education and child quality. This exercise isolates a stylized household block from Galor (2005); it is not the complete dynamic growth model. A parent with potential income $y > \bar{c}$ chooses consumption $c > 0$, fertility $n > 0$, and education time per child $e \geq 0$ to maximize $(1-\gamma)\log c + \gamma \log[nh_a(e)]$, where $h_a(e) = 1 + ae/(\tau + e)$, $0 < \gamma < 1$, $\tau > 0$ is baseline time per child, $a > 0$ indexes the return to education, and $\bar{c} > 0$ is subsistence consumption. The resource and subsistence constraints are $c + yn(\tau + e) \leq y$ and $c \geq \bar{c}$. The function h_a is a closed-form teaching specialization of Galor's general child-human-capital technology; the normalization makes one unit of e one unit of parental time.

Part I. By hand. (a) Write the complete KKT system in (c, n, e), including feasibility, multiplier signs, and complementary slackness. Prove that the resource constraint binds.

(b) Define total child-rearing time $t = n(\tau + e)$. Eliminate c and n, state the feasible set for (t, e), and show that the transformed objective is additively separable in t and e. Write its KKT system. Explain why an inequality can be active with a zero multiplier at a threshold.

(c) Solve globally for (t^*, e^*), then recover (c^*, n^*). Divide the answer into all parameter regions, including the equality cases. A stationary-point calculation alone is not sufficient. Hint: determine the sign of $h_a'(e)(\tau + e) - h_a(e)$.

(d) Let $\tau = 1/4$, $\gamma = 1/2$, and $\bar{c} = 4$. Calculate $(c^*, n^*, e^*, h_a(e^*))$ for every combination $a \in \{1/2, 3\}$ and $y \in \{6, 12\}$. Identify the active inequalities, calculate their transformed-problem multipliers, and verify feasibility and complementary slackness.

(e) Explain the income mechanism while subsistence binds, the quantity--quality response when a rises, and which dynamic elements of Unified Growth Theory this static exercise deliberately leaves out.

Part II. Verified scale-up with Codex and Python. (f) Record four lines before coding: Seed: the four hand calculations; Fixed: preferences, h_a, continuous fertility, and $(\tau, \gamma, \bar{c})$; Scaled: only the number of (a, y) configurations; Risk: threshold misclassification, invalid parameter domains, or mistaking finite computation for proof.

(g) Create a file named work/ugt_kkt.py. On a deterministic rectangular grid containing $a = 1$ and $y = \bar{c}/(1-\gamma)$, compute the closed-form candidate and the transformed-problem KKT multipliers. Reject $a \leq 0$, $\tau \leq 0$, $\bar{c} \leq 0$, $\gamma \notin (0,1)$, and $y \leq \bar{c}$.

(h) At every grid point, check the resource identity, subsistence feasibility, nonnegativity, stationarity, complementary slackness, and agreement with an independently implemented bounded numerical maximization. State the numerical tolerances. Plot e^* and n^* and mark the two analytic threshold lines.

(i) Include tests immediately below, at, and above both thresholds. Report separately what the hand argument proves for every admissible parameter vector, what the program checks only on its finite grid, and what remains an empirical or dynamic modeling claim.

Chapter use case

Seed: Problems 1.1, 1.4. Fixed: all unlisted primitives and assumptions. Scaled: wealth from two hand calculations to a grid, with preferences and prices fixed. Tool: Codex + Python. Verification: Reproduce both hand wealth levels; classify infeasible, boundary, and interior regimes separately for each model; check feasibility, budget exhaustion, and continuity at every valid threshold.

Problem 1.5 [Applied]. Subsistence and quasilinear corners.

The first of two consumers has $u(x) = (x_1 - 1)^{1/3}(x_2 - 2)^{2/3}$ on $x \geq (1,2)$. The second has $u(x) = 4\sqrt{x_1} + x_2$ on $\mathbb{R}^2_+$.

(a) Derive the first consumer's demand above subsistence expenditure. State what happens at and below that threshold.

(b) At $p = (2,1)$, compute the first consumer's demands and shares at wealth 10 and 22. Distinguish homotheticity in discretionary consumption from homotheticity in total consumption.

(c) Normalize the second consumer's numeraire price to one and write $r = p_1$. Derive demand for all $r, w > 0$, including the zero-numeraire region.

(d) Identify where a zero wealth effect for the non-numeraire good is valid. Compare this restriction with the Stone--Geary model.

Chapter 2

Duality, Comparative Statics, and Welfare

Question: How do we compute the model's price responses and welfare implications?

Chapter 1 tells us which bundle the classical rational consumer chooses. This chapter studies the same problem from its value functions. Indirect utility and expenditure minimization yield demand identities, comparative statics, and money-metric welfare measures. Set-valued duality is stated first; differentiability and uniqueness are added only for envelope formulas such as Roy's identity and Shephard's lemma.

2.1 Indirect Utility

The indirect utility function records the maximum utility attainable from a budget:

$$v(p, w) = \max\{u(x) : x \in X,\ p \cdot x \leq w\}.$$

Thus $v : \mathsf{P} \times \mathsf{W} \to \mathbb{R}$ ranks price--wealth situations through the utility they permit.

Every Marshallian maximizer yields this same value: $v(p, w) = u(x^*)$ for every $x^* \in x(p, w)$.

Proposition 2.1. *Suppose that $u(\cdot)$ is a continuous utility function representing a locally nonsatiated preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}^L_+$. The indirect utility function $v(p, w)$ is:*

1. *Homogeneity of degree zero.*
2. *$v(p, w)$ is strictly increasing in w and nonincreasing in p_ℓ.*

3. $v(p, w)$ is quasiconvex in (p, w). The set $\{(p, w)|v(p, w) \leq \overline{v}\}$ is convex.

4. $v(p, w)$ is continuous in p and w.

Proof. Homogeneity follows from $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$. If $p' \geq p$ componentwise, then $B(p', w) \subseteq B(p, w)$, so the value is nonincreasing in every price. Continuity follows from Berge's maximum theorem and the continuity and compact-valuedness of the budget correspondence. To prove strict monotonicity in wealth, let $w' > w$ and choose $x^* \in x(p, w)$. The bundle x^* lies strictly inside $B(p, w')$. Local nonsatiation supplies a bundle arbitrarily close to x^* that is strictly preferred; choosing it close enough keeps it in $B(p, w')$. Hence $v(p, w') > v(p, w)$.

For quasiconvexity, fix $\overline{v}$ and let $L_{\overline{v}} = \{(p, w) : v(p, w) \leq \overline{v}\}$. If $u(x) > \overline{v}$, then every $(p, w) \in L_{\overline{v}}$ must satisfy $p \cdot x > w$, because otherwise x would be affordable and the value would exceed $\overline{v}$. Conversely, this strict inequality for every x with $u(x) > \overline{v}$ rules out an affordable bundle with utility above $\overline{v}$. Hence

$$L_{\overline{v}} = \bigcap_{u(x) > \overline{v}} \{(p, w) : p \cdot x > w\}.$$

Each set in the intersection is a linear half-space in (p, w), so the lower contour set is convex and v is quasiconvex.

2.2 Roy's Identity

Assume the maximizer is unique, denote it by $x(p, w)$, and assume the value function is differentiable. Then $v(p, w) = u(x(p, w))$. The envelope theorem applied to

$$v(p, w) = \max_{x \geq 0,\ p \cdot x \leq w} u(x)$$

gives

$$\partial_w v(p, w) = \lambda(p, w), \qquad \nabla_p v(p, w) = -\lambda(p, w) x(p, w).$$

If demand is unique, the envelope derivatives exist, and $\partial_w v(p, w) = \lambda(p, w) > 0$, Roy's identity is

$$x(p, w) = -\frac{\nabla_p v(p, w)}{\partial_w v(p, w)}.$$

Example 2.2. Quasilinear demand and Roy's identity. Let good L be the numeraire, set

$p_L = 1$, and let utility be $u(q, q_L) = g(q) + q_L$ for $q \in \mathbb{R}^{L-1}_+$. Write $p_{-L} = (p_1, \ldots, p_{L-1})$ for the nonnumeraire price vector. Feasibility of the nonnegative numeraire requires $p_{-L} \cdot q \leq w$. The indirect utility is

$$v(p, w) = w + \max_{q \geq 0,\ p_{-L} \cdot q \leq w} \{g(q) - p_{-L} \cdot q\}.$$

When the unconstrained maximizer of $g(q) - p_{-L} \cdot q$ is affordable and leaves a nonnegative numeraire, the wealth constraint inside the displayed maximum is slack and non-numeraire demands have no income effects. When the maximizer is unique and the envelope theorem applies, Roy's identity gives $x_k(p, w) = -\partial_{p_k} v(p, w)$ for $k < L$, and the numeraire demand is the residual $x_L = w - p_{-L} \cdot x_{-L}(p)$, where $x_{-L}(p)$ collects the demands for the nonnumeraire goods and p_{-L} is their price vector.

2.3 Expenditure Minimization

At prices p, the expenditure minimization problem (EMP) asks for the least expenditure required to attain utility $\overline{u}$:

$$e(p, \overline{u}) = \min\{p \cdot x : x \in X,\ u(x) \geq \overline{u}\}.$$

Proposition 2.3. *Suppose u is a continuous utility function representing a locally nonsatiated preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}^L_+$ and the price vector is $p \gg 0$. Then:*

(i) If x^* is optimal in the UMP when $w > 0$ then x^* is optimal in the EMP when the required level of utility is $u(x^*)$. Moreover, the minimized expenditure level $p \cdot x^* = w$ is exactly the wealth w.

(ii) If x^* is optimal in the EMP when the required utility level is $\overline{u} > u(0)$ then x^* is optimal in the UMP when wealth is $p \cdot x^*$. Moreover, the maximized utility level $u(x^*) = \overline{u}$.

The proof uses local nonsatiation to show that both constraints bind and then applies the definitions of the two optimization problems.

Expenditure function. The function $e(p, \overline{u})$ is the value of the EMP. Whenever a solution exists, $e(p, \overline{u}) = p \cdot x^*$ for every expenditure-minimizing bundle x^*.

Proposition 2.4. *Suppose that u is a continuous utility function representing a locally nonsatiated preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}^L_+$, with $p \in \mathbb{R}^L_{++}$ and the target utility in the attainable range. The expenditure $e(p, \overline{u})$ is*

(i) Homogeneous of degree one in p.

(ii) Nondecreasing in $\overline{u}$ and nondecreasing in p_ℓ for all ℓ. Strict increase in the utility target requires restricting attention to targets above the baseline utility and conditions ensuring that the minimum reaches the target exactly.

(iii) Concave in p

(iv) Continuous in $(p, \overline{u})$.

Proof. (i) Multiplying every price by $\alpha > 0$ does not change the feasible set and multiplies the objective by α; hence $e(\alpha p, \overline{u}) = \alpha e(p, \overline{u})$. For (ii), increasing the utility target shrinks the feasible set, while increasing a price cannot reduce the expenditure on any fixed nonnegative bundle.

(iii) Let $p^\alpha = \alpha p + (1-\alpha)p'$ with $\alpha \in [0,1]$ and choose any $x^\alpha \in h(p^\alpha, \overline{u})$. Feasibility of x^α at each price vector gives

$$\begin{aligned} e(p^\alpha, \overline{u}) &= \alpha p \cdot x^\alpha + (1-\alpha) p' \cdot x^\alpha \\ &\geq \alpha e(p, \overline{u}) + (1-\alpha) e(p', \overline{u}). \end{aligned}$$

Hence the expenditure function is concave in prices.

(iv) Continuity in $(p, \overline{u})$.

Continuity follows from the standard expenditure-function theorem. The unbounded feasible set cannot be inserted directly into Berge's theorem. Instead, around any $(p, \overline{u})$ with strictly positive prices, one first obtains a common expenditure upper bound from one feasible bundle. Strict positivity of nearby prices then confines every optimizer to a common compact box. Berge's theorem applies to this locally compactified problem. Continuity of preferences and local nonsatiation ensure feasibility stability and that an optimum reaches the target utility.

□

2.4 Hicksian Compensated Demand

The Hicksian (compensated) demand correspondence is the solution set of the EMP: $h(p, \overline{u}) = \arg\min\{p \cdot x : x \in X,\ u(x) \geq \overline{u}\}$.

Proposition 2.5. *Suppose $u(\cdot)$ is continuous and represents a locally nonsatiated preference relation $\succeq$, prices satisfy $p \gg 0$, and the attainable target satisfies $\overline{u} > u(0)$. Then $h(p, \overline{u})$ is nonempty and compact-valued and has the following properties:*

(i) It is homogeneous of degree zero in p

(ii) It has no excess utility: For $x^* \in h(p, \overline{u})$, $u(x^*) = \overline{u}$

(iii) Convexity and uniqueness: If $\succeq$ is convex, then $h(p, \overline{u})$ is convex; if $\succeq$ is strictly convex, then Hicksian demand is single-valued.

2.5 KKT Conditions for Expenditure Minimization

Assume u is continuously differentiable and a constraint qualification holds at the candidate optimum. The KKT conditions below are necessary. If u is concave, they are also sufficient.

The Lagrangian for this problem is:

$$\mathcal{L}_{\text{EMP}} = p \cdot x - \lambda(u(x) - \overline{u}).$$

Stationarity and complementary slackness for the nonnegativity constraints give

$$p \geq \lambda \nabla_x u(x^*)$$

$$x^*_\ell [p_\ell - \lambda \nabla u(x^*)_\ell] = 0,$$

for some $\lambda \geq 0$, together with primal feasibility $x^* \geq 0$ and $u(x^*) \geq \overline{u}$, and complementary slackness $\lambda[u(x^*) - \overline{u}] = 0$.

2.6 Shephard's Lemma

Proposition 2.6. *Every Hicksian bundle is a supergradient of the concave expenditure function in prices. If $e(\cdot, \overline{u})$ is differentiable at p, Hicksian demand is unique there and Shephard's identity holds:*

$$\nabla_p e(p, \overline{u}) = h(p, \overline{u}).$$

Proof. Fix $x^h \in h(p, \overline{u})$. For every other price vector p', the same bundle remains feasible, so

$$e(p', \overline{u}) \leq p' \cdot x^h = e(p, \overline{u}) + (p' - p) \cdot x^h.$$

Hence x^h is a supergradient of the concave function $e(\cdot, \overline{u})$. At a differentiability point the superdifferential is the singleton $\{\nabla_p e(p, \overline{u})\}$, so every Hicksian bundle equals that gradient and demand is unique.

□

2.7 Curvature of Hicksian Demand

Assume throughout this subsection that $e(\cdot, \overline{u})$ is twice continuously differentiable, so Hicksian demand is single-valued and differentiable. The Jacobian $D_p h(p, \overline{u})$ has entries $[D_p h(p, \overline{u})]_{k\ell} = \partial_{p_\ell} h_k(p, \overline{u})$.

Proposition 2.7. *The Hicksian price-effect matrix satisfies*

$$D_p h(p, \overline{u}) = D_p^2 e(p, \overline{u}).$$

It is symmetric and negative semidefinite, and $D_p h(p, \overline{u})p = 0$. The first claims follow from Shephard's identity, twice continuous differentiability, and concavity of the expenditure function. The last follows by differentiating the degree-zero homogeneity of Hicksian demand.

Definition 2.8. Negative semidefiniteness. A symmetric $L \times L$ matrix A is negative semidefinite if $z^\top A z \leq 0$ for every vector $z \in \mathbb{R}^L$.

Example 2.9. The following matrix is NSD:

$$A = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$z^\top A z = -2z_1^2 \leq 0$$

for any $z \in \mathbb{R}^2$.

2.8 Consumer Duality

The expenditure function is the value of the EMP: for every $x^h \in h(p, \overline{u})$,$e(p, \overline{u}) = p \cdot x^h$.

Proposition 2.10. *Consumer duality. Suppose that u is continuous and represents a locally nonsatiated preference relation on $X = \mathbb{R}^L_+$. For $p \gg 0$, $w > 0$, and every attainable target $\overline{u} \in u(X)$ with $\overline{u} > u(0)$, the demand and value functions satisfy*

$$h(p, v(p, w)) = x(p, w),$$
$$x(p, e(p, \overline{u})) = h(p, \overline{u}),$$
$$v(p, e(p, \overline{u})) = \overline{u},$$
$$e(p, v(p, w)) = w.$$

The first two equalities are equalities of correspondences; when demand is single-valued, the same notation denotes functions. The restriction $\overline{u} > u(0)$ ensures $e(p, \overline{u}) > 0$, so the Marshallian demand appearing in the second and third identities lies in the declared wealth domain. At the boundary target $u(0)$, the identities require an explicit extension of Marshallian demand to zero wealth.

Proof. Choose $x^m \in x(p, w)$. Local nonsatiation gives $p \cdot x^m = w$. If a bundle delivered utility at least $v(p, w)$ at expenditure strictly below w, local nonsatiation would provide a strictly preferred nearby bundle still inside $B(p, w)$, contradicting the definition of the value function. Thus the expenditure needed to reach $v(p, w)$ is exactly w, and the minimizers are exactly the Marshallian maximizers. For an attainable $\overline{u} > u(0)$, continuity and positive prices ensure that the EMP has a minimizer x^h and that its expenditure is positive. Its utility target binds: if $u(x^h) > \overline{u}$, continuity along the segment from 0 to x^h would produce the target utility at strictly lower expenditure. No bundle affordable at $e(p, \overline{u})$ can yield utility strictly above $\overline{u}$, because the same segment argument would again contradict expenditure minimality. Hence $v(p, e(p, \overline{u})) = \overline{u}$, and budget exhaustion identifies the two demand correspondences. □

2.9 Slutsky Equation

Assume in this section that Marshallian and Hicksian demands are single-valued and continuously differentiable and that the expenditure function is differentiable in prices. The Slutsky matrix is $S : \mathsf{P} \times \mathsf{W} \to \mathbb{R}^{L \times L}$, $S(p, w)_{\ell,k} = \partial_{p_k} x_\ell(p, w) + \partial_w x_\ell(p, w) x_k(p, w)$, $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$.

Differentiate the duality identity $h(p, \overline{u}) = x(p, e(p, \overline{u}))$ with respect to p:

$$D_p h(p, \overline{u}) = D_p x(p, e(p, \overline{u})) + D_w x(p, e(p, \overline{u})) \nabla_p e(p, \overline{u})^\top$$

By Shephard's lemma and the duality identities,

$$D_p h(p, v(p, w)) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top = S(p, w).$$

2.10 Workhorse Demand through Duality

2.10.1 CES Expenditure, Compensation, and Substitution

Keep Chapter 1's degree-one CES normalization. Its unit expenditure index $\mathcal{P}(p) := e(p,1)$ records the minimum cost of one unit of that utility index. Substituting the CES demand into utility, and then using duality, gives

$$\mathcal{P}(p) = \begin{cases} \left(\sum_\ell a_\ell^\sigma p_\ell^{1-\sigma}\right)^{1/(1-\sigma)}, & \sigma \neq 1, \\ \prod_\ell (p_\ell/a_\ell)^{a_\ell}, & \sigma = 1, \end{cases}$$

$$v(p,w) = \frac{w}{\mathcal{P}(p)}, \qquad e(p,\bar{u}) = \bar{u}\mathcal{P}(p), \qquad h_\ell(p,\bar{u}) = \bar{u} a_\ell^\sigma p_\ell^{-\sigma} \mathcal{P}(p)^\sigma.$$

Here $\bar{u} > 0$. The Cobb-Douglas factors a_ℓ in the unit expenditure formula follow from the chosen utility normalization; omitting them changes the utility units. Marshallian demand itself is unaffected by a positive rescaling of utility. Differentiating Hicksian demand gives own-price elasticity $-\sigma(1-b_\ell)$ and, for $k \neq \ell$, cross-price elasticity $\sigma b_k > 0$. CES goods are compensated substitutes even when $\sigma < 1$ makes them gross complements at fixed wealth.

Example 2.11. Exact compensation after a price increase. With equal weights, $\sigma = 2$, $p^0 = (1,2)$, and $w = 120$, Chapter 1 found $x^0 = (80,20)$. Here $\mathcal{P}(p^0) = 8/3$ and initial utility is 45. If $p^1 = (2,2)$, then $\mathcal{P}(p^1) = 4$. Maintaining utility 45 requires wealth 180, so compensation is 60 and the compensated bundle is $(45,45)$. At unchanged wealth 120, the final bundle is $(30,30)$. For good 1, the total change -50 is the compensated change -35 plus the remaining wealth effect -15. This is a finite-change Hicks decomposition; the Slutsky equation describes the corresponding local derivative.

2.10.2 A Homothetic Translog Expenditure Model

The translog family provides a flexible quadratic approximation in logarithms. We use a homothetic expenditure specification; richer translog families also permit wealth-dependent shares. Prices are measured relative to fixed reference units, so each logarithm is dimensionless. Write

$$\log \mathcal{P}_{\mathrm{TL}}(p) = \alpha_0 + \sum_{\ell=1}^{L} \alpha_\ell \log p_\ell + \frac{1}{2}\sum_{\ell=1}^{L}\sum_{k=1}^{L} \gamma_{\ell k} \log p_\ell \log p_k.$$

The intercept is α_0, the linear coefficients are α_ℓ, and the symmetric curvature matrix is

$\Gamma = (\gamma_{\ell k})$. Impose $\sum_\ell \alpha_\ell = 1$ and $\Gamma \mathbf{1}_L = 0$. These restrictions make $\mathcal{P}_{\mathrm{TL}}$ homogeneous of degree one in prices. The matrix Γ here is unrelated to the subsistence vector γ in Stone-Geary utility. The translog approach originates with Christensen, Jorgenson, and Lau; the expenditure formulation and curvature calculation below also follow Diewert's treatment.

[Christensen et al., 1975, Diewert, 2023]

Suppose the price region under study is regular, as specified below. With $e(p, \bar{u}) = \bar{u}\mathcal{P}_{\mathrm{TL}}(p)$ and $v(p, w) = w/\mathcal{P}_{\mathrm{TL}}(p)$, Shephard's lemma gives

$$b_\ell(p) = \frac{\partial \log \mathcal{P}_{\mathrm{TL}}(p)}{\partial \log p_\ell} = \alpha_\ell + \sum_k \gamma_{\ell k} \log p_k,$$
$$h_\ell(p, \bar{u}) = \frac{\bar{u}\mathcal{P}_{\mathrm{TL}}(p)}{p_\ell} b_\ell(p), \qquad x_\ell(p, w) = \frac{w}{p_\ell} b_\ell(p).$$

Adding up and homogeneity follow from the coefficient restrictions, but nonnegative demand and substitution restrictions require more. On an open convex cone $\mathcal{D} \subseteq \mathbb{R}^L_{++}$, require $b_\ell(p) > 0$ and the following matrix to be negative semidefinite at every price in $\mathcal{D}$:

$$H(p) := \Gamma + \boldsymbol{b}(p)\boldsymbol{b}(p)^\top - \mathrm{diag}(\boldsymbol{b}(p)).$$

To see the role of this condition, let $D = \mathrm{diag}(p_1, \ldots, p_L)$. Differentiating $\partial_{p_\ell} \mathcal{P}_{\mathrm{TL}} = \mathcal{P}_{\mathrm{TL}} b_\ell / p_\ell$ yields

$$\nabla_p^2 \mathcal{P}_{\mathrm{TL}}(p) = \mathcal{P}_{\mathrm{TL}}(p) D^{-1} H(p) D^{-1}.$$

Thus $H(p) \preceq 0$ is exactly the local curvature restriction. Requiring it throughout a convex price domain establishes concavity on that domain. Positive fitted shares alone do not establish rationalizability, and checking curvature at one observation does not justify a distant counterfactual. The Slutsky matrix is $S(p, w) = wD^{-1}H(p)D^{-1}$, so symmetry, negativity, and the zero price direction are explicit.

There is a direct optimizing model behind these formulas. Define, for $x \in \mathbb{R}^L_+$,

$$u_{\mathcal{D}}(x) = \inf_{r \in \mathcal{D}} \frac{r \cdot x}{\mathcal{P}_{\mathrm{TL}}(r)}.$$

This is a nonnegative, concave, homogeneous, nondecreasing utility. At $p \in \mathcal{D}$, concavity and homogeneity of the price index imply $r \cdot \nabla \mathcal{P}_{\mathrm{TL}}(p) \geq \mathcal{P}_{\mathrm{TL}}(r)$ for every $r \in \mathcal{D}$, with equality at $r = p$. Hence $u_{\mathcal{D}}(\nabla \mathcal{P}_{\mathrm{TL}}(p)) = 1$. Scaling that bundle attains target utility $\bar{u} > 0$ at expenditure $\bar{u}\mathcal{P}_{\mathrm{TL}}(p)$; any bundle reaching the target costs at least that much. This proves the expenditure

formula, and therefore the displayed demand, at prices in $\mathcal{D}$. It does not extend the translog formula to prices outside its regular region.

The compensated and uncompensated elasticities make the extra flexibility visible. Let $\delta_{\ell k}$ equal one when $\ell = k$ and zero otherwise. Then

$$\frac{\partial \log x_\ell}{\partial \log p_k} = \frac{\gamma_{\ell k}}{b_\ell} - \delta_{\ell k}, \qquad \frac{\partial \log h_\ell}{\partial \log p_k} = \frac{\gamma_{\ell k}}{b_\ell} + b_k - \delta_{\ell k}.$$

Wealth elasticities are one in this homothetic version, while price elasticities vary with the shares. For two goods, the elasticity of substitution is $1 + \gamma_{12}/(b_1 b_2)$ on the regular region. With more goods the same expression is the Allen-Uzawa pairwise elasticity, rather than a universal elasticity of every quantity ratio. Setting $\Gamma = 0$ recovers constant shares and a Cobb-Douglas price index. General translog utility models need not be homothetic.

Example 2.12. A translog model with changing shares. Let $L = 2$, $\alpha_0 = 0$, $\alpha_1 = \alpha_2 = 1/2$, and $\Gamma = \frac{1}{10}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$. Writing $r = p_1/p_2$ gives

$$\mathcal{P}_{\mathrm{TL}}(p) = \sqrt{p_1 p_2}\exp\left(\frac{(\log r)^2}{20}\right), \qquad b_1 = \frac{1}{2} + \frac{\log r}{10}, \quad b_2 = 1 - b_1.$$

At $p = (1, 1)$ and $w = 100$, demand is $(50, 50)$. At $p = (2, 1)$ and the same wealth, shares are approximately $(0.569315, 0.430685)$ and demand is $(28.4657, 43.0685)$. The first quantity falls while its budget share rises. The substitution elasticity changes from 0.6 at equal prices to approximately 0.592162 at $(2, 1)$; a single CES elasticity cannot reproduce that variation throughout the region.

Example 2.13. Positive shares can conceal invalid curvature. In the preceding model, $H = (1/10 - b_1 b_2)\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$. Shares are positive when $|\log r| < 5$, but curvature holds only when $|\log r| \le \sqrt{15}$. Thus $|\log r| < \sqrt{15}$ defines an open convex cone on which the construction above is valid. At $r = \exp(4)$ the shares $(0.9, 0.1)$ still look admissible, yet $1/10 - b_1 b_2 = 0.01 > 0$: compensated own-price responses have the wrong sign. This counterfactual must be rejected for this parameterization.

2.11 What Utility Maximization Implies

Proposition 2.14. *Suppose demand $x(p, w)$ is generated by maximizing a complete, transitive, continuous, and locally nonsatiated preference relation represented by u on each linear budget;*

that is, $x(p,w) = \arg\max_{y \in B(p,w)} u(y)$ *for every* $(p,w) \in \mathsf{P} \times \mathsf{W}$. *Then demand satisfies Walras' law and is homogeneous of degree zero.*

If, in addition, demand is single-valued and continuously differentiable and the expenditure function is twice continuously differentiable, then the Slutsky matrix $S(p,w)$ *is symmetric and negative semidefinite, and it is singular in the price direction:* $S(p,w)p = 0$ *for every* $(p,w) \in \mathsf{P} \times \mathsf{W}$.

This proposition summarizes our previous results.

2.12 Welfare Evaluation

We will compare the wealth needed to achieve a certain level of utility at a given price-wealth pair.

$$u^1 = v(p^1, w)$$

$$u^0 = v(p^0, w).$$

We are interested in comparing the expenditure needed to achieve u^1 or u^0.

For a fixed reference price and a given utility level u, the needed expenditure to reach it is given by

$$e(p^{ref}, u).$$

Thus the comparison in terms of expenditure is given by

$$e(p^{ref}, u^1) - e(p^{ref}, u^0)$$

$$e(p^{ref}, v(p^1, w)) - e(p^{ref}, v(p^0, w)).$$

This expression will be positive when it takes more wealth to achieve u^1 at prices p^{ref} than to achieve u^0. This difference is measured in dollar terms. That is why $e(p^{ref}, v(p,w))$ is often called money metric indirect utility function.

We can construct many money metric indirect utility functions with different reference price.

When $p^{ref} = p^0$ is the original price situation, the expenditure change equals the wealth change that makes the consumer indifferent between the new price with the old wealth and the old price with the new wealth. This asks what wealth change is equivalent to the price change.

Equivalent variation is

$$\mathrm{EV}(p^0, p^1, w) = e(p^0, v(p^1, w)) - e(p^0, v(p^0, w)) = e(p^0, v(p^1, w)) - w,$$

since $e(p^0, v(p^0, w)) = w$.

With this sign convention, equivalent variation is the money change at the reference price that is equivalent to the price change: it is willingness to pay for an improvement, and its negative is willingness to pay to avoid a worsening.

The other case considered is the one where the new price is taken as the reference price, when $p^{ref} = p^1$, the expenditure change equals the wealth change that makes the consumer indifferent between the original situation (p^0, w) and the new situation $(p^1, w + \Delta w)$.

Compensating variation is

$$\mathrm{CV}(p^0, p^1, w) = e(p^1, v(p^1, w)) - e(p^1, v(p^0, w)) = w - e(p^1, v(p^0, w)).$$

The sign convention for compensating variation is analogous: a negative value for a worsening is the compensation required to restore the initial utility.

2.13 Problems

Part I: By hand

Problem 2.1 [Core]. Cobb-Douglas duality.

For $u(x) = x_1^{\alpha} x_2^{1-\alpha}$ with $0 < \alpha < 1$, $p \gg 0$, $w > 0$, and $\overline{u} > 0$, derive the Marshallian demand $x(p, w)$, indirect utility $v(p, w)$, Hicksian demand $h(p, \overline{u})$, and expenditure function $e(p, \overline{u})$.

(a) Verify Walras' law and the identity $e(p, v(p, w)) = w$.

(b) Verify $h(p, v(p, w)) = x(p, w)$.

(c) Verify Roy's identity and Shephard's lemma.

Problem 2.5 [Core]. Translog shares and the regular price region.

Consider a homothetic translog index with $\alpha_0 = 0$, $\alpha = (2/5, 3/5)$, and $\Gamma = -\frac{1}{8}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$.

(a) Derive shares and demands as functions of relative prices and wealth. Find the open interval of p_1/p_2 for which both shares are positive.

(b) Compute $H = \Gamma + \boldsymbol{b}\boldsymbol{b}^\top - \operatorname{diag}(\boldsymbol{b})$. Prove negative semidefiniteness throughout that interval.

(c) At $p = (2, 1)$ and $w = 100$, calculate demand, Marshallian elasticities, and compensated elasticities. Verify adding up and homogeneity.

(d) Explain why setting a negative predicted share to zero while retaining the other formula does not generally give a valid extension.

Part II: Local CES and translog welfare approximation — chapter use case and verified scale-up

Problem 2.2 [Proof]. Differential restrictions on demand.

Let $x : \mathbb{R}^L_{++} \times \mathbb{R}_{++} \to \mathbb{R}^L_+$ be continuously differentiable and satisfy Walras' law.

(a) Differentiate Walras' law with respect to wealth and prices.

(b) Define the Slutsky matrix by $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$. Prove that $p^\top S(p, w) = 0$.

(c) Explain which additional restrictions on S follow from utility maximization and which of them do not follow from Walras' law alone.

Problem 2.3 [Proof]. The Slutsky matrix for Cobb-Douglas demand.

Using the Marshallian demand from Problem 2.1, compute $S(p, w) = D_p x + D_w x\, x^\top$.

(a) Show that S is symmetric.

(b) Show directly that $z^\top S z \leq 0$ for every $z \in \mathbb{R}^2$.

(c) Find the null space of S and interpret it using homogeneity.

Problem 2.4 [Proof]. CES duality and compensated substitution.

Use Chapter 1's CES utility, with positive normalized weights and $\sigma > 0$. For $\sigma \neq 1$, put $D(p) := \sum_k a_k^\sigma p_k^{1-\sigma}$.

(a) Derive $\mathcal{P}(p)$, expenditure, and Hicksian demand. Treat $\sigma = 1$ separately and retain the constants implied by the utility normalization.

(b) Show that the Slutsky matrix has off-diagonal entries $S_{\ell k} = \sigma x_\ell x_k / w$ and diagonal entries $S_{\ell\ell} = -\sigma x_\ell (1 - b_\ell)/p_\ell$. Prove symmetry and negative semidefiniteness using a weighted-variance argument. Identify its null space for strictly positive demand.

(c) Explain how compensated cross-price effects can be positive when uncompensated cross-price effects are negative.

(d) For equal weights, $\sigma = 2$, initial prices $(1, 3)$, wealth 90, and final prices $(2, 3)$, calculate compensation to retain initial utility. Decompose the first quantity's change into a compensated

change and the remaining wealth effect.

Chapter use case

Seed: Problems 2.4, 2.5. Fixed: all unlisted primitives and assumptions. Scaled: log relative-price distance from one reference price. Tool: Codex + Python. Verification: Match the CES value, gradient, and Hessian at the reference price; enforce positive shares and negative semidefiniteness on the translog regular region; report welfare comparisons only on that region.

Problem 2.6 [Applied]. A translog approximation to CES demand.

Fix a two-good CES model, a positive reference price vector p^0, and substitution elasticity $\sigma > 0$. Write b_1^0, b_2^0 for the reference shares and use normalized prices $r_\ell = p_\ell / p_\ell^0$.

(a) Compute the gradient and Hessian of $\log \mathcal{P}_{\text{CES}}$ with respect to log prices at p^0.

(b) Choose translog coefficients in $\log r$ so the log price index agrees with CES through second order at p^0.

(c) Verify the restrictions for homogeneity. Show that demand levels and local price responses agree at the reference observation.

(d) Explain why predictions can disagree for a large price change. State the regularity checks needed before making welfare comparisons.

Chapter 3

Revealed Preference and Recoverability

Question: Could observed behavior have been generated by the benchmark model, and what can the data identify?

The direction of analysis now reverses. Instead of deriving choices from u, we ask whether observed demand could have been generated by the benchmark model and what features of that model the data recover. Local restrictions concern derivatives of a demand function; finite-data restrictions concern revealed-preference comparisons across observed budgets. GARP characterizes classical utility maximization, while WGARP admits the broader coalitional multi-utility model introduced after Afriat's theorem.

3.1 Observed Demand and the Testing Question

After specifying the consumer's environment, we take demand as the observable primitive. A demand function has the typed signature

$$x : \mathbb{R}^L_{++} \times \mathbb{R}_{++} \to X.$$

It must satisfy $x(p, w) \in B(p, w)$ for every price--wealth pair; equivalently, $p \cdot x(p, w) \leq w$.

At prices and wealth (p, w), the coordinate $x_\ell(p, w)$ is the quantity consumed of commodity ℓ.

Demand describes behavior; it does not yet explain that behavior through preferences or optimization. Treating the entire function as observed is an idealization, because an empirical dataset

contains only finitely many price--wealth observations. We therefore distinguish restrictions on a full demand function from tests that can be implemented on finite data.

Example 3.1. Cobb-Douglas Demand Function. $x(p,w) = (\frac{\alpha w}{p_1}, \frac{(1-\alpha)w}{p_2})^\top$ for $\alpha \in (0,1)$.

3.2 Walras' Law, Homogeneity, and WARP

With the environment and demand primitive in place, we turn to consistency in consumption choices. This approach restricts observable behavior before imposing a specific preference representation.

We will impose a set of restrictions or axioms over the Marshallian demand function that we expect to hold in all situations irrespective of the details. Here we understand modeling consumption as restricting the set of allowable behaviors.

These axioms are minimal theoretical requirements, not empirical laws that every dataset must satisfy. Their value is that they make the set of admissible behavior explicit and provide a baseline against which richer models can be compared.

Axiom 3.2. *(Walras' law) A demand function x is said to satisfy Walras' law if and only if $p \cdot x(p,w) = w$ for all price-wealth pairs.*

Walras' law imposes budget exhaustion; it rules out choices that leave unspent wealth when a locally nonsatiated consumer could purchase more.

Axiom 3.3. *(Homogeneity of degree zero, HD0.) A demand function x is homogeneous of degree zero if, for every $\alpha > 0$,$x(\alpha p, \alpha w) = x(p,w)$.*

HD0 says that multiplying all prices and wealth by the same positive scalar does not change demand. It is a normalization of units, not a behavioral claim about intelligence.

Axiom 3.4. *(Weak axiom of revealed preference, WARP.) A demand function x satisfies WARP if, for any two price--wealth pairs (p,w) and (p',w'), whenever $x(p,w) \neq x(p',w')$ and $p \cdot x(p',w') \leq w$, then $p' \cdot x(p,w) > w'$.*

WARP excludes a two-observation reversal: if the bundle chosen at the second budget was affordable at the first, then the distinct bundle chosen at the first must be unaffordable at the second.

Let the observation-index set be $\mathbb{T} = \{1, \ldots, T\}$ and consider the dataset $\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$, where $s, t \in \mathbb{T}$ index observations, $p^t \gg 0$, $x^t \in X \setminus \{0\}$, and observed expenditure is $w^t := p^t \cdot x^t$.

We define direct weak revealed preference by $\succeq^{R,D}$. Thus $x^t \succeq^{R,D} y$ means that the observed bundle cost at least as much as y at the prices under which x^t was chosen. the compositional superscripts and the distinction between WARP and WGARP follow [Aguiar et al., 2026]. Formally,

$$x^t \succeq^{R,D} y \iff p^t \cdot x^t \geq p^t \cdot y.$$

For distinct chosen bundles z and y, WARP is equivalently $z \succeq^{R,D} y \implies \neg(y \succeq^{R,D} z)$. This is a restriction on observed choices; it need not be interpreted as a primitive psychological preference relation.

3.3 Local Implications of WARP

After establishing the axioms of behavior, the next step is to study their implications for the parameters of the model. The parameters of the model in our study of consumer behavior are prices and wealth pairs. Assume throughout this section that the demand function is continuously differentiable on an open domain of strictly positive prices and wealth.

- The price Jacobian is $D_p x(p,w) \in \mathbb{R}^{L \times L}$; its (ℓ, k) entry is $\partial_{p_k} x_\ell(p,w)$.
- We let $D_w x(p,w) = (\partial_w x_\ell(p,w))_{\ell=1}^L$ be the vector of wealth effects.
- The Slutsky matrix is $S(p,w) = D_p x(p,w) + D_w x(p,w) x(p,w)^\top$. It measures the derivative of demand under a price change that adjusts wealth just enough to keep the previously chosen bundle affordable. Its (ℓ, k) entry is $S_{\ell k} = \partial_{p_k} x_\ell(p,w) + \partial_w x_\ell(p,w) x_k$.

Fix a reference bundle q and consider the compensated demand $x(p, p \cdot q)$. The chain rule gives $D_p[x(p, p \cdot q)] = D_p x(p, p \cdot q) + D_w x(p, p \cdot q) q^\top$. At $q = x(p,w)$, Walras' law gives $p \cdot q = w$, so this derivative is exactly the Slutsky matrix: $S(p,w) = D_p x(p,w) + D_w x(p,w) x(p,w)^\top$.

Implications of Walras' Law for the Slutsky Matrix

Fact 3.5. *If x satisfies Walras' law, then (i) $p^\top D_w x(p,w) = 1$, (ii) $p^\top D_p x(p,w) = -x(p,w)^\top$, (iii) $p^\top S(p,w) = 0$.*

Part (i) implies that at least one good has a positive wealth effect. Because prices are strictly positive and $\sum_\ell p_\ell \partial_w x_\ell(p,w) = 1$, there must be some ℓ such that $\partial_w x_\ell(p,w) > 0$. In general, Walras' law restricts the wealth effects, while part (iii) restricts consumption changes under Slutsky-compensated price changes.

Implications of Homogeneity of Degree Zero.

Claim 3.6. If x satisfies Walras' law and homogeneity of degree zero, then $S(p, w)p = 0$.

Proof. For fixed (p, w), homogeneity gives $x(\lambda p, \lambda w) = x(p, w)$ for all $\lambda > 0$. Differentiating with respect to λ at $\lambda = 1$ yields

$$D_p x(p, w)p + D_w x(p, w)w = 0.$$

Walras' law gives $w = p \cdot x(p, w) = x(p, w)^\top p$; therefore the left-hand side is $S(p, w)p$, as required. □

Implications of WARP.

The principal two-budget implication of WARP under Walras' law is the compensated law of demand (CLD).

Definition 3.7. (CLD) We say that x satisfies the Compensated Law of Demand if and only if $(p^* - p) \cdot (x(p^*, w^*) - x(p, w)) < 0$ when $x(p^*, w^*) \neq x(p, w)$ and $w^* = p^* \cdot x(p, w)$.

We next establish that WARP implies the compensated law of demand.

Proposition 3.8. *Let x satisfy Walras' law and WARP. Then x satisfies the CLD.*

Proof. Fix two price--wealth pairs satisfying $w^* = p^* \cdot x(p, w)$ and $x(p^*, w^*) \neq x(p, w)$. Set $z = x(p, w)$ and $y = x(p^*, w^*)$.

Walras' law and the compensation condition give $p \cdot z = w$ and $p^* \cdot z = p^* \cdot y = w^*$.

Because $p^* \cdot z \leq w^*$, WARP and $y \neq z$ imply $p \cdot y > w$, hence $p \cdot (y - z) > 0$.

Combining this strict inequality with $p^* \cdot (y - z) = 0$ gives

$$(p^* - p) \cdot (y - z) < 0,$$

□

The differential version yields the following corollary.

Corollary 3.9. *Suppose x is continuously differentiable and satisfies Walras' law and WARP. Then, for every $a \in \mathbb{R}^L$ and every price-wealth pair, $a^\top S(p, w)a \leq 0$. Equivalently, the symmetric part $[S(p, w) + S(p, w)^\top]/2$ is negative semidefinite.*

Proof. Fix a price-wealth pair (p, w) and a direction $a \in \mathbb{R}^L$. For sufficiently small t, define $p(t) = p + ta$ and compensate wealth by $w(t) = p(t) \cdot x(p, w)$.

$$ta \cdot [x(p(t), w(t)) - x(p, w)] \leq 0.$$

$$\frac{a \cdot [x(p(t), w(t)) - x(p, w)]}{t} \leq 0$$

for every nonzero sufficiently small t after dividing the revealed-preference inequality by $t^2 > 0$. Continuous differentiability and the chain rule imply

$$\lim_{t \to 0} \frac{x(p(t), w(t)) - x(p, w)}{t} = D_p x(p, w)a + D_w x(p, w)\, x(p, w)^\top a = S(p, w)a.$$

Taking limits gives $a^\top S(p, w)a \leq 0$.

□

Marshallian and compensated price effects

The Slutsky decomposition separates the total response of demand to a price change from the response at fixed purchasing power. For a differentiable demand function,

$$\partial_{p_k} x_\ell(p, w) = S_{\ell k}(p, w) - x_k(p, w)\, \partial_w x_\ell(p, w).$$

Under WARP and Walras' law the compensated matrix has a negative-semidefinite symmetric part. Thus a positive Marshallian own-price derivative $\partial_{p_\ell} x_\ell > 0$ is possible only when the income effect is sufficiently adverse: the good is inferior ($\partial_w x_\ell < 0$), and the income effect dominates the nonpositive compensated effect. This is the Giffen mechanism.

The possibility of a Giffen response does not by itself establish a utility representation. Two distinct rationalizability routes are developed later. For a differentiable demand system, local integrability uses Walras' law, homogeneity, Slutsky symmetry and negative semidefiniteness together with explicit domain and regularity hypotheses. For a finite budget dataset, observed expenditure exhaustion and GARP provide the Afriat certificate. The compensated response, rather than the Marshallian derivative alone, is the object restricted by revealed preference.

3.4 Integrability

We now ask for sufficient conditions under which an observed demand function is generated by utility maximization. In the abstract choice setting, WARP and GARP restrict rationalizability. Competitive budget sets are a smaller domain than arbitrary menus, so differentiable demand requires additional restrictions. The relevant restrictions are stated in terms of the Slutsky matrix.

Assume throughout this subsection that the demand function x is continuously differentiable.

Recall that the Slutsky matrix function is $S(p,w) = D_p x(p,w) + D_w x(p,w) x(p,w)^\top$, or in the scalar version $[S(p,w)]_{\ell k} = \partial_{p_k} x_\ell(p,w) + \partial_w x_\ell(p,w) x_k(p,w)$ for all $\ell, k \in \{1,\dots,L\}$.

Remark 3.10. Integrability roadmap. On an open domain, Walras' law, homogeneity, and a symmetric negative-semidefinite Slutsky matrix are the central differential restrictions. They are necessary for every regular utility-generated demand. A sufficiency theorem additionally specifies rank or nondegeneracy conditions, the topology of the price-wealth domain, and boundary behavior for the associated differential system. Because alternative integrability theorems impose different global hypotheses, this text does not present the preceding checklist as a stand-alone global theorem; applications must cite and verify a precise version.

The qualifications are essential. Symmetry and negative semidefiniteness alone, stated on an unspecified domain, do not guarantee a global representation: one must control the domain, boundary behavior, and the differential equation's existence and uniqueness. Conversely, every continuously differentiable demand generated by a regular locally nonsatiated utility satisfies Walras' law, homogeneity, symmetry, and negative semidefiniteness of the Slutsky matrix.

Remark. Some versions of the global integrability theorem impose a bound on wealth elasticity, while others replace it with different domain or boundary conditions. No such bound is a universal consequence of local Slutsky restrictions; the hypotheses of the particular global theorem being used must be checked.

3.5 Examples and Applications of Demand Integrability

Consider two commodities, $L = 2$, and a parameter $0 < \alpha < 1$. Suppose observed demand for good 1 is$x_1(p,w) = \frac{\alpha w}{p_1}$. Normalize $p_2 = 1$, and impose Walras' law. Demand for good 2 is then the budget residual,$x_2 = w - p_1 x_1 = (1-\alpha)w$. The following calculation recovers a normalized indirect utility and then a direct utility representation.

Fix an initial price--wealth pair $(p_1^0, w^0) \in \mathbb{R}^2_{++}$. For $\xi > 0$, let $\chi(\xi; p_1^0, w^0)$ denote the wealth required at prices $(\xi, 1)$ to remain on the indifference curve through the initial pair. The integrability differential equation and its initial condition are

$$\frac{\partial \chi(\xi; p_1^0, w^0)}{\partial \xi} = x_1\big((\xi, 1), \chi(\xi; p_1^0, w^0)\big) = \frac{\alpha \chi(\xi; p_1^0, w^0)}{\xi}, \qquad \chi(p_1^0; p_1^0, w^0) = w^0.$$

The unique positive solution is

$$\chi(\xi; p_1^0, w^0) = w^0 \left(\frac{\xi}{p_1^0}\right)^{\alpha}.$$

Using $(1, 1)$ as the reference price vector gives the normalization

$$v(p_1, w) := \chi(1; p_1, w) = \frac{w}{p_1^{\alpha}}.$$

For any bundle $(x_1, x_2) \gg 0$, invert the two demand equations to recover the normalized price and wealth that generate it:

$$x_1(p, w) = \frac{\alpha w}{p_1}; \quad x_2(p, w) = (1 - \alpha) w$$

$$\frac{x_1}{x_2} = \frac{\alpha}{(1 - \alpha)} \frac{1}{p_1}$$

$$p_1 = \frac{\alpha}{1 - \alpha} \frac{x_2}{x_1}, \quad w = \frac{x_2}{1 - \alpha}.$$

Substituting these inverse-demand expressions into the normalized indirect utility gives

$$u(x_1, x_2) := v\left(\frac{\alpha}{1 - \alpha} \frac{x_2}{x_1}, \frac{x_2}{1 - \alpha}\right).$$

$$u(x_1, x_2) = \frac{\frac{x_2}{1-\alpha}}{\left(\frac{\alpha}{1-\alpha} \frac{x_2}{x_1}\right)^{\alpha}} = \left(\frac{x_2}{1 - \alpha}\right)^{1-\alpha} \left(\frac{x_1}{\alpha}\right)^{\alpha}$$

$$u(x_1, x_2) = \hat{c} x_1^{\alpha} x_2^{1-\alpha}, \quad \hat{c} = \frac{1}{\alpha^{\alpha}(1 - \alpha)^{1-\alpha}}.$$

This is a Cobb-Douglas utility function.

3.6 GARP and Afriat's Theorem

Retain the expenditure-exhausting finite dataset $\mathcal{O} = \{(p^t, x^t)\}_{t\in\mathbb{T}}$ and direct revealed preference $\succeq^{R,D}$ from the preceding section. Let $\succeq^R$ be its transitive closure on the observed bundles. Thus $x^t \succeq^R x^s$ when there is a finite chain of direct weak comparisons from x^t to x^s. Define $x^t \succ^{R,D} x^s$ when $p^t \cdot x^t > p^t \cdot x^s$.

Definition 3.11. Utility rationalization of budget data. A utility function $u : X \to \mathbb{R}$ rationalizes $\mathcal{O}$ if, for every $t \in \mathbb{T}$, $u(x^t) \geq u(x)$ whenever $p^t \cdot x \leq p^t \cdot x^t$. Ties are allowed, so the observed bundle need not be the unique maximizer.

A finite demand dataset $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ satisfies the generalized axiom of revealed preference (GARP) if

$$x^t \succeq^R x^s \quad \Longrightarrow \quad \neg\big(x^s \succ^{R,D} x^t\big)$$

for all observations s, t, where $\succeq^R$ is the transitive closure of direct revealed preference and

$$x^s \succ^{R,D} x^t \iff p^s \cdot x^s > p^s \cdot x^t.$$

Equivalently, a chain of weak revealed-preference comparisons may not be closed by a strict direct comparison in the opposite direction.

Proposition 3.12. *(Afriat's theorem.) Let $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ be a finite dataset with $p^t \gg 0$ and $x^t \in \mathbb{R}^L_+$ for every observation t. The following are equivalent:*

1. There exists a locally nonsatiated utility function u that rationalizes the data in the sense that for all t, $u(x^t) \geq u(x)$ for all x such that $p^t \cdot x^t \geq p^t \cdot x$.

2. The dataset $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ satisfies GARP.

3. There exist numbers $(U^t, \lambda^t) \in \mathbb{R} \times \mathbb{R}_{++}$ satisfying the Afriat inequalities

$$U^t \leq U^s + \lambda^s p^s \cdot (x^t - x^s) \quad \forall s, t \in \{1, \ldots, T\}$$

4. There exists a continuous, concave, and strictly increasing utility function u that rationalizes the data in the sense that for all t, $u(x^t) \geq u(x)$ for all x such that $p^t \cdot x^t \geq p^t \cdot x$.

Proof. (1) $\Longrightarrow$ (2), (2) $\Longrightarrow$ (3), (3) $\Longrightarrow$ (4), (4) $\Longrightarrow$ (1).

(1) implies (2),

Assume that $\mathcal{O}$ is rationalized by a locally nonsatiated utility function and suppose, toward a contradiction, that GARP fails. Then there are observation indices $t_0, \ldots, t_m$ such that

$x^{t_{k-1}} \succeq^{R,D} x^{t_k}$ for $k = 1, \ldots, m$ and $x^{t_m} \succ^{R,D} x^{t_0}$. Rationalization gives $u(x^{t_0}) \geq \cdots \geq u(x^{t_m})$. The strict comparison implies $p^{t_m} \cdot x^{t_0} < p^{t_m} \cdot x^{t_m}$. By local nonsatiation, there is a bundle y with $\|y - x^{t_0}\|_2 < \varepsilon$ and $u(y) > u(x^{t_0})$; choosing $\varepsilon > 0$ sufficiently small preserves strict affordability at prices p^{t_m}. Thus y is affordable at observation t_m and satisfies $u(y) > u(x^{t_m})$, contradicting rationalization.

(2) implies (3)

The implication from GARP to the Afriat inequalities is the finite Afriat feasibility lemma. It is a theorem of alternatives for the displayed linear system, not an immediate consequence of naming Farkas' lemma. We cite the lemma here: for a finite, expenditure-exhausting dataset with strictly positive prices, GARP holds if and only if there exist $(U^1, \ldots, U^T)$ and positive $(\lambda^1, \ldots, \lambda^T)$ satisfying every Afriat inequality. The converse can be seen directly by summing the inequalities around a revealed-preference cycle; the forward direction uses the full finite feasibility lemma.

(3) implies (4)

Let $u(x) = \min_{t \in \{1,\ldots,T\}} \{U^t + \lambda^t p^t \cdot (x - x^t)\}$. As the pointwise minimum of finitely many affine functions, u is continuous and concave. It is strictly increasing in the product order because $\lambda^t > 0$ and $p^t \gg 0$ for every observation t.

To verify rationalization, first observe that $u(x^t) = U^t$ for every observation t. Indeed, the Afriat inequality gives

$$U^t \leq U^m + \lambda^m p^m \cdot (x^t - x^m)$$

for every m, while the term with $m = t$ equals U^t. Therefore the minimum of these affine terms is exactly U^t.

It remains to show that $u(x) \leq u(x^s)$, if $p^s \cdot x^s \geq p^s \cdot x$

If $p^s \cdot x^s \geq p^s \cdot x$, or $p^s \cdot (x^s - x) \geq 0$, then $u(x) = \min_t \{U^t + \lambda^t p^t \cdot (x - x^t)\} \leq U^s + \lambda^s p^s \cdot (x - x^s) \leq U^s = u(x^s)$.

Thus $u(x^s) \geq u(x)$ when $p^s \cdot x^s \geq p^s \cdot x$.

(4) implies (1).

If u is strictly increasing, then it is locally nonsatiated.

□

3.6.1 Out-of-sample predictions

Imagine we have a new data point $T + 1$, where i give the customer a new price out of sample: p^{T+1}, normalize here income to 1.

Then I ask what is the new or set of new demand quantities x^{T+1} such that they are consistent with the the utilitites that rationalized the dataset $\mathcal{O}^T$.

I will form a new system of inequalities:

There exist a candidate bundle $x^{T+1} \in \mathbb{R}^L_+$, utility levels $U^s \in \mathbb{R}$ for $s = 1, \ldots, T+1$, and positive Afriat certificates $\lambda^t > 0$ for $t = 1, \ldots, T$, satisfying the Afriat inequalities

$$U^t \leq U^s + \lambda^s p^s \cdot (x^t - x^s) \quad \forall s, t \in \{1, \ldots, T\}$$

$$U^{T+1} \leq U^t + \lambda^t p^t \cdot (x^{T+1} - x^t)$$

$$U^t \leq U^{T+1} + p^{T+1} \cdot (x^t - x^{T+1})$$

3.7 WGARP and Coalitional Multi-Utility

Afriat's theorem asks whether all observations can be generated by maximization of one utility function. Its answer is GARP. A failure of GARP rejects that joint transitive explanation, but it need not make every pair of observations inconsistent. WGARP asks the weaker pairwise question: can the data avoid every strict two-observation reversal even when a longer revealed-preference cycle remains? Coalitional multi-utility (CMU) is the optimizing model behind that condition[Aguiar et al., 2026]. Crucially, CMU extends the benchmark model: ordinary utility maximization is exactly its singleton special case.

Definition 3.13. Budget WGARP and WARP. The finite dataset $\mathcal{O}$ satisfies WGARP if, for all $s, t \in \mathbb{T}$,

$$p^t \cdot x^t \geq p^t \cdot x^s \quad \Longrightarrow \quad \neg\big(p^s \cdot x^s > p^s \cdot x^t\big).$$

It satisfies WARP if, whenever $x^t \neq x^s$,

$$p^t \cdot x^t \geq p^t \cdot x^s \quad \Longrightarrow \quad p^s \cdot x^s < p^s \cdot x^t.$$

WGARP is the weak/strict condition appropriate for a demand correspondence. WARP also excludes mutual affordability at equality and corresponds to a strict, single-valued version of

the model. GARP and SARP apply the analogous prohibitions after taking revealed-preference closure.

Preference-function demand. A $r : X \times X \to \mathbb{R}$ is a numerical comparison rule, with $x \succeq_r y$ when $r(x, y) \geq 0$. Its demand correspondence is

$$x_r(p, w) = \{x \in B(p, w) : r(x, y) \geq 0 \text{ for every } y \in B(p, w)\}.$$

The function is asymmetric when $r(x, y) \geq 0 \Rightarrow r(y, x) \leq 0$ and $r(x, y) > 0 \Rightarrow r(y, x) < 0$. It is strictly increasing when $x > z \Rightarrow r(x, y) > r(z, y)$ for every y. The strict part of $\succeq_r$ remains the relational strict part; it is not defined merely by $r(x, y) > 0$.

Coherent CMU. Let $C(X)$ be the continuous real-valued functions on X with the compact-open topology. A compact coalition structure Ω is a nonempty compact collection of nonempty compact sets $U \subseteq C(X)$. Its preference function is

$$r_\Omega(x, y) = \max_{U \in \Omega} \min_{u \in U} \big[u(x) - u(y)\big].$$

The structure is coherent if $U \cap V \neq \varnothing$ for every $U, V \in \Omega$. Coherence is pairwise intersection; it does not require one utility to belong to every coalition. A CMU comparison holds when at least one coalition unanimously ranks the first bundle at least as high as the second.

Proposition 3.14. *Classical utility maximization is nested exactly. Let $u \in C(X)$ and set $\Omega_u = \{\{u\}\}$. Then Ω_u is compact and coherent, $r_{\Omega_u}(x, y) = u(x) - u(y)$, and*

$$x_{r_{\Omega_u}}(p, w) = \arg \max_{x \in B(p,w)} u(x).$$

Proof. Both optimization operators in the CMU formula collapse for the singleton coalition, so $r_{\Omega_u}(x, y) = u(x) - u(y)$. Therefore $x \in x_{r_{\Omega_u}}(p, w)$ exactly when $u(x) \geq u(y)$ for every feasible y, which is the ordinary utility-maximization condition. A singleton is compact and intersects itself. □

Theorem 3.15. *Reader theorem ($\star$): WGARP and coherent CMU. For a finite expenditure-exhausting dataset with $L \geq 2$, strictly positive prices, and nonzero nonnegative choices, the following are equivalent: (i) WGARP; (ii) rationalization by an asymmetric, strictly increasing, continuous preference function; (iii) rationalization by a coherent, strictly increasing, continuous*

CMU; and (iv) rationalization by a skew-symmetric, coherent, continuous CMU whose coalitions contain strictly increasing concave utilities.

This is an exact finite-data characterization, not only a necessary condition. The difficult direction uses the fact that every two-observation subdataset of a WGARP dataset satisfies GARP, constructs a pairwise Afriat utility for each pair, and assembles those utilities into a coherent coalition structure. Passing WGARP does not identify nontransitivity: every GARP dataset also passes WGARP and can admit both classical and CMU rationalizations.

Theorem 3.16. *Reader theorem ($\star$): WARP and strict coherent CMU. In the same finite environment, WARP is equivalent to rationalization by an asymmetric, strictly star-concave, continuous preference function; equivalently, by a coherent strictly star-concave CMU; equivalently, by a skew-symmetric coherent CMU whose coalitions contain strictly increasing strictly concave utilities. Here strict star-concavity requires $r(\alpha x + (1-\alpha)y, x) > \alpha r(x,x) + (1-\alpha) r(y,x)$ for $x \neq y$ and $\alpha \in (0,1)$.*

The WARP theorem is separate from the WGARP theorem. WARP is not the finite-data condition for maximization of one ordinary utility function: GARP is the classical weak-rationalizability condition, and SARP is the corresponding strict acyclicity condition. The strict CMU theorem yields at most one demanded bundle on a budget when demand is nonempty, but it does not guarantee nonempty demand on every counterfactual budget.

Example 3.17. A three-good WGARP dataset that violates GARP. Let

$$p^1 = (4,1,5), \quad p^2 = (5,4,1), \quad p^3 = (1,5,4),$$
$$x^1 = (4,1,1), \quad x^2 = (1,4,1), \quad x^3 = (1,1,4).$$

Then

$$[p^t \cdot x^s]_{t,s=1}^3 = \begin{pmatrix} 22 & 13 & 25 \\ 25 & 22 & 13 \\ 13 & 25 & 22 \end{pmatrix}.$$

Hence $x^1 \succ^{R,D} x^2 \succ^{R,D} x^3 \succ^{R,D} x^1$, so GARP fails. Each reverse-budget cost is $25 > 22$, however, so no pair creates a WGARP reversal. This dataset also satisfies WARP because the chosen bundles are distinct and no pair is mutually affordable. With two goods, WGARP and GARP coincide; three goods are essential for this separation under linear budgets.

Proposition 3.18. *Compensated law of demand for preference functions. Let r be asymmetric*

and strictly increasing, let $x \in x_r(p, w)$*, and set* $w' = p' \cdot x$*. If* $x' \in x_r(p', w')$*, then*

$$(p' - p) \cdot (x' - x) \leq 0.$$

Proof. First note that strict increase and asymmetry imply budget exhaustion at every nonempty demand. If a demanded bundle had slack, a strictly larger feasible bundle z would satisfy $r(x, z) \geq 0$ by maximality, while strict increase in the first argument would give $r(z, z) > r(x, z) \geq 0$, contradicting asymmetry at (z, z). Hence $p' \cdot (x' - x) = 0$. Suppose, toward a contradiction, that $p \cdot x' < p \cdot x$. Then some $z > x'$ remains affordable at the original budget. Since x' defeats x at the compensated budget, $r(x', x) \geq 0$. Strict increase gives $r(z, x) > 0$, and asymmetry then gives $r(x, z) < 0$, contradicting $x \in x_r(p, w)$. Therefore $p \cdot (x' - x) \geq 0$, and subtracting this inequality from $p' \cdot (x' - x) = 0$ yields the result. The claim is conditional on both demand sets being nonempty. □

Remark 3.19. What survives, and what does not. Singleton CMU reproduces the UMP exactly. Under asymmetry and strict increase, nonempty CMU demand satisfies budget exhaustion, homogeneity of degree zero, WGARP, and the compensated inequality above. WGARP or coherence alone does not imply one scalar utility, transitivity, GARP, convex preferences, nonempty counterfactual demand, Roy's identity, Shephard's lemma, Slutsky symmetry, or the standard scalar-utility welfare measures. Concavity of the component utilities does not make the induced CMU relation globally convex. An arbitrary noncoherent CMU need not satisfy WGARP.

The star marks a compressed reader theorem, not a weaker truth standard and not a Lean-verification badge. The six-way WGARP statement, the distinct WARP theorem, topology, proof boundary, and formalization record appear in Appendix C.

3.8 SARP and Strict Rationalization

The preceding section used GARP to characterize weak rationalization, allowing ties in observed choice. SARP is the acyclicity condition associated with strict or unique rationalization. The exact equivalence depends on how duplicate observed bundles and strict choice are treated, so this section states the condition without claiming that merely replacing every Afriat inequality by a strict one is sufficient.

Definition 3.20. (Strong axiom of revealed preference, SARP.) The dataset satisfies SARP if

the reflexive-transitive revealed relation $\succeq^R$ has no nontrivial cycle: whenever $x^t \succeq^R x^s$ and $x^t \neq x^s$, it is not the case that $x^s \succeq^R x^t$.

SARP is the natural condition when observed choice is single-valued and ties are ruled out. The weak Afriat theorem above is the maintained finite-data result in this book; strict rationalization results require additional assumptions, including care with duplicate observed bundles.

3.9 Forecasting with Varian's Support Set

Definition 3.21. Given a finite price-choice dataset $\mathcal{O}$ and a new price-wealth pair (p^{T+1}, w^{T+1}), define the augmented dataset associated with a candidate bundle by $\mathcal{O}^+(x) = \mathcal{O} \cup \{(p^{T+1}, x)\}$. Varian's support set is

$$\mathcal{S}_{\mathcal{O}}(p^{T+1}, w^{T+1}) = \left\{ x \in X : \begin{array}{l} \mathcal{O}^+(x) \text{ satisfies GARP}, \\ p^{T+1} \cdot x = w^{T+1} \end{array} \right\}.$$

The definition already gives the exact criterion: a candidate bundle belongs to the support set if it exhausts the new budget and the augmented dataset satisfies GARP. The transitive closure must be checked, not only one-step comparisons. Whenever an old bundle x^t is indirectly revealed preferred to the candidate, GARP requires the inequality

$$p^{T+1} \cdot x \leq p^{T+1} \cdot x^t.$$

3.10 Upper Bounds for Welfare Analysis

Definition 3.22. Money-metric utility at reference prices p is defined by $\mu_p(x) = e(p, u(x))$ where e is the expenditure function associated with u, $e(p, \overline{u}) = \min_{z \in X}\{p \cdot z : u(z) \geq \overline{u}\}$.

For fixed p, $\mu_p(x)$ is a utility function that represents the same preferences as u but is expressed in money units.

For a counterfactual bundle x, extend direct revealed preference to the set $\{x, x^1, \ldots, x^T\}$ by declaring

$$x^t \succeq^{R,D}_{\mathcal{O},x} y \quad \Longleftrightarrow \quad p^t \cdot x^t \geq p^t \cdot y, \qquad y \in \{x, x^1, \ldots, x^T\},$$

and let $\succeq^R_{\mathcal{O},x}$ be its transitive closure. The data-dependent upper bound is

$$m^+_{\mathcal{O}}(p,x) = \min\{p \cdot x^t : x^t \succeq^R_{\mathcal{O},x} x,\ t = 1,\dots,T\},$$

when the index set is nonempty, and $m^+_{\mathcal{O}}(p,x) = +\infty$ otherwise. If u rationalizes $\mathcal{O}$, every revealed chain in the extended relation implies $u(x^t) \geq u(x)$. Monotonicity of expenditure in its utility target and feasibility of x^t then give

$$\mu_p(x) = e(p, u(x)) \leq e(p, u(x^t)) \leq p \cdot x^t.$$

Taking the minimum proves $\mu_p(x) \leq m^+_{\mathcal{O}}(p,x)$ whenever the bound is finite.

3.11 Experimental Tests of GARP

Choi, Fisman, Gale, and Kariv study 93 participants recruited from undergraduate classes and staff at the University of California, Berkeley. Each participant completed 50 portfolio-allocation problems. In every problem the participant chose a point on a linear budget over two Arrow securities, one paying in each of two states. A chosen portfolio can therefore be interpreted as a bundle of state-contingent consumption. The 47 participants in the symmetric treatment faced $\pi_1 = \pi_2 = 1/2$; the remaining 46 faced $\{\pi_1, \pi_2\} = \{1/3, 2/3\}$. The computer varied the budget line across rounds, and each line may be normalized as $p^t \cdot x = 1$. Thus participant i generated a panel $\mathcal{O}_i = \{(p^t_i, x^t_i)\}_{t=1}^{50}$ to which GARP can be applied[Choi et al., 2007].

The exact-GARP tabulation in[Polisson et al., 2020] reports that 12 of 47 participants in the symmetric treatment and 4 of 46 in the asymmetric treatment satisfy GARP. Overall, 16 of 93 participants---approximately 17 percent---pass the exact test. The interpretation is precise but limited: a recorded panel that fails cannot be rationalized exactly by one locally nonsatiated utility function under the maintained budget data. The result alone does not distinguish preference instability from optimization error, recording error, or price misperception. Nor does a binary pass rate measure the economic magnitude of a violation: one strict revealed-preference cycle is sufficient for rejection. The critical cost-efficiency index introduced next provides a graded measure, while a random-choice benchmark helps assess the power of the design.

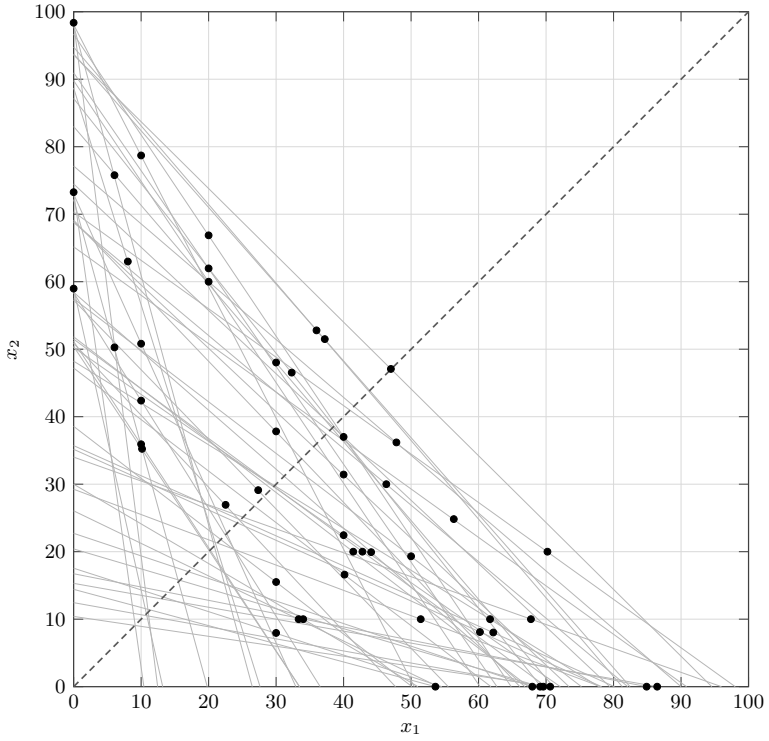


Figure 3.1: Observed portfolio choices of Subject 216 in the symmetric treatment. Gray lines are budget lines, black points are chosen portfolios, and the dashed 45-degree line marks equal consumption in the two states. Redrawn from Quah's survey using the experiment of[Quah, 2015, Choi et al., 2007].

3.12 The Critical Cost-Efficiency Index

When a dataset $\mathcal{O}$ fails GARP, the critical cost-efficiency index measures the size of the revealed-preference violation. For an efficiency level $\alpha \in [0,1]$, write the efficiency-adjusted relations as $\succeq_\alpha^{R,D}, \succ_\alpha^{R,D}, \succeq_\alpha^{R}$. The direct relations are defined by

$$x^t \succeq_\alpha^{R,D} x^s \iff \alpha p^t \cdot x^t \geq p^t \cdot x^s,$$
$$x^t \succ_\alpha^{R,D} x^s \iff \alpha p^t \cdot x^t > p^t \cdot x^s.$$

Let $\succeq_\alpha^{R}$ be the transitive closure of $\succeq_\alpha^{R,D}$. The dataset satisfies α-GARP when every pair of observations s, t obeys

$$x^t \succeq_\alpha^{R} x^s \implies \neg\big(x^s \succ_\alpha^{R,D} x^t\big).$$

Definition 3.23. The Critical Cost Efficiency Index (CCEI) of a dataset $\mathcal{O}$ is $\mathrm{CCEI}(\mathcal{O}) = \sup\{\alpha \in [0,1] : \mathcal{O} \text{ satisfies } \alpha\text{-GARP}\}$. A value of one means that the original data satisfy GARP; a smaller value is the largest efficiency level at which the adjusted revealed-preference test remains

free of GARP violations. This is an expenditure-efficiency normalization, not a claim that the recorded bundle was literally purchased after discarding the remaining expenditure.

The CCEI distribution for the experiment is shown below.

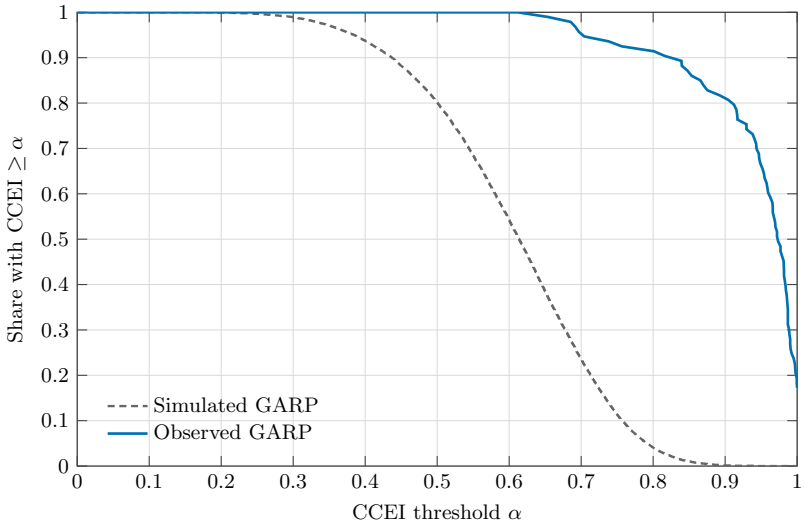


Figure 3.2: Survival functions $\alpha \mapsto \Pr\{\text{CCEI}_i \geq \alpha\}$ for the experimental participants and for Bronars-style simulated participants who choose uniformly on each budget line. Redrawn from Quah's survey; experiment and simulation design from Choi et al., with subsequent tabulation in Polisson, Quah, and Renou[Quah, 2015, Choi et al., 2007, Polisson et al., 2020].

3.13 Measurement Error and Statistical Revealed Preference

Ahn, Choi, Gale, and Kariv collect 50 portfolio choices from each of 154 participants. The assets are three Arrow securities corresponding to three states: one state has a known probability, whereas the other two states are ambiguous. This is a three-state contingent-consumption experiment, not an ordinary three-good demand experiment. Applied directly to the recorded choices and budgets, the deterministic GARP test is passed by 12.99 percent of participants[Ahn et al., 2014, Aguiar and Kashaev, 2021].

A deterministic revealed-preference test treats recorded prices and choices as the economically relevant data. Aguiar and Kashaev instead ask whether the population distribution of recorded panels can be generated by latent utility-maximizing behavior together with a disciplined measurement process[Aguiar and Kashaev, 2021]. For participant i and task t, let (p_i^t, x_i^t) denote the recorded pair and (p_i^{*t}, x_i^{*t}) the latent pair entering the maintained optimizing model. An additive relation such as $x_i^t = x_i^{*t} + \varepsilon_{x,i}^t$ or $p_i^t = p_i^{*t} + \varepsilon_{p,i}^t$ is a schematic illustration, not

the identifying content of the framework. Empirical content comes from restrictions linking observed and latent variables---especially population centering conditions---combined with utility-maximization restrictions. If both latent data and measurement error are unrestricted, observed violations can be rationalized too freely for the exercise to be informative.

In their application, Aguiar and Kashaev study two measurement channels separately. Under the price-misperception specification, recorded choices are exact and perceived prices may differ from displayed prices, subject to agreement between perceived and displayed price vectors on average for each task. Under the trembling-hand specification, displayed prices are exact and chosen allocations may contain mean-zero error. The validation evidence is asymmetric: the paper reports direct evidence for the centering condition on hand errors, whereas the evidence for price misperception is indirect and tied to the graphical design.

The empirical comparison is therefore conditional. At the 5 percent significance level, the authors do not reject the joint null of static utility maximization and mean-centered price misperception. They strongly reject the joint null when the only discrepancy is a centered trembling-hand error in consumption. The data thus favor the price-misperception specification over that particular consumption-error specification. They do not establish that every participant is deterministically GARP-consistent, that price misperception is the unique source of every violation, or that utility maximization has been confirmed. Failure to reject is evidence about a joint model of preferences and measurement, not proof of rationality.

3.14 Problems

Part I: By hand

Problem 3.1 [Core]. A two-observation WARP test.

A consumer chooses $x^1 = (2, 2)$ at $(p^1, w^1) = ((2, 1), 6)$ and $x^2 = (1, 5/2)$ at $(p^2, w^2) = ((1, 2), 6)$.

(a) Verify that each observed bundle exhausts its budget.

(b) Determine whether each bundle was affordable at the other observation.

(c) Does the dataset satisfy WARP for single-valued demand? Identify the two direct revealed-preference comparisons that determine your answer.

Problem 3.2 [Core]. Recovering a quasilinear utility.

Normalize $p_2 = 1$ and suppose an interior demand system satisfies $x_1(p_1, w) = \beta - \alpha p_1$, where $\alpha, \beta > 0$, and Walras' law.

(a) Derive $x_2(p_1, w)$ and state the domain on which both demands are nonnegative.

(b) Compute the normalized scalar Slutsky term for the non-numeraire demand.

(c) Recover a quasilinear utility representation $u(x_1, x_2) = g(x_1) + x_2$ that generates the demand on the interior domain. Distinguish general ordinal uniqueness up to an increasing transformation from the narrower normalization freedom after the coefficient on x_2 is fixed at one.

Problem 3.3 [Core]. Afriat inequalities by hand.

Consider two observations: $p^1 = (1, 1)$, $x^1 = (2, 1)$ and $p^2 = (2, 1)$, $x^2 = (1, 1)$.

(a) Compute observed expenditures and all direct revealed-preference comparisons.

(b) Check GARP.

(c) Verify that $U^1 = 1$, $U^2 = 0$, and $\lambda^1 = \lambda^2 = 1$ satisfy the Afriat inequalities.

(d) Write the corresponding piecewise-linear Afriat utility and verify that it assigns the required utility levels at the observations.

Problem 3.5 [Core]. Classical utility as singleton CMU.

Let $u : X \to \mathbb{R}$ be continuous, let $\Omega_u = \{\{u\}\}$, and define $r_{\Omega_u}(x, y) = \max_{U \in \Omega_u} \min_{v \in U}[v(x) - v(y)]$.

(a) Prove that $r_{\Omega_u}(x, y) = u(x) - u(y)$.

(b) Prove that $x_{r_{\Omega_u}}(p, w) = \arg\max_{x \in B(p,w)} u(x)$.

(c) Explain precisely which conclusions require continuity, strict monotonicity, or concavity of u, and which do not.

Part II: A counterfactual demand support set — chapter use case and verified scale-up

Problem 3.6 [Proof]. WGARP without GARP.

Consider prices $p^1 = (4, 1, 5)$, $p^2 = (5, 4, 1)$, $p^3 = (1, 5, 4)$ and choices $x^1 = (4, 1, 1)$, $x^2 = (1, 4, 1)$, $x^3 = (1, 1, 4)$.

(a) Compute the 3×3 matrix with (t, s) entry $p^t \cdot x^s$.

(b) Show that the data satisfy WGARP and WARP but violate GARP. Identify the revealed three-cycle and explain why no pair forms a reversal.

(c) Suppose an asymmetric, strictly increasing preference function r rationalizes two choices $x \in x_r(p, w)$ and $x' \in x_r(p', p' \cdot x)$. Prove the compensated law of demand $(p' - p) \cdot (x' - x) \leq 0$. State where nonemptiness of demand is used.

Chapter use case

Seed: Problems 3.1, 3.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of observed budgets from two to T, with two goods and the counterfactual budget fixed. Tool: Codex + Python; Lean for candidate certificates. Verification: Recover the hand support interval and verify that added observations can only shrink it. Lean may check GARP or Afriat certificates for candidates; it does not yet certify the full support-set characterization.

Problem 3.4 [Applied]. Demand forecasting with Varian's support set.

A finite dataset $\mathcal{O} = \{(p^t, x^t)\}_{t\in\mathbb{T}}$ satisfies GARP. Fix a counterfactual budget (p^0, w^0) with $p^0 \gg 0$, and define $\mathcal{S}_{\mathcal{O}}(p^0, w^0) = \{x^0 : p^0 \cdot x^0 = w^0,\ \mathcal{O} \cup \{(p^0, x^0)\} \text{ satisfies GARP}\}$.

(a) State a revealed-preference condition under which the augmented data imply $x^t \succeq^R x^0$.

(b) Derive the associated linear restriction on $p^0 \cdot x^0$.

(c) Explain why the restrictions in parts (a)--(b) are necessary but must be combined with every revealed-preference path into and out of the candidate observation to characterize $\mathcal{S}_{\mathcal{O}}(p^0, w^0)$. Explain why this set gives bounds rather than point identification.

Chapter 4

Choice Without Budget Geometry

Question: Which conclusions are properties of choice itself, rather than of linear budget geometry?

4.1 From Choice Data to Preference

The preceding chapters used the geometry of linear budgets. We now abstract from prices and wealth and retain only alternatives, feasible menus, and observed choices. This separation shows which conclusions follow from choice consistency itself and which depend on consumer-budget structure. It also supplies the preference and utility language that underlies the benchmark model used in Chapter 1.

4.2 Choice Environments

Let X be a nonempty set of alternatives and let $\mathcal{A} \subseteq 2^X \setminus \{\emptyset\}$ be the family of feasible menus.

Definition 4.1. Choice correspondence. A choice correspondence is a map $c : \mathcal{A} \rightrightarrows X$ satisfying $\emptyset \neq c(A) \subseteq A$ for every $A \in \mathcal{A}$. A choice structure is the pair $(\mathcal{A}, c)$.

Exact rationalization. A binary relation $\succeq$ rationalizes $(\mathcal{A}, c)$ if $c(A) = \{x \in A : x \succeq y \text{ for every } y \in A\}$ for every $A \in \mathcal{A}$. A rational preference relation is complete and transitive.

Remark 4.2. Replacing equality by the inclusion $c(A) \subseteq \{x \in A : x \succeq y \text{ for every } y \in A\}$ is only a selection-from-maximizers condition. By itself it has no revealed-preference content: universal indifference makes every feasible alternative greatest and therefore accommodates every nonempty

choice correspondence. Whenever the book uses weak rationalization for budget data, it instead imposes the structural conditions stated in Afriat's theorem.

4.3 Preference Relations

Definition 4.3. For a relation $\succeq \subseteq X \times X$, define its strict and symmetric parts by $x \succ y \iff x \succeq y$ and not $y \succeq x$ and $x \sim y \iff x \succeq y$ and $y \succeq x$. If $\succeq$ is complete and transitive, then $\succ$ is irreflexive and transitive, while $\sim$ is an equivalence relation. These elementary implications are the order-theoretic basis for utility representation.

Proposition 4.4. *Finite utility representation. If X is finite, a preference relation on X has a real-valued utility representation if and only if it is complete and transitive. A representation is unique up to a strictly increasing transformation of its range.*

For an infinite commodity or choice space, representation requires additional assumptions such as continuity and a suitable countability condition. The next theorem gives the continuous-utility result for the Euclidean commodity space used in Chapter 1.

4.4 Utility Representation

Definition 4.5. A preference relation $\succeq$ on $X \subseteq \mathbb{R}^L$ is continuous if its upper and lower contour sets are closed: for every $y \in X$,

$$\{x \in X : x \succeq y\} \quad \text{and} \quad \{x \in X : y \succeq x\}$$

are closed in X. Equivalently, if $x^n \to x$, then $x^n \succeq y$ for every n implies $x \succeq y$, and $y \succeq x^n$ for every n implies $y \succeq x$.

Lemma 4.6. *Every total preorder on a countable set has a real-valued utility representation.*

Proof. Let $I_{\succeq}$ be the symmetric part of the preorder. The quotient $X/I_{\succeq}$ is countable, and the relation induced on it is a linear order by Appendix A. Every countable linear order admits an order embedding into $\mathbb{Q}$: enumerate its elements and insert the image of each new element at a rational number between the images of its already embedded lower and upper neighbors. Composing this embedding with the quotient map gives $u : X \to \mathbb{Q} \subseteq \mathbb{R}$ such that $x \succeq y \iff u(x) \geq u(y)$. □

Theorem 4.7. *(Continuous utility representation) If $\succeq$ is a continuous total preorder on $X = \mathbb{R}^L_+$,*

then there exists a continuous function $u : X \to \mathbb{R}$ such that, for every $x, y \in X$,

$$x \succeq y \iff u(x) \geq u(y).$$

Proof. Proof record. This is Debreu's continuous-utility theorem specialized to the Euclidean orthant; see[Debreu, 1959]. The orthant is second countable, separable, and connected, and a continuous total preorder on this space satisfies the theorem's topological hypotheses. The published proof constructs an order-dense countable family of indifference classes, assigns compatible real values to those classes, and extends the assignment continuously to the whole space. We rely on that theorem here rather than reproduce its topological construction. An arbitrary limit of a utility defined only on $\mathbb{Q}^L$ would not suffice, because existence and independence of the approximating sequence would still have to be proved. □

The theorem guarantees existence, not uniqueness. If u represents $\succeq$ and f is strictly increasing on the range of u, then $f \circ u$ represents the same preference relation.

4.5 Revealed Choice and WGARP

Definition 4.8. Direct revealed preference. For a choice structure $(\mathcal{A}, c)$, write $x \succeq^C y$ if there is a menu $A \in \mathcal{A}$ with $x \in c(A)$ and $y \in A$. Write $x \succ^C y$ if, in addition, $y \notin c(A)$. Let $\succeq^{C*}$ be the reflexive-transitive closure of $\succeq^C$.

Weak Generalized Axiom of Revealed Preference (WGARP). The choice correspondence satisfies WGARP when $x \succeq^C y$ implies not $y \succ^C x$. Equivalently, if two menus contain both x and y, choosing x from one and strictly choosing y from the other is ruled out.

Proposition 4.9. *Necessity of WGARP. If a complete and transitive preference relation rationalizes a choice structure, then the structure satisfies WGARP.*

Proof. If $x \in c(A)$ and $y \in A$, rationalization gives $x \succeq y$. If a second menu strictly chose y over x, it would give $y \succ x$, contradicting the first comparison.

4.6 GARP in Abstract Choice and the Budget-Data Specialization

Definition 4.10. Generalized Axiom of Revealed Preference (GARP). A finite dataset satisfies GARP when $x \succeq^{C*} y$ never occurs together with $y \succ^C x$. Thus an indirect revealed-preference

chain may return to an alternative only when the return comparison is not strict.

Remark 4.11. Afriat preview. On finite linear-budget data with positive prices and expenditure exhaustion, GARP is equivalent to rationalization by a continuous, strictly increasing, concave utility function. Chapter 3 states the exact theorem, writes the Afriat inequalities, and constructs the rationalizing utility[Afriat, 1967, Varian, 1982]. This is a result about budget geometry and locally nonsatiated utility, not an equivalence with unrestricted selection from the greatest elements of an arbitrary relation.

WGARP is the one-step restriction for a possibly multivalued choice correspondence. GARP applies the no-strict-reversal requirement to the reflexive-transitive closure of direct revealed preference and is therefore the relevant condition when the data do not contain every pair of alternatives in a common menu. Exact single-valued rationalization can require additional tie-breaking assumptions, so the theorem above is stated for weak rationalization.

Example 4.12. Let $X = \{a, b, c\}$ and suppose $a \succ^C b$, $b \succ^C c$, and $c \succ^C a$. This strict cycle violates GARP and cannot be generated by a transitive preference relation.

The separation between the abstract choice language and the budget-set language is deliberate. In Chapter 3, prices and wealth impose the additional linear-budget structure, and the Afriat inequalities provide a finite certificate that can be checked mechanically.

4.7 Problems

Part I: By hand

Problem 4.1 [Core]. Incomplete preferences and multiple utilities.

Let $X = \{a, b, c\}$ and let R contain all reflexive pairs and the comparison aRb, but no other comparison between distinct alternatives.

(a) Show that R is a preorder but is incomplete.

(b) Prove that no real-valued utility function can represent R in the sense xRy if and only if $u(x) \geq u(y)$.

(c) Construct two utility functions u_1, u_2 such that xRy if and only if $u_k(x) \geq u_k(y)$ for both $k = 1, 2$.

Part II: A menu-design rationality audit — chapter use case and verified scale-up

Problem 4.3 [Proof]. Maximization and WGARP.

Let R be complete and transitive on a finite set X. For each nonempty menu $A \subseteq X$, let $c(A)$ be the set of R-greatest elements of A.

(a) Prove that $c(A)$ is nonempty.

(b) Suppose $x \in c(A)$, $y \in c(B)$, and $x, y \in A \cap B$. Prove that xI_Ry, where xI_Ry means xRy and yRx.

(c) Explain why this conclusion is WGARP, the appropriate weak axiom for a set-valued choice correspondence, and distinguish it from WARP for single-valued demand.

Chapter use case

Seed: Problem 4.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of observed menus on the fixed set $X = \{a, b, c\}$. Tool: Codex + Lean. Verification: Recover the three-menu cycle and the one-observation repair; return an exact cycle certificate or a complete transitive rationalization. The current pairwise-WARP theorem does not exclude this three-cycle; the general checker theorem is a new target.

Problem 4.2 [Core]. Choice cycles on finite menus.

Let $X = \{a, b, c\}$ and suppose a single-valued choice function satisfies $c(\{a, b\}) = a$, $c(\{b, c\}) = b$, and $c(\{a, c\}) = c$. Say that a preference relation R rationalizes these data when the observed alternative is the unique R-greatest element of each observed menu.

(a) Write the direct revealed-preference comparisons generated by these choices.

(b) Show that no transitive preference relation can rationalize the data.

(c) State the smallest change to one observation that restores acyclicity, and exhibit a complete and transitive rationalization of the revised data.

Part II

Risk, Stochastic Choice, and Attention

Chapter 5

Expected Utility and Risk

Expected utility adds two ingredients to the choice framework: objective lotteries and a mixture operation. The chapter first gives the finite von Neumann--Morgenstern model and its affine uniqueness, then studies risk attitudes, stochastic dominance, portfolio choice, and revealed tests. Infinite lottery spaces are treated last because their topology and integrability assumptions are technical extensions rather than part of the finite core.

5.1 Primitives

Let $Z \subseteq \mathbb{R}$ be a finite set of monetary prizes. A lottery is a probability mass function $\mu : Z \to [0,1]$ satisfying $\sum_{z \in Z} \mu(z) = 1$. Write $\Delta(Z)$ for the set of lotteries and δ_z for the degenerate lottery assigning probability one to z. Mixtures are defined pointwise by

$$[\alpha\mu + (1-\alpha)\nu](z) = \alpha\mu(z) + (1-\alpha)\nu(z), \qquad \alpha \in [0,1].$$

For a Bernoulli utility $u : Z \to \mathbb{R}$, expected utility is $U(\mu) = \sum_{z \in Z} u(z)\mu(z)$.

5.2 Expected-Utility Representation

We observe a choice correspondence on a family of subsets of $\Delta(Z)$ and represent the decision maker's preference by a relation $\succeq$ on lotteries.

Definition 5.1. Expected utility representation. The preference relation has an expected utility representation if there is a Bernoulli utility $u : Z \to \mathbb{R}$ such that $\mu \succeq \nu \iff \sum_{z \in Z} u(z)\mu(z) \geq \sum_{z \in Z} u(z)\nu(z)$.

5.3 Independence

Assume throughout this section that $\succeq$ is complete and transitive.

Axiom 5.2. *(Independence Axiom) For all $\mu, \nu, \eta \in \Delta(Z)$ and $\alpha \in [0,1]$, $\mu \succeq \nu$ implies $\alpha\mu + (1-\alpha)\eta \succeq \alpha\nu + (1-\alpha)\eta$.*

The other axiom that we need is more technical

Axiom 5.3. *(Mixture continuity) For all $\mu, \nu, \eta \in \Delta(Z)$, if $\mu \succ \nu \succ \eta$, then the set $\{\alpha \in [0,1] | \alpha\mu + (1-\alpha)\eta \succ \nu\}$ and $\{\alpha \in [0,1] | \nu \succ \alpha\mu + (1-\alpha)\eta\}$ are open in $[0,1]$.*

5.4 The Finite Expected-Utility Theorem

Theorem 5.4. *The rational preference relation $\succeq$ satisfies the Independence Axiom and Mixture continuity if and only if there exists a (Bernoulli) utility function $u : Z \to \mathbb{R}$ such that: $\mu \succeq \nu$ if and only if $\sum_{z \in Z} u(z)\mu(z) \geq \sum_{z \in Z} u(z)\nu(z)$.*

First we prove that there is an affine representation of $\succeq$.

Lemma 5.5. *If the preference relation $\succeq$ satisfies the Independence Axiom and mixture continuity then there is an affine function $U : \Delta(Z) \to \mathbb{R}$ that represents $\succeq$.*

Proof. Choose best and worst degenerate lotteries $\mu^+ = \delta_{z^+}$ and $\mu^- = \delta_{z^-}$ among the finitely many prizes. Repeated use of independence shows $\mu^+ \succeq \mu \succeq \mu^-$ for every lottery μ. If $\mu^+ \sim \mu^-$, all lotteries are indifferent and a constant affine representation works. Otherwise, independence implies

$$\beta\mu^+ + (1-\beta)\mu^- \succ \alpha\mu^+ + (1-\alpha)\mu^- \quad \iff \quad \beta > \alpha.$$

For a fixed μ, mixture continuity makes the two sets of mixture weights giving an outcome weakly above or weakly below μ closed. They cover the connected interval $[0,1]$ and contain opposite endpoints, so they intersect. Strict monotonicity along the best--worst segment gives a unique α_μ with

$$\mu \sim \alpha_\mu \mu^+ + (1-\alpha_\mu)\mu^-.$$

Thus $U(\mu) = \alpha_\mu$ represents $\succeq$. Finally, independence permits substitution of the equivalent best--worst mixtures for μ and ν, yielding

$$U(\beta\mu + (1-\beta)\nu) = \beta U(\mu) + (1-\beta)U(\nu).$$

Hence U is affine on the lottery simplex.

□

The second step is to show that an affine representation on the lottery simplex has the expected-utility form.

Lemma 5.6. *Let $U : \Delta(Z) \to \mathbb{R}$. Then U is affine if and only if there exists a $u : Z \to \mathbb{R}$ such that $U(\mu) = \sum_{z \in Z} u(z)\mu(z)$.*

For finite Z, no functional-analytic representation theorem is needed. Define $u(z) = U(\delta_z)$. Since every lottery has the finite decomposition $\mu = \sum_{z \in Z} \mu(z)\delta_z$, affinity of U gives $U(\mu) = \sum_{z \in Z} u(z)\mu(z)$. The converse follows immediately from distributivity.

Example 5.7. Let $Z = \{0, 10\}$, and let $\mu \in \Delta(Z)$ satisfy $\mu(0) = \mu(10) = \frac{1}{2}$. Then $U(\mu) = \frac{1}{2}u(10) + \frac{1}{2}u(0)$. Normalize the Bernoulli index by $u(0) = 0$ and $u(10) = 1$. The normalized expected utility is therefore $U(\mu) = 1/2$. Any positive affine transformation of this index represents the same preference.

5.5 Affine Uniqueness and Recoverability

Theorem 5.8. *Let $U : \Delta(Z) \to \mathbb{R}$ and $U' : \Delta(Z) \to \mathbb{R}$ be affine maps that represent $\succeq$ on $\Delta(Z)$, then there exist real numbers $\alpha > 0$ and b such that $U'(\mu) = \alpha U(\mu) + b$ for all $\mu \in \Delta(Z)$. If u, v are Bernoulli utilities such that $u \cdot \mu$ and $v \cdot \mu$ represent $\succeq$ then there are real numbers $\alpha > 0$ and b such that $v \cdot \mu = \alpha u \cdot \mu + b$.*

Affine uniqueness is stronger than ordinal uniqueness in deterministic choice. For an ordinary utility representation, every strictly increasing transformation preserves the ranking. Under expected utility, mixture linearity restricts admissible transformations to positive affine maps; a nonlinear increasing transformation generally destroys the independence representation.

5.6 The Allais Paradox

In this example, $Z = \{z_1, z_2, z_3\}$ with the ordered coordinate list $(z_1, z_2, z_3) = (2.5, 0.5, 0)$ million dollars and $N := |Z| = 3$. A lottery vector lists probabilities in this order, and $u_z := u(z)$.

First Prize	Second Prize	Third Prize
2.5 million	0.5 million	0

Two choice sets:

The first,

$L_1 = (0, 1, 0)$ and $L_1' = (0.10, 0.89, 0.01)$.

The second,

$L_2 = (0, 0.11, 0.89)$ and $L_2' = (0.10, 0, 0.90)$.

Some individuals choose $L_1 \succ L_1'$ and $L_2' \succ L_2$.

This is inconsistent with expected utility.

If $L_1 \succ L_1'$ then

$$u_{0.5} > (0.10)u_{2.5} + (0.89)u_{0.5} + (0.01)u_0.$$

Subtracting $(0.89)u_{0.5}$ and adding $(0.89)u_0$ to both sides yields

$$(0.11)u_{0.5} + (0.89)u_0 > (0.1)u_{2.5} + (0.9)u_0,$$

and thus $L_2 \succ L_2'$.

5.7 Attitudes towards Risk

Assume that Bernoulli utility is defined on the convex hull $\text{co}(Z)$ of the finite monetary-prize set. For a lottery $\mu \in \Delta(Z)$, its expected monetary value is $\overline{z}(\mu) = \sum_{z \in Z} z\mu(z) \in \text{co}(Z)$.

Definition 5.9. Risk aversion under expected utility. A decision maker is risk averse on this finite lottery domain if $u(\overline{z}(\mu)) \geq U(\mu)$ for every lottery $\mu \in \Delta(Z)$. By Jensen's inequality, concavity of the Bernoulli utility implies this condition:

$$\sum_{z \in Z} \mu(z)u(z) \leq u\left(\sum_{z \in Z} \mu(z)z\right).$$

If the same comparison is required for every finitely supported lottery with prizes in $\text{co}(Z)$, applying it to arbitrary two-point lotteries also makes concavity necessary.

Certainty equivalent and risk premium. If u is strictly increasing and its range contains $U(\mu)$, the certainty equivalent is the prize $\text{CE}(\mu; u) = u^{-1}(U(\mu))$ and the (monetary) risk premium is $\text{RP}(\mu; u) = \overline{z}(\mu) - \text{CE}(\mu; u)$. A risk-averse expected-utility decision maker has $\text{RP}(\mu; u) \geq 0$.

Probability premium. For prizes $x - \varepsilon, x, x + \varepsilon$ in the domain of u, $\text{PP}(x, \varepsilon; u)$ is defined by

$$u(x) = \left(\frac{1}{2} + \text{PP}(x, \varepsilon; u)\right) u(x + \varepsilon) + \left(\frac{1}{2} - \text{PP}(x, \varepsilon; u)\right) u(x - \varepsilon).$$

It is the change in probability needed to make the symmetric two-point lottery equivalent to the sure prize.

Proposition 5.10. *For a twice differentiable, strictly increasing Bernoulli utility, the local risk premium for a symmetric small spread satisfies*

$$\mathrm{RP}\big(\tfrac{1}{2}\delta_{x-\varepsilon} + \tfrac{1}{2}\delta_{x+\varepsilon}; u\big) = -\frac{u''(x)}{2u'(x)}\varepsilon^2 + o(\varepsilon^2).$$

Thus local concavity corresponds to a nonnegative local risk premium.

5.8 First-Order Stochastic Dominance: Finite Support

Definition 5.11. First-order stochastic dominance. For lotteries $\mu, \nu \in \Delta(Z)$ with cumulative distribution functions $F_\mu(t) = \sum_{z \le t} \mu(z)$ and $F_\nu(t) = \sum_{z \le t} \nu(z)$, write $\mu \succeq^{\mathrm{FOSD}} \nu$ if $F_\mu(t) \le F_\nu(t)$ for every $t \in \mathbb{R}$. The relation is generally incomplete.

Theorem 5.12. *If $\mu \succeq^{\mathrm{FOSD}} \nu$ and u is increasing, then $U(\mu) \ge U(\nu)$ for expected utility U. If the prizes are ordered as $Z = \{z_1 < \cdots < z_N\}$, strict inequality follows whenever $F_\mu(z_k) < F_\nu(z_k)$ and $u(z_{k+1}) > u(z_k)$ for at least one cutoff $k < N$.*

5.9 Portfolio Choice: Demand for a Risky Asset

An asset is a divisible claim to a financial return in the future. Assume that u is continuously differentiable, strictly increasing, and concave on an interval containing every feasible terminal-wealth realization. Assume also that the return and marginal-utility expressions below are integrable.

Let α, β denote the amounts of money invested in a risky and safe asset respectively. The risky asset pays a random gross return $\widetilde{R}$ dollars per dollar invested and the safe asset pays 1 dollar with certainty per dollar invested. The random return has cumulative distribution function $F_R(r)$ We assume that $\mathbb{E}[\widetilde{R}] = \int r\, dF_R(r) > 1$, or that the risky asset expected return exceeds the return of the safe asset. Then the total return is $\alpha\widetilde{R} + \beta$. Feasibility requires $\alpha + \beta = w$.

$$\max_{\alpha,\beta \ge 0} \int u(\alpha r + \beta)\, dF_R(r)$$

$$\text{subject to} \quad \alpha + \beta = w.$$

Substituting $\beta = w - \alpha$ reduces the problem to

$$\max_{0 \le \alpha \le w} \int u(w + \alpha(r-1))\, dF_R(r).$$

Define the derivative of the objective on $[0, w]$ by

$$\phi(\alpha) = \int u'(w + \alpha[r-1])[r-1]\, dF_R(r).$$

The necessary and sufficient first-order conditions for an optimum α^* are

$$\begin{cases} \phi(\alpha^*) \le 0 & \text{if } \alpha^* < w, \\ \phi(\alpha^*) \ge 0 & \text{if } \alpha^* > 0. \end{cases}$$

Because $u'(w) > 0$ and $\int r\, dF_R(r) > 1$, we have $\phi(0) = u'(w) \int (r-1)\, dF_R(r) > 0$. Hence $\alpha^* = 0$ cannot satisfy the first-order conditions.

If a risk is actuarially favorable, then a risk-averse expected-utility maximizer will accept at least a sufficiently small amount under the stated differentiability and interior-wealth assumptions.

5.10 Revealed Expected Utility under Risk Aversion

Let $\mathcal{S}$ be a finite set of states with fixed probabilities $\varpi_s > 0$, satisfying $\sum_{s \in \mathcal{S}} \varpi_s = 1$, and let a contingent consumption plan be $x = (x_s)_{s \in \mathcal{S}} \in \mathbb{R}^{\mathcal{S}}_{++}$. At prices $p \in \mathbb{R}^{\mathcal{S}}_{++}$ and wealth $w > 0$, the consumer solves $\max_x \sum_{s \in \mathcal{S}} \varpi_s u(x_s)$ subject to $\sum_{s \in \mathcal{S}} p_s x_s \le w$, where u is continuous, strictly increasing and concave, with positive marginal utility on the observed consumption range whenever derivatives are invoked.

Proposition 5.13. *Interior first-order conditions. If u is differentiable and an optimum is interior, then there is $\lambda > 0$ such that $\varpi_s u'(x_s) = \lambda p_s$ for every $s \in \mathcal{S}$. Concavity makes these conditions sufficient for a global optimum; boundary observations require the corresponding KKT inequalities.*

Definition 5.14. Risk-averse expected-utility rationalization. A finite dataset $\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$ is rationalized if there is one continuous, strictly increasing and concave Bernoulli utility with positive marginal utility on the observed range u such that each x^t maximizes $\sum_s \varpi_s u(x_s)$ on the budget set $\{x : p^t \cdot x \le p^t \cdot x^t\}$.

Theorem 5.15. *Finite-data test. See[Polisson et al., 2020]. Assume all observations are interior*

and $\varpi_s > 0$. The dataset has a risk-averse expected-utility rationalization if and only if there are numbers $\beta_s^t > 0$ and $\lambda^t > 0$ that satisfy the price equations

$$\varpi_s \beta_s^t = \lambda^t p_s^t \qquad \text{for every } s, t,$$

and the monotonicity inequalities

$$x_s^t > x_{s'}^{t'} \quad \Longrightarrow \quad \beta_s^t \leq \beta_{s'}^{t'}$$

for every pair of observed state-contingent quantities. The monotonicity condition is the finite signature of a single decreasing marginal-utility function; the price equations are the common first-order conditions.

The theorem is an Afriat-style interpolation result for one Bernoulli utility across states and observations. It is stated for interior data to keep the notation transparent; zero quantities are handled by replacing equalities with KKT inequalities.

5.11 Experimental Evidence on Expected Utility

Polisson and Quah develop a revealed preferences test for expected utility. They apply their framework to the famous Choi, Fisman, Gale and Kariv (2007) dataset and find the following passing rates.

- GARP is the benchmark revealed-preference test.
- RDEU denotes rank-dependent expected utility.
- EU denotes expected utility.
- EU* imposes concavity of the Bernoulli utility function in addition to expected utility.

The"drop" column reports how many observations the test permits the researcher to discard.

Observations dropped	GARP	RDEU	EU	EU*
0	12/47 (26%)	2/47 (4%)	2/47 (4%)	0/47 (0%)
1	14/47 (30%)	7/47 (15%)	5/47 (11%)	0/47 (0%)
2	27/47 (57%)	9/47 (19%)	7/47 (15%)	2/47 (4%)
3	32/47 (68%)	12/47 (26%)	10/47 (21%)	2/47 (4%)

Figure 5.1: Share of 47 subjects passing each revealed-preference test after dropping up to three observations [Polisson and Quah, 2013a].

5.12 Measuring Risk Aversion in the Lab

The Holt and Laury multiple-price-list task classifies each subject by the total number of safe (Option A) choices across ten decisions.

Assume constant relative risk aversion (CRRA), with coefficient r: $u(x) = x^{1-r}/(1-r)$. At $r = 1$, use $u(x) = \log x$. A monotone pattern with k initial Option A choices followed by Option B implies an interval for r.

Number of safe choices	CRRA interval for $U(x) = x^{1-r}/(1-r)$	Proportion of choice patterns	
		Low real ($\times 1$)[a]	Real ($\times 20$)
0–1	$r < -0.95$	.01	.01
2	$-0.95 < r < -0.49$	.01	.01
3	$-0.49 < r < -0.15$	.06	.04
4	$-0.15 < r < 0.15$	.26	.13
5	$0.15 < r < 0.41$	.26	.19
6	$0.41 < r < 0.68$	.23	.23
7	$0.68 < r < 0.97$	.13	.22
8	$0.97 < r < 1.37$	.03	.11
9–10	$1.37 < r$	.01	.06

[a] Average over the first and second low-real decisions.

Figure 5.2: Distribution of the number of safe choices in the Holt--Laury task. The final two columns give proportions of observed choice patterns classified by total number of Option A choices under low real ($\times 1$) and real $\times 20$ payoffs. The low-real column averages the first and second low-real decisions [Holt and Laury, 2002].

Individuals choose between A and B in each of the ten rows.

Each option has two monetary outcomes, A_1, A_2 and B_1, B_2; their common high-outcome probability rises by one tenth from one row to the next.

The expected payoff of A exceeds that of B in rows 1--4; the inequality reverses in rows 5--10.

- A risk-neutral expected-utility maximizer chooses A in rows $1, \dots, 4$ and B in rows $5, \dots, 10$.
- Option B has higher variance in each nondegenerate row (rows 1--9).
- Under expected utility and CRRA, five or more initial safe choices imply $r > 0$, and a later single switch to B corresponds to a larger risk-aversion parameter. Choosing A in row 10 violates monotonicity because both options are then certain and $3.85 > 2.00$.

One row is selected at random for payment after all choices are recorded. This random-lottery

incentive system makes every row potentially payoff relevant; interpreting the switching row as a risk-aversion measure additionally requires expected utility, monotonicity, and the stated payoff probabilities.

5.13 Expected Utility on Infinite Outcome Spaces

Let Z be a compact metric prize space and let Σ be its Borel σ-algebra. Recall that a collection of subsets is a σ-algebra when

- $\emptyset \in \Sigma$
- if $E \in \Sigma$ then $E^c \in \Sigma$
- If $\{A_n\}$ is a countable collection of sets such that $A_n \in \Sigma$ then $\cup_n A_n \in \Sigma$.

We say that (Z, Σ) is a measurable space whenever Σ is a σ-algebra.

Definition 5.16. We say that $\mu : \Sigma \to [0, 1]$ is a probability measure if:

- $\mu(Z) = 1$
- if $\{A_n\}$ is a countable collection of pairwise disjoint sets such that $A_n \in \Sigma$, then $\mu(\cup_n A_n) = \sum_n \mu(A_n)$.

We call the collection of probability measures $\Delta(Z, \Sigma)$ for short $\Delta(Z)$ when Σ is fixed.

$\alpha\mu + (1-\alpha)\nu \in \Delta(Z)$ is defined as $(\alpha\mu + (1-\alpha)\nu)(E) = \alpha\mu(E) + (1-\alpha)\nu(E)$.

Equip this space with weak convergence: $\mu^n \Rightarrow \mu$ if $\int f \, d\mu^n \to \int f \, d\mu$ for continuous and bounded functions f.

Axiom 5.17. *(Weak continuity) For any $\mu \in \Delta(Z)$ the sets $\{\nu \in \Delta(Z) : \nu \succ \mu\}$ and $\{\nu \in \Delta(Z) : \mu \succ \nu\}$ are open in the weak topology.*

Theorem 5.18. *Let Z be a compact metric prize space and $\Delta(Z)$ the Borel probability measures with the weak topology. A rational preference relation $\succeq$ on $\Delta(Z)$ satisfies the independence axiom and weak continuity if and only if there exists a $u : Z \to \mathbb{R}$ bounded and continuous such that $U(\mu) = \int u(z) d\mu(z)$ represents $\succeq$.*

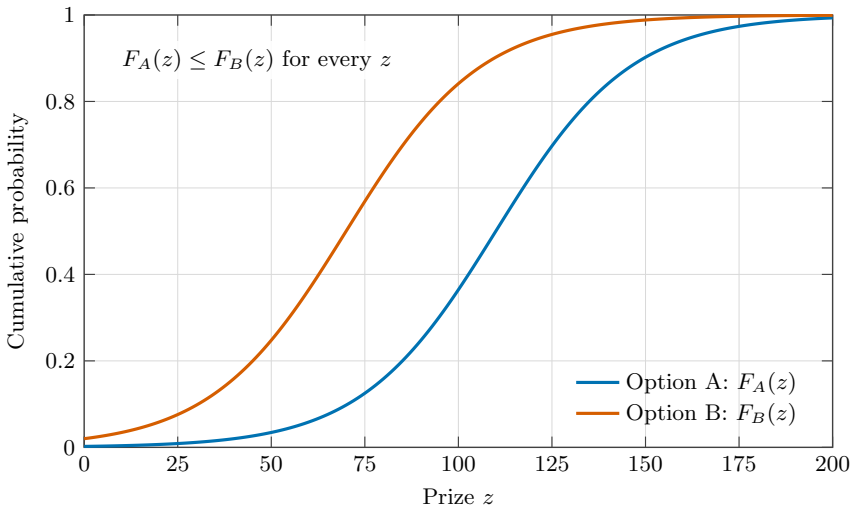


Figure 5.3: First-order stochastic dominance. Option A is a rightward location shift of Option B, so $F_A(z) \le F_B(z)$ for every payoff z.

5.14 First-Order Stochastic Dominance: General Distributions

This section extends the finite-prize result proved earlier in the chapter to probability distributions with integrable utility.

Definition 5.19. Let F and G be cumulative distribution functions on $\mathbb{R}$. Write $F \succeq^{\text{FOSD}} G$ when $F(r) \le G(r)$ for every $r \in \mathbb{R}$; strict dominance means that the inequality is strict at some continuity point.

Theorem 5.20. *First-order stochastic dominance. If $F \succeq^{\text{FOSD}} G$, then $\int u\,dF \ge \int u\,dG$ for every nondecreasing measurable Bernoulli utility u for which both expectations are finite. If $F \neq G$ and u is strictly increasing, the inequality is strict.*

5.15 Problems

Part I: By hand

Problem 5.1 [Core]. Affine expected-utility representations.

Let $Z = \{z_1, \dots, z_n\}$ and suppose $U(\mu) = \sum_j \mu_j u(z_j)$ represents a nontrivial preference on $\Delta(Z)$.

(a) Show directly that U is affine in lotteries.

(b) Prove that $U' = aU + b$ with $a > 0$ represents the same preference.

(c) Show that if two expected-utility representations agree on preferences, their Bernoulli indices differ by a positive affine transformation.

Problem 5.2 [Core]. First-order stochastic dominance.

Lotteries μ and ν on prizes $\{0, 1, 2\}$ have probabilities $\mu = (0.2, 0.3, 0.5)$ and $\nu = (0.3, 0.4, 0.3)$.

(a) Compare their cumulative distribution functions and determine which lottery first-order stochastically dominates the other.

(b) Prove the expected-utility ranking for every increasing Bernoulli utility by discrete summation by parts.

(c) Explain why strict dominance need not imply a strict preference for a weakly increasing utility.

Part II: A multi-asset portfolio under risk — chapter use case and verified scale-up

Problem 5.3 [Proof]. The Arrow-Pratt approximation.

Let $\varepsilon_\sigma = \sigma\xi$, where $\mathbb{E}\xi = 0$, $\mathbb{E}\xi^2 = 1$, and ξ is bounded. An agent with wealth w has utility u that is twice continuously differentiable near w, with $u'(w) > 0$. For all sufficiently small $\sigma \geq 0$, local invertibility gives a unique solution near zero to the indifference equation below. Define the monetary risk premium $\mathrm{RP}(\sigma)$ to be that local branch, characterized by $\mathrm{RP}(\sigma) \to 0$ as $\sigma \downarrow 0$:

$u(w - \mathrm{RP}(\sigma)) = \mathbb{E}[u(w + \varepsilon_\sigma)]$.

(a) Use second-order Taylor expansions to derive the approximation

$\mathrm{RP}(\sigma) = \frac{1}{2}A(w)\sigma^2 + o(\sigma^2)$,

where $A(w) = -u''(w)/u'(w)$.

(b) State the regularity and small-risk qualifications needed for this approximation.

(c) Explain why the formula is invariant to positive affine transformations of u.

Chapter use case

Seed: Problems 5.3, 5.4. Fixed: all unlisted primitives and assumptions. Scaled: the number of jointly normal risky assets from one to K, retaining CARA utility and unconstrained positions. Tool: Codex + Python. Verification: Recover the one-asset formula; verify the first-order condition and concavity using the covariance matrix; report singular-covariance cases explicitly and do not add portfolio constraints silently.

Problem 5.4 [Applied]. CARA-normal portfolio demand.

Terminal wealth is $W_T = wR_f + \theta(\widetilde{R} - R_f)$, where $\widetilde{R} \sim N(m_R, \sigma^2)$, R_f is risk free, $\theta \in \mathbb{R}$ (borrowing and short selling are unrestricted), and utility is $u(W_T) = -\exp(-aW_T)$ with $a > 0$.

(a) Show that maximizing expected utility is equivalent to maximizing a certainty equivalent.

(b) Derive optimal risky investment θ^*.

(c) Give comparative statics with respect to $m_R - R_f$, a, and σ^2.

Chapter 6

Stochastic Choice and Random Utility

Deterministic choice predicts a set of maximizers from each menu; stochastic choice records a probability distribution over that menu. We first define a coherent choice-probability system on a fixed finite universe, then ask when it is a mixture of deterministic rankings. To connect this abstract domain to consumer theory, we next treat complete commodity bundles as the alternatives. A divisible outside good supplies the cost-efficiency bridge of Polisson and Quah; quasilinearity converts each bundle into net surplus; only then do we add preference shocks. Multinomial logit is a final distributional specialization, not the definition of random utility.

6.1 Stochastic Choice on a Finite Universe

Let Y be a finite universe of alternatives and let $\mathcal{B} = 2^Y \setminus \{\varnothing\}$ be its collection of nonempty menus. A stochastic choice rule is a family $P = \{P_B\}_{B \in \mathcal{B}}$ with $P_B \in \Delta(B)$ for every menu. Thus $P_B(a) \geq 0$ for $a \in B$, $\sum_{a \in B} P_B(a) = 1$, and unavailable alternatives receive probability zero. The family, rather than a probability from only one menu, is the empirical object to be rationalized. A deterministic choice function is the special case in which every P_B is degenerate.

6.2 Random Utility Representation

Definition 6.1. Random-utility representation. Let Y be the finite universe, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\boldsymbol{U} : \Omega \to \mathbb{R}^Y$ be a measurable random utility vector. For every nonempty

menu $B \subseteq Y$, the induced choice probabilities are $P_B(a) = \mathbb{P}\{\boldsymbol{U}(a) > \boldsymbol{U}(b) \text{ for every } b \in B \setminus \{a\}\}$. Assume either that ties occur with probability zero or that a measurable tie-breaking rule is specified; under the zero-tie convention the displayed strict-max probabilities sum to one.

When choice probabilities are observed on every nonempty menu, the deterministic types form the finite set $\mathcal{R}_Y$ of strict linear orders on Y. Stacking the probabilities gives $\mathbf{P} = M\lambda$, where $\lambda \in \Delta(\mathcal{R}_Y)$ is a distribution over those orders and $M_{(a,B),r} = \mathbb{I}\{a \text{ is } r\text{-best in } B\}$. The finite feasibility condition $\mathbf{P} = M\lambda$ with $\lambda \geq 0$ and $\mathbf{1}_{|\mathcal{R}_Y|}^{\top}\lambda = 1$ is equivalent to the existence of such a distribution.

6.3 Discrete Bundles, a Numeraire, and Cost Efficiency

Give the abstract alternative universe an economic interpretation. Let $X^{\mathrm{d}} \subseteq \mathbb{R}_+^K$ be a possibly discrete consumption domain and let $Y = \{x^0, \ldots, x^J\} \subseteq X^{\mathrm{d}}$ be the finite catalogue relevant for the choice problem. An alternative is now a complete inside-good bundle x^j, not one coordinate of a bundle. At inside-good prices $p \gg 0$ and wealth $w > 0$, the induced menu is

$$B_X(p, w) := \{x^j \in Y : p \cdot x^j \leq w\}.$$

Budget-generated menus are a structured subclass of the arbitrary menus in $\mathcal{B}$. If the economically possible domain is larger than Y, conclusions drawn from the finite catalogue are conditional on the omitted bundles being unavailable.

Add a divisible outside good $z \in \mathbb{R}_+$ with price $q > 0$. Polisson and Quah study the weakly separable problem

$$\max_{(x,z) \in X^{\mathrm{d}} \times \mathbb{R}_+} \Phi(v(x), z) \quad \text{subject to} \quad p \cdot x + qz \leq w,$$

where Φ is strictly increasing in both arguments; see [Polisson and Quah, 2013b]. The inside subutility $v : X^{\mathrm{d}} \to \mathbb{R}$ need not be differentiable, concave, or increasing.

Proposition 6.2. *Cost efficiency on a discrete domain. If (x^*, z^*) solves the displayed problem, then for every $x \in X^{\mathrm{d}}$:*

$$p \cdot x = p \cdot x^* \quad \Longrightarrow \quad v(x^*) \geq v(x),$$

$$p \cdot x < p \cdot x^* \quad \Longrightarrow \quad v(x^*) > v(x).$$

Consequently, price--bundle observations generated by this problem satisfy GARP even when X^{d} is discrete.

Proof. If an equal-cost bundle had strictly higher subutility, replacing x^* by that bundle while keeping z^* would be feasible and strictly better. If a strictly cheaper bundle x had $v(x) \geq v(x^*)$, the consumer could choose x and increase the outside good to $z^* + p \cdot (x^* - x)/q$, again obtaining strictly higher utility. Along a direct revealed-preference chain these two implications give weakly decreasing subutility, with a strict decrease whenever the closing expenditure comparison is strict; a GARP cycle is therefore impossible. □

This is the deterministic bridge from commodity bundles to discrete alternatives. It uses weak separability and a strictly valued divisible outside good; it does **not** assume quasilinearity and does not yet introduce shocks. Because the outside good is strictly valued, its conditional quantity is $z = (w - p \cdot x)/q$ and the induced deterministic value of an affordable bundle is

$$V(x; p, q, w) = \Phi\left(v(x), \frac{w - p \cdot x}{q}\right).$$

In the general weakly separable model, both wealth and prices can change the ranking of two bundles.

6.3.1 The Quasilinear Specialization

Now impose the stronger cardinal structure $\Phi(r, z) = r + z$. Write normalized inside prices as $\widetilde{p} = p/q$. For each affordable bundle,

$$v(x) + \frac{w - p \cdot x}{q} = \frac{w}{q} + \underbrace{\left[v(x) - \widetilde{p} \cdot x\right]}_{\delta_{\widetilde{p}}(x)}.$$

The common term w/q drops from rankings, so deterministic choice reduces exactly to

$$x^* \in \arg \max_{x \in B_X(p,w)} \delta_{\widetilde{p}}(x).$$

Quasilinearity removes wealth from rankings on a fixed menu; it does not remove the affordability effect of wealth on $B_X(p, w)$. This is the same boundary qualification encountered for quasilinear Marshallian demand in Chapter 1.

The fixed-numeraire assumption is testable. If one function v and normalized prices $\widetilde{p}^t$ generate choices x^t, then every cycle $t_1, \ldots, t_M$, with $t_{M+1} = t_1$, satisfies

$$\sum_{\ell=1}^{M} \widetilde{p}^{t_\ell} \cdot \left(x^{t_\ell} - x^{t_{\ell+1}}\right) \leq 0.$$

This cyclical-monotonicity inequality follows by summing the optimality inequalities around the cycle. It is stronger than GARP. Polisson and Quah recover the more general GARP converse by allowing the outside-good price---equivalently, the Afriat multiplier---to vary across observations.

The next step is cardinal, not merely ordinal. An increasing transformation of v preserves deterministic choices but generally changes utility differences and therefore changes probabilities once additive shocks are introduced. A stochastic specification must normalize the location and scale of utility jointly with the shock distribution.

6.3.2 Add Shocks Only after the Economic Reduction

Let consumers be indexed by i and environments by t. For bundle $x^j \in B_i^t := B_X(p^t, w_i^t)$, define its cost in units of the outside good and its deterministic surplus by

$$c_j^t := \frac{p^t \cdot x^j}{q_t}, \qquad \delta_j^t := v(x^j) - c_j^t.$$

Add an alternative-specific preference shock only now:

$$U_{itj}^* = \delta_j^t + \varepsilon_{ij}.$$

The omitted term w_i^t/q_t is common to every feasible bundle. The resulting conditional probability is

$$P_{ij}^t = \Pr\left\{\delta_j^t + \varepsilon_{ij} \geq \delta_k^t + \varepsilon_{ik} \text{ for every } x^k \in B_i^t\right\}.$$

Assume either atomless pairwise shock differences or a specified measurable tie rule. A joint density is sufficient; continuity of each marginal alone is not sufficient under arbitrary dependence.

If the full shock vector is a persistent preference type and has one distribution independent of (p^t, q_t, w_i^t), the Polisson--Quah argument applies realization by realization: every type follows a cost-efficient deterministic bundle demand, and population shares mix those types. If shocks are redrawn independently at every occasion, a realized person's path need not satisfy GARP. In either interpretation, Polisson--Quah supply the deterministic bundle bridge; stochastic rationalizability still requires the separate random-utility restrictions developed in this chapter.

The familiar differentiated-product index is nested in this construction. If alternative j is the one-unit bundle $x^j = e_j$, then $p^t \cdot x^j = p_j^t$. With attributes g_j and $v(x^j) = g_j^\top \beta$, the latent index becomes $g_j^\top \beta - p_j^t/q_t + \varepsilon_{ij}$. Thus the price term is derived from the consumer's budget rather than inserted as an unexplained attribute.

Example 6.3. Three bundles and two price environments. Let $Y = \{x^0, x^1, x^2\}$ with $x^0 = (0,0)$, $x^1 = (1,0)$, $x^2 = (0,1)$, and $(v_0, v_1, v_2) = (0,3,2)$. Normalize $q = 1$ and take wealth high enough that all three bundles are feasible. At $p^A = (1,1)$, costs are $(0,1,1)$ and deterministic surpluses are $(0,2,1)$, so deterministic choice is x^1. At $p^B = (2,1/2)$, surpluses are $(0,1,3/2)$, so deterministic choice switches to x^2. With independent Type-I extreme-value shocks of scale one, the corresponding probability vectors are approximately $(0.090, 0.665, 0.245)$ and $(0.122, 0.332, 0.547)$. Expected bundle demand is therefore approximately $D(p^A) = (0.665, 0.245)$ and $D(p^B) = (0.332, 0.547)$, and $[D(p^A) - D(p^B)]^\top (p^A - p^B) \approx -0.485 \leq 0$. The shocks smooth the switch; they do not erase the bundle-level opportunity cost that caused it.

6.4 Multinomial Logit and IIA

Suppose the shocks are independent and identically distributed Type-I extreme value with scale $\tau > 0$. Conditional on the feasible bundle menu, the multinomial-logit probabilities are

$$P_{ij}^t = \frac{\exp(\delta_j^t/\tau)}{\sum_{x^k \in B_i^t} \exp(\delta_k^t/\tau)}, \qquad x^j \in B_i^t.$$

Proposition 6.4. *Independence of irrelevant alternatives. Under multinomial logit, for any two alternatives j, k that remain available, $\frac{P_{ij}^t}{P_{ik}^t} = \exp((\delta_j^t - \delta_k^t)/\tau)$ is independent of the other alternatives in the menu.*

Example 6.5. Red-bus/blue-bus. If walking and a red bus initially have equal probabilities, adding a blue bus that is nearly identical to the red bus should split the bus share while leaving walking's share close to its original value. Multinomial logit instead preserves the walking-to-red-bus odds ratio, illustrating the empirical content and limitation of IIA.

Neither IIA nor the extreme-value distribution follows from the Polisson--Quah bundle mapping. They are additional restrictions on the shock distribution. The multinomial probit model replaces the extreme-value shocks with a jointly normal vector. Its pairwise differences are generally correlated, so it does not impose IIA.

6.5 Random Expected Utility

Definition 6.6. Random expected utility. For a finite prize set Z and lotteries $\mu \in \Delta(Z)$, a random Bernoulli utility $\boldsymbol{u} : \Omega \to \mathbb{R}^Z$ induces $\boldsymbol{U}_\omega(\mu) = \sum_{z \in Z} \boldsymbol{u}_\omega(z)\mu(z)$. Choice probabilities are obtained by maximizing this random expected utility on each menu of lotteries.

The representation imposes linearity in mixtures and regularity under menu expansion. Extremeness and weak-star continuity are additional conditions in the Gul--Pesendorfer characterization; see [Gul and Pesendorfer, 2006].

6.6 Attributes, Choice Probabilities, and Consumer Surplus

With observed attributes $g = (g_a)_{a\in A}$, define the mean-utility vector $v(g) = (v(a, g_a))_{a\in A} \in \mathbb{R}^A$. A common specification is

$$U(a, g) = v(a, g_a) + \varepsilon_a,$$

where the distribution of $\varepsilon = (\varepsilon_a)_{a\in A}$ is independent of g. For a d-dimensional linear-index model, let $g_a, \beta \in \mathbb{R}^d$ and write $v(a, g_a) = g_a^\top \beta$. Independence is a maintained exogeneity assumption, not a consequence of the random-utility definition. For any mean-utility vector $v = (v(a))_{a\in A} \in \mathbb{R}^A$, define the conditional choice probability $P_A(a \mid v) := \mathbb{P}\{v(a)+\varepsilon_a > v(b)+\varepsilon_b \text{ for every } b \in A\backslash\{a\}\}$ under the same zero-tie convention as above. Conditional on attributes, $P_A(a \mid g)$ is shorthand for $P_A(a \mid v(g))$.

Definition 6.7. Expected maximum utility. The expected value as a function of the mean-utility vector is

$$\mathcal{V}(v) = \mathbb{E}\left[\max_{a\in A}\{v(a) + \varepsilon_a\}\right], \qquad v \in \mathbb{R}^A.$$

If the deterministic utility contains $-\alpha p_a$ with $\alpha > 0$ as the price coefficient, the publication-safe money-metric object is the change between two scenarios, $\Delta CS = [\mathcal{V}(v^1) - \mathcal{V}(v^0)]/\alpha$. The level $\mathcal{V}(v)/\alpha$ depends on the additive utility normalization. If shocks are merely decision errors rather than preference shocks, this expression is not a welfare measure.

Proposition 6.8. *Williams--Daly--Zachary envelope result. See[McFadden, 1981]. If the shock distribution has finite first moments and ties occur with probability zero, then $\mathcal{V}$ is convex and differentiable in the vector of mean utilities, and $\frac{\partial \mathcal{V}(v)}{\partial v(a)} = P_A(a \mid v)$. If choice probabilities are differentiable in mean utilities (equivalently, if $\mathcal{V}$ is twice differentiable at the point), then the Jacobian of choice probabilities is the Hessian of $\mathcal{V}$, hence symmetric and positive semidefinite. Translation invariance implies that the rows and columns sum to zero.*

The bundle construction turns this mean-utility result into a price-space restriction.

Proposition 6.9. *Stochastic law of demand for quasilinear bundles. Fix one feasible catalogue*

$Y = \{x^0, \dots, x^J\}$ across price environments and write the bundle matrix as $\mathbf{X} = [x^0 \ \cdots \ x^J]$. Let $P(\widetilde{p})$ be the vector of additive-random-utility choice probabilities generated by mean surplus $\delta(\widetilde{p}) = v - \mathbf{X}^\top \widetilde{p}$, and let expected commodity demand be $D(\widetilde{p}) = \mathbf{X}P(\widetilde{p})$. Under the assumptions of the Williams--Daly--Zachary result,

$$\big[D(\widetilde{p}) - D(\widetilde{p}')\big]^\top (\widetilde{p} - \widetilde{p}') \le 0.$$

More generally, for every price cycle $\widetilde{p}^1, \dots, \widetilde{p}^T$ with $\widetilde{p}^{T+1} = \widetilde{p}^1$,

$$\sum_{t=1}^{T} D(\widetilde{p}^t)^\top (\widetilde{p}^{t+1} - \widetilde{p}^t) \ge 0.$$

Proof. Convexity of $\mathcal{V}$ makes its gradient P monotone and cyclically monotone in mean surplus. Since $\delta(\widetilde{p}) - \delta(\widetilde{p}') = -\mathbf{X}^\top(\widetilde{p} - \widetilde{p}')$, substituting $D = \mathbf{X}P$ gives the two displayed inequalities. □

The envelope identity is the basis for likelihood and welfare calculations in discrete-choice models. Additional sign restrictions such as monotone substitution patterns require additional assumptions on the shock distribution and utility index; they do not follow from random utility alone. The stochastic law of demand above uses the quasilinear bundle index, a stable shock distribution, and a common feasible catalogue. If wealth or prices change availability, menu variation must be modeled explicitly.

6.7 Problems

Part I: By hand

Problem 6.1 [Core]. Logit odds and IIA.

Choice probabilities satisfy $P(j \mid A) = \exp(\delta_j/\tau)/\sum_{k\in A}\exp(\delta_k/\tau)$, where $\tau > 0$ and δ_j is deterministic net surplus.

(a) Derive the odds ratio $P(j \mid A)/P(k \mid A)$.

(b) Prove the independence-of-irrelevant-alternatives property.

(c) Explain precisely which part of IIA is an empirical restriction and which part is an algebraic implication of the model.

Problem 6.3 [Core]. A finite random-utility mixture.

There are alternatives a, b, c. A population consists of three strict preference types: $a \succ b \succ c$ with mass 0.5, $c \succ a \succ b$ with mass 0.2, and $b \succ a \succ c$ with mass 0.3.

(a) Compute choice probabilities from menus $\{a, b\}$ and $\{a, b, c\}$.

(b) Let the columns of M correspond to all six strict orders on $\{a, b, c\}$, with zero masses allowed. Stack the observed probabilities as $\mathbf{P}$ and write $\mathbf{P} = M\lambda$.

(c) Explain why nonnegativity of λ and $\mathbf{1}_6^\top \lambda = 1$ are the central finite random-utility feasibility restrictions.

Part II: Red bus, blue bus, and duplicated products — chapter use case and verified scale-up

Seed: Problems 6.1, 6.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of observationally identical bus labels under independent-logit shocks. Tool: Codex + Python; Lean for the IIA core. Verification: Recover the one-bus and two-bus probabilities, verify that probabilities sum to one, and separate the verified logit odds result from any shared-shock comparison model.

Problem 6.2 [Applied]. The red-bus blue-bus problem.

Walking and a red bus initially have systematic utilities normalized so that each has probability one half under logit. A blue bus with the same systematic utility as the red bus is added.

(a) Compute the new logit probabilities.

(b) Compare the total bus share before and after entry.

(c) Explain why a model with highly correlated bus-specific shocks would predict a different substitution pattern.

Chapter 7

Limited Attention and Costly Information

The benchmark chapters assume that the decision maker observes the relevant environment and optimizes without cognitive friction. This chapter separates two departures from that benchmark. Sparse-max demand takes distorted or incomplete price perception as given. Rational inattention instead makes information acquisition a choice: the decision maker trades the value of a more informative signal against its cost. Keeping these mechanisms distinct clarifies both their empirical content and their connection to the stochastic-choice models of the preceding chapter.

7.1 The Sparse-Max Decision Rule

Let $X = \mathbb{R}^L_+$, true prices $p \gg 0$, perceived prices $\widehat{p} \gg 0$, and wealth $w > 0$. Assume u is differentiable and that the tangency system below has a solution. A sparse-max choice is obtained by first selecting a bundle that satisfies the perceived-price first-order condition

$$\nabla u(x) = \lambda \widehat{p}, \qquad \lambda > 0,$$

and then choosing λ so that the selected bundle satisfies the actual budget equation $p \cdot x = w$. At a boundary solution, replace the equality by the corresponding KKT inequalities. This two-stage rule is written $x^{\mathrm{SM}}(p, \widehat{p}, w)$. It is not the ordinary utility-maximization problem with $\widehat{p}$ in the budget constraint.

Suppose instead that u is strictly concave and differentiable and that the relevant solutions

are interior. For a sparse-max tangency bundle x, set $\widetilde{w} = \widehat{p} \cdot x$. The perceived-price first-order condition is then sufficient for the unique global optimum, so

$$x = x^{\mathrm{FI}}(\widehat{p}, \widetilde{w}).$$

Conversely, any full-information bundle $x^{\mathrm{FI}}(\widehat{p}, \widetilde{w})$ satisfying $p \cdot x^{\mathrm{FI}}(\widehat{p}, \widetilde{w}) = w$ obeys the sparse-max rule. If the map $\widetilde{w} \mapsto p \cdot x^{\mathrm{FI}}(\widehat{p}, \widetilde{w})$ is continuous, strictly increasing, and has w in its range, this $\widetilde{w}$ is unique. This is a change of representation, not an assertion that the consumer faces $\widehat{p}$ in the market.

7.2 Cobb--Douglas Sparse-Max Demand

Example 7.1. Let $u(x) = \sum_{\ell=1}^{L} \alpha_\ell \log x_\ell$ with $\alpha_\ell > 0$ and $\sum_\ell \alpha_\ell = 1$. The perceived-price first-order conditions give

$$x_\ell = \frac{\alpha_\ell}{\lambda \widehat{p}_\ell}.$$

Imposing the actual budget yields $\lambda = \frac{1}{w} \sum_{k=1}^{L} \alpha_k \frac{p_k}{\widehat{p}_k}$ and therefore

Example 7.2.

$$x_\ell^{\mathrm{SM}}(p, \widehat{p}, w) = \frac{\alpha_\ell w / \widehat{p}_\ell}{\sum_{k=1}^{L} \alpha_k p_k / \widehat{p}_k}.$$

Indeed, $\sum_\ell p_\ell x_\ell^{\mathrm{SM}} = w$. If $\widehat{p} = p$, this reduces to the ordinary Cobb--Douglas demand $x_\ell = \alpha_\ell w / p_\ell$.

7.3 Perceived Prices and Limited Attention

One parsimonious attention rule combines the true price with a default price $p^d \gg 0$:

$$\widehat{p}_\ell(p) = a_\ell p_\ell + (1 - a_\ell) p_\ell^d, \qquad a_\ell \in [0, 1].$$

The vector $a = (a_1, \ldots, a_L)$ measures attention to price components. Full attention is $a_\ell = 1$ for every ℓ; setting $a_\ell = 0$ makes that component equal to its default. With $A = \mathrm{diag}(a_1, \ldots, a_L)$, $\widehat{p}(p) = Ap + (I_L - A)p^d$.

Proposition 7.3. *Comparative statics under perceived prices. Suppose the tangency bundle can be written as a differentiable map $r(\widehat{p}, \lambda)$ and $x^{\mathrm{SM}}(p, w) = r(\widehat{p}(p), \lambda(p, w))$, where $\lambda(p, w)$ is*

determined by $p \cdot r(\widehat{p}(p), \lambda) = w$. *Then the chain rule gives*

$$D_p x^{\mathrm{SM}} = D_{\widehat{p}} r \, A + \partial_\lambda r \, (\nabla_p \lambda)^\top, \qquad D_w x^{\mathrm{SM}} = \partial_\lambda r \, \partial_w \lambda.$$

The second terms are the response of the multiplier needed to restore the actual budget. Define the sparse-max and full-information Slutsky matrices by $S^{\mathrm{SM}} = D_p x^{\mathrm{SM}} + D_w x^{\mathrm{SM}} (x^{\mathrm{SM}})^\top$ and $S^{\mathrm{FI}} = D_p x^{\mathrm{FI}} + D_w x^{\mathrm{FI}} (x^{\mathrm{FI}})^\top$. Consequently, an identity such as $S^{\mathrm{SM}} = S^{\mathrm{FI}} A$ requires additional compensation and normalization assumptions; it is not a generic implication of the attention rule.

Because the default price is fixed, scaling (p, w) by a common positive factor generally changes $\widehat{p}(p)$ and therefore can violate homogeneity of degree zero. WARP, Slutsky symmetry, and welfare theorems must likewise be checked for the specified behavioral demand rather than imported from the rational benchmark.

7.4 Rational Inattention: Choosing an Information Structure

Sparse-max demand begins with perceived prices. Rational inattention explains why a decision maker may choose to observe only a noisy summary of the state. Let $\mathcal{S}$ be a finite state space, let $\mathcal{A}$ be a finite action set, and let $\mu_0 \in \Delta(\mathcal{S})$ have full support. The payoff from action a in state s is $v(a, s)$. An information-and-decision strategy can be written directly as a stochastic choice rule $q(a \mid s)$, with $q(\cdot \mid s) \in \Delta(\mathcal{A})$ for every state. This direct formulation absorbs both the signal and the action chosen after each signal.

Definition 7.4. Shannon information cost. The unconditional action probability induced by q is $q(a) := \sum_{s \in \mathcal{S}} \mu_0(s) q(a \mid s)$. The mutual information between the state and the chosen action is

$$\mathcal{I}(q) = \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} \mu_0(s) q(a \mid s) \log\left(\frac{q(a \mid s)}{q(a)} \right),$$

The sum may equivalently be taken only over actions with $q(a) > 0$. A term with $q(a \mid s) = 0 < q(a)$ is defined as zero; because the prior has full support, $q(a) = 0$ implies $q(a \mid s) = 0$ for every state, so that inactive action contributes zero. The cost of the channel is $\kappa \mathcal{I}(q)$, where $\kappa > 0$ is the marginal cost of information.

The canonical rational-inattention problem is

$$\max_{q(\cdot\mid s)\in\Delta(\mathcal{A})} \left\{ \sum_{s\in\mathcal{S}} \sum_{a\in\mathcal{A}} \mu_0(s) q(a \mid s) v(a,s) - \kappa \mathcal{I}(q) \right\}.$$

The first term is expected payoff and the second is the cost of making actions state-dependent. If $q(a \mid s) = q(a)$ for every (a, s), then actions reveal no information about the state and $\mathcal{I}(q) = 0$. This formulation follows the Shannon-cost rational-inattention model[Sims, 2003, Matějka and McKay, 2015].

Proposition 7.5. *Rational-inattention choice probabilities. At an optimum, every action with* $q(a) > 0$ *satisfies*

$$q(a \mid s) = \frac{q(a)\exp\{v(a,s)/\kappa\}}{\sum_{b\in\mathcal{A}} q(b)\exp\{v(b,s)/\kappa\}} \qquad \text{for every } s \in \mathcal{S}.$$

Conversely, a feasible channel satisfying these equations together with the consistency conditions $q(a) = \sum_s \mu_0(s) q(a \mid s)$ *solves the concave program on its active support.*

Proof. Attach a multiplier to $\sum_a q(a \mid s) = 1$ for each state and differentiate the objective with respect to a positive $q(a \mid s)$. Mutual information has derivative $\mu_0(s)\log(q(a \mid s)/q(a))$. The first-order condition therefore makes $\log(q(a \mid s)/q(a))$ equal to $v(a,s)/\kappa$ plus a state-specific normalizing constant. Exponentiating and imposing that conditional probabilities sum to one gives the displayed formula. Concavity supplies sufficiency on the active support. □

The formula resembles multinomial logit, but its baseline weight $q(a)$ is endogenous and must satisfy a fixed-point condition. Thus random utility and rational inattention can produce similar choice probabilities for different structural reasons: the former randomizes tastes, whereas the latter optimally limits state information. The distinction matters for welfare and counterfactual information policies.

7.5 A Gaussian Precision-Choice Illustration

Let the unknown scalar state be $\theta \sim \mathcal{N}(\theta^d, \sigma_\theta^2)$ and let the signal be $Y = \theta + \varepsilon$ with independent $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. Under quadratic loss $\mathscr{L}(a,\theta) = \frac{1}{2}(a-\theta)^2$, the optimal action is the posterior mean

$$a^*(y) = \mathbb{E}[\theta \mid Y = y] = \gamma y + (1-\gamma)\theta^d, \qquad \gamma = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2} \in [0,1].$$

The posterior variance is $(1-\gamma)\sigma_\theta^2$, so expected loss at the optimal action is $\frac{1}{2}(1-\gamma)\sigma_\theta^2$. If attention itself is costly, represent the cost by an increasing function $\mathcal{C}(\gamma)$ and solve

$$\max_{0\le\gamma\le 1} -\frac{1}{2}(1-\gamma)\sigma_\theta^2 - \mathcal{C}(\gamma).$$

For example, with $\mathcal{C}(\gamma) = \kappa\gamma^2/2$, the solution is $\gamma^* = \min\{1, \sigma_\theta^2/(2\kappa)\}$. Here $\gamma = 1$ denotes the limiting noiseless signal $\sigma_\varepsilon^2 = 0$. Cost specifications that diverge as $\gamma \uparrow 1$, such as the logarithmic specification below, instead guarantee an interior precision. The comparative statics therefore depend on the cost specification.

A capacity formulation is also transparent. If the information cost is $-\frac{1}{2}\log(1-\gamma)$ and capacity is bounded by K, then $-\frac{1}{2}\log(1-\gamma) \le K$ implies $\gamma \le 1 - e^{-2K}$. Since the value of information is increasing in γ, the capacity-constrained optimum attains this upper bound.

7.6 Group Thresholds under Noisy Signals

Consider a firm deciding whether to hire an applicant. Productivity is Q and hiring yields payoff Q, while rejecting yields 0. Before observing an applicant-specific signal, the firm observes $G \in \{0,1\}$ and has prior $Q \mid G = g \sim \mathcal{N}(q_g, \sigma_q^2)$ with $q_0 < q_1$.

The firm observes $Y = Q + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is independent. The posterior mean is

$$\mu_g(y) = \mathbb{E}[Q \mid Y = y, G = g] = \gamma y + (1-\gamma)q_g, \qquad \gamma = \frac{\sigma_q^2}{\sigma_q^2 + \sigma_\varepsilon^2}.$$

Proposition 7.6. *Optimal hiring rule. Conditional on (y, g), hire if and only if $\mu_g(y) > 0$; otherwise reject. For $\gamma > 0$ the signal threshold is $y_g = -(1-\gamma)q_g/\gamma$.*

If both group means are negative, the lower-mean group has the higher signal threshold. Hiring rates must nevertheless be computed from the group-conditional distributions because $Y \mid G = g$ has mean q_g; the groups share a noise variance, not a common signal distribution. If information is chosen, parameterize the decision by precision or by the informativeness index $\gamma \in [0,1]$ and solve $\max_{0\le\gamma\le 1} \mathbb{E}[\max\{\mu_g(Y), 0\}] - \mathcal{C}(\gamma)$ with an increasing attention cost $\mathcal{C}$. Greater noise variance means less attention, so a cost increasing in σ_ε^2 would reverse the intended economic tradeoff. Group-specific attention requires the group priors and a fully specified cost function.

7.7 Problems

Part I: By hand

Problem 7.1 [Core]. Sparse-max quasilinear demand.

Let $u(x_1, x_2) = \sqrt{x_1} + x_2$. The consumer chooses $x \in \mathbb{R}^2_+$, uses perceived prices $\widehat{p} \gg 0$ in the first-order condition, and must satisfy the actual budget $p \cdot x = w$, where $p \gg 0$ and $w > 0$.

(a) Derive the interior sparse-max choice.

(b) State the condition under which the implied x_2 is nonnegative.

(c) Show that ordinary Marshallian demand is recovered when $\widehat{p} = p$, including the corner when the interior condition fails.

Problem 7.2 [Core]. CES demand under perceived prices.

Let $u(x) = [\alpha x_1^\rho + (1-\alpha)x_2^\rho]^{1/\rho}$, where $0 < \alpha < 1$ and $\rho < 1$, $\rho \neq 0$. Consider an interior choice $x \in \mathbb{R}^2_{++}$ with perceived prices $\widehat{p} \gg 0$, actual prices $p \gg 0$, and wealth $w > 0$.

(a) Use perceived prices $\widehat{p}$ to derive the chosen ratio x_1/x_2.

(b) Scale the bundle so that it exhausts the actual budget $p \cdot x = w$.

(c) Identify separately the perceived-price composition effect of changing $\widehat{p}$ and the feasibility effect of changing p. Explain why the first is not automatically a Hicksian substitution effect.

Part II: Paying for information before choosing — chapter use case and verified scale-up

Seed: Problem 7.3. Fixed: all unlisted primitives and assumptions. Scaled: the common number N of equally likely states and matching actions, beginning at $N = 2$. Tool: Codex + Python. Verification: Recover the binary optimum, enforce conditional-probability and information-cost identities, and label any symmetry restriction used in the N-state extension. Problems 7.1 and 7.2 are comparison models, not derivations of rational inattention.

Problem 7.3 [Applied]. A binary rational-inattention problem.

There are two equally likely states and two actions. The decision maker receives payoff one from the correct action and zero from the incorrect action. A symmetric strategy is correct with probability $q \in [1/2, 1]$ and pays Shannon-information cost $\kappa[\log 2 - H(q)]$, where $H(q) = -q \log q - (1-q)\log(1-q)$ and $\kappa > 0$.

(a) Write expected net payoff as a function of q.

(b) Derive the interior optimum and show that $q^* = \exp(1/\kappa)/(1 + \exp(1/\kappa))$.

(c) Describe the limits as $\kappa \downarrow 0$ and $\kappa \uparrow \infty$.

Part III

Aggregation and Markets

Chapter 8

Aggregate Demand and Representative Consumers

8.1 Three Questions About Aggregation

- When does aggregate demand depend on prices and total wealth rather than on the entire wealth distribution?
- When does aggregate demand satisfy a demand restriction such as WARP?
- When can aggregate demand be given a welfare interpretation?

Let consumers be indexed by $i = 1, \dots, I$. Consumer i has Marshallian demand $x_i(p, w_i)$ at common prices $p \in \mathbb{R}^L_{++}$ and wealth $w_i > 0$. Aggregate demand is

$$x^A(p, \mathbf{w}) = \sum_{i=1}^{I} x_i(p, w_i),$$

where $\mathbf{w} = (w_1, \dots, w_I)$. Total wealth is $W = \sum_i w_i$. Without additional structure, two wealth distributions with the same W can generate different aggregate bundles.

8.2 Gorman Aggregation

Definition 8.1. Gorman form. Consumer i has Gorman-form indirect utility if there are a consumer-specific intercept and a common positive wealth coefficient, $a_i(p), b(p)$ with $b(p) > 0$,

such that

$$v_i(p, w_i) = a_i(p) + b(p)w_i.$$

Proposition 8.2. *Gorman aggregation. See[Gorman, 1953, Mas-Colell et al., 1995]. Suppose each v_i has the Gorman form and is differentiable. Roy's identity gives*

$$x_i(p, w_i) = -\frac{\nabla_p a_i(p) + w_i \nabla_p b(p)}{b(p)}.$$

Therefore aggregate demand takes the form

$$x^A(p, \mathbf{w}) = A(p) + B(p)W,$$

where $A(p) = -\sum_i \nabla_p a_i(p)/b(p)$ and $B(p) = -\nabla_p b(p)/b(p)$. Aggregate demand consequently depends on (p, W) and not on the distribution of wealth.

Differentiating the displayed Gorman form yields the common wealth derivative

$$D_{w_i} x_i(p, w_i) = B(p) \qquad \text{for every consumer } i.$$

This equality is necessary for the displayed Gorman representation; no global converse is asserted without separate integrability and domain hypotheses.

8.3 Aggregate Demand and WARP

Even if every individual demand satisfies WARP, aggregate demand need not do so. Aggregate WARP compares two price--total-wealth pairs while holding fixed the rule that assigns total wealth across consumers; the corresponding compensation need not leave each consumer's wealth unchanged.

Fix a wealth-allocation rule $\mathbf{w}(p, W) \in \mathbb{R}^I_+$ satisfying $\sum_i w_i(p, W) = W$, and define the induced price--total-wealth demand by $x^A(p, W) := x^A(p, \mathbf{w}(p, W))$. Every use of $x^A(p, W)$ below holds this rule fixed.

Definition 8.3. Aggregate WARP. A single-valued aggregate demand $x^A(p, W)$ satisfies WARP if, for distinct observations (p, W) and (p', W'),$p \cdot x^A(p', W') \leq W$ implies $p' \cdot x^A(p, W) > W'$ whenever $x^A(p, W) \neq x^A(p', W')$.

Proposition 8.4. *Sufficient condition for aggregate WARP. Suppose wealth shares are fixed and*

strictly positive, $w_i = \alpha_i W$, where $\alpha_i > 0$ and $\sum_i \alpha_i = 1$. Suppose every individual demand is single-valued, homogeneous of degree zero, and satisfies Walras' law. Assume also that, for every normalized price pair $q, q' \gg 0$,

$$(q - q') \cdot \big[x_i(q, \alpha_i) - x_i(q', \alpha_i)\big] \leq 0,$$

with strict inequality whenever the two individual demanded bundles differ. Then the aggregate demand formed with these fixed shares satisfies aggregate WARP.

Proof. Set $q = p/W$ and $X(q) = \sum_i x_i(q, \alpha_i)$. Homogeneity gives $x^A(p, W) = X(q)$, while Walras' law gives $q \cdot X(q) = 1$. Summing the individual inequalities yields

$$(q - q') \cdot [X(q) - X(q')] \leq 0,$$

and the inequality is strict whenever $X(q) \neq X(q')$. If $q \cdot X(q') \leq 1$ and the aggregate bundles differ, then

$$0 > (q - q') \cdot [X(q) - X(q')] = 2 - q \cdot X(q') - q' \cdot X(q),$$

so $q' \cdot X(q) > 2 - q \cdot X(q') \geq 1$. This is aggregate WARP on the normalized domain and hence on the original price--wealth domain. Fixed shares are essential: if shares change with prices or after compensation, the summation argument does not apply. □

8.4 Representative Consumers and Welfare

Definition 8.5. Positive representative consumer. A preference relation $\succeq^{rep}$ is a positive representative consumer for x^A if its Walrasian demand equals $x^A(p, W)$ for every price--wealth pair in the domain. This is a statement about demand representation, not about welfare.

Normative representative consumer. Fix a social welfare function $\mathcal{W} : \mathbb{R}^I \to \mathbb{R}$ and individual indirect utilities v_i. A nonnegative wealth allocation $\mathbf{w}(p, W)$ is welfare-optimal if it solves

$$\max_{\mathbf{w} \in \mathbb{R}^I_+} \mathcal{W}\big(v_1(p, w_1), \ldots, v_I(p, w_I)\big) \quad \text{subject to} \quad \sum_i w_i \leq W.$$

. A positive representative consumer is normative relative to $\mathcal{W}$ only when the wealth rule used to form x^A is welfare-optimal at every (p, W).

Gorman aggregation can deliver a positive representative consumer, but it does not select a social welfare function or justify interpersonal utility comparisons. Normative claims require the

welfare criterion and the wealth-allocation rule to be stated explicitly.

A representative consumer can hide an important distinction: spending may concentrate within each household while becoming less concentrated across the whole market. Chapter 12 returns to this issue after general equilibrium, using product space, shopping baskets and capacity constrained assignment to connect the earlier consumer models.

8.5 Problems

Part I: By hand

Problem 8.1 [Core]. Gorman aggregation.

Suppose consumer i has indirect utility $v_i(p, w_i) = a_i(p) + b(p)w_i$, with the same positive function b for every consumer. Assume a_i and b are differentiable on the relevant positive-price domain and that each indirect utility satisfies the regularity conditions for Roy's identity.

(a) Apply Roy's identity to derive individual demand.

(b) Show that aggregate demand depends on the wealth distribution only through total wealth $W = \sum_i w_i$.

(c) Construct a representative indirect utility that generates aggregate demand.

Problem 8.2 [Core]. Aggregation under common homothetic preferences.

All consumers have the same homothetic preferences, strictly positive wealth $w_i > 0$, and a single-valued demand. Normalize the unit-wealth demand as $x(p, 1)$.

(a) Show that $x_i(p, w_i) = w_i x(p, 1)$.

(b) Derive aggregate demand and identify a representative consumer.

(c) Explain why common homotheticity is sufficient but not necessary for exact aggregation.

Part II: Aggregation when household wealth moves with prices — chapter use case and verified scale-up

Seed: Problems 8.1, 8.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of consumers, with each wealth rule and preference class fixed. Tool: Codex + Python; Lean for the finite-sum core. Verification: Verify the finite aggregation identity and isolate every wealth-revaluation term. Do not describe the derivative or Slutsky bridge as Lean verified.

Problem 8.3 [Proof]. Why uncompensated aggregate price derivatives need not be

symmetric.

Let $x^A(p) = \sum_i x_i(p, w_i(p))$, where individual wealth may vary with prices through endowments or profit shares.

(a) Differentiate x^A with respect to prices.

(b) Separate compensated substitution terms from wealth-revaluation terms.

(c) Explain why symmetry and negative semidefiniteness of each individual Slutsky matrix do not generally imply the same properties for the uncompensated derivative $D_p x^A$.

Part IV

Production and Equilibrium

Chapter 9

Theory of the Firm

This chapter treats technology as a feasible set of net-output vectors. Profit maximization and cost minimization are dual ways of valuing that set at prices; aggregation and efficiency follow from the same support-function geometry.

9.1 Production sets

There are $L \geq 1$ commodities. A production plan $y \in \mathbb{R}^L$ records net outputs: $y_\ell > 0$ is an output and $y_\ell < 0$ is an input. The production set $Y \subseteq \mathbb{R}^L$ contains exactly the technologically feasible plans.

The benchmark assumptions are: Y is nonempty and closed; $0 \in Y$ (inaction); and free disposal, $y \in Y$ and $y' \leq y$ imply $y' \in Y$. A no-free-lunch condition rules out $y \in Y \cap \mathbb{R}^L_+$ other than 0. Convexity of Y is the convex-technology assumption. Together with $0 \in Y$, it guarantees feasibility of proportional contractions, but it does not impose decreasing returns to scale for expansions.

If a differentiable transformation function satisfies $Y = \{y : F(y) \leq 0\}$ and $F(\overline{y}) = 0$ with $F_k(\overline{y}) \neq 0$, the implicit-function theorem gives the local slope of the transformation frontier:

$$\frac{dy_k}{dy_\ell} = -\frac{F_\ell(\overline{y})}{F_k(\overline{y})}.$$

The minus sign is essential: increasing one net output requires a compensating reduction in the other when the frontier is written as $F(y) = 0$.

9.2 Production functions and input substitution

With M outputs and $L-M$ inputs, write outputs as $q \in \mathbb{R}^M_+$ and input use as $\boldsymbol{z} \in \mathbb{R}^{L-M}_+$. In the single-output case a production function f describes free disposal through $q \le f(\boldsymbol{z})$. If f is differentiable, the marginal rate of technical substitution between inputs ℓ and k is

$$\mathrm{MRTS}_{\ell k}(\boldsymbol{z}) = \frac{f_\ell(\boldsymbol{z})}{f_k(\boldsymbol{z})},$$

the amount of input k needed to offset a marginal reduction in input ℓ while holding output fixed. Concavity of f makes the input requirement set $\{\boldsymbol{z} : f(\boldsymbol{z}) \ge q\}$ convex.

9.3 Profit maximization

Definition 9.1. For a price vector $p \in \mathbb{R}^L$, the profit function is

$$\pi_Y(p) = \sup_{y \in Y} p \cdot y,$$

and the supply correspondence is $\mathcal{Y}_Y(p) = \arg\max_{y \in Y} p \cdot y$ whenever the supremum is finite and attained. If Y is closed and unbounded, finiteness and attainment require separate assumptions.

Proposition 9.2. *Profit-function properties. See[Hotelling, 1932, Mas-Colell et al., 1995]. The function π_Y is convex and homogeneous of degree one in prices. The supply correspondence is homogeneous of degree zero; if Y is convex, it is convex-valued. At a differentiability point, Hotelling's lemma gives the unique supply vector $y_Y(p) = \nabla_p \pi_Y(p)$. If π_Y is twice differentiable, $D_p y_Y(p) = D_p^2 \pi_Y(p)$ is symmetric positive semidefinite and $D_p y_Y(p) p = 0$.*

The non-differentiable law of supply is the monotonicity inequality

$$(p - p') \cdot (y - y') \ge 0$$

for $y \in \mathcal{Y}_Y(p)$ and $y' \in \mathcal{Y}_Y(p')$. It follows by adding the two profit-maximization inequalities.

For a single-output technology, $q = f(\boldsymbol{z})$, let $p_q > 0$ be the scalar output price and $\boldsymbol{w} \gg 0$ the input-price vector. Profit maximization is $\max_{\boldsymbol{z} \ge 0} \{p_q f(\boldsymbol{z}) - \boldsymbol{w} \cdot \boldsymbol{z}\}$. At a differentiable interior optimum, $p_q \nabla f(\boldsymbol{z}^*) = \boldsymbol{w}$; with nonnegative inputs the KKT conditions are

$$p_q f_\ell(\boldsymbol{z}^*) \le w_\ell, \qquad z^*_\ell \big(w_\ell - p_q f_\ell(\boldsymbol{z}^*)\big) = 0.$$

9.4 Cost minimization

Definition 9.3. Given input prices $\boldsymbol{w} \gg 0$ and a target $q > 0$, the cost function is

$$C(\boldsymbol{w}, q) = \min_{\boldsymbol{z} \geq 0} \{\boldsymbol{w} \cdot \boldsymbol{z} : f(\boldsymbol{z}) \geq q\}.$$

Assume the minimum is finite and attained. The conditional factor-demand correspondence is the argmin set $\mathcal{Z}(\boldsymbol{w}, q)$. At a differentiable optimum satisfying a constraint qualification, the complete KKT system is

$$\begin{gathered} \boldsymbol{z}^* \geq 0, \quad f(\boldsymbol{z}^*) \geq q, \quad \lambda \geq 0, \\ w_\ell - \lambda f_\ell(\boldsymbol{z}^*) \geq 0, \quad z_\ell^* \big(w_\ell - \lambda f_\ell(\boldsymbol{z}^*)\big) = 0, \\ \lambda \big(f(\boldsymbol{z}^*) - q\big) = 0. \end{gathered}$$

Proposition 9.4. *Cost-function properties. See[Shephard, 1953, Mas-Colell et al., 1995]. The cost function is homogeneous of degree one and concave in input prices, nondecreasing in the output target, and—when f is concave—convex in q. If the cost function is differentiable, Shephard's lemma gives the unique conditional factor demand $\boldsymbol{z}(\boldsymbol{w}, q) = \nabla_{\boldsymbol{w}} C(\boldsymbol{w}, q)$. If, in addition, the output constraint binds, the relevant constraint qualification holds, and the value function is differentiable in q, then the envelope theorem gives $\partial_q C(\boldsymbol{w}, q) = \lambda$. If twice differentiable, $D_{\boldsymbol{w}} \boldsymbol{z} = D_{\boldsymbol{w}}^2 C$ is symmetric negative semidefinite and $D_{\boldsymbol{w}} \boldsymbol{z}\, \boldsymbol{w} = 0$ by price homogeneity.*

The firm's output choice can be written as $\max_{q \geq 0} \{p_q q - C(\boldsymbol{w}, q)\}$. If the cost function is differentiable, an interior optimum satisfies $p_q = \partial_q C(\boldsymbol{w}, q)$; at $q = 0$ the corresponding one-sided inequality applies.

9.5 Aggregation and efficiency

For firms $j = 1, \ldots, J$ with production sets Y_j, the aggregate technology is the Minkowski sum

$$Y = \sum_{j=1}^{J} Y_j = \left\{ \sum_{j=1}^{J} y_j : y_j \in Y_j \right\}.$$

Proposition 9.5. *If each firm's profit maximum is finite and attained, then*

$$\pi_Y(p) = \sum_{j=1}^{J} \pi_{Y_j}(p), \qquad \mathcal{Y}_Y(p) = \sum_{j=1}^{J} \mathcal{Y}_{Y_j}(p)$$

where the second equality is understood as equality of attainable aggregate supply sets. The first equality follows directly from separability of the maximization over the Minkowski sum.

Definition 9.6. A feasible production plan $y \in Y$ is efficient if there is no $y' \in Y$ with $y' \geq y$ and $y' \neq y$.

Proposition 9.7. *If $p \gg 0$ and $y \in \mathcal{Y}_Y(p)$, then y is efficient. If Y is convex and closed and satisfies free disposal, every efficient plan is supported by some nonzero $p \geq 0$; this supporting-price statement is a separating-hyperplane theorem and may allow zero prices.*

9.6 Modern production: what optimization lets us learn

The preceding theory starts from a technology and derives optimal choices. Empirical production theory asks the reverse question: which features of technology can observed choices or values recover? Start with the dual when physical output is difficult to measure or has several components: optimizing values can discipline a production set without first constructing a scalar physical output index. We first explain that advantage and define productivity as an expansion of production possibilities. We then specialize to a physical output model, where a flexible input's optimality condition restricts expenditure shares. Both routes require explicit measurement and behavioral assumptions; a good fit to revenue is not itself recovery of technology.

9.6.1 Why the dual matters when output is hard to measure

A hospital supplies many services; a developer supplies housing of different sizes, locations and qualities. Adding the numbers of these services does not produce a natural physical output measure. Choosing a scalar index before studying technology can build the desired conclusion into the measurement. The production set avoids this step: it records which vectors of goods can be produced with which vectors of inputs.

Let $Y(e) \subseteq \mathbb{R}^{K+d}$ be the feasible net plans of a technology of type e, including disposal. For plans with nonnegative output and input use, write $y = (\boldsymbol{q}, -\boldsymbol{z})$, where $\boldsymbol{q} \in \mathbb{R}^K_+$ lists outputs and $\boldsymbol{z} \in \mathbb{R}^d_+$ lists inputs. At common prices $p = (\boldsymbol{p}_q, \boldsymbol{w}) \geq 0$, the profit function is

$$\pi(p, e) = \sup_{y \in Y(e)} p \cdot y, \qquad p \cdot (\boldsymbol{q}, -\boldsymbol{z}) = \boldsymbol{p}_q \cdot \boldsymbol{q} - \boldsymbol{w} \cdot \boldsymbol{z}.$$

Profit is a scalar value, not a scalar physical output. Prices value distinct outputs in a common monetary unit without assuming that their quantities can be added. Each optimized

value supplies a supporting bound $p \cdot y \leq \pi(p, e)$ on the technology. With enough price variation and the maintained shape restrictions, those restrictions recover the production set; with less variation, they bound it. The support recovery proposition below states exactly when the converse works.

The same logic accommodates fixed choices. For a common fixed input bundle $\boldsymbol{z}$, define $Q(\boldsymbol{z}, e) := \{\boldsymbol{q} \geq 0 : (\boldsymbol{q}, -\boldsymbol{z}) \in Y(e)\}$. Maximized revenue and the cost of a prescribed output vector are, respectively,

$$r(\boldsymbol{p}_q, \boldsymbol{z}, e) = \sup_{\boldsymbol{q} \in Q(\boldsymbol{z}, e)} \boldsymbol{p}_q \cdot \boldsymbol{q},$$

$$C(\boldsymbol{w}, \boldsymbol{q}, e) = \inf_{\substack{\boldsymbol{z} \geq 0: \\ (\boldsymbol{q}, -\boldsymbol{z}) \in Y(e)}} \boldsymbol{w} \cdot \boldsymbol{z}.$$

Restricted profit with those inputs fixed is $r - \boldsymbol{w} \cdot \boldsymbol{z}$. Fixing the output vector instead turns maximizing negative input expenditure into $-C$. Under free disposal, producing the prescribed output is equivalent to being able to produce at least that vector. An infeasible target has cost $+\infty$; a value function need not be finite or attained without additional assumptions. These are conditional optimization problems, not permission to treat a fixed choice as freely adjustable.

This is the empirical power of the dual: quantities of flexible goods can be latent while values and sufficiently informative prices or price proxies still restrict technology. But the dual does not make an ill defined commodity well defined. One must declare the goods and qualities, use compatible prices and units, and justify why observed accounting values equal, or are a specified noisy measure of, the relevant globally optimized value. A hospital may face administered prices and maximize a nonfinancial objective. Its decision problem must be justified, rather than presumed to be profit maximization. Price taking, constraints and the observation model must match the application. The identification argument in Aguiar, Kashaev and Allen makes these additional steps explicit; we return to it after the worked examples.

9.6.2 Productivity as an expansion of production possibilities

Definition 9.8. Set expansion productivity. In a common commodity space, call Y' weakly more productive than Y in the set expansion sense if $Y \subseteq Y'$. It can reproduce every plan of Y and may offer additional ones. This comparison makes sense with many outputs and does not require a scalar production function. At any common fixed input bundle, it expands the feasible output set; at any common feasible output target, it weakly lowers minimum input cost.

Containment is a partial order on technologies, not generally a complete ranking. An ordered

scalar type e satisfying $e \le e' \Rightarrow Y(e) \subseteq Y(e')$ assumes that the technologies are nested. It rules out some patterns of specialization. The labels are ordinal: replacing e by a strictly increasing transformation changes no ranking, and twice the index does not mean twice the output. Multiplying a scalar production function by a productivity factor is one special nested model, not the definition of productivity with several outputs.

Proposition 9.9. *The dual test for a productivity ordering. Inclusion $Y \subseteq Y'$ implies $\pi_Y(p) \le \pi_{Y'}(p)$ at every common price. If both sets are nonempty, closed, convex and freely disposable, the converse holds when the value inequalities are known at every $p \in \mathbb{R}^{K+d}_+$, with infinite values allowed. Comparing values at only the observed prices generally does not establish set inclusion.*

Proof. For the forward direction, every plan available to the first technology remains available to the second, so taking a supremum preserves the inequality. For the converse, any $y \in Y$ satisfies $p \cdot y \le \pi_Y(p) \le \pi_{Y'}(p)$ for all nonnegative prices. The support recovery proposition below then gives $y \in Y'$. The forward direction needs no convexity assumption; the converse uses the stated technology class and all supporting prices. □

Strict expansion need not raise value strictly at a particular price: a newly available plan may not improve the optimum there. Nor does a value ordering rank every chosen input or output coordinate. Common prices and common restrictions are essential. Differences in revenue or profit across firms facing different prices or fixed inputs cannot, by themselves, be read as productivity differences.

Example 9.10. A larger output set need not raise every output. Hold the same input bundle fixed and consider two feasible output sets in $\mathbb{R}^2_+$:

$$Q_A = \{\boldsymbol{q} \ge 0 : q_1 + q_2 \le 1\}, \qquad Q_B = \{\boldsymbol{q} \ge 0 : q_1 + q_2/2 \le 1\}.$$

At this fixed input bundle, $Q_A \subsetneq Q_B$: the second technology can produce everything the first can, and $(0, 2)$ is newly feasible. Maximizing a linear revenue over each triangle gives $r_A(\boldsymbol{p}_q) = \max\{p_1, p_2\}$ and $r_B(\boldsymbol{p}_q) = \max\{p_1, 2p_2\}$ at nonnegative output prices. At $(p_1, p_2) = (3, 1)$ both values are 3. At $(3/2, 1)$, however, the unique revenue maximizing output switches from $(1, 0)$ to $(0, 2)$, and value rises from $3/2$ to 2. Output 1 falls to zero despite the technological improvement. The common fixed input cost cancels in the profit comparison.

For contrast, let $Q_C = \{\boldsymbol{q} \ge 0 : q_1/2 + q_2 \le 1\}$. Neither Q_B nor Q_C contains the other: their exclusive output possibilities include $(0, 2)$ and $(2, 0)$, respectively. At prices $(3, 1)$, their

maximized revenues are 3 and 6; at $(1, 3)$, they are 6 and 3. These technologies specialize in different directions, so no price independent set expansion ranking exists. A scalar productivity index would conceal that distinction.

The dual therefore handles multiple outputs by valuing feasible vectors, not by declaring each output proportional to one latent scalar. An empirical scalar type model adds rankability and, when needed for identification, strict value separation at the maintained prices. Those assumptions must be stated rather than inferred from the existence of a profit function.

9.6.3 A specialization with one physical output

Now specialize to one physical output. Keep f for the production function, $\boldsymbol{z}$ for physical inputs, p_q for output price and $\boldsymbol{w}$ for input prices. Revenue $R = p_q q$ is a value, not an output quantity; R denotes realized revenue, whereas r above denotes maximized revenue conditional on fixed inputs. If output quality or product prices differ across firms, dividing every revenue observation by a common price index need not recover q. Likewise $w_m z_m$ is expenditure on input m, not its quantity. The measurement question comes before differentiation.

For an explicit timing model, index firms and dates by (i, t) and suppose $q_{it} = f(\boldsymbol{z}_{it}) \exp\{\omega_{it} + \varepsilon_{it}\}$. Only in this section, ω_{it} denotes productivity known to the firm before its current choices; it is not an endowment. The shock ε_{it} is realized afterward. Assume it is independent of current information $\mathcal{I}_{it}$, with $\mathbb{E}\varepsilon_{it} = 0$ and $\kappa_\varepsilon := \mathbb{E}\exp\{\varepsilon_{it}\} < \infty$. The prices relevant to the choice are known and taken as given. Mean zero in logarithms does not imply $\kappa_\varepsilon = 1$.

$$\log q_{it} = \log f(\boldsymbol{z}_{it}) + \omega_{it} + \varepsilon_{it}, \qquad \mathbb{E}[q_{it} \mid \mathcal{I}_{it}] = f(\boldsymbol{z}_{it}) e^{\omega_{it}} \kappa_\varepsilon.$$

A productive firm can choose more inputs precisely because it knows ω_{it}. Regressing log output on log inputs therefore does not, by itself, isolate technological elasticities. The observed association combines technology with the selection of inputs. Identification asks whether distinct primitives can generate the same observable distribution; estimation asks how to learn an identified object from a sample.

9.6.4 A flexible input turns a first order condition into a share equation

Hold the other inputs fixed and let $z_m > 0$ be a flexible input: it has no adjustment cost, dynamic consequence or binding capacity constraint in this decision. Assume an interior optimum, differentiability, and concavity of f in z_m. Conditional expected profit is maximized when marginal

expected revenue equals w_m. Define the physical output elasticity $\vartheta_m(\boldsymbol{z}) := z_m f_m(\boldsymbol{z})/f(\boldsymbol{z})$, where $f > 0$. Multiplying the optimality condition by input use gives

$$p_q e^{\omega} \kappa_\varepsilon f_m(\boldsymbol{z}) = w_m, \qquad w_m z_m = \vartheta_m(\boldsymbol{z})\, \mathbb{E}[R \mid \mathcal{I}].$$

Expenditure divided by conditional expected revenue equals the elasticity, but a realized share contains the output shock. Write $s^m := w_m z_m / R > 0$. In the maintained model,

$$s^m = \vartheta_m(\boldsymbol{z}) \kappa_\varepsilon e^{-\varepsilon}, \qquad \log s^m = \log \vartheta_m(\boldsymbol{z}) + \log \kappa_\varepsilon - \varepsilon.$$

The distinction is economically useful. A positive output surprise raises realized revenue without changing the input that was already purchased, so the measured share falls even though the production elasticity has not changed. With capacity, adjustment costs, market power or distorted input prices, the displayed equation must be rederived from the actual decision problem.

There is a constructive recovery step. On the observed positive input support define $h_s(\boldsymbol{z}) := \mathbb{E}[\log s^m \mid \boldsymbol{z}]$. Since inputs are chosen before the independent mean zero log shock, $h_s = \log \vartheta_m + \log \kappa_\varepsilon$. The observable share residual then recovers the shock, and its exponential moment recovers the level correction:

$$\begin{aligned} \varepsilon &= h_s(\boldsymbol{z}) - \log s^m, \\ \kappa_\varepsilon &= \mathbb{E} \exp\{h_s(\boldsymbol{z}) - \log s^m\}, \\ \vartheta_m(\boldsymbol{z}) &= \frac{\exp\{h_s(\boldsymbol{z})\}}{\kappa_\varepsilon}. \end{aligned}$$

These are population identities conditional on the maintained model. In a sample, estimating h_s and the moment is a statistical task. They identify this flexible input derivative on the input support, not every production derivative or an unrestricted counterfactual technology.

Example 9.11. A transparent power technology. Let $f(z) = az^b$ with $a > 0$ and $0 < b < 1$, set both shocks $\omega = \varepsilon = 0$, and take $p_q, w > 0$. The coefficient a now captures the known productivity level. Strict concavity and the derivative at zero give the unique global optimum $z^* = (p_q a b / w)^{1/(1-b)}$. At that choice, $wz^* = bp_q q^*$ and $\pi = (1-b)p_q q^*$. A technological improvement raises profit and changes input use, but the input share remains b. A higher observed share is not automatically evidence of higher productivity.

A stochastic output shock changes the measured share even when the production elasticity remains fixed.

Example 9.12. Two output surprises, one elasticity. Let $f(z) = 4\sqrt{z}$, $p_q = 4$, $w = 5$, $\omega = 0$, and let ε equal $\log 2$ or $-\log 2$ with probability $1/2$ each. Then $\kappa_\varepsilon = 5/4$ and expected profit $20\sqrt{z} - 5z$ is uniquely maximized at $z^* = 4$. The two realized shares are $5/16$ and $5/4$; their geometric mean is $5/8$, so the recovery formula gives $(5/8)/(5/4) = 1/2$, the true elasticity. If instead capacity imposes $z \leq 1$, the multiplier is 5 and applying the unmodified recovery formula gives $1/4$: it mistakes a binding capacity wedge for lower technological elasticity.

Gandhi, Navarro and Rivers use flexible input optimality to recover an input elasticity, then combine integration with further dynamic restrictions to recover the remaining production function. A productivity proxy in that approach is not the same object as a proxy for an unobserved price in the value function approach below.

[Gandhi et al., 2020]

To see why an elasticity is only a first step, use log coordinates $\xi_j = \log z_j$ and define $\mathfrak{f}(\boldsymbol{\xi}) := \log f(e^{\xi_1}, \ldots, e^{\xi_d})$. These symbols denote logarithms only here; $\boldsymbol{z}$ still denotes levels. If ϑ_m were known on a suitable input region, the fundamental theorem of calculus gives

$$\mathfrak{f}(\boldsymbol{\xi}) = \int_{\xi_m^0}^{\xi_m} \vartheta_m(e^{\boldsymbol{\xi}_{-m}}, e^u)\, du + a(\boldsymbol{\xi}_{-m}).$$

The notation inside the integrand means that coordinate m is e^u and every other coordinate is held fixed. The unknown function a is not a single constant: its arguments are all the other inputs. Learning one input derivative therefore does not recover the entire technology. Nor can a flexible input condition be imposed on predetermined capital without a separate argument.

9.6.5 Profits reveal supporting inequalities, not just a scalar outcome

Now suppose a firm has a fixed net output technology Y and chooses globally at several prices. At each observed price p^t, let $\pi^t = \pi_Y(p^t)$ be maximum profit. Every technologically feasible plan obeys $p^t \cdot y \leq \pi^t$. Thus values alone give the outer restriction

$$Y \subseteq Y_{\mathcal{P}}^{\text{out}} := \bigcap_{p \in \mathcal{P}} \{y : p \cdot y \leq \pi_Y(p)\}.$$

If the chosen plans y^t are also observed, global profit maximization adds the producer revealed preference inequalities $p^t \cdot (y^t - y^s) \geq 0$ for every s, t. These compare feasible alternatives, not slopes at one local optimum. Observing profit alone does not reveal the plan that generated it; observing both prices and quantities is strictly more informative.

Proposition 9.13. *Technology recovery from all supporting prices. Let $Y \subseteq \mathbb{R}^L$ be nonempty, closed, convex and freely disposable. With $\pi_Y(p) = \sup_{y \in Y} p \cdot y$, allowing $+\infty$ where appropriate, $Y = \bigcap_{p \in \mathbb{R}^L_+} \{y : p \cdot y \leq \pi_Y(p)\}$. An infinite right hand side gives no restriction at that price. This is an all supporting prices statement, not a claim that a finite or narrow observed price domain identifies Y.*

Proof. The forward inclusion follows from the definition of a supremum. If $y \notin Y$, separation gives a nonzero vector p with $p \cdot y > \sup_{v \in Y} p \cdot v$. The separating vector cannot have a negative coordinate: free disposal would allow that coordinate of v to fall without bound, making $p \cdot v$ arbitrarily large. Thus $p \geq 0$, and one of the displayed inequalities excludes y. □

The qualifications matter. A support function cannot distinguish a technology from its closed convex hull. Free disposal lets nonnegative separating prices suffice. Restricted price variation generally leaves unobserved directions undetermined. Homogeneity adds rays through observed prices, not new relative prices; a small neighborhood is not automatically enough for global recovery. For a fixed input or output, apply the same support logic to the corresponding feasible slice and its restricted value function.

9.6.6 A complete example of partial identification

Consider a single input with price $w = 1$ and output price $p_q > 0$. The teaching technology $f_A(z) = 2\sqrt{z}$, $z \geq 0$, has maximum profit $\pi_A(p_q, 1) = p_q^2$. Suppose the analyst sees only profits 1 and 4 at output prices 1 and 2. The two supporting inequalities imply

$$q - z \leq 1, \qquad 2q - z \leq 4, \qquad q \leq \min\{z + 1, z/2 + 2\}.$$

They do not identify the frontier. Another continuous, concave, nondecreasing technology with $f_B(0) = 0$ is $f_B(z) = \min\{2z,\, 2z/3 + 4/3,\, 4\}$. It has the same observed maximum profits: $z = 1, q = 2$ is optimal at price 1, and $z = 4, q = 4$ is optimal at price 2. Yet at $p_q = 3/2$ its profit is 2, whereas technology A gives $9/4$. Distinct technologies fit every observed value and disagree on a counterfactual.

The original technology's observed optimizing plans also give a useful lower bound at the counterfactual price. Reusing either $(z, q) = (1, 2)$ or $(4, 4)$ yields profit 2. Convexity of the profit function gives an upper bound by interpolation. Therefore any rationalizing technology containing those observed plans satisfies

$$2 \leq \pi_Y(3/2, 1) \leq \tfrac{1}{2}\pi_Y(1, 1) + \tfrac{1}{2}\pi_Y(2, 1) = \tfrac{5}{2}.$$

These are valid bounds, not a point prediction. With only profit values, rather than observed optimizing plans, the lower bound needs a separate justification and must not be smuggled in from quantities the analyst did not observe. Additional price variation, shape restrictions or quantity observations can tighten the identified set. A numerical fit cannot manufacture that information.

9.6.7 From productivity types to price proxies

The set expansion ordering in Section 9.6.2 supplies the economic interpretation of a nested productivity type. Identification requires more than this weak ordering: the analyst must distinguish types in the observed values and recover any prices used to interpret them.

Aguiar, Kashaev and Allen study recovery from optimizing values with possibly latent quantities and prices. Their discrete type argument uses strict value rankings, monotone conditional type presence, and specified measurement error and separation conditions. Their price proxy argument additionally uses excluded proxy coordinates, an observed price, an anchor normalization and a rank condition. Neither ordinary profit duality nor correlation with a price supplies those assumptions.

[Aguiar et al., 2023]

An elementary calculation shows what homogeneity contributes. Suppose there is one output and one flexible input, output price p_q is observed, and the unobserved input price is $w = g(v) > 0$, where v is an observed price proxy. Hold all other determinants fixed. Assume g is continuously differentiable, $g' \neq 0$, and the reduced form $\widetilde{\pi}(p_q, v) := \pi(p_q, g(v))$ has already been identified on a product of open intervals. Euler's identity and the chain rule give

$$\begin{aligned} p_q \pi_{p_q} + w\pi_w &= \pi, \\ \partial_v \widetilde{\pi} &= \pi_w g'(v), \\ \frac{g'(v)}{g(v)} &= \frac{\partial_v \widetilde{\pi}}{\widetilde{\pi} - p_q \partial_{p_q} \widetilde{\pi}}. \end{aligned}$$

The ratio is used only where its denominator is nonzero. At a differentiability point with positive input demand, Hotelling's lemma makes that denominator $w\pi_w = -wz < 0$. If the right hand side is $H(v)$ and a price anchor $g(v_0) = w_0$ is known, integration yields

$$g(v) = w_0 \exp\left\{\int_{v_0}^{v} H(u)\,du\right\}.$$

This is a conditional teaching derivation, not a general identification theorem. If the calculated ratio varies with p_q at fixed v, the proposed single price proxy representation fails its own restriction. With several unobserved prices, the analogous equations must have enough independent variation; arbitrary proxies cannot simply be relabeled as observed prices. Identifying the reduced form from noisy values and heterogeneous firms is itself an earlier, separate problem.

The modern lesson is to preserve the classical optimization model while becoming explicit about what is observed. Choose the quantity route when timing and flexible input conditions are credible. Choose a value function route when optimizing values and price information are credible. In either case, state the maintained technology class, distinguish identification from fitting, and report a set of possibilities when the data do not select one.

Advanced Problem 9.A6 develops the housing application. An original teaching specialization inspired by Epple, Gordon and Sieg and the housing discussion in Prices, Profits, Proxies, and Production connects developer optimization, a latent housing price and the rental value of scarce land. It then closes the goods and land markets. Keep recovery across comparable markets distinct from a technology counterfactual that changes the price proxy relationship.

[Epple et al., 2010]

9.7 Problems

Part I: By hand

Problem 9.1 [Core]. Cobb-Douglas production and cost.

A firm has production function $f(\boldsymbol{z}) = z_1^{\alpha} z_2^{\beta}$ with $\alpha, \beta > 0$.

(a) Classify returns to scale as a function of $\alpha + \beta$.

(b) Derive conditional factor demands and the cost function for output $q > 0$ and input prices $\boldsymbol{w} \gg 0$.

(c) Verify that cost is homogeneous of degree one and concave in $\boldsymbol{w}$.

Problem 9.3 [Core]. Constant returns and unbounded profit.

Assume Y is a cone and $0 \in Y$.

(a) Prove that either $\pi_Y(p) = 0$ or $\pi_Y(p) = +\infty$.

(b) Characterize the price vectors for which profit is zero.

(c) Explain why a competitive constant-returns firm can operate at any scale only when the zero-profit condition holds.

Advanced exam practice

These problems build toward integrated graduate exam arguments. The suggested time applies to the hand questions, not the optional tool extension. First draw a diagram or state a solution plan, define the feasible set, and identify possible corners. Explain why each candidate is globally optimal and check the economic accounting. Attempt the hand questions before using tools. For an optional extension, state Seed, Fixed, Scaled and Risk before coding and enlarge one feature at a time.

Difficulty guide. Problems 9.A1 to 9.A4 provide new optimization and regime practice. Problem 9.A5 restores the Western 2017 final examination question on indivisible production, including its original endowments and global equilibrium reasoning. Problem 9.A6 adds a simple housing production case connecting developer optimization, latent prices, land rents and general equilibrium.

Problem 9.A1 [Advanced]. Automation, production lines and the value of capacity.

Suggested time: 35 minutes. Total: 30 points. The optional tool extension is not included in the suggested time.

A competitive manufacturer sells a homogeneous output at the fixed price $p = 8$. It operates two production lines. Their outputs are $q_1, q_2 \geq 0$, and their combined capacity is $q_1 + q_2 \leq 6$. Costs are $C_1(q_1) = 2q_1 + \frac{1}{2}q_1^2, \qquad C_2(q_2; a) = aq_2 + \frac{1}{2}q_2^2$, where $a \geq 0$ is a technology parameter, not a choice. The manufacturer maximizes revenue less both production costs. There are no fixed costs.

(a) Write the optimization problem and its KKT conditions. Use a nonnegative multiplier λ for total capacity. Explain how the condition for a line with zero output differs from the equality for a line with positive output. [5]

(b) For $a = 4$, determine both outputs, profit and λ. Check every KKT condition rather than only solving the interior equations. [5]

(c) An automation improvement reduces a from 4 to 1, without changing output price or capacity. Solve the new problem. Explain why output on the first line can fall even though total technology has improved. [5]

(d) Keeping capacity equal to 6, solve for all $a \geq 0$. Identify the value at which the second line shuts down. At that value, distinguish a binding capacity constraint from a strictly positive

shadow value. [5]

(e) Prove global optimality and uniqueness of your proposed allocation. You may use strict concavity and KKT sufficiency, but state the premises. Alternatively give an exact expression for the profit loss from any feasible deviation. [5]

(f) Let $V(K, a)$ be maximum profit when total capacity is K. Interpret $\partial V/\partial K$ at $K = 6$ for $a = 4$ and $a = 1$. Explain why the automation improvement raises the value of capacity in this example. [5]

Optional tool extension. Extend the manufacturer to ten lines with costs $c_j q_j + \kappa_j q_j^2/2$, where $\kappa_j > 0$. Sort the entry thresholds, solve the capacity equation, and test primal feasibility, multiplier signs, complementary slackness and every excluded line. Compare direct optimization with the exact active set method. A numerical solver is a check, not the proof requested in part (e).

Problem 9.A2 [Advanced]. Human input, machine capacity and cost duality.

Suggested time: 45 minutes. Total: 36 points. The optional tool extension is not included in the suggested time.

A firm uses labor $\ell \geq 0$ and machine services $r \geq 0$. It can produce output $q \geq 0$ satisfying $q \leq \sqrt{\ell} + \sqrt{ar}, \qquad r \leq R$. Here $a > 0$ is machine productivity and $R > 0$ is the available quantity of machine services. Labor and machine prices are $w > 0$ and $v > 0$. The firm pays $w\ell + vr$. All inputs and outputs are continuously divisible, and there is no fixed cost.

(a) For a required output $q > 0$, write the cost minimization problem and show that its production constraint binds. Using $x = \sqrt{\ell}$ and $y = \sqrt{ar}$, reduce it to a strictly convex quadratic allocation problem. [5]

(b) Find the unique conditional input demands and cost function $C(w, v, q; a, R)$ for every $q > 0$. Identify the output threshold at which the machine capacity starts binding, and verify that cost and marginal cost match across the threshold. [8]

(c) At a fixed output price $p > 0$, derive optimal labor, machine use, output and profit. Give the complete price partition, including the boundary. [7]

(d) Verify Hotelling's lemma with respect to p in both regimes. Verify Shephard's lemma with respect to w and v in the conditional cost problem, stating which objects are held fixed. [6]

(e) Take $p = 3$, $w = v = 1$ and $R = 1$. Compare $a = 1$ with $a = 4$. Calculate output, both inputs, profit and the machine capacity multiplier in each case. Explain why greater productivity need not make the capacity constraint less valuable. [5]

(f) Explain why the proposed solution is globally optimal. Distinguish improving machine

productivity a from increasing available machine services R. Which model feature prevents the firm from using an unlimited amount of the improved technology? [5]

Optional tool extension. Draw the two dimensional regime map in (p, R) for several values of a. Check continuity and envelope identities at the boundary using exact formulas before numerical differentiation. Report the analytical statement separately from any graph or solver output.

Problem 9.A3 [Advanced]. A common emissions budget and the limits of firm aggregation.

Suggested time: 40 minutes. Total: 32 points. The optional tool extension is not included in the suggested time.

Two production units sell the same output at the fixed competitive price $p = 6$. Their costs are $C_1(q_1) = q_1 + \frac{1}{2}q_1^2, \qquad C_2(q_2) = 2q_2 + \frac{1}{2}q_2^2$. Outputs satisfy $q_1, q_2 \geq 0$. Initially each unit of output from unit 1 uses one emissions permit and each unit from unit 2 uses two. A manufacturer owns both units and has a total emissions budget $B > 0$, so $q_1 + 2q_2 \leq B$. It maximizes total revenue less production costs. There are no other constraints or fixed costs.

(a) Derive the KKT system and explain the units of the common constraint multiplier. Why is adding the two unconstrained supplies generally incorrect? [5]

(b) Solve the problem for every $B > 0$, finding all thresholds at which a unit enters or the emissions budget ceases to have a positive shadow value. Check the boundary values. [8]

(c) At $B = 4$, compute outputs, gross revenue less production costs and the emissions multiplier. Prove that the allocation is globally optimal and unique. [5]

(d) Suppose instead that independent firms operate the units and permits can be traded at the competitive price $\tau \geq 0$. Each firm pays τ times its emissions. Show that an appropriate permit price implements the allocation from part (c). Explain why the initial ownership of permits affects income but not these output choices under the stated assumptions. [5]

(e) An improvement reduces the emissions requirement of unit 2 from two permits to one, holding $p = 6$ and $B = 4$ fixed. Solve the new problem and compare each output, their total, production profit before permit payments, and the permit price. Explain why the price of the scarce permit can rise when a production unit becomes cleaner. [6]

(f) Explain the difference between this manufacturer problem and a social planner problem that values consumer surplus or environmental damage. State one missing primitive needed to make a welfare claim. [3]

Optional tool extension. Add heterogeneous units with costs $c_j q_j + q_j^2/(2e_j)$ and emissions

coefficients $b_j > 0$. Construct $q_j(\tau) = e_j[p - c_j - \tau b_j]_+$ and solve the permit clearing equation. Check the entry inequalities and distinguish output changes from transfers of permit rents.

Problem 9.A4 [Advanced]. A fixed software cost, entry and a missing competitive equilibrium.

Suggested time: 45 minutes. Total: 36 points. The optional tool extension is not included in the suggested time.

A firm must activate a production system before producing. Activation is $d \in \{0, 1\}$ and output satisfies $0 \leq q \leq Kd$, where $K > 0$. Its profit at the fixed output price $p \geq 0$ is $pq - Fd - cq - \frac{q^2}{2e}$, where $F > 0$ is the activation cost, $c \geq 0$ is unit operating cost and $e > 0$ is productivity. Inactivity has $d = 0$, $q = 0$ and zero profit. All parameters are fixed when the firm chooses (d, q).

(a) Conditional on $d = 1$, solve for output using KKT, including the lower and upper output constraints. Why is this conditional calculation not enough to find a global optimum of the original problem? [6]

(b) Derive the entry price, allowing for the possibility that capacity is already binding at entry. State the entire output supply correspondence below, at and above that price. Do not discard tied optima. [9]

(c) Set $c = 2$, $e = 1$, $K = 4$ and $F = 2$. Calculate supply and maximum profit at $p = 3$, $p = 4$ and $p = 5$. Explain the jump in supply. [5]

(d) At the fixed price $p = 7/2$, an AI software improvement reduces F from 2 to $1/2$, without changing c, e or K. Compare activity, output and profit before and after the change. For prices at which the firm is active under both costs, does reducing F change its conditional output? [5]

(e) Return to $F = 2$. Suppose this firm is the only producer, consumers have aggregate demand $D(p) = \max\{5 - p, 0\}$, and all agents are price takers. Prove that there is no competitive market clearing price. Does the reduction to $F = 1/2$ restore an equilibrium? If so, find it and verify that the firm prefers to operate. [7]

(f) Explain which convexity property fails and why a smooth first order condition cannot rule out the failure in part (e). Distinguish the claim that no competitive equilibrium exists from the claim that no feasible or optimal allocation exists. [4]

Optional tool extension. For a population of such firms, vary the distribution of fixed costs and capacities. Track the supply correspondence, allowing for ties at entry. Compare an atomic firm population with a continuum. A plot that crosses interpolated supply points is not proof of exact market clearing.

Problem 9.A5 [Advanced]. Indivisible production and two endowment configurations.

Restored from the Western University Fall 2017 final examination. The indivisible technology, literal utility and two original endowment configurations are retained.

Suggested time: 40 minutes. Total: 40 points.

There is one consumer and one firm, with one physical input and one output good. The consumer owns the firm. A consumer bundle is $(c, d) \in \mathbb{R}^2_+$, where c is retained input and d is consumption of the output. Utility is $u(c, d) = 2 - c + 2d$. Thus the utility of retained input is decreasing, exactly as in the original displayed formula; do not replace it by a leisure utility.

The firm has the production set $Y = \bigcup_{a \in \mathbb{Z}_+} \{(-a, a) - v : v \in \mathbb{R}^2_+\}$, where $\mathbb{Z}_+ = \{0, 1, 2, \ldots\}$. It can choose an integer activity a that uses a units of input and produces a units of output, and it can freely dispose of either commodity. Normalize the output price to 1 and denote the input price by $r \geq 0$. The firm maximizes $(r, 1) \cdot y$ over $y \in Y$. If this maximum is finite and attained, denote it by $\pi(r)$.

For endowment $\omega = (\omega_c, \omega_d)$, the consumer budget is $rc + d \leq r\omega_c + \omega_d + \pi(r)$. A competitive equilibrium requires both global optimization problems and exact clearing $(c, d) = \omega + y$.

(a) Characterize the firm's profit and its complete optimizing production correspondence for all $r \geq 0$. Explain why free disposal and integer activity matter, and why the firm must not insert the economy's total input endowment as an additional private constraint. [12]

(b) Begin with the original endowment $\omega = (2, 0)$. Find every competitive equilibrium price, consumer bundle and production plan. Prove that no other price or integer activity can be an equilibrium. [12]

(c) Change only the endowment to the other original case, $\omega = (0, 2)$. Find the full equilibrium set and compare the input price determination with part (b). [10]

(d) Exhibit two plans in Y whose midpoint is not in Y. Explain why zero equilibrium profit does not make this technology a convex cone, and why first order calculus in a cannot replace the global production comparison. [6]

Problem 9.A6 [Advanced]. Housing values, latent prices and land rents.

This is an original teaching specialization of the housing production and value proxy ideas in Epple, Gordon and Sieg (2010) and Aguiar, Kashaev and Allen (2023). It is not a reproduction of either paper's general identification theorem.

Part I. By hand.

Suggested time: 50 minutes. Total: 50 points. The optional tool extension is not included in

the suggested time.

Housing services are a homogeneous output. Developers use land $L \geq 0$ and materials $M \geq 0$ to produce $H = 2A\sqrt{LM}$, where $A > 0$ is a common technology parameter. Materials are units of the numeraire and cost 1 each. The price of one unit of housing is $p_q > 0$ and the rent of one unit of land is $\rho > 0$. Every developer takes both prices as given and chooses its own scale. There is no fixed cost, market power or uncertainty. All prices and revenues refer to the same single period.

For a developer using positive land, define materials per unit of land $m = M/L$, housing per unit of land $q = H/L$, and observed housing revenue per unit of land $v = p_q q$. The conditional operating value per unit of land, before paying for land, is $\Pi(p_q; A) := \max_{m \geq 0}\{2Ap_q\sqrt{m} - m\}$. Net developer profit also subtracts ρL.

(a) Find the unique conditional materials intensity, housing intensity, revenue and operating value. Provide a global argument, using either KKT after a change of variable or an exact algebraic bound. Explain why a land payment that is fixed conditional on L does not change the choice of m. [8]

(b) Consider otherwise comparable markets with the baseline technology $A = 1$. Housing quantity q and its price p_q are not separately observed. The analyst observes v and land rent $\rho(v) = v/2$ for every $v > 0$, maintains zero net developer profit, and represents the latent price by $p_q = g(v)$. The units of housing are anchored by $g(2) = 1$. Use the conditional operating value from part (a), the observed rent and zero profit to recover the price, housing intensity and materials intensity by algebra. Check the anchor and explain why this parametric inversion does not identify an unrestricted unknown technology. [8]

(c) Return to any $A > 0$ and let a developer choose both L and M. Characterize its maximum net profit and all optimal input plans according to the relation between ρ and the conditional operating value from part (a). Which relation permits positive land use and free entry without unbounded profit? Give a global certificate, not only two first order equalities. [6]

(d) Close a genuine general equilibrium. A single household has utility $u(C, H) = C + B \log H$, where $C \geq 0$ is numeraire consumption, $H > 0$ is housing and $B > 0$ is fixed. It owns a numeraire endowment $W > B$, all developers, and the fixed total land $\overline{L} > 0$. It receives land rent, and firms have zero net profit in a free entry equilibrium. Its budget is $C + p_q H \leq W + \rho\overline{L}$. Materials used by developers come from the numeraire endowment. Write household KKT conditions taking prices as given. Derive equilibrium prices and aggregate H, M, C. Verify household global optimality, developer global optimality, land clearing, housing clearing and the numeraire resource

identity $C + M = W$. Distinguish uniqueness of aggregate equilibrium from the division of production among firms. [12]

(e) Set $W = 10$, $B = 4$, $\overline{L} = 2$ and initially $A = 1$. Calculate all prices, aggregate quantities and household income. Now double A without changing preferences or endowments. Calculate the new equilibrium and the change in v. Explain how positive land rents can coexist with zero developer profit. [8]

(f) For general $W > B$ and $\overline{L} > 0$, compare A and $2A$. Compute the household's utility gain and distinguish changes in physical housing from changes in housing expenditure. Can the analyst reuse the baseline price proxy and its anchor from part (b) after the technology changes? Explain what stays fixed in a valid comparison across markets and what fails when technology moves. State one reason this single location model does not by itself describe the spatial structure of a city. [8]

Part II. Codex and verification.

Optional analytic bridge. Now regard the common production function per unit land as unknown rather than impose the stated Cobb Douglas form. Maintain interior conditional optima, differentiability, material price 1, the full observed rent curve $\rho(v) = v/2$, zero net profit and the anchor $g(2) = 1$. Use the envelope theorem and $v = p_q q$ to derive a differential equation for g. Recover the implied price and production relationship and explain which assumptions make this different from the algebraic inversion in part (b).

Seed: the two equilibria in part (e). Fixed: $W = 10$, $B = 4$, the technology class, ownership, material price normalization, preferences, and the strictly positive domains. Scaled: aggregate land $\overline{L} > 0$ over a grid, first at $A = 1$ and then under the same stated doubling of A. This changes one market feature at a time while repeating the fixed technology comparison. Risk: confusing a conditional land choice with fixed aggregate land, counting rent as an additional resource cost, or using the old price proxy outside its technology regime.

Build an equilibrium verifier rather than only plotting formulas. Check household KKT, the developer profit certificate, all three market identities, income accounting and the welfare difference. Reject invalid inputs. Include a test that deliberately reuses the old price proxy after the technology change and explain its failure. An optional Lean extension may certify the algebraic profit bound and the finite accounting identities from explicitly supplied positive price and equilibrium equations. Do not describe the empirical price proxy interpretation or a general existence theorem as Lean verified.

Part II: Input substitution in a multi-input firm — chapter use case and verified scale-up

Problem 9.2 [Proof]. Dual properties of the profit function.

Let $Y \subseteq \mathbb{R}^L$ be nonempty, closed, and convex. Assume that profit $\pi_Y(p) = \sup_{y \in Y} p \cdot y$ is finite on a neighborhood of the price under study and that the supremum is attained.

(a) Prove that π_Y is convex and homogeneous of degree one in prices.

(b) If supply is unique at p, use a subgradient argument to prove Hotelling's lemma and explain why the profit function is differentiable there.

(c) State what changes when supply is set-valued.

Chapter use case

Seed: Problem 9.1. Fixed: all unlisted primitives and assumptions. Scaled: the input dimension from two to L, with the output target and technology family fixed. Tool: Codex + Python; Lean for finite algebra. Verification: Recover the two-input cost benchmark; check Shephard derivatives, Hessian symmetry, negative semidefiniteness, and the homogeneity null direction; state the analytic assumptions not encoded in Lean.

Problem 9.4 [Proof]. Shephard's lemma and substitution.

Suppose the minimum in $C(\boldsymbol{w}, q) = \min\{\boldsymbol{w} \cdot \boldsymbol{z} : f(\boldsymbol{z}) \geq q, \boldsymbol{z} \geq 0\}$ exists, conditional factor demand $\mathcal{Z}(\boldsymbol{w}, q)$ is single-valued, and C is twice differentiable in $\boldsymbol{w}$.

(a) Prove Shephard's lemma.

(b) Show that $D_{\boldsymbol{w}}\boldsymbol{z} = D^2_{\boldsymbol{ww}}C$ is symmetric and negative semidefinite.

(c) Use homogeneity to show $D_{\boldsymbol{w}}\boldsymbol{z}(\boldsymbol{w}, q)\boldsymbol{w} = 0$.

Chapter 10

Competitive Markets and Partial Equilibrium

This chapter collects the competitive-equilibrium definitions used later and develops a transparent partial-equilibrium model with a numeraire. The partial-equilibrium model is deliberately specialized: its quasilinear preferences remove income effects for the good of interest.

10.1 Competitive equilibrium

There are I consumers, J firms, and L goods. Consumer i has consumption set $X_i \subseteq \mathbb{R}^L$, preference relation $\succeq_i$, and endowment $\omega_i \in \mathbb{R}^L$. Firm j has production set $Y_j \subseteq \mathbb{R}^L$, where positive coordinates are net outputs and negative coordinates are inputs. Let $\omega = \sum_i \omega_i$.

Definition 10.1. Feasibility. An allocation $(x_1, \ldots, x_I; y_1, \ldots, y_J)$ is feasible when $x_i \in X_i$, $y_j \in Y_j$, and

$$\sum_{i=1}^{I} x_i \leq \omega + \sum_{j=1}^{J} y_j.$$

Equality is imposed when all goods have strictly positive prices and free disposal is not used.

Competitive (Walrasian) equilibrium. A nonzero nonnegative price vector $p \geq 0$ and an allocation (x^*, y^*) form an equilibrium. Consumer i owns share $\theta_{ij} \geq 0$ of firm j, with $\sum_i \theta_{ij} = 1$ for every firm. Every firm solves

$$y_j^* \in \arg\max_{y_j \in Y_j} p \cdot y_j$$

and every consumer weakly prefers their assignment to every affordable bundle:

$$x_i^* \succeq_i x_i \quad \text{for every } x_i \in X_i \text{ with } p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*.$$

Markets clear exactly: $\sum_i x_i^* = \omega + \sum_j y_j^*$.

Proposition 10.2. *Price normalization and Walras' law. If (p, x^*, y^*) is a competitive equilibrium, then so is $(\lambda p, x^*, y^*)$ for every $\lambda > 0$. If consumers exhaust their budgets, summing the budget equalities and using market clearing gives*

$$p \cdot \left(\sum_i x_i^* - \omega - \sum_j y_j^* \right) = 0.$$

Therefore, if all but one market clear, the last market clears as well provided its price is strictly positive.

10.2 Partial equilibrium with a numeraire

There are two goods. Good x is the good of interest and m is a numeraire with price one. Consumer i has

$$u_i(m_i, x_i) = m_i + \phi_i(x_i), \qquad X_i = \mathbb{R} \times \mathbb{R}_+,$$

where ϕ_i is increasing, strictly concave, twice continuously differentiable, and bounded above. Let $p > 0$ be the price of x and let ω_{mi} denote consumer i's initial numeraire endowment.

Firm j produces $q_j \geq 0$ units of x at numeraire cost $c_j(q_j)$. Its net-output set is

$$Y_j = \{(-z_j, q_j) : q_j \geq 0,\ z_j \geq c_j(q_j)\}.$$

Assume c_j is increasing, strictly convex, twice continuously differentiable, and $c_j'(q) \to \infty$ as $q \to \infty$.

Definition 10.3. Partial competitive equilibrium. A price p and quantities $(x_1, \ldots, x_I; q_1, \ldots, q_J)$ form an equilibrium when

$$\phi_i'(x_i) \leq p, \quad x_i > 0 \Rightarrow \phi_i'(x_i) = p,$$

$$p \leq c_j'(q_j), \quad q_j > 0 \Rightarrow p = c_j'(q_j),$$

and the market for x clears:

$$\sum_i x_i = \sum_j q_j.$$

These are the KKT conditions for consumer and firm optimization plus market clearing.

Equivalently, the unique choices are characterized by

$$x_i(p) = \arg\max_{x \geq 0} \{\phi_i(x) - px\}, \qquad q_j(p) = \arg\max_{q \geq 0} \{pq - c_j(q)\}.$$

The associated aggregate excess demand is $z(p) = \sum_i x_i(p) - \sum_j q_j(p)$.

Proposition 10.4. *Existence and uniqueness in the partial model. Suppose* $0 < \min_j c_j'(0) < \max_i \phi_i'(0)$ *and the preceding curvature and tail assumptions hold. Then* z *is continuous on the displayed positive-price interval, demand is nonincreasing, and supply is nondecreasing. A clearing price* $p^* \in [\min_j c_j'(0), \max_i \phi_i'(0)]$ *exists. The maintained strict curvature makes the aggregate crossing strict whenever a side is active, so the equilibrium price is unique.*

To see the sign change, at a price no larger than $\min_j c_j'(0)$ all firms choose zero output while at least one consumer demands a positive amount. At a price at least $\max_i \phi_i'(0)$ all consumers choose zero while at least one firm supplies a positive amount. The intermediate value theorem then gives a clearing price.

The numeraire allocation is recovered from the budget identity

$$m_i = \omega_{mi} + \sum_j \theta_{ij} \big(pq_j - c_j(q_j)\big) - px_i.$$

Partial equilibrium deliberately suppresses the wealth effects that would reappear with general preferences or additional markets.

10.3 A continuum of consumers and heterogeneous firms

The finite model is a useful starting point, but a market problem must account for distributions of tastes, productivity and wealth. A continuum replaces a sum by a type integral. It does not remove heterogeneity or allow a conditional distribution to stand in for a population measure. This section develops the method used in the historical final and problem set cases at the end of the chapter.

10.3.1 Individual optimization before integration

Consumer mass is $\eta > 0$, with taste law F on $[0, \bar{\alpha}]$. Firm mass is $\nu > 0$, with productivity law G on $(0, \bar{e}]$. Both laws are probability distributions. Assume $A_F = \int \alpha \, dF(\alpha) > 0$ and $E_G = \int e \, dG(e) > 0$. Population measures are ηF and νG; the masses need not be one.

A type α consumer maximizes $2\sqrt{\alpha x} + m$ subject to $px + m \leq w(\alpha)$, with $x \geq 0$ and $m \in \mathbb{R}$. Wealth is exogenous to this decision. A type e producer chooses $q \geq 0$ at cost $C(q; e, a) = q^2/(2ae)$, with common technology $a > 0$. The numeraire price is one. The square root benefit is unbounded but sublinear, unlike the earlier bounded benefit sufficient setup: subtracting px still makes the objective tend to negative infinity at every $p > 0$. Its marginal benefit at zero is a limit, not a finite derivative.

Budget substitution reduces the consumer problem to maximizing $2\sqrt{\alpha x} - px$. Strict concavity and the positive derivative near zero give $x(\alpha, p) = \alpha/p^2$ for $\alpha > 0$; type zero chooses no consumption good. The strictly concave producer problem gives $q(e, p) = aep$ and $\pi(e, p) = aep^2/2$, with $\partial\pi/\partial p = q$. These global arguments justify the individual formulas before integration.

10.3.2 Population moments determine this particular market

$$D(p) = \eta \int x(\alpha, p) \, dF(\alpha) = \frac{\eta A_F}{p^2}, \qquad S(p) = \nu \int q(e, p) \, dG(e) = \nu a E_G p.$$

Demand decreases continuously from infinity to zero and supply increases continuously from zero to infinity. There is exactly one positive clearing price. Average demand is not total demand: the consumer mass multiplies the former, and firm mass multiplies average output.

$$p^* = \left(\frac{\eta A_F}{\nu a E_G} \right)^{1/3}, \qquad Q^* = (\eta A_F)^{1/3} (\nu a E_G)^{2/3}.$$

This moment reduction belongs to these benefits and costs, not to a general representative consumer theorem. Total output is Q^*, average consumption is Q^*/η and its cross sectional variance is $\mathrm{Var}_F(\alpha)/(p^*)^4$. More consumers can raise total output while lowering average consumption and variance. Dispersion alone does not establish a welfare improvement.

10.3.3 Destruction, selection and replacement

Suppose firms with $e < c$ disappear, with no atom at the cutoff. The original survivor measure remains $\nu \, dG(e)$ on $[c, \bar{e}]$. Its mass is $\nu[1 - G(c)]$, and supply is

$$S_{\text{loss}}(p) = \nu a p \int_c^{\bar{e}} e \, dG(e).$$

This is surviving mass times conditional mean productivity times ap. Dividing by $1 - G(c)$ without multiplying by surviving mass silently creates additional firms. A different experiment replaces the original producers by a new mass ν drawn from the conditional law. It gives

$$S_{\text{replacement}}(p) = \nu a p \frac{\int_c^{\bar{e}} e \, dG(e)}{1 - G(c)}.$$

Both experiments raise conditional mean productivity. Actual destruction removes productive mass, while replacement preserves it and raises its average productivity. Their equilibrium price effects can have opposite signs. A statement about a conditional mean is not a statement about total productive capacity.

Example 10.5. A worked distribution different from the exam calibration. Let $\eta = \nu = a = 1$, let tastes be uniform on $[0, 2]$ and let productivity have density $g(e) = 2e$ on $(0, 1)$. Then $A_F = 1$ and $E_G = 2/3$. Removing $e < 1/2$ eliminates firm mass $1/4$ but only the productivity moment $1/12$, leaving $\int_{1/2}^1 e \, dG(e) = 7/12$. A new unit mass of selected firms instead has conditional mean productivity $7/9$.

Supply coefficients are $2/3$ originally, $7/12$ after destruction and $7/9$ after replacement. Equating each supply to $1/p^2$ gives prices $(3/2)^{1/3}$, $(12/7)^{1/3}$ and $(9/7)^{1/3}$, respectively. The price ordering follows directly from the supply ordering.

10.3.4 Welfare and the constraint that breaks the shortcut

At fixed wealth, indirect utility is $v(\alpha, p) = w(\alpha) + \alpha/p$, so a lower price benefits every positive taste consumer. Total surplus instead adds producer interests and subtracts real input costs. At clearing, gross consumption benefit is $2\eta A_F/p^*$ and production cost is $\nu a E_G (p^*)^2/2 = \eta A_F/(2p^*)$. Total surplus, excluding initial numeraire, is therefore $3\eta A_F/(2p^*)$. Consumer payments and producer receipts are transfers, not additional resources lost.

If consumers own firms, individual wealth includes distributed profit. Under unrestricted numeraire this does not change goods demand, but it changes utility and money holdings. If money must be nonnegative or borrowing is limited, the demand formula itself can fail. Revisit household KKT conditions and integrate the resulting type regimes before claiming that wealth distribution is irrelevant.

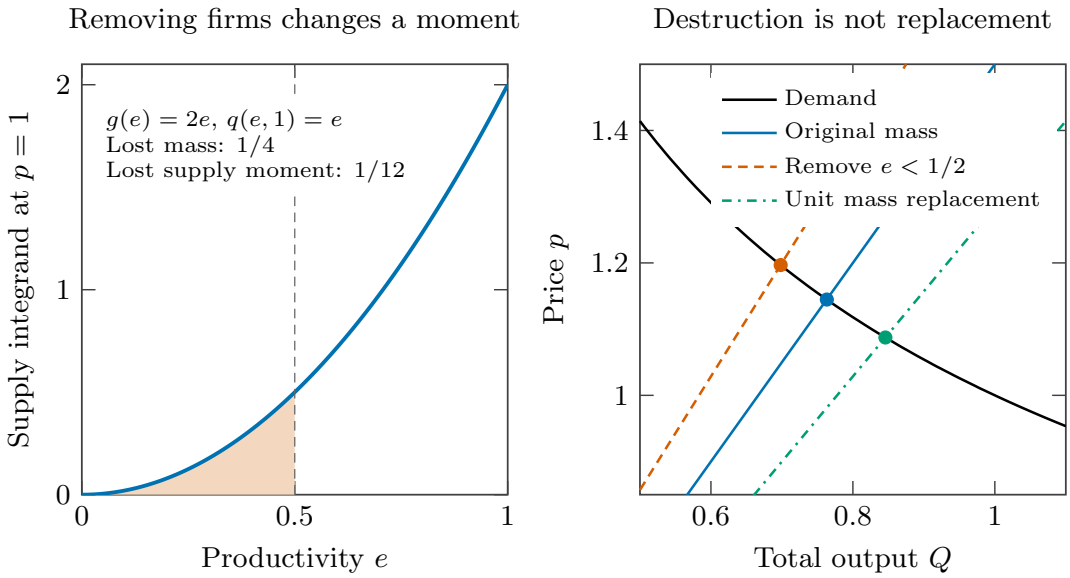


Figure 10.1: Population measures enter equilibrium. The shaded area is the lost productivity moment, not the mass of removed firms. Actual destruction shifts supply inward, whereas unit mass replacement shifts supply outward. The diagram uses the worked distributions, not the exam calibration.

The historical final and historical heterogeneous elasticity problem at the chapter end retain these difficulties. They require population measures, selection, dispersion and welfare, or an aggregation identity with profit ownership. The shorter finite type questions are preparatory practice.

10.4 Welfare in the partial model

Definition 10.6. Pareto efficiency. A feasible allocation is Pareto efficient if no other feasible allocation makes every consumer weakly better off and at least one consumer strictly better off.

Proposition 10.7. *First welfare theorem for the quasilinear model. See[Mas-Colell et al., 1995]. A competitive allocation is Pareto efficient under the stated convexity and local nonsatiation assumptions.*

The planner's problem is

$$\max_{x_i, q_j \geq 0} \left\{ \sum_i \phi_i(x_i) - \sum_j c_j(q_j) \right\} \quad \text{subject to} \quad \sum_i x_i \leq \sum_j q_j.$$

Its KKT multiplier μ satisfies $\phi_i'(x_i) \leq \mu$ with equality for $x_i > 0$ and $\mu \leq c_j'(q_j)$ with equality

for $q_j > 0$. Thus a competitive equilibrium satisfies the planner conditions with $\mu = p$.

Proposition 10.8. *Second welfare theorem for the quasilinear model. See[Mas-Colell et al., 1995]. Under concavity, continuity, and the cheaper-bundle condition stated in the general theorem below, any solution of the planner's problem can be decentralized at a supporting price $p = \mu$ by lump-sum transfers of the numeraire whose sum is zero. The theorem concerns allocations or utility vectors attainable by the planner; an arbitrary utility vector need not be feasible.*

10.5 Problems

Part I: By hand

Problem 10.1 [Core]. Linear partial equilibrium.

Demand and supply are $Q^d(p) = a - bp$ and $Q^s(p) = d(p - c)$ on their economically relevant ranges, with $a, b, d > 0$, $c \geq 0$, and $a > bc$.

(a) Solve for equilibrium price and quantity.

(b) Compute consumer and producer surplus using the choke price a/b of inverse demand and the price intercept c of inverse supply, so that the usual triangular-surplus geometry applies.

(c) Verify that the competitive quantity maximizes total surplus.

Advanced exam practice

These problems build toward integrated graduate exam arguments. The suggested time applies to the hand questions, not the optional tool extension. First draw a diagram or state a solution plan, define the feasible set, and identify possible corners. Explain why each candidate is globally optimal and check the economic accounting. Attempt the hand questions before using tools. For an optional extension, state Seed, Fixed, Scaled and Risk before coding and enlarge one feature at a time.

Difficulty guide. Problems 10.A1 to 10.A4 are preparatory practice. Problems 10.A5 and 10.A6 restore the harder continuum questions from the Western 2019 final examination and the SFU 2024 Problem Set 7. Their source notes distinguish necessary corrections from new extensions. Use the restored questions as the main timed exam preparation. Problem 10.A7 restores the historical equilibrium and demand estimation exercise, with explicit corrections and a labelled time series extension. Problem 10.A8 connects discrete choice demand to pricing and endogeneity before a BLP style computational extension.

Problem 10.A1 [Advanced]. Dispatch, capacity rents, and a technology improvement.

Suggested time: 40 minutes. Total: 40 points.

A competitive electricity market lasts one period. A representative buyer chooses electricity $Q \geq 0$ and money $c \geq 0$ to maximize $c + 14Q - Q^2/2$ subject to $c + pQ \leq 100$, where $p \geq 0$ is the electricity price. The buyer takes prices as given. There are two price taking suppliers. Supplier 1 chooses $q_1 \in [0, 4]$ and pays $C_1(q_1) = 4q_1 + q_1^2/2$. Supplier 2 chooses $q_2 \geq 0$ and initially pays $C_2(q_2) = 8q_2 + q_2^2$. Both suppliers can choose zero output at zero cost. Equilibrium requires $Q = q_1 + q_2$. Ignore externalities. All money payments are transfers between the buyer and the suppliers, and total surplus counts electricity benefits less real production costs.

(a) Derive the buyer's demand and both suppliers' supply functions for every $p \geq 0$. For supplier 1, write a KKT system including the capacity constraint and identify the price at which capacity begins to bind. Explain why each proposed choice is globally optimal. [10]

(b) Find the initial competitive equilibrium. Report p, Q, q_1, q_2, and the capacity multiplier of supplier 1. Check market clearing and the inequalities implied by your proposed regimes. [8]

(c) A software improvement changes only supplier 2's cost to $C_2^A(q_2) = 2q_2 + q_2^2$. There is no adoption cost. Find the new equilibrium and determine whether supplier 1's capacity still binds. Do not retain the old regime without checking it. [10]

(d) Calculate consumer surplus, each supplier's profit, and total surplus before and after the improvement. Does the improvement benefit every market participant? Explain why a supplier can lose profit even when total surplus rises. Interpret supplier 1's capacity multiplier in both equilibria as the local marginal social value of additional capacity, holding the technology regime fixed. [12]

Optional tool extension. Replace the capacity 4 by $K > 0$ and supplier 2's linear cost by $b \geq 0$. Use a script to enumerate candidate active sets, solve their market clearing equations, and reject candidates that violate output or capacity inequalities. Test the original and improved cases exactly using rational arithmetic. Plot equilibrium price, outputs, and the capacity value over a parameter grid. A graph is a diagnostic, not a proof of the regime boundaries.

Problem 10.A2 [Advanced]. A corrective tax that changes who participates.

Suggested time: 45 minutes. Total: 40 points.

Two consumers $i \in \{A, B\}$ choose $x_i \geq 0$ and money $c_i \geq 0$. Their utilities are $u_A = c_A + 12x_A - x_A^2/2$ and $u_B = c_B + 8x_B - x_B^2/2$, and each initially has 100 units of money. The consumer price is $p \geq 0$. A competitive producer chooses $Q \geq 0$ with cost $C(Q) = 2Q + Q^2/2$. A

unit tax $t \in [0, 6]$ creates the wedge between the consumer price p and the producer's receipt $p - t$. The government returns the revenue as lump sum money transfers, which are not conditional on a consumer's own purchase. Wealth is sufficient that money remains positive in every case considered. Total purchases $Q = x_A + x_B$ generate real external damage $D(Q) = Q^2/2$, measured in money and imposed outside these two consumers' choice problems. Social welfare is their gross consumption benefit less production cost and damage. There are no other market failures.

(a) Derive individual demands and producer supply. Find the equilibrium at $t = 0$ and verify whether both consumers buy positive quantities. [8]

(b) Find equilibrium price and quantities as functions of t on $[0, 6]$. Identify all tax thresholds at which a consumer's choice changes regime. Then calculate the equilibrium at $t = 4$. Check the omitted consumer's entry inequality whenever you claim a zero purchase. [12]

(c) Formulate the planner's maximization over $x_A, x_B \geq 0$ and find the unique efficient allocation. Find a unit tax that implements it. Explain why simply assuming that both consumers remain active can give an incorrect tax. [12]

(d) Calculate social welfare at $t = 0$, $t = 4$, and at your efficient tax. For $t = 4$, reconcile welfare with consumer surplus, producer profit, government revenue, and external damage. Is it possible here for a welfare improving tax to reduce one consumer's market consumer surplus? Distinguish that surplus from the consumer's final welfare after any revenue transfer. [8]

Optional tool extension. Replace damage by $\delta Q^2/2$, where $\delta \geq 0$, and for this extension allow any unit tax $t \geq 0$ rather than retaining the upper bound 6. Enumerate the efficient active sets and derive the value of δ at which consumer B exits. Write exact tests for primal feasibility, all zero purchase inequalities, and the equality $t = \delta Q$ at the proposed implementing allocation. Numerical optimization can check examples but does not replace the regime proof.

Problem 10.A3 [Advanced]. Free entry, smaller plants, and the limit of marginal conditions.

Suggested time: 40 minutes. Total: 40 points.

A market has demand $D(p) = \max\{a - p, 0\}$, with $a > 6$. It is generated by a buyer who chooses $c, Q \geq 0$ to maximize $c + aQ - Q^2/2$ subject to $c + pQ \leq M$, where $p \geq 0$ and money wealth $M > a^2/4$. Each potential price taking plant may stay closed and pay zero, or open, choose $q > 0$, and pay $C_F(q) = F + q^2$, where $F > 0$ is an avoidable real setup cost. Thus $C_F(0) = 0$. Begin with $F = 9$ and $a = 12$. In the first model the active mass n of identical atomistic plants may be any nonnegative real number. A free entry competitive equilibrium requires individual global profit maximization, market clearing, and zero maximal profit when

$n > 0$, since further identical plants may enter. Ignore financing constraints and externalities.

(a) For every $p \geq 0$, find a plant's globally optimal supply correspondence. Explain why solving a first order condition conditional on opening does not determine whether the plant should open. Include the price at which the plant is indifferent between opening and staying closed. [10]

(b) For $F = 9$ and $a = 12$, find the equilibrium price, output per active plant, active mass, and total output. Verify all the equilibrium conditions. [8]

(c) A production improvement reduces F from 9 to 1 without changing variable cost. Find the new equilibrium. Calculate the change in consumer surplus and total surplus, counting every active plant's setup cost. Explain how output per plant, total output, and the active mass can move differently. [10]

(d) Now change the model. There are countably many potential plants, the number n of active plants must be an integer, and every active plant remains a price taker. Retain the equilibrium requirement that no inactive plant can earn strictly positive profit at the prevailing price. Set $F = 9$ and $a = 13$. Does such a competitive free entry equilibrium exist? Prove your answer. Explain which divisibility assumption in the first model matters and why an integer entry game would be a different model. [12]

Optional tool extension. For a grid of a and F, compute the divisible mass equilibrium and then test whether it can be implemented by an integer number of identical plants under the exact prevailing price definition in part d. Keep this model separate from a strategic entry game in which an entrant anticipates a new price. Use exact arithmetic when a and $\sqrt{F}$ are rational.

Problem 10.A4 [Advanced]. Two local markets, a capacity bottleneck, and investment.

Suggested time: 45 minutes. Total: 40 points.

Two locations A and B consume a homogeneous service. Their inverse demands are $P_A(q_A) = 10 - q_A$ and $P_B(q_B) = 14 - 2q_B$ over nonnegative quantities, with zero purchases at prices at or above their respective choke prices. They arise from utilities $c_A + 10q_A - q_A^2/2$ and $c_B + 14q_B - q_B^2$, with money sufficiently abundant that its nonnegativity never binds. All production occurs at A. A price taking producer chooses total output $Q \geq 0$ and pays $C(Q) = 2Q + Q^2/2$. An existing transporter buys t units at price p_A and resells them at p_B, at no variable resource cost, subject to $0 \leq t \leq k$, where $k > 0$. It takes both prices as given. Clearing requires $Q = q_A + t$ and $t = q_B$. The producer and transporter maximize profits. There are no taxes or externalities.

(a) Write the surplus maximization problem over $q_A, q_B \geq 0$ with $q_B \leq k$. Derive a KKT

certificate, prove global optimality, and find the allocation as a function of $k > 0$. Identify the exact capacity threshold beyond which additional capacity is unused. [12]

(b) Find prices and profits that decentralize your allocation as a competitive equilibrium. Interpret $p_B - p_A$ and verify the transporter's choice in both the binding and slack capacity regimes. Give all quantities, prices, and the transporter's profit at $k = 1$. [10]

(c) Compare $k = 1$ with $k = 3$. Calculate the change in total surplus and in each location's consumer surplus. Does greater connectivity necessarily make both locations' consumers better off? Explain how this can coexist with a total surplus gain. [8]

(d) A planner can expand the original $k = 1$ link by $e \geq 0$ at real cost e^2, so $k = 1 + e$. Find the globally optimal expansion and verify the regime used. Relate the expansion condition to the capacity price gap. Explain why the transporter's lost or gained rent is not an additional resource cost of the project. [10]

Optional tool extension. Extend the planner model to three locations and two directed links. List flow conservation and link capacity constraints explicitly. Enumerate active links, derive nodal prices from marginal benefits and production costs, and check every zero flow or capacity inequality. First reproduce the two location solution exactly. A solver's allocation is not sufficient without those checks.

Problem 10.A5 [Advanced]. The original continuum market: population, productive capacity and welfare.

Restored from the Western University Fall 2019 final examination. The original model and five part structure are retained. The two interpretations of producer selection are stated separately, and the welfare comparison holds initial wealth fixed.

Suggested time: 60 minutes. Total: 60 points. For a timed rehearsal, attempt parts (a) to (d) in 45 minutes and complete part (e) afterward.

Consider a competitive market for a consumption good x and a numeraire m. There is a mass $\eta \geq 1$ of consumers. A consumer has taste $\alpha \sim U[0, 1]$, initial numeraire wealth $\omega \geq 0$, and utility $u_\alpha(x, m) = 2\sqrt{\alpha x} + m$. The joint distribution F of (α, ω) is held fixed across comparisons, and its mean wealth is $\Omega < \infty$. The consumer chooses $x \geq 0$ and unrestricted $m \in \mathbb{R}$ subject to $px + m \leq \omega$. Thus borrowing is allowed, the initial endowment of the consumption good is zero, and initial wealth is a fixed partial equilibrium primitive. Do not add producer profit to this consumer budget.

Initially there is a unit mass of competitive firms. Their productivity e is uniformly distributed on $(0, 1)$, and a type e firm chooses output $q \geq 0$ with numeraire cost $C(q; e) = q^2/(2e)$. The

output price is $p > 0$ and the numeraire price is 1. There is no fixed cost or entry margin.

(a) Find each consumer's goods demand and each firm's supply. Aggregate over both populations and solve for the equilibrium price and total quantity. Distinguish a mean quantity from a population total, and justify individual optimality and equilibrium uniqueness. [14]

(b) Derive the individual profit function $\pi(p, e)$ and verify Hotelling's lemma by differentiating with respect to the output price. [8]

(c) Consider the original question's earthquake interpretation: all firms with $e < 1/2$ disappear without replacement. State the mass of surviving firms, derive aggregate supply, and find the new equilibrium price and quantity. Now consider a different experiment: replace the initial firms by a unit mass whose productivity is uniformly distributed on $[1/2, 1]$. Solve this experiment as well. Compare both prices with the initial equilibrium and explain why a higher conditional mean productivity does not by itself determine the effect of an earthquake. [16]

(d) At the initial equilibrium, derive the cross sectional variance of an individual consumer's goods consumption. How does it change with η when the producer population and the joint type and wealth distribution are held fixed? This part uses the absolute variance, not a normalized relative inequality measure. [10]

(e) Derive average indirect utility and average remaining numeraire consumption at the initial equilibrium. How do they change with η? Reconcile the welfare comparison with part (d), and explain why holding initial wealth fixed is different from holding remaining numeraire consumption fixed. [12]

Problem 10.A6 [Advanced]. Heterogeneous elasticities, a hidden aggregation identity and ownership income.

Restored from the first exercise of the SFU Fall 2024 Problem Set 7, also used at Western in 2020. The continuum model and ownership accounting are retained. The endpoint utility is normalized, and the price limit and resource accounting are corrected. Part (d) and the optional tool extension are new practice extensions.

Suggested time: 60 minutes. Total: 60 points. Work by hand.

There is a unit mass of consumers indexed by (α, ω). Their joint probability distribution is F, its taste marginal is $\alpha \sim U[1, 2]$, and the initial numeraire endowment $\omega \geq 0$ has finite mean Ω. A consumer chooses a consumption good $x > 0$ and numeraire $m \in \mathbb{R}$. The numeraire can be negative, so no borrowing or nonnegative money constraint is imposed. For a fixed demand parameter $A \in (0, 1]$, utility is $u_\alpha(x, m) = \phi_\alpha(x; A) + m$, where $\phi_\alpha(x; A) = A^{1/\alpha} \frac{\alpha}{\alpha - 1}(x^{1-1/\alpha} - 1)$ for $\alpha > 1$ and $\phi_1(x; A) = A \log x$.

There is also a unit mass of competitive firms with type $\beta \sim U[1,2]$. A type β firm chooses $q \geq 0$ and uses the numeraire input $C_\beta(q) = q^{1+1/\beta}/(1+1/\beta)$. The consumption good price is $p > 0$ and the numeraire price is 1. There are no fixed costs or entry choices. Every firm is owned by consumers. The ownership kernel is a measurable function $\theta(\alpha, \omega, \beta) \geq 0$ satisfying $\int \theta(\alpha, \omega, \beta) dF(\alpha, \omega) = 1$ for every β. The consumer's income is $y(\alpha, \omega; p) = \omega + \int_1^2 \theta(\alpha, \omega, \beta)\pi_\beta(p) d\beta$, and the budget is $px + m \leq y$. Assume the displayed integrals are finite. The initial endowment of the consumption good is zero.

(a) Verify the utility limit at $\alpha = 1$. Derive the consumer's unique demand and the firm's unique output and profit at every $p > 0$. Give global optimality arguments, not only first order equations. Explain why ownership income does not affect the goods demand under the stated budget domain. [10]

(b) Define $H(p) = \int_1^2 p^t dt$. Derive its expression for $p \neq 1$, its value at $p = 1$, and $H'(1)$. Express aggregate goods demand $D(p)$ and supply $S(p)$ using H. Find an identity relating $D(p)$, $S(p)$ and p that remains valid at $p = 1$. Explain which feature of the two type distributions makes this identity possible. [14]

(c) Find the unique equilibrium price and total quantity for every $A \in (0, 1]$. Verify the case $A = 1$ directly instead of canceling terms containing $\log p$. Derive each type's equilibrium quantity and show how consumer and producer types can be paired to check market clearing. [10]

(d) Compute the price elasticity with respect to A. Express the elasticity of total equilibrium output with respect to A as a weighted average of producer types, and derive meaningful bounds. Evaluate its limiting value as A approaches 1. Explain why individual constant elasticities do not imply a constant aggregate quantity elasticity along this equilibrium path. Do not infer a welfare ranking merely from changing a taste parameter. [12]

(e) Derive mean numeraire consumption, including distributed profits. Prove that average consumption of the numeraire plus the numeraire used in production equals its initial aggregate endowment. At $A = 1$, evaluate total profit, total production cost and mean numeraire consumption. Explain precisely which assumption supports the claim that the distribution of initial wealth and ownership does not change the equilibrium goods price, and why that claim can fail if consumers must have $m \geq 0$. [14]

Optional tool extension. First replace the uniform laws by a consumer elasticity distribution G on $[1,2]$ and a producer distribution equal to the law of $3 - \alpha$ when $\alpha \sim G$. Derive the new aggregate identity and price for consumer mass $M_c > 0$ and firm mass $M_f > 0$. State how the ownership normalization must change when the consumer measure has mass M_c. Then choose

a producer distribution that is not this reflection and test whether the same closed form still holds. Verify individual budgets, ownership totals, both aggregate resource equations and the $p = 1$ limits. Numerical integration is a check, not a substitute for the aggregation proof.

Problem 10.A7 [Advanced]. How market clearing creates endogeneity.

Restored from Exercise 2 of the SFU Fall 2024 Problem Set 7, also used at Western in 2019. The original static market model and four identification tasks are retained. Slope terminology, stochastic assumptions and the excluded supply shifter are corrected. Part (e) and Part II are new time series extensions.

Suggested time: 50 minutes for Part I. Total: 60 points. Work by hand before using a program.

Part I. By hand.

At each date t, a competitive market clears after a demand shock u_t and a supply shock v_t are realized. Good 2 is the numeraire; p_t is the price of good 1 in units of good 2, and Q_t is its traded quantity. Aggregate demand and supply on their interior ranges are $Q_t^d = \alpha_0 + \alpha_1 p_t + u_t$ and $Q_t^s = \beta_0 + \beta_1 p_t + v_t$, where $\alpha_1 < 0 < \beta_1$ and all four coefficients are fixed. There is no storage or choice linking dates. Equilibrium requires $Q_t = Q_t^d = Q_t^s$. Assume a bounded shock support on which the clearing price and quantity are strictly positive; state the required inequalities in part (a). Do not truncate negative computed observations or replace them by zero.

Initially, (u_t, v_t) are independent and identically distributed across dates, have mean zero, satisfy $\mathrm{Cov}(u_t, v_t) = 0$, and have finite variances σ_u^2, σ_v^2 with $\sigma_u^2 + \sigma_v^2 > 0$. The shocks need not be independent within a date. Write $D = \beta_1 - \alpha_1 > 0$. An analyst observes only equilibrium (p_t, Q_t) and regresses Q_t on a constant and p_t. Let $\widehat{\gamma}_{1,T}$ denote the ordinary least squares slope from T observations, using sample covariances with sample means removed.

(a) Derive the unique clearing price and quantity for every shock realization. State the two inequalities defining the strictly positive equilibrium region and verify both original market equations. Explain why observations from different dates do not generally trace one fixed demand curve. [12]

(b) Derive $\mathrm{Cov}(p_t, u_t)$, $\mathrm{Cov}(p_t, v_t)$, $\mathrm{Var}(p_t)$ and the probability limit of $\widehat{\gamma}_{1,T}$. Express that limit as a weighted average of the demand and supply slopes. Calculate its difference from α_1. Explain the economic source of the correlation rather than calling it only an omitted variable problem. Distinguish a slope from the price elasticity of demand at an interior observation. [16]

(c) Characterize exactly when this regression identifies the demand slope under the maintained assumptions, and when it instead identifies the supply slope. What if neither shock varies? For

$\alpha_1 = -2$, $\beta_1 = 1$, $\sigma_u^2 = 4$ and $\sigma_v^2 = 1$, determine the sign and value of the limiting regression slope. Can the fitted relationship slope upward even though every demand curve slopes downward? [8]

(d) Replace the supply shock by $v_t = -\delta z_t + \eta_t$, where $\delta > 0$ and z_t is an observed cost shifter. For this part, u_t, η_t, z_t are mutually independent, centered and independent across dates, with finite variances and $\mathrm{Var}(z_t) > 0$. Retain the positive equilibrium support restriction. Derive the covariance of z_t with equilibrium price and quantity. Show which slope is identified by their ratio and justify consistency of the corresponding sample covariance estimator. Give a plausible economic cost shifter and state its exclusion restriction. Explain why a shifter that directly enters demand is not generally an excluded instrument for demand. Which slope would a pure demand shifter identify under the symmetric exclusion assumptions? [12]

(e) New time series extension. Keep the static market and the cost shifter formulation, but let $u_t = \rho_u u_{t-1} + \varepsilon_t$, $\eta_t = \rho_\eta \eta_{t-1} + \xi_t$ and $z_t = \rho_z z_{t-1} + \omega_t$, where $|\rho_u|, |\rho_\eta|, |\rho_z| < 1$. Each process is initialized in its stationary law. The three innovation sequences are mutually independent across processes and dates, centered, bounded and have finite variance. The three stationary processes are independent of one another; their supports and intercepts ensure positive clearing prices and quantities. You may use the stationary moment law of large numbers for these processes. Does contemporaneous z_t remain a valid demand instrument? Derive $\mathrm{Cov}(p_{t-1}, u_t)$ and assess whether lagged price is automatically a valid and relevant instrument. Explain why correcting standard errors for serial correlation cannot repair an invalid instrument. Does the simultaneity problem here require stochastic trends or dynamic optimization? [12]

Part II. Codex and verification.

Seed: the clearing equations and population moments from Part I. Fixed: the static economic model, coefficients, marginal shock variances, independence across the three shock processes, sample length and estimator definitions. Scaled: a common persistence parameter $\rho_u = \rho_\eta = \rho_z = \rho$ over $0, 1/4, 1/2, 3/4, 4/5$. Risk: confusing serial dependence with identification, using a weak covariance denominator, or silently changing the economic model by clipping observations.

Use $\alpha_0 = 100$, $\beta_0 = 0$, $\alpha_1 = -2$, $\beta_1 = 1$, $\delta = 1$, and stationary marginal variances $\mathrm{Var}(u_t) = \mathrm{Var}(\eta_t) = \mathrm{Var}(z_t) = 1$. One bounded implementation uses independent innovations taking the two values $\pm\sqrt{1-\rho^2}$ with equal probability for each process. Document stationary initialization or a controlled burn period. Keep $T = 20000$ retained observations per run and use several recorded random seeds.

Compute clearing prices and quantities directly, verify both market equations and prove that the prescribed support keeps them positive. Compare ordinary least squares, the current cost

instrument and the lagged price instrument with the population limits derived by hand. At $\rho = 0$, report a population first stage that vanishes as irrelevance, not as a numerical estimate of a structural slope. Report covariance denominators and sampling variation; if reporting standard errors, use a procedure valid for dependent moment observations.

Include a deliberately invalid instrument $z_t^{bad} = z_t + u_t$ and explain which moment restriction fails. Distinguish the finite stationary experiment from a spurious regression between unrelated stochastic trends. Your program must reproduce the independent date benchmark and the analytical moment identities before presenting graphs. A simulation does not establish an exclusion restriction or a continuum of parameter claims.

Problem 10.A8 [Advanced]. Prices, quality and a small differentiated product market.

This original teaching case connects stochastic choice to firms that set prices. It is inspired by Berry, Levinsohn and Pakes, Automobile Prices in Market Equilibrium, Econometrica 1995, DOI 10.2307/2171802. The hand calculation is a homogeneous logit benchmark, not the full BLP model or its automobile application.

Part I. By hand.

Allow 50 minutes. There are 50 practice points. In each market t, a unit mass of consumers chooses at most one of two inside products $j = 1, 2$, or the outside option $j = 0$. Product j has an observed scalar characteristic x_{jt}, coefficient β, and quality ξ_{jt} that consumers and firms know but an analyst may not observe. Write $\delta_{jt} = \beta x_{jt} + \xi_{jt}$. Consumer i obtains $u_{ijt} = \delta_{jt} - \alpha p_{jt} + \varepsilon_{ijt}$ from an inside product and $u_{i0t} = \varepsilon_{i0t}$ from the outside option. Here $p_{jt} \geq 0$ is a price, $\alpha > 0$ is the marginal utility cost of a unit of expenditure, and all ε_{ijt} are independent across consumers, products and markets with distribution function $F(e) = \exp\{-\exp(-e)\}$. The outside option has price and deterministic utility zero. Consumers take prices as given; income constraints do not bind in this quasilinear demand specification.

You may use the logit formula: if $v_{jt} := \delta_{jt} - \alpha p_{jt}$, then $s_{jt} = e^{v_{jt}}/(1 + e^{v_{1t}} + e^{v_{2t}})$ and $s_{0t} = 1/(1+e^{v_{1t}}+e^{v_{2t}})$. These shares are probabilities over the whole market, including consumers who buy neither inside product.

(a) Derive $\log(s_{jt}/s_{0t})$ and all own and cross price derivatives of the shares, holding qualities fixed. For $k \neq j$, including $k = 0$, define the diversion ratio from j to k as $D_{j\to k} := (\partial s_{kt}/\partial p_{jt})/(-\partial s_{jt}/\partial p_{jt})$. Derive it and explain what the formula does, and does not, say about products that are near one another in characteristic space. [12 points]

(b) Each product is sold by a different firm. Firm j has constant marginal cost $c_j > 0$, no

fixed cost or capacity constraint, and chooses $p_j \geq 0$ to maximize $\pi_j = (p_j - c_j)s_j$. It observes qualities and takes the rival price as given. Derive its interior pricing condition and express the markup $p_j - c_j$ using α and its share. Then prove that the condition identifies the unique global best response, not merely a stationary price. You may first exclude prices below cost and then study how the sign of the profit derivative changes. Do not assume that the profit function is globally concave. [14 points]

(c) Set $\alpha = 1$, $\beta x_{jt} = 0$ and $c_1 = c_2 = 1/2$ in both markets. In market 1, both qualities are $\xi_{j1} = 2$ and a proposed price profile is $p_{11} = p_{21} = 2$. In market 2, both qualities are $\xi_{j2} = 9/4 + \log 3$ and a proposed profile is $p_{12} = p_{22} = 9/4$. Compute all three shares, each firm's markup and profit in each market. Verify that both profiles are Nash equilibria using the global best response argument. Within each market, hold its supplied qualities fixed when considering a price deviation. [12 points]

(d) An analyst pools the four product observations and runs an ordinary least squares regression with an intercept of $y_{jt} := \log(s_{jt}/s_{0t})$ on p_{jt}, omitting quality. Calculate the fitted price slope exactly. Compare it with the structural price coefficient in utility. Explain how a higher price and a higher share can occur together across these markets without violating downward sloping demand. State an economically plausible excluded cost shifter, the exclusion and relevance conditions it would need, and why two constructed markets do not establish those conditions. [12 points]

Part II. Codex and verification.

Scale one feature: replace the common price coefficient by two known consumer types, with $\alpha_1 = 1/2$ and $\alpha_2 = 3/2$, each of mass $1/2$. Hold the two products, two markets, recorded prices, qualities, outside option and shock distribution fixed. This is a conditional demand inversion exercise. Do not assume that the old prices remain pricing equilibria after the preference distribution changes.

Let $\bar{\alpha} = 1$ and define the mean utility net of the mean price effect by $v_{jt} = \delta_{jt} - \bar{\alpha}p_{jt}$. Type h has deterministic inside utility $v_{jt} - (\alpha_h - \bar{\alpha})p_{jt}$. Compute its logit probabilities, including the outside option, and average over types to obtain aggregate shares $s_{jt}(v_t, p_t)$. First generate exact model target shares S_{jt} from the supplied qualities and prices. Keep the known type distribution fixed throughout inversion.

Write a small program at `work/blp_case.py` that recovers v_t from the target inside shares and the recorded prices. You may use the iteration $v_{jt}^{r+1} = v_{jt}^{r} + \log S_{jt} - \log s_{jt}(v_t^r, p_t)$, where r is an iteration count. State initialization, stopping rule, tolerance and maximum iteration count.

Compute probabilities stably and keep the outside deterministic utility fixed at zero. Report a failure rather than a recovered answer if the stopping conditions are not met.

Run `python3 work/blp_case.py --check`. The check must compare recovered mean utilities with the values used to generate the targets, reconstruct all inside and outside shares, verify probability adding up, and test several distinct initial vectors. Require maximum share error below 10^{-10} and maximum mean utility error below 10^{-8}. Include the homogeneous limit in which both price coefficients equal one, one deliberately invalid target with zero outside share that the input checks reject, and a demonstration that the homogeneous log odds formula need not invert the mixed model. Numerical success is not a proof of a global contraction on all utility vectors.

Before coding, record the hand seed, fixed assumptions, the single heterogeneity extension and its new failure risk. Submit source, material prompts, the validation command and output, and one generated step checked independently. Finally explain why share inversion conditional on known taste parameters does not itself identify unknown taste parameters or solve price endogeneity. Discuss what additional markets, excluded cost variation and rank conditions an actual estimation design would require. Estimating the full BLP model, simulating an automobile industry and proving a general identification theorem are not part of this exercise.

Part II: Music sales and concerts — chapter use case and verified scale-up

Problem 10.2 [Applied]. A per-unit tax.

Introduce a per-unit tax $t > 0$ paid by sellers into the market of Problem 10.1, so buyers pay p_b and sellers receive $p_s = p_b - t$. Assume t is small enough that equilibrium quantity remains nonnegative.

(a) Solve for p_b, p_s, and quantity.

(b) Derive tax incidence. Express the local burden shares using the demand and supply slopes, and interpret them using point elasticities evaluated at the common no-tax equilibrium.

(c) Compute deadweight loss and its derivative with respect to t.

Problem 10.3 [Proof]. Planner and competitive allocation.

A representative consumer has quasilinear utility $u(q)+m$, with enough transferable numeraire that the optimum is not constrained by a lower bound on m. The functions u and C are differentiable on $(0, \infty)$, have the required right derivatives at zero, u is increasing and strictly concave, and C is increasing and convex. Assume the surplus maximum over $q \geq 0$ is attained.

(a) State and solve the planner's problem using Kuhn-Tucker conditions.

(b) Show that any interior competitive equilibrium satisfies the same condition.

(c) State the assumptions required for the converse decentralization claim.

Chapter use case

Seed: Problems 1.2, 9.1, 10.1, 10.3. Fixed: all unlisted primitives and assumptions. Scaled: the music-volume or royalty share τ from the two benchmark values to a grid. Tool: Codex + Python. Verification: Reproduce the demand and cost formulas and both endpoint comparisons; bracket every stationary point, check both price boundaries, establish the global maximizer, and keep the volume-share and royalty-share scenarios distinct.

Problem 10.4 [Applied]. The Economics of Superstars: Music Sales and Concerts.

Taylor S. is a stylized superstar whose fans consume music services and concerts. This is the graduate version of the music-and-concerts case, with explicit preferences, technology, and market assumptions. All numbers are hypothetical; the model is not an empirical description of Taylor Swift's contracts. Quantities are divisible, and the mass of identical fans is one.

Part I. Fan demand. Write q_M, q_C for music and concert services, the two coordinates of consumption. A fan has wealth $w > 0$, prices $p_M, p_C > 0$, and utility $u(q_M, q_C) = (\frac{1}{2}q_M^\rho + \frac{1}{2}q_C^\rho)^{1/\rho}$, where $0 < \rho < 1$. The letters M, C are local commodity labels.

(a) Formulate the budget problem and derive demands using $\sigma = 1/(1-\rho)$. For the remaining parts set $\rho = 1/2$, $w = 1$, and $p_M = 1$. Explain why music demand changes when the concert price changes.

Part II. Production. For $j \in \{M, C\}$ the artist's technology is $y_j = A_j\sqrt{n_j}$, where $n_j \geq 0$ is labor and $A_j > 0$ productivity. Labor costs $\nu > 0$ per hour, distinct from fan wealth w. The processes are independent. This stylized music-service cost does not describe the cost of creating an original recording as a function of its streams.

(b) Derive $C_j(\nu, y_j)$, average cost, and marginal cost. Explain decreasing returns. Set $A_M = A_C = \nu = 1$ in the numerical parts.

Part III. A share of volume and receipts. An exogenous $\tau \in (0, 1]$ is the fraction of music consumption attributable to this artist. Her output is $y_M = \tau q_M$ and her music receipts are $\tau p_M q_M$. Concert output is $y_C = q_C$. She takes p_M, τ as given and chooses $p_C > 0$. There is no shared labor constraint or fixed cost.

(c) Write total revenue and profit after substituting demand and costs. Distinguish a revenue derivative with respect to ticket price from marginal revenue with respect to output. Account for

changes in both revenue streams and production costs.

(d) Derive the first-order condition for the optimal concert price. Examine profit as $p_C \downarrow 0$ and $p_C \to \infty$. Establish global optimality and uniqueness, analytically or with a justified numerical method accounting for every stationary point and boundary.

(e) Compare $\tau = 1$ and $\tau = 1/2$. In a table report optimal concert prices, both fan quantities, both delivered outputs, revenues, costs, and profits. State numerical accuracy. Does the concert price rise or fall when the music share falls?

(f) Use the first-order condition or implicit differentiation to analyze the price response for $\tau \in [1/2, 1]$. Interpret the roles of substitution and production costs. Do not assume the same sign for every technology or parameter value.

Part IV. A royalty share. Now τ changes only the fraction of music revenue paid to the artist: music output is $y_M = q_M$, while receipts remain $\tau p_M q_M$. All other assumptions are unchanged.

(g) Write the revised profit function and repeat the comparison between $\tau = 1$ and $\tau = 1/2$. Identify the changed cost channel and determine whether the earlier price response survives.

(h) Explain what evidence distinguishes a fall in music volume from a fall in the royalty fraction. Discuss the roles of CES substitution, decreasing returns, and control of concert prices. Contrast the artist's decision with price-taking supply in the main chapter.

Chapter 11

General Equilibrium

General equilibrium connects individual choices through common prices and common resource constraints. Begin with the Edgeworth box: one picture makes feasibility, gains from trade and endowment wealth visible. Then ask whether a clearing price exists, what welfare properties the allocation has, whether prices are unique, and what observed equilibria can reveal about preferences.

The chapter follows one economic model rather than restarting it under separate headings. Exchange supplies the first benchmark; production adds transformation and ownership income. Existence, welfare, uniqueness, adjustment and identification are different questions. Each needs its own assumptions, even when the same example illustrates several of them.

For a first reading, work through the box, aggregate excess demand, the Brouwer proof and the welfare arguments. Then study regularity, gross substitutes, coalition blocking and the empirical restrictions. The technical appendix gives the boundary details and the Kakutani extension when demand is not unique. The chapter ends with one problem program organized into three topic groups, preserving the released problem identifiers.

11.1 Pure exchange economies: the Edgeworth box

A market connects individual optimization to a common resource constraint. How can trade improve both consumers' welfare without creating more goods, and what must prices achieve? The Edgeworth box makes these questions visible. Every point gives two consumption bundles. Individual rationality asks whether each consumer accepts the trade; efficiency asks whether a Pareto improvement remains; equilibrium additionally requires a common price and the correct

wealth for each consumer. Keep these tests separate when solving an exchange problem.

11.1.1 One point, two consumers

Consider two consumers $i = 1, 2$ and two goods $\ell = 1, 2$. Consumer i has consumption set $X_i = \mathbb{R}^2_+$, preferences $\succeq_i$ and endowment $\omega_i = (\omega_{i1}, \omega_{i2})$. Aggregate endowment is $\omega = \omega_1 + \omega_2 \gg 0$. An allocation $x = (x_1, x_2)$ is feasible when $x_1 + x_2 \leq \omega$ componentwise. The rectangle represents allocations with no disposal: its horizontal length is $\omega_1^{\text{tot}} = \omega_{11} + \omega_{21}$ and its vertical height is $\omega_2^{\text{tot}} = \omega_{12} + \omega_{22}$. The superscript distinguishes a total of a good from a consumer's endowment vector.

Use the lower left origin for consumer 1. A point with coordinates (a, b) assigns $x_1 = (a, b)$ and $x_2 = (\omega_1^{\text{tot}} - a, \omega_2^{\text{tot}} - b)$. Consumer 2 therefore reads quantities from the upper right origin, increasing to the left and downward. Never read both consumers' bundles from the lower left origin. Under strictly increasing preferences a Pareto optimum cannot discard either good; otherwise the rectangle does not describe every feasible allocation.

The endowment point gives $x_1 = \omega_1$ and $x_2 = \omega_2$. Trading moves the allocation, not the box: pure exchange changes who holds each good but cannot change the totals. Production will change this resource accounting later in the chapter.

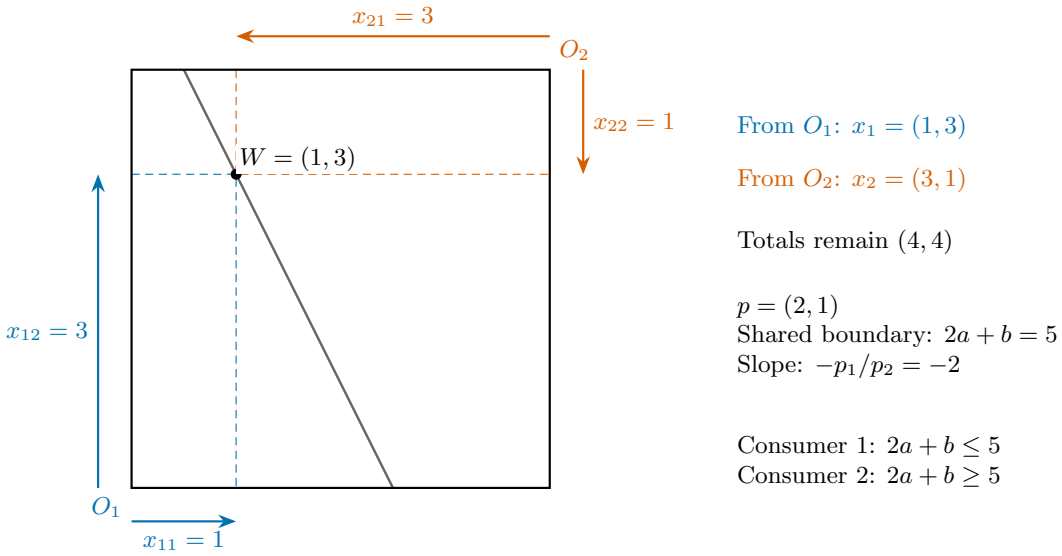


Figure 11.1: One point, two bundles. The endowment W gives consumer 1 $(1, 3)$ from the lower left origin and consumer 2 $(3, 1)$ from the upper right origin. At the illustrative prices $(2, 1)$ the shared boundary has slope -2, and the affordable sides are opposite. These prices illustrate budgets, not equilibrium.

11.1.2 Gains from trade before prices

An allocation is individually rational relative to the endowment if $x_i \succeq_i \omega_i$ for both consumers. Draw each consumer's indifference curve through the endowment and shade the intersection of their weakly preferred sets. This is the trade lens. With continuous, strictly increasing utilities, every interior point of the lens strictly improves both consumers' utility. The lens is a set of acceptable trades, not a demand curve and not a set of equilibria.

Example 11.1. A box we can calculate. Let $\omega_1 = (1, 3)$, $\omega_2 = (3, 1)$ and $u_i(x_i) = \sqrt{x_{i1}x_{i2}}$. Total resources are $(4, 4)$ and both initial utilities equal $\sqrt{3}$. In consumer 1 coordinates, the lens is described by $ab \geq 3$ and $(4 - a)(4 - b) \geq 3$, together with $0 \leq a, b \leq 4$. Thus its lower boundary is $b = 3/a$ and its upper boundary is $b = 4 - 3/(4 - a)$ wherever these expressions lie in the box. Both formulas refer to one drawing but to different consumers.

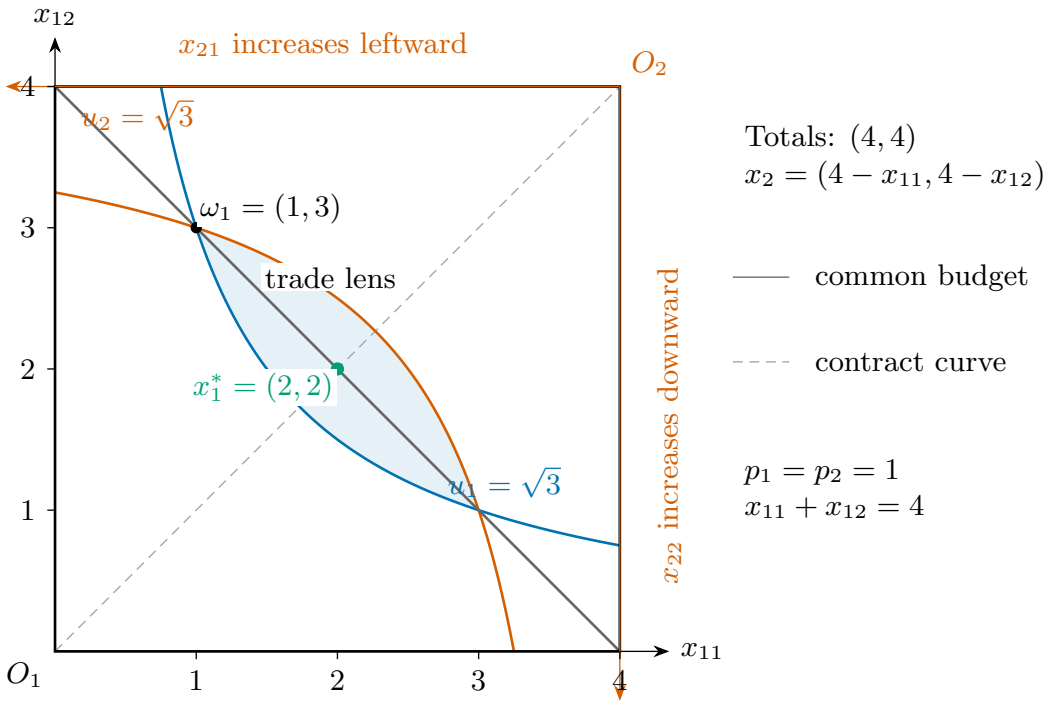


Figure 11.2: Reading an exchange economy. The blue and orange curves are the initial indifference curves of consumers 1 and 2. The shaded lens contains allocations weakly preferred by both to the endowment. The descending line is their common budget boundary at $p_1 = p_2 = 1$; their affordable sides are opposite. The diagonal is the contract curve and $(2, 2)$ is the equilibrium bundle of consumer 1.

11.1.3 Efficiency is not a tangency formula

Definition 11.2. Pareto efficiency. A feasible allocation x is Pareto efficient if there is no feasible x' with $x'_i \succeq_i x_i$ for every consumer and $x'_i \succ_i x_i$ for at least one. The Pareto set depends on preferences and aggregate resources. Individual endowments matter for voluntary participation and wealth, not for this definition.

In a smooth interior allocation with strictly positive marginal utilities, define the marginal rate of substitution by $\text{MRS}_i = (\partial u_i/\partial x_{i1})/(\partial u_i/\partial x_{i2})$. If $\text{MRS}_1 > \text{MRS}_2$, consumer 1 is willing to give more of good 2 for an additional unit of good 1 than consumer 2 requires to give it up. A sufficiently small trade can therefore benefit both. Interior efficiency requires equal marginal rates of substitution. This is a necessary local test, not a universal definition or an unconditional sufficiency theorem.

The contract curve is the set of efficient allocations in the box, including efficient boundary points. With kinks, use supporting slopes or solve the resource constrained problem directly. With corners, the missing good imposes a one sided condition. With nonconvex preferences, equal slopes can describe a nonoptimal stationary allocation. For a global test, impose a lower bound on one consumer's utility and maximize the other's over feasible allocations. If the maximum equals the candidate utility, also check whether a maximizing tie can strictly improve the first consumer. A weak Pareto improvement cannot be ruled out by a tangency or a single objective value alone.

Example 11.3. For the preceding square root utilities, the contract curve is $a = b = t$, $0 \le t \le 4$. To verify the entire set, not just interior tangencies, note that $\sqrt{ab} \le (a+b)/2$ and $\sqrt{(4-a)(4-b)} \le 4 - (a+b)/2$. Replacing an off diagonal point by (t,t) with $t = (a+b)/2$ strictly increases both utilities. On the diagonal, retaining utility at least t for consumer 1 requires $a + b \ge 2t$, so consumer 2 cannot obtain more than $4 - t$. Strictly raising consumer 1's utility above t forces $a + b > 2t$ and lowers consumer 2's maximum utility below $4 - t$. This proves efficiency of every diagonal point, including both endpoints.

For this two consumer economy, the individually rational part of the Pareto set is the strong core under the strong blocking convention: a coalition blocks when every member is weakly better off and at least one is strictly better off. Here the strong core is the diagonal segment $\sqrt{3} \le t \le 4 - \sqrt{3}$. Individual rationality rules out singleton blocking and efficiency rules out blocking by the grand coalition. The ordinary core uses a different strict blocking convention in these notes; do not silently exchange the two definitions.

11.1.4 Prices select an allocation through wealth

At prices $p \gg 0$, consumer i has wealth $p \cdot \omega_i$ and budget set $B_i(p) = \{x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i\}$. The price vector is common but wealth is not generally the same. On the no disposal rectangle, the two budget boundaries are the same geometric line through the endowment, with slope $-p_1/p_2$. Consumer 1 can afford the side toward the lower left origin; consumer 2 can afford the side toward the upper right origin. A feasible point is not automatically affordable to both.

Definition 11.4. Walrasian equilibrium in the exchange economy. A pair (p^*, x^*) with $p^* \gg 0$ is an equilibrium if $x_1^* + x_2^* = \omega$ and, for each i, $x_i^* \in B_i(p^*)$ and $x_i^* \succeq_i x_i$ for every $x_i \in B_i(p^*)$. This section works with positive prices; models requiring zero prices need the more general boundary treatment. Multiplying all prices by the same positive number changes neither budgets nor choices.

Example 11.5. In the square root example, demands are $x_{i1}(p) = (p \cdot \omega_i)/(2p_1)$ and $x_{i2}(p) = (p \cdot \omega_i)/(2p_2)$. Summing good 1 demands gives $2 + 2p_2/p_1$. Equating this to total supply 4 yields $p_1/p_2 = 1$. Normalize $p_2 = 1$. Both wealths equal 4, both bundles equal $(2, 2)$, both budgets bind and both markets clear. A common tangency at another diagonal point would not satisfy these original wealth constraints.

Cobb Douglas exchange with a parameter. Suppose $u_i(x_i) = x_{i1}^{\alpha} x_{i2}^{1-\alpha}$, $0 < \alpha < 1$, and endowments are $\omega_1 = (1, 2)$ and $\omega_2 = (2, 1)$. Then $x_1(p) = (\alpha(p_1 + 2p_2)/p_1, (1-\alpha)(p_1 + 2p_2)/p_2)$ and $x_2(p) = (\alpha(2p_1 + p_2)/p_1, (1-\alpha)(2p_1 + p_2)/p_2)$. Clearing good 1 yields $p_1/p_2 = \alpha/(1-\alpha)$. After normalizing $p_2 = 1$, substitute this ratio into both demands and verify good 2 as well. This check distinguishes a complete equilibrium from a price candidate.

11.1.5 Welfare and redistribution

Proposition 11.6. *First welfare theorem for the exchange box. If preferences are locally nonsatiated and (p^*, x^*) is a Walrasian equilibrium, then x^* is Pareto efficient.*

The economic argument is a budget contradiction. Any bundle weakly preferred to an optimum must cost at least the consumer's wealth, and a strictly preferred bundle must cost strictly more. A Pareto improvement would therefore have aggregate expenditure strictly above $p^* \cdot \omega$, although feasibility permits no more than $p^* \cdot \omega$. Local nonsatiation supplies the weak expenditure inequality. Convex preferences are not needed for this direction.

Proposition 11.7. *Second welfare theorem in a smooth interior case. Suppose utilities are concave, differentiable at a strictly positive Pareto optimum x^*, and have strictly positive marginal*

utilities there. Then there are common supporting prices $p \gg 0$ and constants $\lambda_i > 0$ with $\nabla u_i(x_i^*) = \lambda_i p$. Set $T_i = p \cdot (x_i^* - \omega_i)$. Feasibility without disposal gives $T_1 + T_2 = 0$, and x_i^* maximizes utility on $p \cdot x_i \leq p \cdot \omega_i + T_i$. Concavity turns the supporting gradient conditions into global budget optimality.

This statement separates price support from affordability. The general convex preference version later in the chapter first obtains a price quasi equilibrium; an additional condition, such as a strictly cheaper feasible bundle for each consumer, is needed to obtain full budget optimality. Boundary allocations and zero prices require care. A tangency picture alone does not establish the general second welfare theorem.

Example 11.8. An efficient allocation need not be an equilibrium at the original endowment. In the $(4, 4)$ box, consider $x_1^* = (3/2, 3/2)$ and $x_2^* = (5/2, 5/2)$. Both marginal rates of substitution equal 1. Prices $(1, 1)$ support it after transfers $T_1 = -1$ and $T_2 = 1$, which give wealths 3 and 5. Consumer 1's utility falls from $\sqrt{3}$ to $3/2$, so this allocation is outside the initial strong core. Efficiency does not imply voluntary acceptance of redistribution.

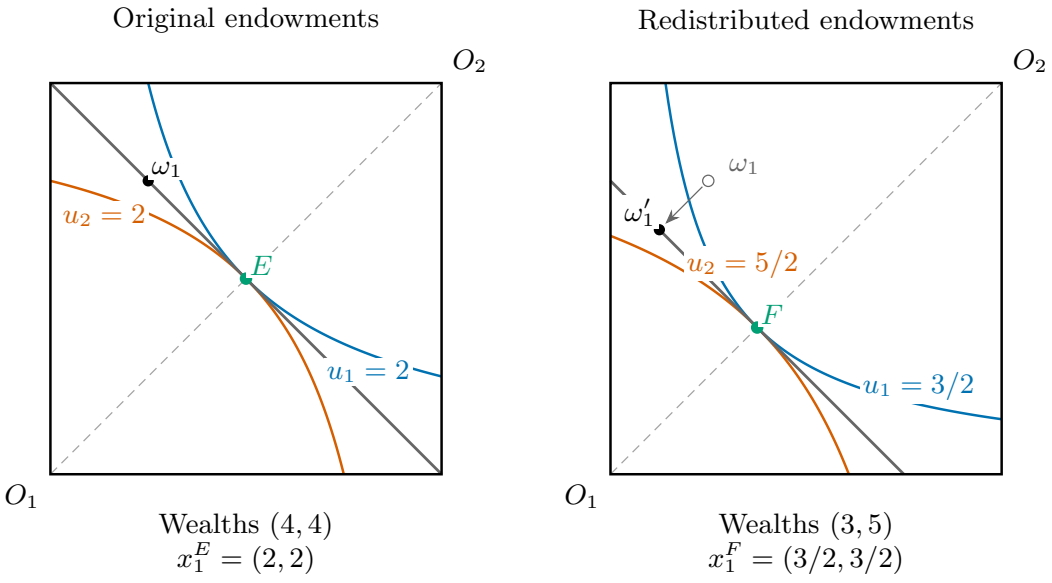
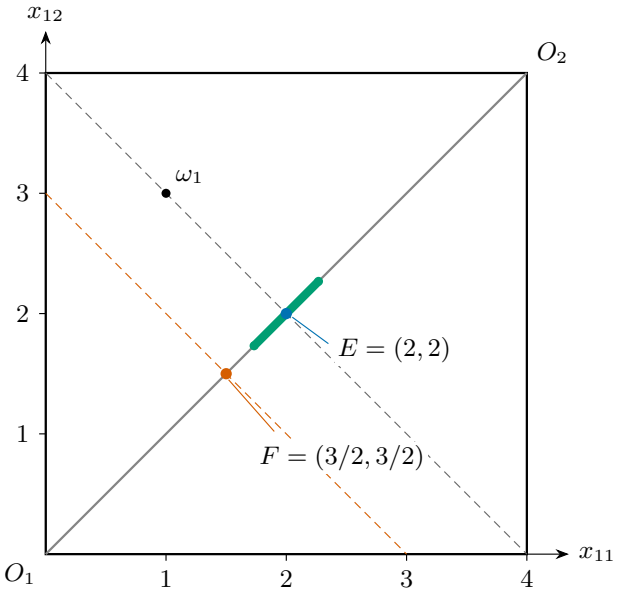


Figure 11.3: Redistribution changes wealth within a fixed box. Transferring half a unit of each good from consumer 1 to consumer 2 changes wealth from $(4, 4)$ to $(3, 5)$. At prices $(1, 1)$ equilibrium moves from E to F. Each colored curve is tangent to its panel's budget line. The relative price stays fixed in this example, not as a general rule.



Gray diagonal: Pareto set
Green segment: strong core

$\sqrt{3} \leq x_{11} = x_{12} \leq 4 - \sqrt{3}$

E: equilibrium at original wealth
F: efficient, not individually rational
F needs $T_1 = -1$, $T_2 = 1$

Figure 11.4: Three distinct sets. The full diagonal is efficient; only the green segment is individually rational and in the strong core. The original competitive equilibrium is E. Point F is efficient and can be supported after a transfer, but is not voluntary relative to the original endowment. The two parallel budget lines show that a transfer changes wealth, not the supporting relative price in this example.

11.1.6 A solution plan for an unfamiliar box

First compute totals and state which consumer's coordinates the diagram uses. Second derive each budget problem at arbitrary prices, retaining any demand correspondence at a tie. Third normalize one positive price and impose market clearing. Fourth check the candidate against both original budgets, both optimization problems and every market. Finally distinguish efficiency, individual rationality, coalition blocking and any redistribution. Use tangencies only after checking that the proposed choices are interior and differentiable. The advanced problems at the end of this chapter ask you to carry this plan through kinks, wealth effects, ownership and policy.

The welfare results and their more general assumptions follow [Mas-Colell et al., 1995]. The examples and diagrams here connect those arguments to budget calculations.

11.2 From individual choices to aggregate excess demand

Consider finitely many consumers $i = 1, \ldots, I$ and $L \geq 2$ commodities. Consumer i has consumption set $\mathbb{R}^L_+$, endowment $\omega_i \in \mathbb{R}^L_+$ and preferences represented by u_i. Aggregate resources are $\omega = \sum_i \omega_i \gg 0$. At strictly positive prices $p \gg 0$, wealth is $w_i(p) = p \cdot \omega_i$, not an independently fixed income. Consumer demand is

$$D_i(p, w) = \arg\max_{x \geq 0} \{u_i(x) : p \cdot x \leq w\}.$$

A positive price vector is a candidate trading rule, not yet an equilibrium. The consumer's desired net trade is $x_i - \omega_i$ for some $x_i \in D_i(p, p \cdot \omega_i)$. A positive coordinate means a desired net purchase; a negative coordinate means a desired net sale. These plans need not be jointly feasible away from equilibrium. Writing down demands does not mean that the corresponding transactions can all occur.

Definition 11.9. Aggregate excess demand. With possibly multivalued demand, define the Minkowski sum correspondence $Z(p) = \sum_i D_i(p, p \cdot \omega_i) - \{\omega\}$. A sum of sets consists of every vector obtained by choosing one demand from each set and adding them. If every demand is single valued, write $x_i(p)$ for the chosen bundle and $z(p) = \sum_i x_i(p) - \omega$ for the excess demand function.

Pure exchange equilibrium. A pair (p^*, x^*) with $p^* \gg 0$ is a Walrasian equilibrium if $x_i^* \in D_i(p^*, p^* \cdot \omega_i)$ for every consumer and $\sum_i x_i^* = \omega$. Thus $0 \in Z(p^*)$. When demand is single valued, this is the equation $z(p^*) = 0$.

11.2.1 Three accounting restrictions, not three equilibrium conditions

Proposition 11.10. *Walras' law, homogeneity and the resource lower bound. Suppose consumers are locally nonsatiated and their positive price budget problems attain maxima. For every selection $z \in Z(p)$, $p \cdot z = 0$. Moreover $Z(\lambda p) = Z(p)$ for $\lambda > 0$, and every component satisfies $z_\ell \geq -\omega_\ell$. These statements specialize to the excess demand function when demand is unique.*

Proof. Local nonsatiation makes every chosen budget bind: $p \cdot x_i = p \cdot \omega_i$. Summing gives Walras' law. Multiplying all prices by $\lambda > 0$ scales both expenditure and endowment wealth, leaving every feasible budget set unchanged, which proves homogeneity. Finally, $\sum_i x_{i\ell} \geq 0$ implies $z_\ell = \sum_i x_{i\ell} - \omega_\ell \geq -\omega_\ell$. □

Walras' law is value accounting at every positive price. It does not say that every market already clears. In two goods with $p = (q, 1)$, it reads $z_2(q, 1) = -q z_1(q, 1)$. A shortage of good 1 is matched by excess supply of good 2 in value, not necessarily in equal physical units. If $L - 1$ markets clear and the remaining price is positive, the final market must clear too. The positivity qualification is essential.

Under continuous, strictly convex and strongly monotone preferences, demand is a continuous function at strictly positive prices and positive wealth. If demands are multivalued, use an upper

hemicontinuous correspondence and a clearing selection instead of treating an arbitrary selected bundle as the unique prediction. At zero prices a desirable good may have no finite demanded quantity, so interior demand cannot simply be evaluated on the boundary.

11.2.2 A worked excess demand system

Example 11.11. Return to the two consumer box with $\omega_1 = (1, 3)$, $\omega_2 = (3, 1)$ and $u_i(x_i) = \sqrt{x_{i1}x_{i2}}$. At $p = (q, 1)$, each consumer spends half of wealth on each good. Total wealth is $4q + 4$, aggregate demand is $(2 + 2/q, 2q + 2)$ and total endowment is $(4, 4)$. Hence

$$z_1(q, 1) = \frac{2}{q} - 2, \qquad z_2(q, 1) = 2q - 2.$$

At $q = 1/2$, the desired imbalances are $(2, -1)$: consumers want two more units of good 1 and one fewer unit of good 2. Their values cancel because $(1/2)2 - 1 = 0$. At $q = 2$, the imbalances are $(-1, 2)$ and again values cancel. Only $q = 1$ makes both coordinates zero. It gives the allocation $(2, 2)$ to each consumer. The diagram displays incompatible plans on either side of the clearing price, not physical shortages that an auctioneer can actually deliver.

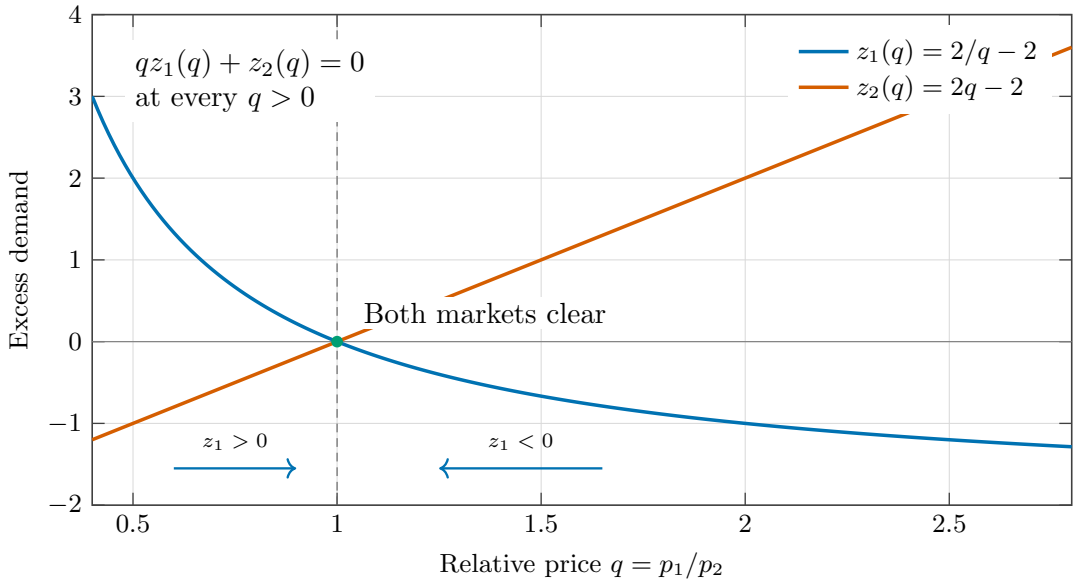


Figure 11.5: Desired trade and market clearing in the worked exchange economy. Both excess demand components vanish at $q = 1$, but Walras' law holds at every positive price. The arrows indicate the direction of a proposed adjustment of the relative price of good 1; such a rule is an additional assumption, not part of the definition of equilibrium.

11.2.3 Price normalization and the missing equation

Only relative prices determine the budgets. We may use the simplex $\Delta = \{p \in \mathbb{R}^L_+ : \mathbf{1}_L^\top p = 1\}$, or the chart $p_L = 1$. The corresponding strictly positive price descriptions are equivalent. In two goods, simplex prices $(s, 1-s)$ correspond to $q = s/(1-s)$, with $s = q/(1+q)$. The closed simplex is useful for existence arguments, but interior demand remains defined only on its strictly positive part until the boundary is treated.

When differentiability holds, two derivative identities make the redundancy explicit:

$$D_p z(p)p = 0, \qquad [D_p z(p)]^\top p = -z(p).$$

Differentiate $z(\lambda p) = z(p)$ with respect to λ to obtain the first identity. Differentiate $p \cdot z(p) = 0$ with respect to each price to obtain the second. At an equilibrium, p is both a right and a left null vector of the full Jacobian. Its determinant must therefore be zero. Testing that full determinant for regularity would be a mistake: one price scale and one clearing equation are redundant.

Price effects also include endowment wealth effects. If the underlying Marshallian demands $x_i(p, w)$ are differentiable, the total derivative of the exchange economy demand is

$$D_p z(p) = \sum_i \left[D_p x_i(p, w_i(p)) + D_w x_i(p, w_i(p))\, \omega_i^\top \right].$$

The first term holds nominal wealth fixed; the second records the change in the value of the endowment. This is one reason that the compensated demand restrictions of an individual consumer do not automatically give a monotone aggregate equilibrium response.

11.3 Existence through a continuous price map

11.3.1 From a crossing to a fixed point

With two goods write $f(q) = z_1(q, 1)$. Continuity and $f(a) > 0 > f(b)$ at positive $a < b$ give a clearing price by the intermediate value theorem; Walras' law then clears the other market. In our worked economy $f(q) = 2/q - 2$, so existence is immediate. Its negative derivative proves uniqueness separately. With many goods there is no single horizontal axis on which to look for a crossing. A fixed point supplies the corresponding argument.

Theorem 11.12. *Brouwer fixed point theorem. Let $K \subseteq \mathbb{R}^m$ be nonempty, compact and convex. Every continuous function $T : K \to K$ has a point $v^* \in K$ such that $T(v^*) = v^*$.*

11.3.2 A sufficient exchange economy theorem

Proposition 11.13. *Walrasian existence via Brouwer. Consider finitely many consumers with consumption sets $\mathbb{R}^L_+$, endowments $\omega_i \gg 0$, and complete, transitive, continuous, strictly convex and strongly monotone preferences. Then there are normalized prices $p^* \gg 0$ and utility maximizing bundles x_i^* with $\sum_i x_i^* = \omega := \sum_i \omega_i$. Here strong monotonicity means that $x \geq y$ and $x \neq y$ imply $x \succ_i y$. Strict convexity is a property of preferences, not convexity of a utility function.*

Strict convexity makes demand a function, which lets us use Brouwer directly. It is not necessary for equilibrium existence. Appendix B proves the corresponding result with convex preferences using Kakutani and supplies the technical lemmas used below. Both statements are sufficient versions for finite pure exchange, not claims about every production or zero price model.

11.3.3 Build the map, then read its fixed point

First make demand well defined at every candidate price. Normalize prices to the closed simplex $\Delta = \{p \geq 0 : \sum_\ell p_\ell = 1\}$ and temporarily cap every individual's consumption at 2ω. Let $d_i^B(p)$ be the unique best bundle in $B = [0, 2\omega]$ satisfying $p \cdot x \leq p \cdot \omega_i$. The cap is only a proof device: an allocation that clears cannot give anyone more than ω. Positive individual endowments make wealth positive even at a zero candidate price. The maximum theorem therefore makes d_i^B continuous on the whole closed simplex; the appendix checks this step.

$$z^B(p) = \sum_i d_i^B(p) - \omega, \qquad a_\ell(p) = \max\{z_\ell^B(p), 0\}, \qquad S(p) = \sum_\ell a_\ell(p).$$

Now add the positive excess demand to each price weight and normalize. The positive part a_ℓ is the desired purchase in excess of supply when that imbalance is positive, and zero otherwise. This gives the continuous function

$$T_\ell(p) = \frac{p_\ell + a_\ell(p)}{1 + S(p)}, \qquad T : \Delta \longrightarrow \Delta.$$

Every coordinate is nonnegative and the coordinates sum to one. Brouwer gives $p^* = T(p^*)$. To interpret it, suppress the argument p^* and rearrange the fixed point identity as $a_\ell = Sp_\ell$. If $S > 0$, every good with positive price has $z_\ell^B = Sp_\ell > 0$. Consequently

$$p^* \cdot z^B = S \sum_\ell (p_\ell^*)^2 > 0 \quad \text{if } S > 0.$$

That is impossible: adding the individual budget inequalities gives $p^* \cdot z^B \leq 0$. Thus $S = 0$ and $z^B \leq 0$. Notice that we used affordability, not expenditure equality: the artificial cap could have prevented spending the whole budget before the fixed point was found.

Finally remove the proof device. Since aggregate demand does not exceed ω, each chosen bundle lies strictly below 2ω. A zero priced good could now be increased within the cap and would strictly improve utility, so every price is positive. With room below the cap, optimality makes every budget bind. Hence $p^* \cdot z^B = 0$; positive prices and $z^B \leq 0$ force $z^B = 0$. If an unrestricted affordable bundle were strictly preferred, a small mixture with the capped optimum would still be inside the cap and would be strictly preferred by strict convexity. This contradiction proves unrestricted optimality. We have an equilibrium of the original economy.

The proof has three jobs: make a continuous map on a compact price set, use the budget contradiction to rule out positive excess demand at its fixed point, and show that the artificial bounds do not affect the answer. The map is an existence construction, not a model of actual trading or a convergence theorem.

11.3.4 A fixed point we can draw

Example 11.14. For the worked Cobb Douglas economy use simplex prices $(s, 1-s)$ with $s \in [1/10, 9/10]$. Then $z_1(s) = 2/s - 4$. The illustrative scalar function $T(s) = s + z_1(s)/32 = s + 1/(16s) - 1/8$ maps this interval into itself: its minimum is $3/8$ at $s = 1/4$ and its maximum is $38/45$. Its fixed point has $z_1(s) = 0$, giving $s^* = 1/2$. This scalar map is a transparent illustration, not a replacement for checking the general price map's hypotheses.

Cobb Douglas preferences on the closed orthant have zero utility edges, so they do not literally satisfy every global strictness assumption of the conservative theorem above. Their displayed equilibrium was derived directly. For further function based existence proofs see the following references; the correspondence extension and broader economic existence results are distinguished in the appendix.

[Geanakoplos, 2003, Tonin, 2018, Arrow and Debreu, 1954, Debreu, 1959]

11.4 One consumer and one producer

There is one consumer with labor-time endowment $\overline{N}$ of time and no initial endowment of the consumption good. The consumption set is $X = [0, \overline{N}] \times \mathbb{R}_+$, and $u : X \to \mathbb{R}$ is the consumer's utility function. Let x_1 be leisure, x_2 consumption, and $f(z)$ output from labor $z \geq 0$. At output

price $p > 0$ and wage $w > 0$, the firm solves

$$\pi_f(p, w) := \max_{z \geq 0} \{pf(z) - wz\}.$$

The consumer owns the firm, receives profit $\pi_f(p, w)$, and solves

$$\max_{(x_1, x_2) \in X} u(x_1, x_2) \quad \text{subject to} \quad px_2 \leq w(\overline{N} - x_1) + \pi_f(p, w).$$

Definition 11.15. A competitive equilibrium is a price pair (p^*, w^*) and choices (x^*, z^*) satisfying both optimization problems and the clearing conditions

$$x_2^* = f(z^*), \qquad z^* = \overline{N} - x_1^*.$$

Proposition 11.16. *Under continuity, convexity, local nonsatiation, and the usual concavity and attainment assumptions, a competitive allocation solves the central-planner problem*

$$\max_{x_1, x_2} u(x_1, x_2) \quad \textit{subject to} \quad x_2 \leq f(\overline{N} - x_1), \quad 0 \leq x_1 \leq \overline{N}.$$

Conversely, an interior planner solution can be supported by prices equal to the relevant marginal rates.

11.4.1 Reading production and household choice in one diagram

Example 11.17. Let time endowment be $\overline{N} = 3$, output be $f(z) = 2\sqrt{z}$ and utility over leisure ℓ and consumption c be $\sqrt{\ell c}$. Normalize $p = w = 1$. The firm maximizes $2\sqrt{z} - z = 1 - (\sqrt{z} - 1)^2$, so its unique optimum uses labor one, produces output two and earns profit one.

The owner household has wealth $w\overline{N} + \pi = 4$ and budget $\ell + c \leq 4$, with $0 \leq \ell \leq 3$. The inequality $\ell c \leq [(\ell + c)/2]^2 \leq 4$ gives its unique optimum $(\ell, c) = (2, 2)$. Labor supply $3 - \ell = 1$ equals labor demand, and consumption equals output. The prices decentralize a globally optimal allocation.

The budget line is not the production frontier. Other affordable household plans need not be compatible with firm choices. At equilibrium the frontier slope, the marginal rate of substitution and the wage relative to output price agree. Production determines what can be available; prices and ownership determine what is affordable.

11.5 Production, ownership, and feasibility

Consider I consumers, J firms, and L commodities. Consumer i has consumption set X_i, preference $\succeq_i$ represented by $u_i : X_i \to \mathbb{R}$, and endowment $\omega_i \in \mathbb{R}^L$; firm j has production set $Y_j \subseteq \mathbb{R}^L$. Let $\omega = \sum_i \omega_i$. An allocation (x, y) is feasible when

$$\sum_{i=1}^{I} x_i \leq \omega + \sum_{j=1}^{J} y_j.$$

In a private-ownership economy, consumer i owns share $\theta_{ij} \in [0,1]$ of firm j, with $\sum_i \theta_{ij} = 1$ for each j.

Definition 11.18. Walrasian equilibrium. A price vector $p \gg 0$ and allocation (x^*, y^*) form an equilibrium when, for every firm,

$$y_j^* \in \arg\max_{y_j \in Y_j} p \cdot y_j,$$

and, for every consumer,

$$x_i^* \in \arg\max_{x_i \in X_i} \{u_i(x_i) : p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*\},$$

with market clearing $\sum_i x_i^* = \omega + \sum_j y_j^*$.

Definition 11.19. Price equilibrium with transfers. Replace each consumer's ownership wealth by an assigned w_i satisfying $\sum_i w_i = p \cdot \omega + \sum_j p \cdot y_j^*$, and require $x_i^* \in \arg\max\{u_i(x_i) : p \cdot x_i \leq w_i\}$ while firms maximize profit and markets clear.

11.6 Welfare theorems

Proposition 11.20. *First Fundamental Welfare Theorem. See[Debreu, 1959, Mas-Colell et al., 1995]. If preferences are locally nonsatiated and (p, x^*, y^*) is a price equilibrium with transfers, then (x^*, y^*) is Pareto efficient. A Walrasian equilibrium is the special case in which wealth is generated by endowments and ownership shares.*

The proof is the supporting-price argument. Consumer optimality and local nonsatiation imply that every bundle weakly preferred to x_i^* costs at least w_i; a strictly preferred bundle costs strictly more. A Pareto improvement therefore has weakly greater expenditure for every consumer and strictly greater expenditure for at least one consumer. Summing and combining this strict inequality with firm profit maximization contradicts feasibility and market clearing.

Definition 11.21. Price quasi-equilibrium with transfers. A nonzero price vector, allocation, and assigned wealths form a price quasi-equilibrium with transfers when firms maximize profit, markets clear, $p \cdot x_i^* = w_i$ for every consumer, the assigned wealths add to aggregate endowment and profit value, and every bundle strictly preferred to x_i^* has cost at least w_i.

Proposition 11.22. *Second Fundamental Welfare Theorem. See[Debreu, 1959, Mas-Colell et al., 1995]. Suppose every production set is closed and convex, every preference is convex, continuous, and locally nonsatiated, and the aggregate attainable set is closed. Every Pareto-efficient allocation can be supported by a nonzero price vector as a price quasi-equilibrium with transfers. If, for every consumer, there is a bundle $\widehat{x}_i \in X_i$ with $p \cdot \widehat{x}_i < w_i$, continuity and convexity upgrade the quasi-equilibrium to a price equilibrium with transfers.*

The result follows by separating the convex aggregate set of bundles that make every consumer strictly better off from the convex aggregate production-plus-endowment set. The existence of a strictly cheaper feasible bundle for each consumer supplies the final upgrade from weak support to full budget maximization.

11.7 Local uniqueness, regularity and comparative statics

Definition 11.23. Normalized local uniqueness. A normalized equilibrium p^* is locally unique if a neighborhood contains no other normalized equilibrium. Global uniqueness means that there is no other normalized equilibrium anywhere in the relevant price domain.

Work in the chart $p_L = 1$ and define $\widehat{z}(p_1, \ldots, p_{L-1}) = (z_1, \ldots, z_{L-1})$. The missing equation follows from Walras' law. This is a square system of $L - 1$ equations in $L - 1$ relative prices.

Definition 11.24. Regular equilibrium. Suppose $\widehat{z}$ is continuously differentiable near an equilibrium. The equilibrium is regular when the reduced Jacobian $D\widehat{z}(p^*)$ is nonsingular. With two goods, this means $f'(q^*) \neq 0$.

Proposition 11.25. *Regularity implies local uniqueness. A nonsingular reduced Jacobian makes the equilibrium an isolated zero by the inverse function theorem. This is a local statement and does not rule out another zero far away. See [Mas-Colell et al., 1995].*

The converse is false. A scalar function can cross zero with a horizontal tangent and still have exactly one root. The local shapes $-r$, $-r^3$ and $-r^2$, where $r = q - q^*$, distinguish a regular crossing, a singular but isolated crossing, and a singular tangency. For the last shape, adding a small positive constant creates two nearby roots and adding a small negative constant removes

both. A zero derivative removes the inverse function theorem guarantee; it does not by itself prove multiplicity.

For an economic parameter θ, write the reduced system as $F(\widehat{p}, \theta) = 0$. If F is continuously differentiable and $D_{\widehat{p}}F$ is nonsingular, the implicit function theorem supplies a locally unique differentiable equilibrium branch with

$$\frac{d\widehat{p}^*}{d\theta} = -[D_{\widehat{p}}F]^{-1} D_\theta F.$$

The sign depends on both the equilibrium response matrix and the way the parameter enters the model. A parameter cannot simply be called income or technology without specifying which primitives change. If several equilibrium branches exist, this formula describes one selected branch locally, not an unqualified response of the entire economy.

If every normalized equilibrium is regular and the equilibrium set is compact inside the price domain, there are only finitely many equilibria. Otherwise an infinite set would have an accumulation point; continuity makes that point an equilibrium, contradicting its local isolation. Regularity without the compactness and boundary step does not establish finiteness.

11.8 Multiplicity and the limits of aggregate restrictions

11.8.1 An exact three crossing illustration

Consider the illustrative system

$$f(q) = -\frac{(q-1)(q-2)(q-3)}{q(1+q^2)}, \qquad z(q,1) = (f(q), -qf(q)), \qquad q > 0.$$

For arbitrary positive prices define $z(p_1, p_2) = z(p_1/p_2, 1)$. Homogeneity and Walras' law then hold exactly. Unlike a polynomial drawn without a domain check, this example also has finite commodity lower bounds:

$$f(q) + 1 = \frac{6q^2 - 10q + 6}{q(1+q^2)} > 0, \qquad -qf(q) = q - 6 + \frac{10q}{1+q^2} > -6.$$

The numerator in the first expression is $6(q - 5/6)^2 + 11/6$. As q approaches zero, good 1 excess demand diverges upward while good 2 excess demand tends to -6. As q tends to infinity, good 1 excess demand tends to -1 and good 2 excess demand diverges upward. The signs fit the two good boundary accounting.

There are three clearing prices, $q = 1, 2, 3$, with derivatives $f'(1) = -1$, $f'(2) = 1/10$ and $f'(3) = -1/15$. Every equilibrium is regular and locally isolated, but the system is not globally unique. No underlying finite collection of preferences and endowments is constructed here. Satisfying necessary identities and lower bounds is not a substitute for a rationalization proof.

11.8.2 The signed count

In the chart $p_L = 1$, use the index convention $\operatorname{sign}\det[-D\widehat{z}(p^*)]$ at a regular equilibrium. In two goods this is $\operatorname{sign}[-f'(q^*)]$. For the three crossing illustration the indices sum to 1. A continuous scalar curve that begins above zero and ends below zero, with finitely many simple crossings, crosses downward one more time than upward. Its number of crossings is therefore odd.

The many good index theorem is a global result. Under the appropriate compactness, regularity and inward boundary hypotheses, the sum of equilibrium indices is one. It does not follow from the inverse function theorem alone, does not select an equilibrium, and does not certify the stability of a numerical algorithm. See [Mas-Colell et al., 1995].

11.8.3 What Sonnenschein, Mantel and Debreu establish

Proposition 11.26. *Sonnenschein, Mantel and Debreu aggregation results. On compact normalized price regions strictly inside the positive price domain, aggregate excess demand has very flexible possible shapes subject to the maintained regularity and accounting restrictions of the relevant realization theorem. Individual rationality alone does not generally imply unique prices, stable adjustment or uniformly signed equilibrium comparative statics. See [Sonnenschein, 1972, Mantel, 1974, Debreu, 1974].*

The result concerns constructed exchange economies, not arbitrary vector fields without restrictions. Homogeneity, Walras' law, finite endowment lower bounds and the stated continuity conditions still matter. Global versions also require the appropriate boundary behavior. With several prices approaching zero, one must not require every vanishing price component to have divergent demand along every sequence; the boundary condition must match the theorem being invoked.

The lesson is not that equilibrium theory says nothing. Additional preferences, technologies, ownership structures or repeated observations impose further restrictions. The lesson is that those additional primitives must do the work. One cannot transfer an individual compensated demand theorem to an aggregate endowment economy without checking the aggregation and wealth terms.

11.9 Adjustment is different from comparative statics

11.9.1 Continuous adjustment requires its own rule

Suppose, as an additional behavioral rule, $\dot{q} = \kappa f(q)$ with $\kappa > 0$. The price of good 1 rises when its desired purchases exceed supply. Linearization near a regular root gives $\dot{h} = \kappa f'(q^*)h$. A negative derivative gives local asymptotic stability; a positive derivative gives instability. Thus the first and third roots of the illustration are locally stable under this particular rule and the middle root is unstable. None of these dynamic claims follows merely from consumer rationality or equilibrium existence.

11.9.2 Even a unique equilibrium can defeat a price iteration

Example 11.27. Use the actual worked Cobb Douglas excess demand $f(q) = 2/q - 2$, whose only clearing price is $q^* = 1$. A proposed discrete algorithm is $q_{n+1} = q_n + h(2/q_n - 2)$, with step size $h > 0$. Its derivative at the fixed point is $1 - 2h$. Local convergence is guaranteed by $0 < h < 1$; step sizes between $1/2$ and 1 produce local alternating adjustments. A large step may leave the positive price domain, so positivity must also be checked.

There is an exact positive two cycle when $h = 9/8$: starting at $q_0 = 3/4$ gives $q_1 = 3/2$, then $q_2 = 3/4$, and repeats forever. A unique equilibrium therefore does not imply convergence of this solver. This is not a second economic equilibrium: it is a failure of a specified computational procedure.

Comparative statics instead changes the equation itself. For the three root illustration, set $F(q, \theta) = f(q) + \theta/(1 + q)$. At each fixed price, increasing θ raises good 1 excess demand. The three local branch derivatives at $\theta = 0$ are $1/2, -10/3, 15/4$. The middle branch moves in the opposite direction from the other two. Calling this perturbation an actual taste or endowment shock would require an economic model connecting that primitive to $D_\theta F$.

11.10 Gross substitutes: an extra restriction that restores uniqueness

Definition 11.28. Strict gross substitutes. If one commodity's price rises while all other prices are held fixed, excess demand for every other commodity strictly rises. Weak gross substitutes uses weak inequalities. This is a global finite comparison property on the positive price region, stronger than homogeneity and Walras' law.

In the worked Cobb Douglas system, $z_1(p) = 2p_2/p_1 - 2$ and $z_2(p) = 2p_1/p_2 - 2$. Both cross effects are positive everywhere. In contrast, the illustrative three root system has $f'(2) > 0$. At that equilibrium, raising one price has the wrong cross effect for strict gross substitutes. Favorable local slopes at the outer roots cannot replace a global assumption.

Proposition 11.29. *Gross substitutes uniqueness theorem. If aggregate excess demand is homogeneous, satisfies Walras' law and has strict gross substitutes on the relevant positive price domain, there is at most one normalized strictly positive equilibrium price vector. Continuity and a separate existence argument provide existence where required. See [Mas-Colell et al., 1995].*

Proof. Suppose p and r are positive equilibrium prices and are not proportional. Divide r by the smallest ratio r_ℓ/p_ℓ. The rescaled vector is no smaller than p in every component and equals it in at least one component, say k. At least one other price strictly rises. Raise those other prices one at a time, holding price k fixed. Strict gross substitutes makes z_k strictly larger at the rescaled r than at p. Yet homogeneity and market clearing make both values zero, a contradiction. Therefore equilibrium prices must be proportional. □

With weak gross substitutes the same comparisons give only weak inequalities, so the contradiction may disappear. There can be an interval of clearing prices when consumers remain at the same corners. Stronger conclusions about the geometry of a weak substitutes equilibrium set require their own additional hypotheses; weak substitution alone is not a universal price uniqueness theorem.

11.10.1 A production qualification

For a constant returns production cone $Y \subseteq \mathbb{R}^L$ containing zero, a positive price vector supports the technology if $p \cdot y \leq 0$ for every $y \in Y$. If consumer excess demand satisfies $z(p) \in Y$ and Walras' law, choosing production $y = z(p)$ clears markets and earns the maximal profit of zero. This separates the household condition from technology support. It does not establish a gross substitutes property or a uniqueness theorem for every economy with production.

11.11 Coalitions, the core and the discipline of replication

11.11.1 What resources does a blocking coalition own?

Return to a finite pure exchange economy. A feasible allocation satisfies $\sum_i x_i = \sum_i \omega_i$. A nonempty coalition $S \subseteq \{1, \ldots, I\}$ may redistribute exactly its own aggregate endowment $\sum_{i \in S} \omega_i$.

It cannot start from the proposed allocation and seize the other consumers' goods. The question is whether its members can do better by trading only with each other, using their original resources.

Definition 11.30. Blocking and the core. A coalition blocks x if it can choose $(y_i)_{i\in S}$ with $\sum_{i\in S} y_i = \sum_{i\in S} \omega_i$ and $y_i \succ_i x_i$ for every member. The core is the set of feasible allocations blocked by no coalition.

Strong blocking and the strong core. A coalition strongly blocks x if the same resource equality holds, every member satisfies $y_i \succeq_i x_i$, and at least one satisfies $y_j \succ_j x_j$. The strong core permits no such deviation. Every ordinary block is a strong block, so the strong core is contained in the core. Under monotone preferences and free disposal, weak resource inequalities can be completed to exact feasibility without losing the improvement.

A singleton coalition tests whether a consumer prefers the original endowment to the assignment. The grand coalition tests whether a joint improvement is feasible. Intermediate coalitions add constraints that neither individual rationality nor aggregate Pareto efficiency alone can capture. For two consumers, the strong core is exactly the individually rational part of the Pareto set, because these are the only coalitions. With more consumers, that shortcut fails.

Proposition 11.31. *Walrasian allocations belong to the strong core and hence to the core. Suppose preferences are locally nonsatiated and (p^*, x^*) is an equilibrium with $p^* \gg 0$. A weakly improved coalition member needs expenditure at least $p^* \cdot \omega_i$, and a strictly improved member needs strictly more. Summing gives $p^* \cdot \sum_{i\in S} y_i > p^* \cdot \sum_{i\in S} \omega_i$, contradicting coalition feasibility. See [Debreu and Scarf, 1963, Mas-Colell et al., 1995].*

Prices prevent a coalition deviation through a common value certificate. No utility levels need be added in that general proof. The converse fails in a finite economy: some allocations survive all available coalitions without being competitive for the original endowment distribution.

11.11.2 A concrete noncompetitive allocation and its challengers

Example 11.32. There are two types, A and B, with endowments $(4, 1)$ and $(1, 4)$ and utility $\sqrt{x_1 x_2}$. In a one copy economy, consider $x^A = (t, t)$ and $x^B = (5 - t, 5 - t)$. The diagonal is efficient by the same product argument used for the Edgeworth box. Both consumers obtain utility 2 at their endowments, so the strong core on this diagonal is $2 \le t \le 3$. The original competitive allocation is $t = 5/2$, obtained at equal prices. The other points in this interval show why the finite core need not coincide with competitive allocations.

In an r copy replica, there are r consumers of each type. An equal treatment allocation assigns (t, t) to every A consumer and $(5-t, 5-t)$ to every B consumer. A coalition with m A members and n B members has resources $(4m+n, m+4n)$. To weakly match the current utilities, it needs a total utility of at least $C = mt + n(5-t)$. Its maximum feasible sum of square root utilities is $D = \sqrt{(4m+n)(m+4n)}$.

For necessity, the Cauchy Schwarz inequality gives $\sum_{j \in S} \sqrt{y_{j1} y_{j2}} \leq D$. A strong block would make this sum strictly above C. For sufficiency, if $D > C > 0$, give each member the fraction u_j/C of each coalition resource, where u_j is that member's current utility. Its new utility is $u_j D/C > u_j$. Thus this coalition strongly blocks exactly when $D > C$. The test compares complete feasible alternatives, not merely a slope at a proposed allocation.

Take $t = 14/5$. This allocation survives the one copy economy. In a two copy economy, one A and two B consumers can leave together. Their endowment totals $(6, 9)$. Give the A member $(12/5, 18/5)$ and each B member $(9/5, 27/10)$. Both goods add exactly to $(6, 9)$. The A member's squared utility rises by $4/5$ and each B member's by $1/50$. All three strictly gain. Replication supplies a coalition composition that was previously unavailable.

11.11.3 The shrinking intervals can be computed, not just sketched

For this particular example let $d = t - 5/2$, $N = m + n$ and $k = (m-n)/N$. The capacity comparison reduces to

$$\frac{D^2 - C^2}{N^2} = -\frac{k}{4}\left[20d + (9 + 4d^2)k\right].$$

For $d > 0$, a challenging coalition has more B than A members, so $k < 0$; for $d < 0$ reverse the roles. With $0 \leq m, n \leq r$, the smallest nonzero absolute composition imbalance is $1/(2r-1)$, attained by $(r-1, r)$ or its reverse. Checking that composition and all larger imbalances gives the exact interval of equal treatment allocations that survive every coalition:

$$\left| t - \frac{5}{2} \right| \leq h_r, \qquad h_r = \frac{5s - \sqrt{25s^2 - 9}}{2}, \qquad s = 2r - 1.$$

At $r = 1$ this is $[2, 3]$; at $r = 2$ it is approximately $[2.3485, 2.6515]$. As r increases, h_r tends to zero and only the competitive split survives in the limit. These exact intervals concern the stated utility, endowments and equal treatment family. They are not a universal finite replica rate. Because every target utility here is positive, the strict capacity test also constructs an ordinary block whenever a strong block is possible within this family.

11.11.4 The general core convergence statement

A conservative sufficient version of the Debreu and Scarf theorem uses finitely many fixed types, consumption sets $\mathbb{R}^L_+$, strictly positive individual endowments, and continuous, strictly convex, strongly monotone preferences. Replicate every type r times and identify equal treatment allocations with their per type bundles. Every limit point of core allocations as r tends to infinity is Walrasian; every Walrasian allocation belongs to the core of every replica. The same limiting inclusion applies to the strong core because it is a subset of the core. See [Debreu and Scarf, 1963].

We state stronger sufficient hypotheses rather than quietly omit boundary qualifications. With weaker desirability or endowment assumptions, a limiting support price may have zero components or require a quasi equilibrium formulation. The square root replica example was proved directly by a capacity inequality, so its conclusion does not rest on asserting strict convexity on zero utility edges of the closed orthant.

11.12 General equilibrium with inattention

An inattention model must distinguish true prices from the prices or signals used by a consumer's decision rule. Fix $\underline{m}_i \in (0,1]$ and let the admissible attention class be $\mathcal{M}_i = \{\operatorname{diag}(m_{i1}, \ldots, m_{iL}) : m_{i\ell} \in [\underline{m}_i, 1]\}$. Thus $M_i p \gg 0$ whenever $M_i \in \mathcal{M}_i$ and $p \gg 0$. Write $D_i^{\mathrm{B}}(p, M_i)$ for the demand correspondence generated by the fully specified behavioral model, while the unadorned $D_i(p, w_i)$ remains ordinary Walrasian demand at true prices and wealth. The model must state whether M_i changes perceived prices, the budget, utility, or information.

For a price-perception benchmark, $\widetilde{p}_i = M_i p$ and a possible behavioral budget are written explicitly, for example

$$D_i^{\mathrm{B}}(p, M_i) = \arg\max_{x \in X_i} \{u_i(x) : \widetilde{p}_i \cdot x \leq \widetilde{p}_i \cdot \omega_i\}.$$

This is not a Walrasian budget unless $M_i = I_L$ or the model separately establishes equivalence with the true budget.

Definition 11.33. Inattention equilibrium. A pair (p, x) is an inattention equilibrium when each $x_i \in D_i^{\mathrm{B}}(p, M_i)$ satisfies the model's actual feasibility restrictions and

$$\sum_{i=1}^{I} x_i = \sum_{i=1}^{I} \omega_i.$$

Proposition 11.34. *Compatibility with Pareto efficiency. Suppose preferences are differentiable, strictly increasing, and concave and x^* is an interior Pareto-efficient allocation. Then there is a common supporting price vector $q \gg 0$ and multipliers $\mu_i > 0$ such that $\nabla u_i(x_i^*) = \mu_i q$ for every i. If the behavioral first-order condition is $\nabla u_i(x_i^*) = \lambda_i M_i p$ with $\lambda_i > 0$, then each $M_i p$ must be collinear with the same q.*

This is a necessary compatibility condition, not an if-and-only-if statement about attention matrices. Identical matrices can support efficiency in a regular model, while different matrices can also be compatible at special allocations. Perceived-price optimality alone does not justify either welfare theorem.

11.13 What equilibrium observations can reveal

The same preferences must now explain several observed equilibria, not one market at one price. This adds empirical discipline: the latent individual choices must satisfy both rationality across observations and resource clearing within each observation. Keep this question distinct from the flexibility of aggregate excess demand in one economy.

Fix a finite set of consumers $i = 1, \ldots, I$, goods $\ell = 1, \ldots, L$, and observations $t = 1, \ldots, T$. At observation t, let $p^t \in \mathbb{R}^L_{++}$ denote the price vector, let $w_i^t > 0$ denote consumer i's income, and let $\omega^t \in \mathbb{R}^L_+$ denote the aggregate endowment. The accounting restriction is $\sum_i w_i^t = p^t \cdot \omega^t$. Prices are normalized because only their positive scale is economically relevant.

Definition 11.35. (Pure-trade rationalization.) A finite dataset is rationalized by a pure-trade economy if there are continuous, locally nonsatiated, concave utility functions u_i and bundles $x_i^t \in \mathbb{R}^L_+$ such that, for every i, t, $x_i^t \in \arg\max\{u_i(x) : p^t \cdot x \leq w_i^t\}$, and, for every t, $\sum_i x_i^t = \omega^t$.

(Equilibrium manifold.) For a market excess-demand function $z(\hat{\omega}, p)$, where $\hat{\omega} = (\omega_1, \ldots, \omega_I)$, the equilibrium manifold is $\mathcal{M} = \{(\hat{\omega}, p) : z(\hat{\omega}, p) = 0\}$. It records pairs of latent individual endowment profiles and prices that can clear markets for fixed underlying preferences. Without differentiability and an appropriate rank condition this is only an equilibrium set; the smooth manifold terminology requires those additional regularity assumptions. The Brown--Matzkin observable set below is a projection of this theoretical manifold onto prices, individual incomes, and aggregate endowments.

11.14 Afriat inequalities and elimination of unobservables

For a finite collection of observed or latent bundles with strictly positive prices and expenditure exhaustion, Afriat's inequalities are

$$U_i^t \le U_i^s + \lambda_i^s p^s \cdot (x_i^t - x_i^s) \qquad \text{for all } s, t,$$

with $U_i^t \in \mathbb{R}$ denoting a utility-level certificate and $\lambda_i^s > 0$ denoting a marginal-utility-of-wealth certificate. They are equivalent to the existence of a monotone, concave utility representation for the finite choice data.

The equilibrium restrictions combine these inequalities with the budget equalities and market-clearing equalities

$$p^t \cdot x_i^t = w_i^t, \qquad \sum_i x_i^t = \omega^t.$$

The unobserved variables are

$$(x_i^t, U_i^t, \lambda_i^t)_{i,t}.$$

Prices, individual incomes, and aggregate endowments are the data. Because λ_i^s multiplies latent x_i^t, the joint system is polynomial but not linear in all unknowns.

(Tarski--Seidenberg theorem.) The projection of a semialgebraic set is semialgebraic. Consequently, eliminating the unobserved variables from any finite polynomial system yields a quantifier-free semialgebraic condition in the observables. This theorem supplies an elimination principle; it does not say that the resulting restrictions are easy to compute or that every model has a nontrivial restriction.

11.15 Brown--Matzkin theorem

Brown and Matzkin (1996) apply this principle to pure-trade equilibrium. Under their regularity and invariance assumptions, finite observations on prices, individual incomes, and aggregate endowments are rationalizable if and only if they satisfy a finite family of semialgebraic restrictions obtained after eliminating the latent utility levels, marginal utilities of income, and consumption bundles. The restrictions are nontrivial: some finite datasets cannot lie on the equilibrium manifold of any economy in the maintained class. See[Brown and Matzkin, 1996]; its quantifier-elimination construction is not reproved in this text.

The theorem should not be confused with the Sonnenschein--Mantel--Debreu result. The latter

concerns the flexibility of aggregate excess demand when individual preferences are allowed to vary; Brown--Matzkin hold individual preferences fixed across observations and exploit variation in observed incomes and aggregate endowments.

11.16 A unified solution plan

Begin with resources, preferences and ownership. Derive individual choices at announced prices, keeping the endowment wealth terms. Normalize one price and use Walras' law to remove one redundant equation. Establish existence with the hypotheses of the relevant theorem, then verify every candidate against optimization, budgets and all markets. Distinguish a welfare comparison from affordability and distinguish a regular local branch from a globally unique equilibrium.

For a policy response, specify which primitive changes and which equilibrium branch is followed. For a proposed price update, check the update's own domain and derivative; the Brouwer construction is not a convergence guarantee. For a coalition deviation, use the coalition's original resources. For equilibrium data, keep each consumer's preference fixed across observations and separate passing a rationalization test from identifying a counterfactual.

The problem program below follows these connections. The 11 series develops exchange, production, ownership and welfare; the 12 series studies excess demand, uniqueness and coalitions; the 13 series studies empirical restrictions and counterfactual identification. These are permanent catalog identifiers, retained to keep released assignments and private answer keys consistent, not three separate printed chapters.

11.17 Problems

Exchange, production and welfare

Part I: By hand

Problem 11.1 [Core]. Cobb-Douglas exchange equilibrium.

There are two consumers with common utility $u_i(x_{i1}, x_{i2}) = x_{i1}^{\alpha} x_{i2}^{1-\alpha}$, $0 < \alpha < 1$, and endowments $\omega_1 = (1, 2)$, $\omega_2 = (2, 1)$.

(a) Derive individual Walrasian demands.

(b) Normalize $p_2 = 1$ and solve for the equilibrium relative price.

(c) Compute equilibrium allocations and verify both market-clearing equations.

Advanced exam practice

These problems build toward integrated graduate exam arguments. The suggested time applies to the hand questions, not the optional tool extension. First draw a diagram or state a solution plan, define the feasible set, and identify possible corners. Explain why each candidate is globally optimal and check the economic accounting. Attempt the hand questions before using tools. For an optional extension, state Seed, Fixed, Scaled and Risk before coding and enlarge one feature at a time.

Difficulty guide. Problems 11.A1 to 11.A6 develop geometric and regime reasoning. Problem 11.A7 restores the Western 2018 final examination question on the empirical content of equilibrium with production. It requires a full mathematical certificate and both directions of a rationalization argument, not only a computed equilibrium.

Problem 11.A1 [Advanced]. The lens, the strong core and the price line.

Suggested time: 40 minutes for the hand part. Total: 50 points. Work by hand.

There are two consumers A and B, two goods, consumption sets $\mathbb{R}^2_+$, and utilities $u^i(x) = \sqrt{x_1 x_2}$. Endowments are $\omega^A = (5, 1)$ and $\omega^B = (1, 5)$. Prices are strictly positive. For this question, a coalition strongly blocks an allocation if it can redistribute its own endowment so that every member is weakly better off and at least one is strictly better off; the strong core admits no such coalition.

(a) Draw the box, mark both origins and the endowment point, and express B's consumption when A receives (a, b). State the resource equations and the inequalities defining the individually rational lens. [8]

(b) Derive the complete Pareto set. Give an explicit mutually beneficial feasible trade from every allocation with $a \neq b$, rather than relying only on tangency. [10]

(c) Determine the strong core exactly. Explain why its characterization uses both singleton coalitions and the coalition containing both consumers. [8]

(d) Normalize $p_2 = 1$. Derive demands, solve for the equilibrium relative price and allocation, and check both budgets and both markets. Mark the common budget line in your diagram. [10]

(e) Can every allocation in the strong core be a competitive allocation with the original endowments? Justify your answer. Find transfers with $T_A + T_B = 0$ that support $x^A = (7/2, 7/2)$ and the complementary bundle for B. [8]

(f) Explain which of the box, Pareto set, individually rational lens and competitive allocation can change when physical endowments are redistributed while their aggregate is held fixed. [6]

Optional tool extension. Keep utilities and aggregate resources fixed and vary only the scalar

$s \in [0, 6]$ in endowments $(s, 6 - s)$ and $(6 - s, s)$. Use Codex and Python to draw the lens and strong core and recover the equilibrium for a rational grid including $s = 0, 3, 6$. Verify both budgets and resources exactly, and verify individual rationality through squared utilities. Distinguish these finite checks from the hand proof.

Problem 11.A2 [Advanced]. A kink, an indifferent consumer and one clearing allocation.

Suggested time: 40 minutes for the hand part. Total: 50 points. Work by hand.

Consumption sets are $\mathbb{R}^2_+$. Consumer A has $u^A(x) = \min\{x_1, x_2\}$ and endowment $(4, 0)$. Consumer B has $u^B(x) = 2x_1 + x_2$ and endowment $(0, 4)$. Normalize $p = (q, 1)$ with $q > 0$. A competitive equilibrium requires individual utility maximization and exact clearing in both goods.

(a) Derive A's demand and B's full demand correspondence for every $q > 0$. Explain why applying a differentiable tangency condition to A or selecting a single arbitrary demand for B can give a wrong answer. [12]

(b) Show that no equilibrium can have $q < 2$ or $q > 2$. At the remaining relative price, select demands that clear both markets. Verify budgets and identify whether prices, equilibrium allocations and individual demand are unique. [12]

(c) Draw the box, both origins, the endowment, the equilibrium and the common budget line. At $q = 1$ and $q = 3$, compute aggregate excess demand and explain the direction in which relative prices would have to move to reach equilibrium. [8]

(d) Derive the entire Pareto set using a direct argument. Which resources can A give away without lowering utility when $x^A_1 \neq x^A_2$? [10]

(e) Under the book's strong blocking convention, determine the strong core. Compare it with the competitive allocation and explain what this example teaches about kinks, corner choices and market clearing. [8]

Optional tool extension. Vary only q on a positive rational grid containing $1, 2, 3$ and values immediately on both sides of 2. Compute the full demands and check $qz_1 + z_2 = 0$ exactly. At $q = 2$, represent B's full demand segment and recover the clearing selection rather than inserting an arbitrary bundle. A plot of excess demand is not a proof of price adjustment convergence.

Problem 11.A3 [Advanced]. An endowment threshold makes a consumer enter.

Suggested time: 45 minutes for the hand part. Total: 50 points. Work by hand.

There are two goods, denoted (x, y), and two consumers with nonnegative consumption. Preferences are $u^A(x, y) = \log(1 + x) + y$ and $u^B(x, y) = 2\log(1 + x) + y$. Endowments are

$\omega^A = (Q, 0)$ and $\omega^B = (0, 10)$, where $Q > 0$. Normalize prices to $(q, 1)$ with $q > 0$. Consumers take prices as given.

(a) Derive both demands for every $q > 0$, including A's possible zero consumption of y and zero consumption of x. Do not assume the numeraire is positive before checking it. [12]

(b) Find all competitive equilibria for every $Q > 0$. Give the exact threshold at which A starts consuming good x, and report all four quantities and the relative price on both sides of the threshold, including equality. [14]

(c) Verify both budgets and both markets. Give a global optimality argument for each consumer and evaluate the entry condition for A's good x at the threshold and on either side. [10]

(d) Compute the equilibrium for $Q = 1/2$ and $Q = 4$. Draw the corresponding boxes, with good x horizontal. Mark the equilibrium corner in the first box and the interior allocation in the second. [8]

(e) Is the equilibrium relative price continuous and differentiable at the entry threshold? Explain why a zero choice at an equality threshold is compatible with a unique optimum, and why quasilinearity alone does not justify ignoring the nonnegative numeraire constraint. [6]

Optional tool extension. Vary only $Q > 0$ on a rational grid containing $1/2, 1, 4$ and values immediately around 1. Implement the full capped demands and the equilibrium formulas independently. Check nonnegativity, both budgets, both market equations, entry inequalities and complementary products exactly. Plot the entry threshold and explain the different one sided derivatives.

Problem 11.A4 [Advanced]. Redistribution moves prices but does not resize the box.

Suggested time: 40 minutes for the hand part. Total: 50 points. Work by hand.

Two consumers have $u^A(x) = x_1^{2/3} x_2^{1/3}$ and $u^B(x) = x_1^{1/3} x_2^{2/3}$ on $\mathbb{R}^2_+$. Endowments are $\omega^A = (t, 3)$ and $\omega^B = (6 - t, 3)$, where $0 \leq t \leq 6$. Normalize prices to $(q, 1)$ with $q > 0$. Increasing t transfers physical units of good 1 from B to A before trade.

(a) Derive both demands as functions of q and t. Solve for the unique equilibrium relative price and all four consumption quantities. Check both resource equations. [16]

(b) Compute the equilibria at $t = 3$ and $t = 6$. Explain why a transfer of a good can raise that good's relative price even though aggregate resources have not changed. [8]

(c) Using A's coordinates (a, b), derive the interior contract curve and include its endpoints with a valid boundary argument. Draw the two equilibria on the same box. [10]

(d) A policymaker proposes the equal allocation $x^A = x^B = (3, 3)$. Can any lump sum transfers and common strictly positive prices decentralize it as an equilibrium for these preferences? Give an explicit Pareto comparison rather than relying only on unequal marginal rates of substitution. [10]

(e) Distinguish a redistribution of initial physical endowments from a change in the aggregate endowment. Explain which objects remain fixed here and why calculating a price effect while holding both consumers' wealths fixed would answer a different question. [6]

Optional tool extension. Vary only $t \in [0, 6]$ on a rational grid containing $0, 3, 6$. Verify the two budget identities, both resource equations and membership of the reported allocation in the contract curve exactly. Plot $q(t)$ and the equilibrium path on a single unchanged box. Do not introduce a second taste or resource parameter.

Problem 11.A5 [Advanced]. Who owns the firm and who receives its gains.

Suggested time: 45 minutes for the hand part. Total: 50 points. Work by hand.

There are two goods and two consumers, each endowed with $(2, 0)$ and each having utility $u^i(x) = \sqrt{x_1 x_2}$ on $\mathbb{R}^2_+$. A firm can use $z \geq 0$ units of good 1 to produce $2\sqrt{z}$ units of good 2. A owns share $\theta \in [0, 1]$ of its profit and B owns share $1 - \theta$. Normalize prices to $(q, 1)$ with $q > 0$. The firm and consumers take prices as given.

(a) Solve the firm's profit maximization problem for every $q > 0$, prove global optimality and report input, output and profit. Explain why diminishing returns can generate positive profit without monopoly power. [8]

(b) Write both consumers' wealths and derive their demands. Find the equilibrium price, production and consumption for every $\theta \in [0, 1]$. [14]

(c) Verify both consumer budgets and the two resource equations, including the input used by the firm. Explain why good 1 consumption does not sum to its initial endowment. [8]

(d) Compare $\theta = 0$ and $\theta = 1$. In this model, which real decisions depend on ownership and which do not? Identify the preference feature responsible for the aggregate invariance, without claiming that ownership never matters for equilibrium production. [6]

(e) A planner uses a fixed weight $\gamma \in (0, 1)$ and maximizes $\gamma(\log x_1^A + \log x_2^A) + (1-\gamma)(\log x_1^B + \log x_2^B)$ subject to the production and resource constraints. Solve first for the allocation conditional on z, then for z. Find the weight supporting the competitive allocation for a given θ. [10]

(f) After production is fixed, draw the feasible exchange box for the output and remaining input. Explain why using the original 4 by 0 endowment rectangle as the final consumption box would be misleading. [4]

Optional tool extension. Vary only $\theta \in [0,1]$ on a rational grid containing $0, 1/2, 1$. Check both ownership shares, both private budgets and both resource equations, retaining $\sqrt{3}$ symbolically or stating numerical tolerances. Verify the corresponding planner weight and distinguish distributional changes from the invariant production and price choices.

Problem 11.A6 [Advanced]. A tax wedge remains even when every dollar is rebated.

Suggested time: 45 minutes for the hand part. Total: 50 points. Work by hand.

Two consumers have utility $u^i(x,y) = \sqrt{xy}$ on $\mathbb{R}^2_+$. Endowments are $\omega^A = (4,0)$ and $\omega^B = (0,4)$. The producer prices are $(q,1)$ with $q > 0$. Consumer A pays an ad valorem tax $\tau \geq 0$ on every unit of good y consumed. Consumer B is not taxed and receives all tax revenue as a lump sum transfer T. The government consumes no goods. Each consumer takes (q,T) as given. Their budgets are $qx_A + (1+\tau)y_A \leq 4q$ and $qx_B + y_B \leq 4 + T$. Government budget balance requires $T = \tau y_A$.

(a) Derive both demands conditional on (q,T). State the exact equations for clearing the goods markets and balancing the government budget. [8]

(b) Solve for q, T and all four consumption quantities as functions of τ. Verify both goods markets and both private budgets. Do not count tax revenue as destroyed physical output. [14]

(c) Draw the equilibria at $\tau = 0$ and $\tau = 2$ on the same Edgeworth box. Explain why A and B generally do not face the same budget slope even though there is one producer price vector. [8]

(d) Compare both consumers' welfare as τ rises from zero. Explain why a policy can benefit its rebate recipient but still generate an inefficient allocation. [6]

(e) For every $\tau > 0$, construct a feasible allocation that makes both consumers strictly better off than the taxed equilibrium. Verify the gains algebraically, including nonnegative consumption. You may look for a trade that moves each consumer toward equal quantities of the two goods. [10]

(f) Which hypothesis of the competitive welfare argument fails? Distinguish a transfer of purchasing power from the marginal tax wedge, and explain why rebating every dollar does not eliminate the latter. [4]

Optional tool extension. Vary only $\tau \geq 0$ on a rational grid containing $0, 2$ and small positive taxes. Check both private budgets, government balance, both resource equations and the exact squared utility gain identity for the proposed Pareto improvement. Keep the rebate rule fixed and label $\tau = 0$ separately, where the strict gain becomes zero.

Problem 11.A7 [Advanced]. Observed production and the empirical content of gen-

eral equilibrium.

Restored from the Western University Fall 2018 final examination. The two original tasks are retained. The supporting inequality indices, income accounting and fixed consumer identity are made explicit.

Suggested time: 60 minutes. Total: 60 points. You may use Afriat's theorem, but must explain how it is applied and how the market and ownership restrictions enter.

There are L goods, a finite set of market observations $t = 1, \ldots, T$, and the same finite set of consumers $i = 1, \ldots, I$ in each observation. Consumer i has one preference across all observations. Rationalization means that there exist continuous, concave and strictly increasing utilities $u_i : \mathbb{R}^L_+ \to \mathbb{R}$ for which the observed market conditions are competitive equilibria.

For every observation t, the data include positive prices $p^t \in \mathbb{R}^L_{++}$, positive individual incomes $b^t_i > 0$, aggregate endowment $\omega^t \in \mathbb{R}^L_+$, and the net production plan $y^t \in \mathbb{R}^L$ of the unique local firm. The common technology $Y \subseteq \mathbb{R}^L$ is known, nonempty, closed, convex and satisfies free disposal. Observed plans must belong to Y and maximize profit at their observed prices. Individual consumption bundles are unobserved. Individual nonnegative endowments summing to ω^t and nonnegative ownership shares summing to 1 may vary across observations; incomes include endowment value and the individual's share of firm profit. There are no unobserved outside transfers.

(a) Write a necessary and sufficient rationalization certificate using latent bundles x^t_i, utility levels U^t_i and positive multipliers λ^t_i. Include consumer budgets, all cross observation consumer inequalities, goods market clearing, firm profit maximization and aggregate income accounting. Prove sufficiency by constructing utilities from the certificate. Also show that the aggregate income identity allows a feasible allocation of individual endowments and ownership shares consistent with the stated incomes. Distinguish what is a finite latent consumer system from what must be checked against the known technology. [35]

(b) Now specialize to the same single consumer observed in every market and to the original free disposal technology $Y = \mathbb{R}^L_-$. The observed data are $(p^t, b^t, \omega^t)^T_{t=1}$. Prove that these data are rationalizable if and only if $b^t = p^t \cdot \omega^t$ in every observation and the price and bundle data $(p^t, \omega^t)^T_{t=1}$ satisfy GARP. State GARP precisely. Explain why the result would not be valid if each observation belonged to a different consumer with an unrelated preference. [25]

Part II: A consumer-owned firm — chapter use case and verified scale-up

Problem 11.2 [Proof]. First Fundamental Welfare Theorem.

Consider a private-ownership economy with locally nonsatiated preferences and profit-maximizing firms.

(a) State the definition of Pareto efficiency used in the theorem.

(b) Prove that a competitive equilibrium allocation is Pareto efficient.

(c) Identify exactly where local nonsatiation, budget feasibility, profit maximization, and market clearing enter the proof.

Problem 11.3 [Proof]. Supporting a Pareto optimum.

Suppose consumption sets and production sets are closed and convex, preferences are continuous, convex, and locally nonsatiated, and a feasible allocation is Pareto efficient. Assume free disposal and that the aggregate net-resource-requirement set used below is closed, convex, and upward comprehensive. Assume the remaining separation hypotheses stated in the chapter.

(a) Formulate the aggregate set-separation argument and show how it supports the allocation as a price quasi-equilibrium with transfers.

(b) Explain why the initial endowment distribution generally must be changed by lump-sum transfers.

(c) State the positive-wealth or boundary condition needed to upgrade the quasi-equilibrium to a price equilibrium with transfers, and distinguish that object from a competitive equilibrium for the original ownership economy.

Chapter use case Seed: Problems 11.1, 11.4. Fixed: all unlisted primitives and assumptions. Scaled: the production exponent from one-half to $\beta \in (0, 1)$. Tool: Codex + Python. Verification: Recover the square-root economy; verify planner and decentralized allocations coincide; check second-order conditions, profit finiteness, both market-clearing equations, and limiting behavior in β.

Problem 11.4 [Applied]. One consumer and one producer.

There are two goods. Good 1 can be consumed or used as input z to produce good 2 according to $y_2 = \sqrt{z}$. The consumer owns the firm and has an endowment $\bar{\omega}_1 > 0$ of good 1 and no good 2. Utility is $\log x_1 + \log x_2$.

(a) Solve the planner's problem.

(b) Find supporting relative prices from marginal conditions.

(c) Verify that the consumer and firm choices clear both markets.

Existence, uniqueness and the core

Part I: By hand

Problem 12.1 [Core]. Differential identities for excess demand.

Let aggregate excess demand $z : \mathbb{R}^L_{++} \to \mathbb{R}^L$ be continuously differentiable, homogeneous of degree zero, and satisfy Walras' law.

(a) Differentiate homogeneity to show $D_p z(p)p = 0$.

(b) Differentiate Walras' law to show $[D_p z(p)]^\top p = -z(p)$.

(c) Specialize the second identity to an equilibrium and interpret the left null vector.

Advanced exam practice

These problems build toward integrated graduate exam arguments. The suggested time applies to the hand questions, not the optional tool extension. First draw a diagram or state a solution plan, define the feasible set, and identify possible corners. Explain why each candidate is globally optimal and check the economic accounting. Attempt the hand questions before using tools. For an optional extension, state Seed, Fixed, Scaled and Risk before coding and enlarge one feature at a time.

Problem 12.A1 [Advanced]. One allocation and a whole interval of equilibrium prices.

Suggested time: 35 minutes for the hand part. Total: 50 points. Work by hand.

Two consumers have $u^A(x) = 2x_1 + x_2$ and $u^B(x) = x_1 + 2x_2$ on $\mathbb{R}^2_+$. Endowments are $\omega^A = (4, 0)$ and $\omega^B = (0, 4)$. Normalize prices to $(q, 1)$ with $q > 0$.

(a) Derive both full demand correspondences, including the prices at which a consumer is indifferent across a budget line. Find every normalized equilibrium price and every equilibrium allocation. [14]

(b) Give a KKT or direct budget certificate supporting the equilibrium allocation at every price you found. Why would imposing equality of the two marginal rates of substitution incorrectly reject the allocation? [10]

(c) Draw the entire Pareto set and determine the strong core under the book's strong blocking convention. Explain whether the original endowment permits any mutually acceptable strict gain from trade. [12]

(d) At an equilibrium price strictly inside the equilibrium price interval, calculate the derivative of normalized aggregate excess demand. Is this equilibrium regular? Distinguish uniqueness of an

allocation, local uniqueness of normalized prices and the single valuedness of individual demand. [8]

(e) Does this example contradict the chapter's strict gross substitutes uniqueness proposition? Identify the missing strictness rather than treating linear utility as evidence of irrational behavior. [6]

Optional tool extension. Vary only $q > 0$ on a rational grid including $1/2, 2$, points between them and points just outside them. Implement full demand correspondences at the two equality prices. Verify the clearing selection and corner entry inequalities exactly, and check the reduced derivative inside the interval. Explain why sampling many clearing prices does not by itself prove the full interval claim.

Part II: A coalition challenges a competitive allocation — chapter use case and verified scale-up

Problem 12.2 [Proof]. Regularity and local uniqueness.

Normalize prices by $p_L = 1$ and let $\widehat{z}$ be the first $L - 1$ excess-demand equations.

(a) Define a regular equilibrium.

(b) Use the inverse function theorem to prove that a regular equilibrium is locally isolated.

(c) Suppose every normalized equilibrium is regular. Explain why compactness and boundary conditions then imply that there are only finitely many equilibria.

Problem 12.3 [Proof]. Gross substitutes and uniqueness.

Assume aggregate excess demand satisfies strict gross substitutes on the relevant price domain: increasing one price strictly raises excess demand for every other good whenever the comparison is nondegenerate. Suppose two strictly positive normalized equilibrium price vectors exist.

(a) Compare the largest relative price change across goods.

(b) Use gross substitutes and Walras' law to derive a contradiction unless the two price vectors are proportional.

(c) Explain why weak gross substitutes alone need not imply uniqueness and where strictness (or an irreducibility substitute) enters the proof.

Chapter use case Seed: Problem 12.4. Fixed: all unlisted primitives and assumptions. Scaled: the blocking predicate from all-strict improvement to weak improvement for all members and strict improvement for one witness. Tool: Codex + Lean. Verification: Compile the Lean strong-coalition price-support declaration and match each hypothesis to the economic proof. Lean

verifies the finite price-summing contradiction, not the derivation of price support from Walrasian optimization or the replica-economy convergence claim. This case closes the price-support and core thread; Problems 12.2 and 12.3 are separate regularity and uniqueness extensions.

Problem 12.4 [Core]. Competitive allocations and the strong core.

Use the strong-blocking convention in a finite pure-exchange economy with locally nonsatiated preferences. A nonempty coalition S strongly blocks x when it can choose $(y_i)_{i\in S}$ such that $\sum_{i\in S} y_i = \sum_{i\in S} \omega_i$, $y_i \succeq_i x_i$ for every $i \in S$, and $y_j \succ_j x_j$ for at least one $j \in S$. Define the associated strong core.

(a) Prove that every Walrasian allocation belongs to the strong core.

(b) Give an economic reason why the converse can fail with finitely many consumers.

(c) State the content, but not a proof, of the core-convergence theorem for replica economies.

Empirical restrictions and equilibrium identification

Part I: By hand

Problem 13.1 [Core]. Equilibrium Afriat feasibility.

At each observation t, prices $p^t \gg 0$, individual incomes $(w_i^t)_i$, and aggregate endowment ω^t are observed, while individual allocations are latent.

(a) Introduce latent consumptions x_i^t and write individual budget equalities $p^t \cdot x_i^t = w_i^t$ and aggregate market clearing $\sum_i x_i^t = \omega^t$.

(b) Write Afriat inequalities for every consumer across observations.

(c) Explain why joint feasibility is a finite semialgebraic revealed-equilibrium test, why the latent multipliers and bundles make the full system bilinear, and how quantifier elimination projects the restrictions onto the observables.

Part II: A counterfactual equilibrium identified set — chapter use case and verified scale-up

Problem 13.2 [Proof]. What equilibrium data identify.

Suppose only equilibrium prices and aggregate endowments are observed.

(a) List the preference, allocation, and ownership objects that remain unobserved.

(b) Explain why market clearing alone cannot identify individual demand.

(c) Describe how additional observations, known individual endowments, or parametric restrictions can turn an underidentified model into a testable one.

Chapter use case Seed: Problems 13.1, 13.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of equilibrium observations from two to T, with the maintained preference class and counterfactual fixed. Tool: Codex + Python. Verification: Verify feasibility and market clearing for every retained candidate, show how each added observation changes the bounds, and never replace set identification by a point estimate.

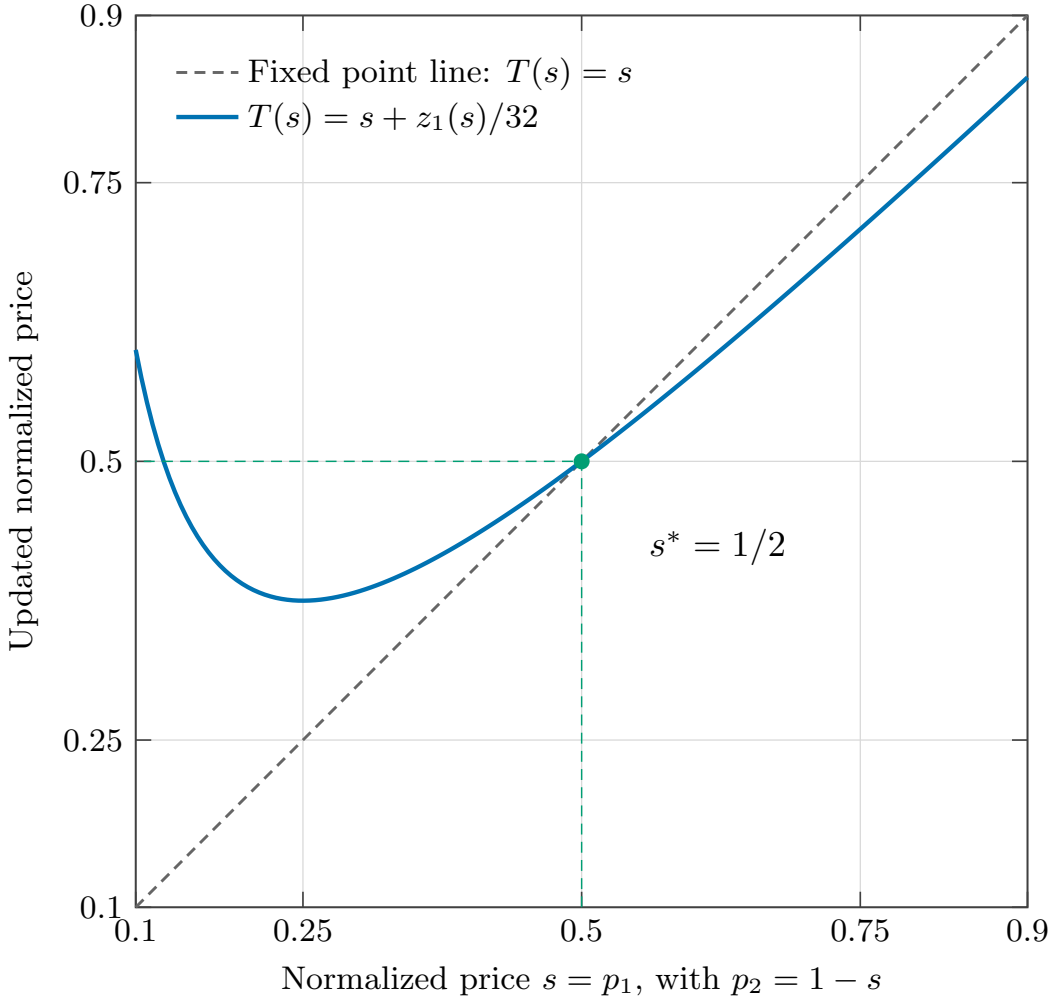
Problem 13.3 [Applied]. Counterfactual discipline.

A dataset passes a revealed-equilibrium feasibility test under a maintained class of preferences.

(a) Explain why passing the test does not point-identify a counterfactual equilibrium.

(b) Define an identified set for a counterfactual price or allocation.

(c) State two economically meaningful restrictions that could tighten this set, and explain the cost of each restriction.



$$T : [1/10, 9/10] \to [1/10, 9/10]$$
$$T(s) = s \iff z_1(s) = 0$$

Figure 11.6: A continuous self map and its fixed point. The crossing with the diagonal is the normalized clearing price in the worked economy. Existence of the crossing does not show that every price iteration converges.

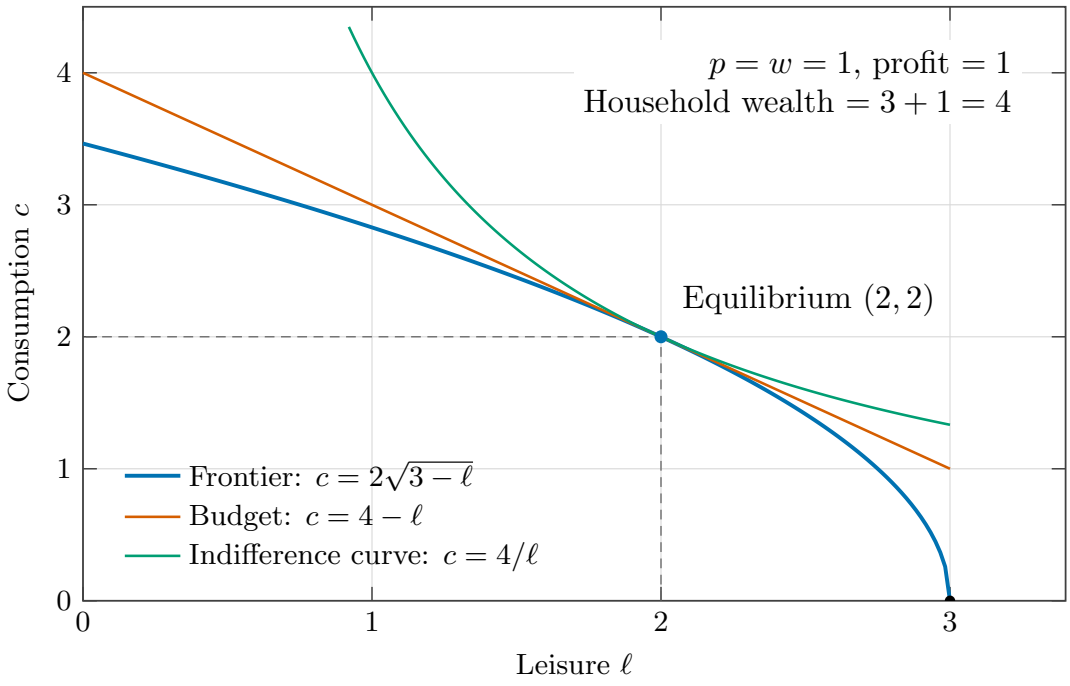


Figure 11.7: The physical frontier, a supporting price budget and an indifference curve meet at equilibrium. Distributed profit makes wealth four, so the original time endowment $(3, 0)$ is not on the household budget line. Profit changes purchasing power, not the physical frontier.

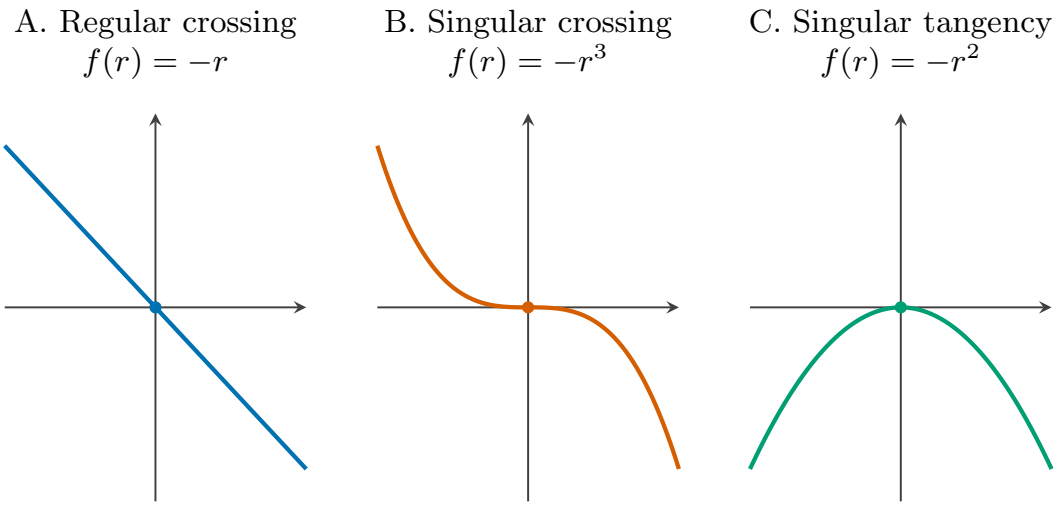


Figure 11.8: Three local geometries of an equilibrium equation. Only panel A has a nonzero reduced derivative. Panel B still has a unique local root. Panel C can gain or lose roots under a vertical perturbation. These are local geometric illustrations, not claims that each displayed polynomial is globally the excess demand of a specified finite economy.

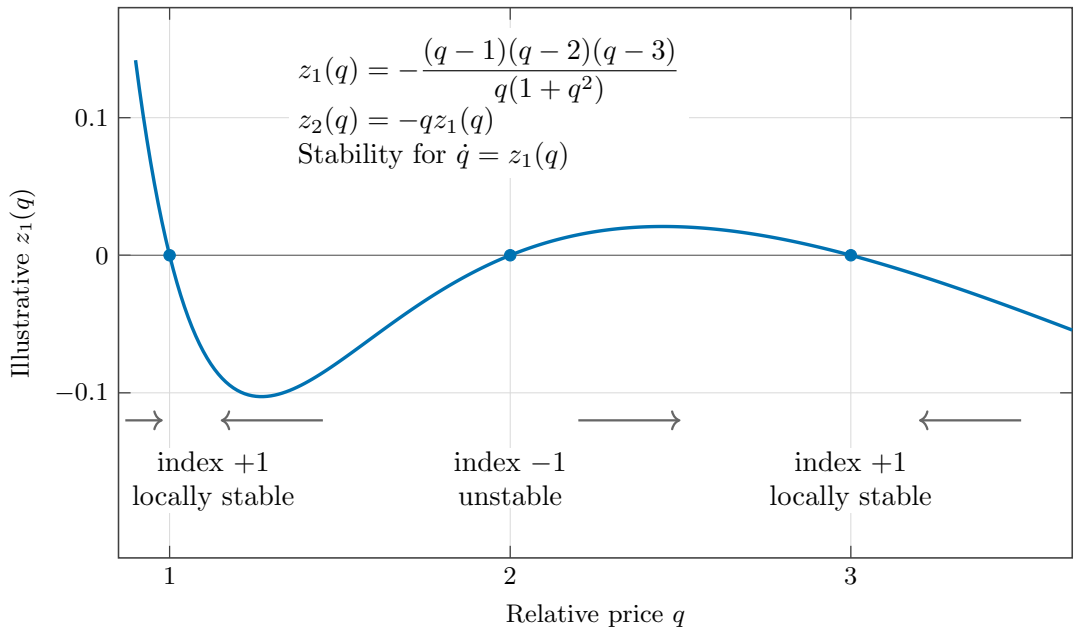


Figure 11.9: Local isolation does not imply global uniqueness. The illustrative system has three regular clearing prices. Its indices are $+1, -1, +1$. Stability labels and arrows refer only to the explicitly imposed scalar adjustment law $\dot{q} = \kappa f(q)$ with $\kappa > 0$. A positive index is not a general stability theorem in many goods.

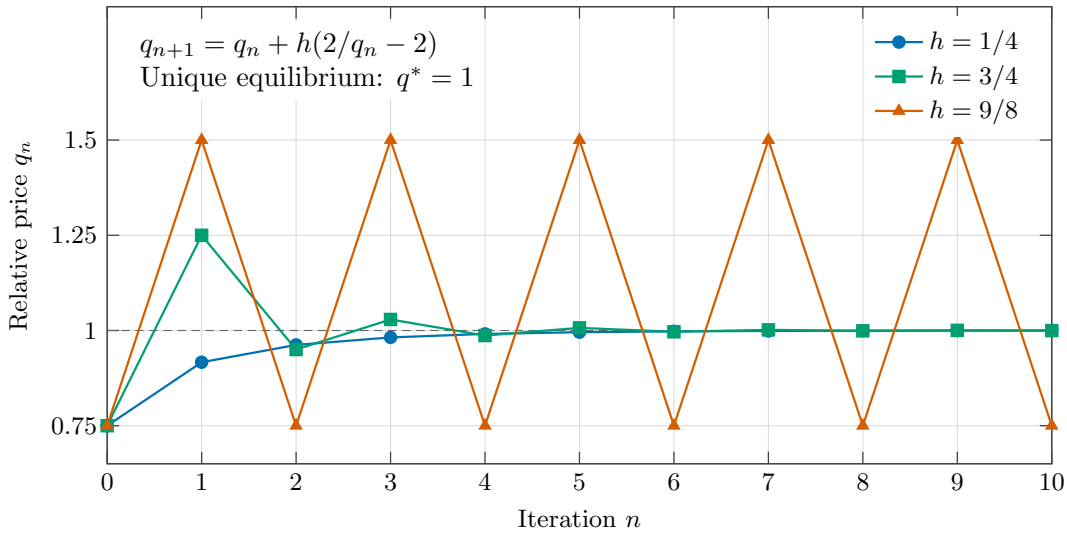


Figure 11.10: The same economic equilibrium, three numerical paths. All paths start at $q_0 = 3/4$. The smaller steps approach the unique clearing price; the step $9/8$ cycles exactly between $3/4$ and $3/2$. Algorithm design and economic uniqueness are separate issues. The displayed convergent paths illustrate behavior; the local convergence claim uses the derivative condition in the text.

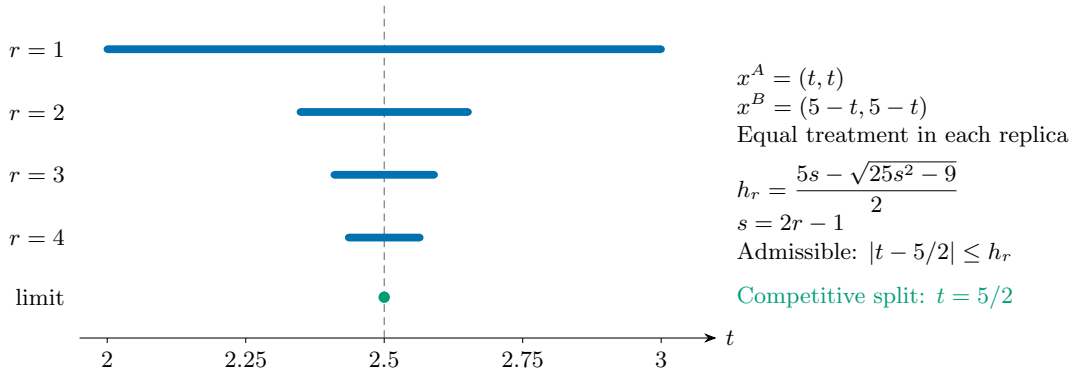


Figure 11.11: Replication creates more coalition compositions. The bars are exact equal treatment core intervals for the stated two type example, obtained by checking the complete coalition capacity criterion. Their limit is the original competitive allocation. The convergence rate and interval formula belong to this example, not to every economy.

Part V

Advanced Equilibrium Applications

Chapter 12

Niche Consumption and Product Space

The preceding chapter connects households and firms through prices and resource constraints for a fixed finite list of commodities. We now ask how that description should change when products are numerous, differentiated and continually replaced. Neiman and Vavra's evidence motivates the question: household concentration and aggregate concentration need not move together. The task is to choose the right commodity and demand objects before drawing welfare or substitution conclusions.

This is a research synthesis, not another equilibrium existence theorem. A continuum of differentiated goods requires its own topological and economic assumptions; the finite commodity theorem in Chapter 11 does not automatically establish a continuum equilibrium. The chapter separates individual product choice, basket demand and assignment under capacity, then returns to the empirical evidence and its interpretation.

12.1 The Many-Product Consumer

The fixed commodity vector $x \in \mathbb{R}^L_+$ is exact when the list of goods is fixed. It becomes a poor organizing device when products enter continuously, differ along many characteristics, and are filtered by search or recommendation before a consumer sees them. Amazon's current public description, for example, reports more than 300 million products across more than 35 categories [Amazon, 2026]. Exact catalog counts change; the economic point is the scale, not a claim that the count is one billion.

Let characteristics live in a compact metric product space $(\mathcal{G}, d)$. Three objects must remain distinct: the available catalog $\mathcal{G}_0 \subseteq \mathcal{G}$, consumer i's consideration set $A_i \subseteq \mathcal{G}_0$, and the purchased support $\operatorname{supp} x_i \subseteq A_i$. Availability is therefore not attention, and attention is not positive demand. At the individual level, an unseen option and a disliked option generate the same zero purchase [Abaluck and Adams-Prassl, 2021]. At the market level, an available product can receive zero observed empirical share, requiring methods that do not treat zero share as proof of unavailability [Gandhi et al., 2023].

12.2 One item: ideal points and local diversion

To isolate product-space geometry, suppose every consumer buys exactly one item, the ideal-point distribution μ is a probability measure, and all products receive full consideration:$A_i = \mathcal{G}_0$. Participation can instead be modeled by adding an outside option. Product j is located at $g_j \in \mathcal{G}$, has quality intercept δ_j and price p_j, and consumer i has ideal point $b_i \in \mathcal{G}$. A quadratic mismatch specification is

$$U_{ij}^* = \delta_j - p_j - \kappa_d d(b_i, g_j)^2, \qquad \kappa_d > 0.$$

The types choosing product j form the cell

$$C_j(p) = \left\{ b \in \mathcal{G} : \delta_j - p_j - \kappa_d d(b, g_j)^2 \geq \delta_k - p_k - \kappa_d d(b, g_k)^2 \text{ for every } k \right\}.$$

Demand share is $s_j(p) = \mu(C_j(p))$ for the distribution μ of ideal points. Assume utility-tie sets have μ-measure zero; otherwise fix a tie-breaking rule and define the cells accordingly. These are generalized weighted Voronoi cells; with squared Euclidean distance they are power cells. Equal prices and qualities imply nearest-product choice. With unequal prices or qualities, the nearest product need not be chosen. This formulation joins spatial competition, the characteristics approach, and discrete choice [Hotelling, 1929, Lancaster, 1966, Anderson et al., 1989, McFadden, 1981].

Proposition 12.1. *Local diversion on a line. Let $\mathcal{G} = [\underline{g}, \overline{g}] \subset \mathbb{R}$ with Euclidean metric $d(b, g) = |b - g|$, let active products satisfy $g_1 < \cdots < g_J$, and let ideal points have distribution function F with continuous density f that is strictly positive at the active boundaries. Fix an interior product j and an open neighborhood of prices in which its nonempty choice interval has only products $j - 1$ and $j + 1$ as active neighbors, all other alternatives remain strictly slack at*

its boundaries, and no boundary has a triple tie. The boundary between j and $j+1$ is

$$\bar{b}_{j,j+1} = \frac{g_j + g_{j+1}}{2} + \frac{\delta_j - \delta_{j+1} - p_j + p_{j+1}}{2\kappa_d(g_{j+1} - g_j)},$$

and

$$s_j = F(\bar{b}_{j,j+1}) - F(\bar{b}_{j-1,j}).$$

At every price vector preserving this active adjacency,

$$\begin{aligned}
\frac{\partial s_j}{\partial p_{j+1}} &= \frac{f(\bar{b}_{j,j+1})}{2\kappa_d(g_{j+1} - g_j)} > 0, \\
\frac{\partial s_j}{\partial p_{j-1}} &= \frac{f(\bar{b}_{j-1,j})}{2\kappa_d(g_j - g_{j-1})} > 0, \\
\frac{\partial s_j}{\partial p_j} &= -\frac{f(\bar{b}_{j,j+1})}{2\kappa_d(g_{j+1} - g_j)} - \frac{f(\bar{b}_{j-1,j})}{2\kappa_d(g_j - g_{j-1})} < 0,
\end{aligned}$$

whereas $\partial s_j/\partial p_k = 0$ locally for every nonneighbor $k \notin \{j-1, j, j+1\}$. Equivalently, an increase in p_j diverts the lost share only to the two active neighbors:

$$\frac{\partial s_{j+1}}{\partial p_j} = \frac{f(\bar{b}_{j,j+1})}{2\kappa_d(g_{j+1} - g_j)}, \qquad \frac{\partial s_{j-1}}{\partial p_j} = \frac{f(\bar{b}_{j-1,j})}{2\kappa_d(g_j - g_{j-1})},$$

all other product shares are locally unchanged, and these two gains sum to $-\partial s_j/\partial p_j$.

Proof. Equating the two adjacent utilities and expanding the squares gives the displayed boundary. Under stable adjacency, product j's cell is exactly the interval between its left and right boundaries, so its share is the difference of the two distribution-function values. Differentiate the right boundary with respect to p_j and p_{j+1}, and the left boundary with respect to p_{j-1} and p_j. A nonneighbor's price enters neither active boundary while the cell adjacency is unchanged. □

The proposition is local and conditional: a sufficiently large price change can alter the active-neighbor graph. It also shows why a product metric alone is insufficient. The utility specification and stable cell geometry create local diversion. By contrast, conditional logit without an outside option and with price coefficient $\alpha_p > 0$ satisfies

$$s_j = \frac{\exp(\delta_j - \alpha_p p_j)}{\sum_k \exp(\delta_k - \alpha_p p_k)}, \qquad \frac{\partial s_j}{\partial p_k} = \alpha_p s_j s_k > 0 \quad (j \neq k),$$

so every product substitutes for every other product regardless of distance. Random coefficients allow richer substitution, but distance decay is still a restriction to be modeled and tested rather than a theorem of discrete choice [Berry et al., 1995, Berry and Pakes, 2007, Feenstra and

Levinsohn, 1995, Pinkse et al., 2002].

12.3 Many items: bundles over product space

A continuum of alternatives is not the same object as a bundle drawn from a continuum of goods. For bundle demand, let $\mathcal{M}_+(\mathcal{G})$ be the finite nonnegative Radon measures on $\mathcal{G}$. A basket is $x \in \mathcal{M}_+(\mathcal{G})$, a price schedule is $p \in C_{++}(\mathcal{G})$, and expenditure is the dual pairing

$$\langle p, x \rangle := \int_{\mathcal{G}} p(g)\, x(dg).$$

The budget is $\{x \in \mathcal{M}_+(\mathcal{G}) : \langle p, x \rangle \le w\}$. An observed finite basket is included exactly as the atomic measure $x = \sum_{j=1}^{J} q_j \delta_{g_j}$, for which expenditure is $\sum_j p(g_j) q_j$. A density $q(g)$ is the special case $x(dg) = q(g)\nu(dg)$ relative to a reference measure ν. The measure formulation therefore contains both finite baskets and genuinely diffuse consumption [Mas-Colell, 1975, Bewley, 1972].

Neiman and Vavra use a continuum of products for tractability, but their consumer chooses quantities of many varieties in an endogenous subset and incurs a utility cost of expanding that subset. Their heterogeneity is a mixture of common and idiosyncratic product ranks. It is not a metric product geometry: adjacent ranks need not be nearby in characteristics. Their framework therefore supplies selection and variety mechanisms, not a theorem that a price change diverts demand to nearby products [Neiman and Vavra, 2023].

Definition 12.2. Neighborhood compensated elasticity. The radius-r neighborhood of product point g is

$$B_r(g) := \{g' \in \mathcal{G} : d(g, g') \le r\}.$$

Let locally unique Hicksian demand be a measure, and let $h(A; p, \overline{u}) > 0$ denote its mass in measurable region A. Fix a family of continuous log-price shock profiles $\varphi_B : \mathcal{G} \to [0, 1]$ indexed by closed regions B, normalized by $\varphi_B = 1$ on B and by a common support rule, for example $\operatorname{supp} \varphi_B \subseteq \{g : d(g, B) \le \rho\}$ for a fixed bandwidth ρ. Set $p_\varepsilon(g) = p(g) \exp\{\varepsilon \varphi_B(g)\}$. If the scalar path $\varepsilon \mapsto h(A; p_\varepsilon, \overline{u})$ is differentiable at zero, define

$$\mathcal{E}^H(A; \varphi_B) := \left. \frac{d \log h(A; p_\varepsilon, \overline{u})}{d\varepsilon} \right|_{\varepsilon = 0}.$$

In a finite catalog with the discrete topology, one may take $\varphi_B(g) = \mathbb{I}\{g \in B\}$.

This statistic replaces an unmanageable product-by-product elasticity matrix with a region-

to-region response. It also avoids an inadmissible discontinuous point shock when prices must remain continuous; for diffuse baskets, a singleton additionally has zero mass. For fixed-radius separated closed regions and profiles with the same amplitude and bandwidth, a precise locality restriction uses a nonincreasing envelope $\Psi : [0,\infty) \to [0,\infty)$ satisfying

$$\left|\mathcal{E}^H(A;\varphi_B)\right| \le \Psi(d(A,\operatorname{supp}\varphi_B))\,.$$

where $d(A,C) := \inf\{d(a,c) : a \in A,\ c \in C\}$. The absolute value permits either substitution or complementarity; the sign records which one obtains. Such decay is an additional economic restriction. It does not follow from continuity, utility maximization, or the existence of a continuum. With bundles, some neighborhoods may instead be complements; compensated and Marshallian cross effects must also be kept distinct.

12.4 Finite-data Afriat logic is dimension-free

The finite-data logic of Afriat's theorem survives this change of commodity space. What the data use is not a coordinate count but the finite matrix of observed expenditures. The functional-analytic assumptions below supply the global utility and demand-existence claims that are automatic in a bounded finite-dimensional presentation.

Proposition 12.3. *Afriat rationalization over a continuum product space. Let $(\mathcal{G},d)$ be compact metric, let $\mathcal{M}_+(\mathcal{G})$ be the cone of finite nonnegative Radon measures in $C(\mathcal{G})^*$ equipped with the weak-star topology $\sigma(C(\mathcal{G})^*, C(\mathcal{G}))$, let bundles lie in $\mathcal{M}_+(\mathcal{G})$, and let each observed price schedule $p^t \in C_{++}(\mathcal{G})$. Suppose the finite dataset $\mathcal{O} = \{(p^t,x^t)\}_{t=1}^T$ has through-observation budget*

$$B_t := \{x \in \mathcal{M}_+(\mathcal{G}) : \langle p^t, x\rangle \le \langle p^t, x^t\rangle\}.$$

Define $x^t\ R^D\ x^s$ when $x^s \in B_t$, define $x^t\ P^D\ x^s$ when $\langle p^t, x^s\rangle < \langle p^t, x^t\rangle$, and let R be the transitive closure of R^D. Say that the dataset satisfies GARP if, for every s,t, $x^t\ R\ x^s$ implies $\neg(x^s\ P^D\ x^t)$. GARP holds if and only if there exist numbers $(U^t,\lambda^t)_{t=1}^T$ satisfying the Afriat inequalities

$$U^s \le U^t + \lambda^t\langle p^t, x^s - x^t\rangle, \qquad \lambda^t > 0, \qquad s,t = 1,\dots,T.$$

For any such certificate,

$$U(x) := \min_{1\le t\le T}\left\{U^t + \lambda^t\langle p^t, x - x^t\rangle\right\}$$

is concave, weak-star continuous, and strictly increasing in the measure order, and it rationalizes every observation. Moreover, for every $p \in C_{++}(\mathcal{G})$ and $w > 0$, U attains a maximum on $\{x \geq 0 : \langle p, x\rangle \leq w\}$.

Proof. Apply the finite-expenditure-matrix form of Afriat's lemma to $a_{ts} := \langle p^t, x^s\rangle$. This algebraic lemma says that GARP for the finite comparison graph is equivalent to the displayed certificate; it does not use the dimension of the ambient commodity space. At observation x^s, every affine term in the displayed minimum is at least U^s and term s equals U^s. Hence $U(x^s) = U^s$. If $\langle p^s, x\rangle \leq \langle p^s, x^s\rangle$, term s gives $U(x) \leq U^s = U(x^s)$, which proves rationalization. A finite minimum of continuous affine functionals is concave and weak-star continuous. If $y - x$ is a nonzero positive measure, compactness and strict positivity give $\min_{g\in\mathcal{G}} p^t(g) > 0$ and therefore $\lambda^t\langle p^t, y - x\rangle > 0$ for every t. Since T is finite,

$$\delta := \min_{1\leq t\leq T} \lambda^t\langle p^t, y - x\rangle > 0.$$

Consequently $U(y) \geq U(x) + \delta > U(x)$. Finally, if $m = \min_g p(g) > 0$, every budget-feasible measure satisfies $x(\mathcal{G}) \leq w/m$. The positive budget set is weak-star closed and is contained in the closed dual norm ball of radius w/m in $C(\mathcal{G})^*$. That ball is weak-star compact by Banach--Alaoglu, so the budget set is weak-star compact as a closed subset. Continuity of U gives attainment. □

Remark 12.4. The proposition is a dimension-free corollary of the ordinary Afriat construction, not a claim that all infinite-dimensional consumer theory is automatic. It uses divisible measure-valued bundles, continuous strictly positive linear prices, compact product space, and finitely many observations. Indivisible unit demand, arbitrary nonlinear budgets, equilibrium existence, or differentiable Slutsky operators require separate arguments [Afriat, 1967, Varian, 1982, Mas-Colell, 1975, Bewley, 1972].

12.5 When product choice becomes matching

With unlimited replication at posted prices, consumers independently choosing products is a choice model with assignment geometry; it is not yet a matching market. Matching becomes literal when products, sellers, or qualities have capacity. Let μ and ν be probability measures on compact spaces $\mathcal{B}$ of consumer types and $\mathcal{G}$ of products, respectively, let $S(b, g)$ be continuous

transferable surplus, and define

$$\Pi(\mu, \nu) := \{\pi : \pi_1 = \mu,\ \pi_2 = \nu\}.$$

The planner's assignment problem is

$$\max_{\pi \in \Pi(\mu,\nu)} \int_{\mathcal{B}\times\mathcal{G}} S(b, g)\, d\pi(b, g).$$

Compactness and continuity give existence. The fixed second marginal exhausts supply; a model with upper capacities or unmatched agents instead needs marginal inequalities, partial transport, or dummy outside types. This is the planner problem underlying transferable-utility hedonic matching. A market interpretation additionally requires participation conditions and Kantorovich dual potentials that support the assignment as utilities and prices. If $\mathcal{B} \subseteq \mathbb{R}^m$ and μ is absolutely continuous, additional twist and regularity conditions can yield an almost-everywhere pure match; without them the optimal plan need not be a function. This capacity-constrained problem is conceptually distinct from the one-to-one deferred-acceptance model studied later in the book [Chiappori et al., 2010].

12.6 Evidence and the modern research map

Neiman and Vavra's Nielsen Homescan evidence covers 2004--2016. Within Nielsen product groups, each household's annual UPC spending becomes more concentrated while aggregate UPC spending becomes less concentrated: households buy narrower but increasingly different baskets. The model infers roughly 70 percent growth in effective variety. In a counterfactual that holds taste disagreement fixed and attributes that growth entirely to catalog variety, welfare rises by 0.56 percent per year in the covered categories; 0.52 percentage points, or about 93 percent of that gain, comes from improved selection rather than conventional love of variety. That variety-only exercise does not reproduce the observed decline in the number of varieties per household. Specifications that target all three trends imply annual welfare gains of 0.45--0.47 percent. These are structural model implications rather than direct causal estimates. In particular, household concentration, aggregate concentration, and the observed average number of varieties consumed jointly identify the dispersion parameter η and effective variety $\widetilde{N} = N\alpha/(1-\alpha)$, not catalog size N separately from taste disagreement α [Neiman and Vavra, 2023].

The literature divides naturally by the economic object:

- Characteristics and ideal points explain how a dense catalog acquires geometry; pure-characteristics and local-competition models turn that geometry into substitution restrictions [Lancaster, 1966, Berry and Pakes, 2007, Feenstra and Levinsohn, 1995, Pinkse et al., 2002].

- Random-coefficients demand estimates heterogeneous tastes and flexible diversion among a large but finite set of products [Berry et al., 1995].

- Variety and long-tail models distinguish ordinary love of variety, access costs, and better matching to household tastes [Brynjolfsson et al., 2003, 2011, Broda and Weinstein, 2010, Li, 2021, Neiman and Vavra, 2023].

- Consideration-set methods distinguish products that are available from products that enter the consumer's consideration set [Abaluck and Adams-Prassl, 2021].

- Zero-share methods address available products that receive zero observed empirical market share; they do not by themselves identify whether a consumer saw the product [Gandhi et al., 2023].

- Hedonic matching and optimal transport add capacity, fixed supply marginals, and market-clearing assignments to the product-space geometry; endogenous quality design additionally requires producer technology and costs [Chiappori et al., 2010].

The model-selection rule is therefore simple. If the consumer buys one item, use discrete choice on product space. If the consumer buys a basket, use measure-valued demand. If product capacities constrain who can receive what, use matching or optimal transport. In every case, product nearness must be given economic content before it can discipline cross-price effects.

12.7 Reading and synthesis

Begin with the distinction between the available catalog, consideration and positive purchases. Then compare the local diversion example with logit and recognize a finite basket as an atomic nonnegative measure within the broader formulation. The matching interpretation applies only after capacities or fixed supply marginals are imposed. These are different economic models, not interchangeable names for a large catalog.

This chapter creates no additional weekly assignment. The companion practice book connects the synthesis to existing exercises on revealed preference, logit, aggregation, redistribution,

equilibrium data and matching. Finite computations can illustrate the continuum formulation, but they do not prove its analytic claims. Problem identifiers remain the permanent labels of the original bank.

Chapter 13

Sequential Trade

13.1 Arrow--Debreu benchmark

There are $S \geq 1$ states, indexed by $\mathcal{S} = \{1, \ldots, S\}$, and L goods in each state. Consumers are indexed by $i \in \{1, \ldots, I\}$. Consumer i has state-contingent endowment $\omega_i = (\omega_{i1}, \ldots, \omega_{iS})$ and preference $\succeq_i$ over contingent consumption plans $x_i = (x_{i1}, \ldots, x_{iS})$.

Definition 13.1. (Arrow--Debreu equilibrium.) Prices $p = (p_1, \ldots, p_S) \in (\mathbb{R}^L_{++})^S$ and an allocation $x^* = (x_1^*, \ldots, x_I^*)$ form an Arrow--Debreu equilibrium if, for each consumer, x_i^* maximizes $\succeq_i$ over $B_i^{AD}(p) = \{x_i : \sum_{s=1}^S p_s \cdot x_{is} \leq \sum_{s=1}^S p_s \cdot \omega_{is}\}$, and markets clear state by state: $\sum_i x_{is}^* = \sum_i \omega_{is}$ for every s.

13.2 Spot markets and sequential trade

Suppose contingent commodity markets are unavailable at date $t = 0$. There are K tradable assets with payoff matrix $R \in \mathbb{R}^{S \times K}$: one unit of asset k pays R_{sk} units of the first good in state s. Asset prices are $q \in \mathbb{R}^K$, and spot-good prices in state s are $p_s \in \mathbb{R}^L_{++}$.

Definition 13.2. (Radner budget set.) Given asset holdings $z_i \in \mathbb{R}^K$ and state-contingent consumption x_i, consumer i satisfies $q \cdot z_i \leq 0$ and $p_s \cdot x_{is} \leq p_s \cdot \omega_{is} + p_{s1}(Rz_i)_s$ for every s. The zero on the date-0 constraint is the normalization of zero initial asset endowment.

(Radner equilibrium.) A price system $(q, p_1, \ldots, p_S)$ and plans $(z_i^*, x_i^*)_{i=1}^I$ form a Radner equilibrium when every consumer maximizes their preference over the Radner budget set, asset markets clear $\sum_i z_i^* = 0$, and goods markets clear in every state: $\sum_i x_{is}^* = \sum_i \omega_{is}$ for all s.

Proposition 13.3. *(Arrow--Radner equivalence.) See[Radner, 1972, Mas-Colell et al., 1995]. Suppose consumption lies in* $\mathbb{R}_+^{LS}$*, preferences are continuous and strictly monotone, the payoff matrix has full row rank* $\operatorname{rank}(R) = S$*, and asset prices satisfy no arbitrage. Full spanning and no arbitrage yield a strictly positive state-price vector* ψ *with* $q = R^\top \psi$*. For positive spot prices, define Arrow--Debreu commodity prices by* $\widetilde{p}_s = (\psi_s / p_{s1}) p_s$*. Then the consumer budget sets project to the same contingent-consumption budget, so the sets of consumption allocations supported by Arrow--Debreu and Radner equilibria coincide. If* $\operatorname{rank}(R) < S$*, some contingent trades are unattainable and the equivalence need not hold.*

The full-rank condition is the precise meaning of complete asset markets in this finite-state model. The proposition is an allocation-equivalence statement conditional on equilibrium prices; it does not by itself assert equilibrium existence.

13.2.1 A payoff span is not a budget set

Example 13.4. With two states and one good per state, a single bond has payoff $(1, 1)^\top$. Its portfolios produce only vectors on the diagonal line; no position can produce the net trade $(1, -1)$. Add a security paying one unit only in state 1. The matrix $R = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has rank two, and portfolio $z = (-1, 2)$ gives $Rz = (1, -1)$. Short positions are allowed in this example.

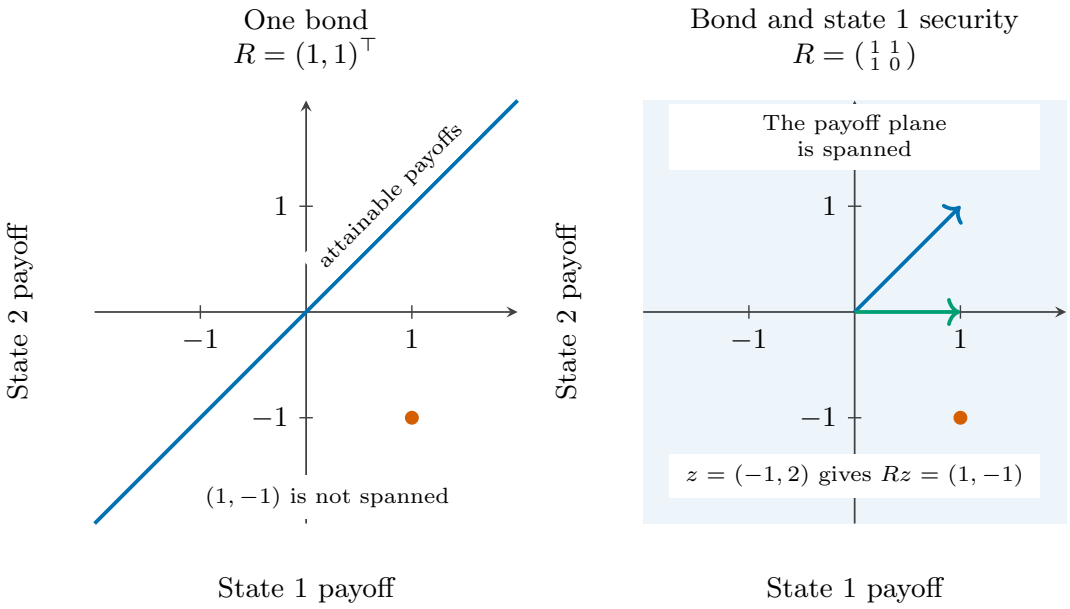


Figure 13.1: Spanning is a linear representation question before affordability. One bond spans a line, while two independent payoff vectors span the whole two state payoff plane. Portfolio cost and nonnegative final consumption must still be checked. Negative payoffs are delivery obligations.

Positive state prices $\psi = (\psi_1, \psi_2)$ assign bond price $\psi_1 + \psi_2$ and state 1 security price ψ_1. The displayed portfolio costs $\psi_1 - \psi_2$, the state price value of $(1, -1)$. Full rank permits representation of every payoff vector, not affordability of every vector. The date zero budget and final consumption constraints remain in force.

13.3 Problems

Part I: By hand

Problem 14.1 [Core]. Spanning contingent consumption.

There are S states tomorrow and K traded assets with payoff matrix $R \in \mathbb{R}^{S \times K}$ and date-zero asset-price vector $q \in \mathbb{R}^K$. Portfolios are unrestricted, $\theta \in \mathbb{R}^K$, so short sales are allowed. Assume no arbitrage.

(a) State the rank condition under which every contingent net trade can be implemented by an asset portfolio.

(b) If $S = 2$ and the assets have payoff vectors $(1, 1)$ and $(1, 0)$, construct the portfolio that delivers any target payoff (c_1, c_2).

(c) Let $\psi \in \mathbb{R}^S_{++}$ be state prices satisfying $q = R^\top \psi$. Explain how the date-zero portfolio budget maps into an Arrow-Debreu present-value budget.

Problem 14.2 [Core]. Incomplete markets.

Let $S = 3$ and suppose the only assets are a bond with payoff $(1, 1, 1)$ and an asset with payoff $(1, 0, 0)$.

(a) Characterize the attainable payoff subspace.

(b) Determine whether the claim $(0, 1, -1)$ is attainable.

(c) Explain why competitive equilibrium with these assets need not be Pareto efficient relative to the full contingent-commodity economy.

Part II: Complete versus incomplete sequential markets — chapter use case and verified scale-up

Seed: Problems 14.1, 14.2. Fixed: all unlisted primitives and assumptions. Scaled: the asset span from rank $S - 1$ to rank S at a fixed number of states. Tool: Codex + Python; Lean target. Verification: Check portfolio replication, the state-price units map, and present-value budgets;

include a rank-deficient claim that cannot be implemented. Arrow-Radner equivalence remains a future Lean target.

Problem 14.3 [Proof]. Arrow-Debreu and sequential equivalence.

There are S states, K assets with payoff matrix $R \in \mathbb{R}^{S\times K}$ of rank S, date-zero asset prices q, and state-s spot-price vectors p_s. Asset payoffs are denominated in the state numeraire. Consumer i chooses a portfolio z_i subject to $q\cdot z_i \le 0$, then chooses x_{is} subject to $p_s\cdot x_{is} \le p_s\cdot\omega_{is}+p_{s1}(Rz_i)_s$. Assume no arbitrage and strictly monotone continuous preferences.

(a) Map an Arrow-Debreu budget into date-zero asset purchases and state-by-state spot trades.

(b) Prove the reverse mapping from a sequential budget to a present-value budget.

(c) Identify which step fails under incomplete markets or binding portfolio constraints.

Chapter 14

Matching

14.1 One-to-one matching

Let M, W be finite disjoint sets of agents on the two sides of the market. For each $m \in M$, $\succ_m$ is a strict order on $W \cup \{m\}$; for each $w \in W$, $\succ_w$ is a strict order on $M \cup \{w\}$. Ranking oneself represents remaining unmatched. A matching is a map $\varphi : M \cup W \to M \cup W$ that assigns each agent either an acceptable partner on the other side or themselves, and satisfies $\varphi(\varphi(a)) = a$.

Definition 14.1. (Stability.) A matching is individually rational if no agent is matched with an unacceptable partner. A pair (m, w) not matched to one another blocks φ if $w \succ_m \varphi(m)$ and $m \succ_w \varphi(w)$. A matching is stable when it is individually rational and has no blocking pair.

14.2 Deferred acceptance

In the men-proposing deferred-acceptance algorithm, each unmatched man proposes to the most-preferred woman who has not rejected him. Each woman tentatively holds her most-preferred acceptable proposal received so far and rejects the rest. The process stops when no proposal is rejected.

Theorem 14.2. *(Gale--Shapley theorem.) See[Gale and Shapley, 1962]. For every finite marriage problem, deferred acceptance terminates and returns a stable matching. Among all stable matchings, the outcome is weakly best for every man and weakly worst for every woman.*

Termination is finite because a man proposes to each woman at most once. The stability and proposer-optimality statements are the Gale--Shapley theorem.

14.3 Housing markets and top trading cycles

In a housing market, each agent initially owns one house and has a strict preference over houses. The strong core consists of allocations for which no coalition can make every member weakly better off and at least one member strictly better off using only the houses owned by that coalition.

The top-trading-cycles algorithm directs each agent to point to their favorite remaining house and each house to point to its owner. Every directed cycle is executed, the participating houses and agents are removed, and the procedure is repeated.

Theorem 14.3. *(Top-trading-cycles theorem.) See[Shapley and Scarf, 1974]. With strict preferences and one-house ownership, top trading cycles terminates, is Pareto efficient, and returns the unique allocation in the strong core; the same allocation is supported as the competitive allocation of the housing market.*

14.4 Problems

Part I: By hand

Problem 15.1 [Core]. Deferred acceptance by hand.

Three men have rankings $m_1 : w_1 \succ w_2 \succ w_3$, $m_2 : w_1 \succ w_2 \succ w_3$, and $m_3 : w_2 \succ w_3 \succ w_1$. The women rank $w_1 : m_2 \succ m_1 \succ m_3$, $w_2 : m_1 \succ m_3 \succ m_2$, and $w_3 : m_3 \succ m_2 \succ m_1$.

(a) Run men-proposing deferred acceptance round by round.

(b) Verify stability of the resulting matching by checking potential blocking pairs.

(c) Explain why termination is guaranteed in every finite market.

Part II: Housing allocation and coalitional stability — chapter use case and verified scale-up

Problem 15.2 [Proof]. Proposer optimality.

Consider a finite balanced one-to-one matching market in which every participant has a complete strict ranking of the other side and all partners are acceptable.

(a) Prove that if a receiver rejects a proposer during deferred acceptance, that rejected proposer-receiver pair cannot be matched to each other in any stable matching.

(b) Use this rejection lemma to prove proposer optimality among stable matchings.

(c) Explain why the receiver side obtains its least-preferred stable matching.

Chapter use case

Seed: Problem 15.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of agents and houses from three to n. Tool: Codex + Python; Lean target. Verification: Recover the three-agent top-trading-cycles allocation; check permutation, individual rationality, Pareto comparisons, and coalition blocking by enumeration. Top trading cycles and housing-core claims remain future Lean targets.

Problem 15.3 [Core]. Top trading cycles.

Agents $1, 2, 3$ initially own houses h_1, h_2, h_3. Their rankings are $1 : h_2 \succ h_1 \succ h_3$, $2 : h_1 \succ h_2 \succ h_3$, and $3 : h_1 \succ h_3 \succ h_2$.

(a) Run top trading cycles.

(b) Show that the outcome is individually rational and Pareto efficient.

(c) Verify directly that no coalition blocks the outcome.

Chapter 15

Computable General Equilibrium

15.1 From accounts to counterfactuals

Computable general equilibrium (CGE) analysis uses a calibrated microeconomic model and a social accounting matrix to evaluate policy counterfactuals. Calibration chooses parameters so that the model reproduces a benchmark allocation and the associated accounting identities before the policy is changed.

15.2 A two-sector small open economy

Let sectors be indexed by $j \in \mathcal{J} := \{1, \ldots, J\}$ and households by $h \in \mathcal{H} := \{1, \ldots, H\}$; the benchmark application has $J = 2$ goods. Sector j uses labor $n_j \geq 0$ and sector-specific capital $K_j \geq 0$ to produce output Q_j. The corresponding aggregate factor endowments are $\bar{n}$ and $\bar{K}_j$. Let $P_j > 0$ denote the domestic price of good j, $\boldsymbol{P} := (P_1, \ldots, P_J)$ the domestic price vector, W the wage, and r_j the rental rate on capital specific to sector j. A small open economy takes world prices as given. For sector j, a constant-elasticity-of-substitution production function is

$$Q_j = A_j \left[\alpha_j K_j^{-\rho_j} + (1 - \alpha_j) n_j^{-\rho_j} \right]^{-1/\rho_j},$$

with $A_j > 0$, $0 < \alpha_j < 1$, and $\rho_j > -1$, $\rho_j \neq 0$. The associated elasticity parameter is $\sigma_j = 1/(1 + \rho_j) > 0$.

Definition 15.1. (Cost and conditional factor demand.) Given output $\bar{Q}_j$ and factor prices

(r_j, W), the unit-consistent cost function is

$$c_j(r_j, W, \bar{Q}_j) = \min_{K_j, n_j \geq 0} \{r_j K_j + W n_j : Q_j(K_j, n_j) \geq \bar{Q}_j\}.$$

When differentiable, conditional factor demands satisfy

$$K_j^D = \frac{\partial c_j}{\partial r_j}, \qquad n_j^D = \frac{\partial c_j}{\partial W}.$$

This derivative formulation avoids confusing the price P_j with the CES parameter ρ_j.

15.3 Social accounting and calibration

The social accounting matrix records production, factor payments, household income, and expenditure flows that must balance at the benchmark. Let Y_h denote household h's nominal income and let $\beta_{hj} \in [0,1]$ be its expenditure share on good j, with $\sum_{j \in \mathcal{J}} \beta_{hj} = 1$. Under Cobb--Douglas preferences, household demand $x_{hj}(\boldsymbol{P}, Y_h)$ satisfies the expenditure equation $P_j x_{hj}(\boldsymbol{P}, Y_h) = \beta_{hj} Y_h$, so the observed benchmark share identifies $\beta_{hj} = P_j x_{hj}(\boldsymbol{P}, Y_h)/Y_h$. The calibrated model is checked by substituting the parameter values back into every accounting identity and market-clearing condition.

15.4 Equilibrium closure

A computable model is not defined by technology and accounting identities alone. Let $\bar{n}_h$ be household h's labor endowment and $\bar{K}_{hj}$ its endowment of capital specific to sector j, where $\sum_{h \in \mathcal{H}} \bar{n}_h = \bar{n}$ and $\sum_{h \in \mathcal{H}} \bar{K}_{hj} = \bar{K}_j$. Household income must specify factor ownership, taxes, transfers, and the disposition of profits and tariff revenue. Writing T_h for transfers, Π_h for distributed profits, and Tax_h for direct taxes, a representative identity is

$$Y_h = W \bar{n}_h + \sum_{j \in \mathcal{J}} r_j \bar{K}_{hj} + T_h + \Pi_h - \mathrm{Tax}_h \,.$$

Demand equations $x_{hj} = x_{hj}(\boldsymbol{P}, Y_h)$ then close the household side.

For each good $j \in \mathcal{J}$, let M_j and E_j denote imports and exports, Inv_j investment demand, and G_j government final demand. Let $a_{jk} \geq 0$ denote the amount of good j used as an intermediate

input per unit of sector k's output. The resource balance is

$$Q_j + M_j = \sum_{h \in \mathcal{H}} x_{hj}(\boldsymbol{P}, Y_h) + G_j + \mathrm{Inv}_j + E_j + \sum_{k \in \mathcal{J}} a_{jk} Q_k.$$

Factor markets satisfy

$$\sum_{j \in \mathcal{J}} n_j = \bar{n} = \sum_{h \in \mathcal{H}} \bar{n}_h, \qquad K_j = \bar{K}_j = \sum_{h \in \mathcal{H}} \bar{K}_{hj} \quad \text{for sector-specific capital.}$$

Constant returns and competitive pricing impose zero profit, so the domestic producer price equals unit cost whenever the sector operates.

The government budget identifies the use of tax and tariff receipts, and the external account identifies the financing of net imports. Let $P_j^W > 0$ denote the world price of traded good j and let $\mathrm{ER} > 0$ denote the exchange rate measured in units of domestic currency per unit of foreign currency. A balanced-trade closure, for example, imposes

$$\sum_{j \in \mathcal{J}} P_j^W M_j = \sum_{j \in \mathcal{J}} P_j^W E_j,$$

whereas a fixed foreign transfer replaces the right-hand side by exports plus that transfer. The model must choose one nominal normalization, such as $W = 1$ or $\mathrm{ER} = 1$, and must not impose both as independent equilibrium equations.

A benchmark replication check solves this entire system at the calibrated parameters and reports the maximum absolute residual in every household, firm, market, government, and external-account equation. Counterfactual welfare must also be declared, for example equivalent variation relative to the benchmark utility and price system.

15.4.1 Following the physical and monetary circuits

To read a small social accounting matrix, first specialize the model to a closed benchmark with no government, taxes, transfers, saving, foreign trade or intermediate inputs. Keep both households and both sectors, with sector specific capital. Write $Y = \sum_h Y_h$ for aggregate household income, $\Pi = \sum_j \pi_j = \sum_h \Pi_h$ for distributed profit and $\mathcal{E} = \sum_{h,j} P_j x_{hj}$ for consumption spending. The script letter distinguishes expenditure from the exports E_j of the full model.

Before calibrating preferences or technology, follow one transaction around the accounts. The household receives factor income and any distributed profit, and its consumption purchases generate firms' sales receipts. Firms pay for the factor services used in production. Under the

simplified closed benchmark in the figure, household expenditure $\mathcal{E}$ equals aggregate income Y, and both equal the value of final output. Physical quantities travel in the opposite direction to the associated payment. Neither a wage payment nor a transfer consumes a unit of the final good by itself.

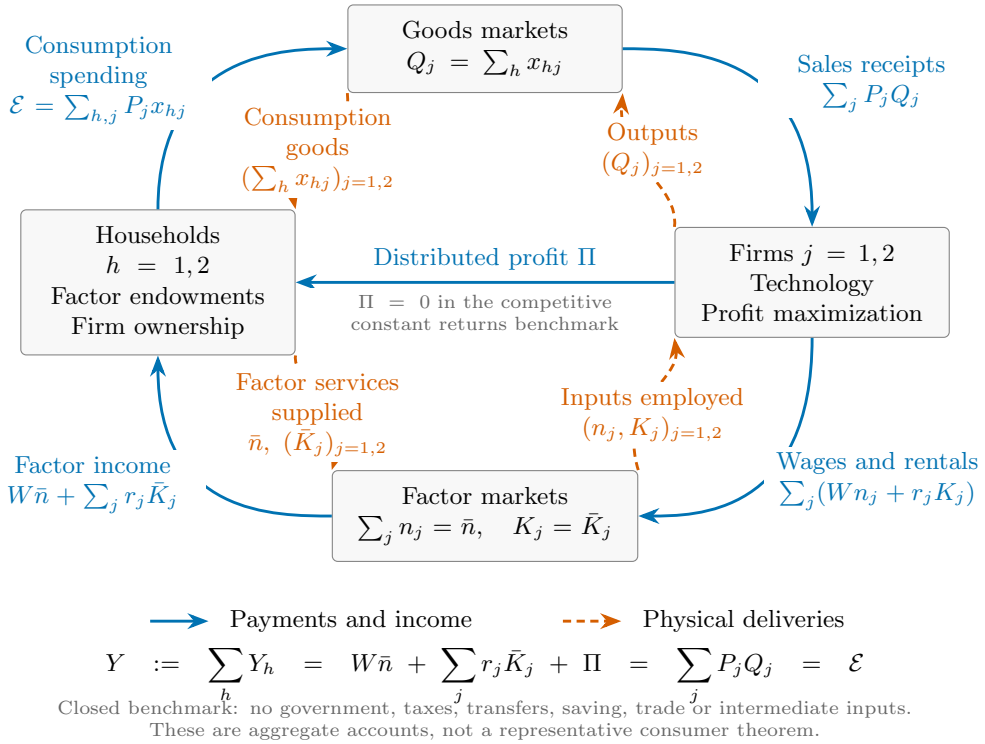


Figure 15.1: Physical deliveries and monetary payments close different circuits. Dashed orange arrows carry goods and factor services; solid blue arrows carry expenditures, wages, rentals and distributed profit. The profit arrow is zero in the competitive constant returns CES benchmark but shows where ownership income belongs in the accounts. This illustration specializes to no government, foreign sector, saving or intermediate inputs. It sums household accounts without assuming a representative consumer.

The factor account must respect units and ownership. Labor services and the two sector specific capital services are different objects, so their physical quantities are not added. Their payments can be added after multiplication by W and the appropriate r_j. The return to scarce capital is a rental payment, not an omitted profit. With the chapter's constant returns CES technologies and competitive producer prices, profit is zero when each sector pays for all its inputs.

A social accounting matrix records each monetary arrow once, with the recipient in its row and the payer in its column. Row totals are receipts and column totals are outlays. Balance checks whether these accounts fit together. It does not establish that households optimize, that

firms minimize cost, that a chosen elasticity is correct, or that a counterfactual equilibrium exists. Those are separate behavioral and mathematical restrictions.

The figure is a deliberately small closure, not the entire small open economy. Restoring taxes, transfers, government demand or foreign trade requires the corresponding government and external accounts, as well as the physical resource equations above. A tax payment is a monetary transfer; it is not destroyed output. Government purchases, unlike the tax receipt itself, use goods. The full model must choose how receipts are spent and how net imports are financed.

15.5 Counterfactual policies

For an imported good $j \in \mathcal{J}$ that is a perfect substitute for the domestic good, the law of one price with an ad valorem tariff $\tau_j \geq 0$ changes the domestic price according to

$$P_j = \mathrm{ER}\, P_j^W (1 + \tau_j).$$

Free trade is the baseline $\tau_j = 0$; a tariff counterfactual changes τ_j and resolves the full equilibrium system. Other counterfactuals change tax, subsidy, transfer, or technology equations while leaving the accounting identities and calibrated preference and technology parameters in place.

15.6 Problems

Part I: By hand

Problem 16.1 [Core]. Calibrating a CES technology.

A constant-returns technology is $Q = A[\alpha K^{-\rho} + (1-\alpha)n^{-\rho}]^{-1/\rho}$, with $\rho > -1$ and $\rho \neq 0$. A strictly positive interior cost-minimizing benchmark observation gives (Q, K, n, r, W) and an elasticity of substitution $\sigma > 0$, $\sigma \neq 1$. (The Cobb-Douglas limit $\sigma = 1$ is outside the displayed parameterization.)

(a) Recover ρ from σ.

(b) Use the cost-minimizing first-order condition to recover α.

(c) Recover A and verify that the benchmark is exactly replicated.

Problem 16.2 [Core]. Social-accounting consistency.

A two-sector social accounting matrix records sectoral sales, intermediate purchases, labor and capital payments, household income, taxes, savings, and imports.

(a) State the row-balance and column-balance conditions.

(b) Explain why every market-clearing equation has a corresponding accounting identity but not every accounting identity is a behavioral equation.

(c) Design a numerical residual check that would detect an incorrectly calibrated benchmark.

Advanced exam practice

These problems build toward integrated graduate exam arguments. The suggested time applies to the hand questions, not the optional tool extension. First draw a diagram or state a solution plan, define the feasible set, and identify possible corners. Explain why each candidate is globally optimal and check the economic accounting. Attempt the hand questions before using tools. For an optional extension, state Seed, Fixed, Scaled and Risk before coding and enlarge one feature at a time.

Advanced case guide. Problems 16.A1 and 16.A2 restore both the theory and Arabasta and Baltigo modeling parts of the SFU 2024 final examination. Problem 16.A3 is a new general equilibrium and dynamic extension of the Galor household problem 1.6. Each separates hand benchmarks from computational verification and states its equilibrium closure explicitly. Separate student packets contain these same questions.

Problem 16.A1 [Advanced]. CES trade, market clearing and the limits of a fixed point argument.

Restored from the theory problem in the SFU Fall 2024 final examination. The original four stage argument and CES weighting are retained. The symmetry and fixed point claims are corrected. The following timing and points are for practice, not the original exam allocation.

Suggested time: 70 minutes. Total: 60 points for Part I.

Part I. By hand.

There are $N \geq 2$ countries, indexed by i for origins and j for destinations. Country i owns a fixed quantity $Q_i > 0$ of its unique variety, has no other goods endowment, and has factory price $p_i > 0$. Its income is $y_i = p_i Q_i$. All varieties are traded competitively. Delivering one unit from i to j uses $t_{ij} \geq 1$ origin units, with $t_{ii} = 1$. Delivered prices are $p_{ij} = p_i t_{ij}$. There are no outside transfers or trade deficits. Destination j chooses nonnegative delivered quantities c_{ij} with budget $\sum_i p_i t_{ij} c_{ij} \leq y_j$ and utility $U_j(c) = [\sum_i \beta_i^{(1-\sigma)/\sigma} c_{ij}^{(\sigma-1)/\sigma}]^{\sigma/(\sigma-1)}$, where $\beta_i > 0$ and $\sigma > 0$, $\sigma \neq 1$. For $0 < \sigma < 1$, define utility to be zero when any component is zero, its continuous boundary value. You may use the standard concavity of this CES aggregator. Bilateral expenditure is $x_{ij} = p_i t_{ij} c_{ij}$, including domestic expenditure when $i = j$. Let $Y = \sum_i y_i$ and $\theta_i = y_i / Y$.

(a) Derive the optimal bilateral quantities, bilateral expenditures and destination unit expenditure index P_j, step by step. Explain why the optimum exhausts the budget, is interior and is globally optimal. Your argument must preserve the displayed β_i weighting. [16]

(b) State physical market clearing, including iceberg use, and its nominal form. Use your expenditure formula to isolate $\beta_i p_i$ as a function of y_i, the trade costs and the destination indices. Explain why the income identity $y_i = p_i Q_i$ must still hold when interpreting the result as the original fixed supply economy. [10]

(c) Define outward multilateral resistance by $\Pi_i^{1-\sigma} = \sum_j \theta_j (t_{ij}/P_j)^{1-\sigma}$. Eliminate $\beta_i p_i$ to derive the structural gravity equation and the inward resistance system. Show the reciprocal scale freedom of the two resistance systems and state a normalization. Take logs to identify the exporter effect, importer effect and coefficient on $\log t_{ij}$. Distinguish the conditional index normalization from changing nominal currency units in the underlying economy. [16]

(d) Suppose $t_{ij} = t_{ji}$. You may use the following result without proof: for a strictly positive finite trade cost matrix and positive shares summing to one, the joint inward and outward resistance system has a positive index pair, and any two such pairs differ only by reciprocal scalar rescaling. Use this result to explain when a normalization with $\Pi_i = P_i$ is available, and why symmetry alone does not make every unnormalized pair equal.

For the rest of this part use exactly the original diagnostic values $N = 2$, $\sigma = 1/2$, $t_{ij} = 1$ and $\theta_i = 1/2$. After choosing the symmetric normalization, write the inward equations as $P = g(P)$. Determine whether g maps $\{P : P_i \geq 1\}$ into itself. Then use $K = [1/2, 2]^2$: prove that g maps K into itself and apply Brouwer to establish a fixed point. Determine all positive fixed points directly. Compute Dg at $(1, 1)$ and decide whether an induced matrix norm can be strictly below one there. Finally calculate two iterations from $P^{(0)} = (2, 2)$ and explain what they show about repeated substitution, existence and uniqueness. [18]

Part II. Codex and verification.

Construct a synthetic economy with at least ten countries, positive β_i, Q_i, finite iceberg costs and $\sigma > 1$. Keep fixed physical supplies rather than silently treating both factory prices and nominal incomes as independent data. Solve for positive factory prices in log variables with one numeraire normalization. Compute incomes, delivered consumption, expenditure, both resistance systems and the physical shipment totals. Check every consumer budget, both row and column expenditure sums, all physical goods markets, positivity and the omitted redundant equation. Repeat under a change of currency units and verify how prices, incomes, nominal trade and physical quantities transform. Compare a normalized resistance iteration with a root solver

conditional on the resulting income shares, align their index gauges before comparing them, and report full residuals as well as stopping criteria. For the synthetic fixed effect regression, require nonadditive bilateral variation in $\log t_{ij}$ and report the design rank after choosing a dummy normalization. Recover the theoretical cost coefficient. Reproduce the two country diagnostic separately and do not infer convergence from an equation having a unique solution.

Problem 16.A2 [Advanced]. Arabasta and Baltigo: building the original gravity model of trade.

Restored from Problem 2 of the SFU December 2024 final examination. The original six modeling tasks, Arabasta and Baltigo, unequal country incomes, transport technology and distance comparisons are retained. Part II is a new computational extension.

Part I. By hand.

Suggested time: 75 minutes. Total: 60 points.

There are two countries, Arabasta a and Baltigo b, and one good from each origin. Quantity c_{ij} is consumption in destination j of origin i's good. Nominal income and expenditure are $y_a, y_b > 0$. Origin prices are $p_a, p_b > 0$. Domestic delivery has no resource loss, while bilateral trade has the symmetric iceberg factor $\tau \geq 1$: $p_{aa} = p_a$, $p_{bb} = p_b$, $p_{ab} = p_a\tau$ and $p_{ba} = p_b\tau$. Delivering one unit abroad requires τ units at origin. Bilateral expenditure is $x_{ij} = p_{ij}c_{ij}$ and origin revenue clearing requires $y_i = x_{ia} + x_{ib}$.

Use equal origin weights and common CES preferences. Write the substitution elasticity as $\sigma > 1$ and use $\sigma = 2$ for the closed form equilibrium calculations. The original key's power utility with exponent $1/2$ is ordinally equivalent to this case; its exponent is not the substitution elasticity.

Normalize $p_b = 1$. In Part I, y_a and y_b are fixed nominal incomes measured in these units. Origin supply quantities are not independently fixed: after solving the revenue equations they are implied by $Q_i = y_i/p_i$. State this conditional nominal income closure when interpreting a counterfactual. It is not a comparison that holds both nominal incomes and physical endowments fixed.

(a) Original task 1.1. Specify a one parameter CES utility family and Arabasta's budget. Explain which goods, prices and income enter this problem. [10]

(b) Original task 1.2. Write the Lagrangian and derive both consumption quantities in Arabasta. Relabel carefully to obtain Baltigo's demands. Give a global optimality argument and state where your solution is interior. [10]

(c) Original task 2.1. Write both countries' revenue clearing conditions. Specialize to $\sigma = 2$,

impose $p_b = 1$, and solve explicitly for the unique positive origin price p_a as a function of y_a, y_b, τ. Reject any algebraic root that is not an admissible price, and check the case $\tau = 1$ directly. [10]

(d) Original task 2.2. Let $d \geq 0$ be distance. Propose a one parameter engineering transport technology $d = f(k)$ with nonincreasing returns in ship capital services k. Normalize the rental cost of these services to one in the origin good resource units used to define iceberg shipping. Derive the per delivered unit resource requirement $C(d)$ and set $\tau(d) = 1 + C(d)$. Verify $\tau(0) = 1$ and that larger distance increases the barrier. Explain why you must not add a second monetary transport bill to the country budget. The historical key's constant returns choice is admissible; a concave power technology is also admissible. [10]

(e) Original task 3.1. Eliminate the origin price from Arabasta's equilibrium import quantity c_{ba} and nominal imports x_{ba}. Derive the corresponding exports c_{ab}, x_{ab} and verify both budgets and both physical resource equations, including iceberg use. As a common numerical check, use $y_a = 5$, $y_b = 2$, $d = 1$ and the admissible technology $f(k) = \sqrt{k}$. This check is part of the existing task, not an additional scored question. [10]

(f) Original task 3.2. Prove that doubling both nominal incomes doubles bilateral nominal trade in this closure. Then prove that increasing distance strictly reduces bilateral nominal imports, holding those incomes fixed. Separate the response of import quantities from nominal import values, and explain why this is not yet a welfare theorem for a fixed physical endowment economy. [10]

Part II. Codex and verification.

This is a new extension, separate from the restored final question. Begin only after completing the hand arguments.

1. Use the numerical seed from part (e), fix the engineering parameter, and vary only distance over $d \geq 0$. Ask Codex to implement your analytical price, expenditure and quantity formulas, and independently solve the two revenue equations. Compare both methods. Test positivity, budget exhaustion, balanced bilateral expenditure, the domestic expenditure residuals, and every origin resource equation. Include zero distance without evaluating a formula that divides by zero. Plot bilateral expenditure and both implied origin supply quantities. Explain explicitly which physical quantity changes when nominal GDP is held fixed.

2. Optional different closure. Freeze physical endowments at the values you derived from the seed, and retain $p_b = 1$. Now income must be endogenous, $y_a = p_a Q_a$ and $y_b = Q_b$. Derive the positive price equation under this closure, establish uniqueness of its positive root, and solve it as distance varies. Verify the original consumer problems, physical market clearing and income

accounting independently. Compare the two closures at the seed and away from it. Any utility comparison must use the same CES preference and the full fixed endowment closure, not interpret the fixed nominal GDP family as the same physical economy.

Problem 16.A3 [Advanced]. From household KKT to a land constrained growth economy.

Suggested time: 90 minutes for the hand benchmarks. Total: 80 points. The optional program is not included in this time. This teaching case extends Problem 1.6 to competitive factor prices and a sequence of generations. It is inspired by Galor's quantity versus quality mechanism, not a reproduction of his full Unified Growth Theory.

There is one consumption good, whose price is one, fixed land $X > 0$, and a mass $N > 0$ of identical current adults with skill $h > 0$. A competitive constant returns firm rents land $Z \geq 0$ and raw adult labor hours $L \geq 0$ to produce $Y = AZ^{1-\alpha}(hL)^{\alpha}$, where $A > 0$ and $0 < \alpha < 1$. The land market must clear at $Z = X$. The wage $w > 0$ is paid per raw hour, and R is rent per unit of land. Every adult owns an equal land share and receives $\rho = RX/N$. Firm profit is distributed but is zero at a competitive supporting price.

Each adult has one unit of time. It chooses consumption $c > 0$, continuous fertility $n > 0$ and education time per child $e \geq 0$ to maximize $(1-\gamma)\log c + \gamma \log[nh_a(e)]$, where $0 < \gamma < 1$, $\tau > 0$, $a > 0$, $\bar{c} > 0$ and $h_a(e) = 1 + ae/(\tau + e)$. Total child time is $t = n(\tau + e)$. Constraints are $c + wt \leq w + \rho$, $t \leq 1$ and $c \geq \bar{c}$. Land rent is nonlabor income: child time sacrifices wages, not rent. Each household takes w and ρ as given. The logarithm excludes zero fertility in this model.

Part I. By hand.

(a) Set $E = ne$ and $Q(n, E) = n + anE/(\tau n + E)$. State the transformed feasible set and write the complete KKT system, including the time bound, subsistence and the zero education entry inequality. Establish concavity and explain why a feasible KKT certificate gives a global optimum. [12]

(b) At fixed prices, write $I = w + \rho > \bar{c}$. Solve globally for education, child time and consumption, including all three possible bounds on child time and the equality case $a = 1$. Explain why replacing w by I in the cost of child time would be an economic error. [12]

(c) Define $B = Ah^{\alpha}(X/N)^{1-\alpha}$. A symmetric general equilibrium has $L = N(1-t)$ and $Nc = Y$. Derive factor prices, zero profit and the resource identity. Find all positive equilibria for every $B > \bar{c}$, identify the subsistence threshold and verify the household KKT certificate at the equilibrium. Explain what happens to feasibility when $B \leq \bar{c}$. [16]

Dynamic benchmarks. Continue by hand. The optional program at the end checks these

calculations; it does not replace the arguments. At the end of a generation all children become next generation adults, so $N' = nN$. Skill is inherited and augmented by education: $h' = hh_a(e)$. Land and A stay fixed, and land ownership is shared equally among the next cohort. Parents value child quantity and quality through the displayed utility but do not solve a forward looking dynamic program. Thus $n = 1$ means replacement of the current adult mass, not a literal demographic birth rate.

(d) Derive the one dimensional law for B'. Set $\alpha = 3/4$, $\gamma = 1/2$, $\tau = 1/4$ and $\overline{c} = 4$. With $a = 1/2$, find a subsistence steady state with replacement fertility and constant skill. Calculate the derivative of the local B map there and explain the stability conclusion, including its scope. [12]

(e) Start at that steady state and permanently raise a to 3. Calculate the impact education, child time, fertility and skill multiplier. Does the next generation cross the threshold into slack subsistence? Prove your comparison exactly. Compute the subsequent fertility and asymptotic growth factors of skill, adult mass, B and consumption. [18]

(f) Explain which feedbacks close the economy, why rents alter fertility incentives without being sacrificed per hour, and why the model does not deliver exact zero fertility or a full demographic transition. Distinguish an exogenous education return shock from endogenous technological progress, and distinguish the impact fertility response from its later level. [10]

Part II. Codex and verification.

Optional tool extension. Keep $A = 8$, $X = h_0 = 1$ and all calibrated parameters fixed, with the permanent $a = 1/2$ to $a = 3$ change at date zero. Vary only initial adult mass $N_0 \in (0, 16)$, including the steady state and regime boundary populations obtained by hand and points immediately around them. State Seed, Fixed, Scaled and Risk before coding. Simulate twenty generations using independently implemented household and equilibrium routines. Check all budgets, time and subsistence bounds, raw labor and land clearing, zero profit, aggregate goods clearing, KKT signs and complementary products at every date. Compare the household formula with an independent bounded optimization at selected dates and state tolerances. Plot the paths and mark the subsistence switch. Report what is proved analytically, what is checked on this finite grid and what is only a maintained demographic assumption.

Part II: A tariff counterfactual from partial to general equilibrium — chapter use case and verified scale-up

Seed: Problems 16.1, 16.2, 16.3. Fixed: all unlisted primitives and assumptions. Scaled: model closure from fixed factor prices to one endogenous shared labor market linking import-competing and export sectors. Tool: Codex + Python. Verification: Replicate the partial-equilibrium benchmark, then report factor-market, goods-market, trade-balance, government-budget, and welfare residuals. Identify which change comes from the single added general-equilibrium channel.

Problem 16.3 [Applied]. A tariff counterfactual.

A small open economy takes the world price P^W and exchange rate ER as given. A tariff τ implies domestic price $P = \mathrm{ER}\, P^W(1+\tau)$. Let differentiable domestic demand and supply be $D(P)$ and $S(P)$, imports be $M(P) = D(P) - S(P) > 0$, and tariff revenue be returned lump-sum, $\mathrm{TR}(\tau) = \tau\, \mathrm{ER}\, P^W M(P(\tau))$. Interpret D as quasi-linear (or otherwise free of income effects) and S as competitive supply generated by marginal cost, so consumer and producer surplus are valid partial-equilibrium welfare measures.

(a) Derive the first-order change in the domestic price from a small tariff change.

(b) Derive the symbolic first-order decomposition $d\mathcal{W} = d\,\mathrm{CS} + d\,\mathrm{PS} + d\,\mathrm{TR}$ and simplify it at free trade. Do not assume numerical functional forms.

(c) Explain which general-equilibrium channels are omitted by that approximation and must be restored in a CGE counterfactual.

Additional Practice Reserve

These question-only problems are cleaned and independently checked descendants of older Western/UWO and SFU assessments. Each begins with hand-solvable work and then supplies a controlled formal or computational scale-up. They supplement, rather than replace, the end-of-chapter problem sets. Solutions and source provenance are kept in the private instructor handbook.

Revealed Preference and Recoverability

Problem R001 [Proof]. Quasilinear Afriat inequalities and welfare bounds.

Suggested weekly placement: PS03.

Let $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^{T}$, where $p^t \gg 0$ and $x^t \in \mathbb{R}^L_+$.

Part I — by hand. Prove that the observations can satisfy $x^t \in \arg\max_{x \in \mathbb{R}^L_+}\{u(x) - p^t \cdot x\}$ only if there are numbers $v^1, \ldots, v^T$ such that $v^t - v^s \geq p^t \cdot (x^t - x^s)$ for every s, t. Prove the converse by studying $u(x) = \min_t\{v^t + p^t \cdot (x - x^t)\}$. Establish continuity, concavity, strict monotonicity, and rationalization rather than merely citing Afriat's theorem.

For $V(p) = \max_{x \in \mathbb{R}^L_+}\{u(x) - p \cdot x\}$, suppose optimizers $x(p)$ and $x(q)$ exist. Derive upper and lower bounds on $V(p) - V(q)$ using the two observed demands. Extend the argument to a finite path of price vectors and state precisely when a componentwise price increase gives a strict welfare loss.

Part II — verified scale-up. Express the inequalities as a difference-constraints graph. With Codex and Python, implement a feasibility test for arbitrary T and L, return one normalized vector v, and report a violating positive cycle when the system is infeasible. Unit-test the code on a small rationalizable data set and on a perturbed data set. Verify every returned inequality directly. As an optional Lean task, formalize the telescoping path bound conditional on the finite system of inequalities.

Abstract Choice and Order Foundations

Problem R002 [Proof]. Sequential satisficing with a default rule.

Suggested weekly placement: PS04.

Let X be finite, let $\triangleright$ be a strict linear (total) search order, and let $S = \{x : u(x) \geq \bar{u}\}$. From a nonempty menu A, inspect alternatives in the order $\triangleright$ and choose the first member of $A \cap S$. If $A \cap S$ is empty, choose the $\triangleright$-first member of A.

Part I — by hand. Construct a single strict priority order $\succ^*$ whose maximal element agrees with this procedure on every nonempty menu. Use that representation to prove WARP and SARP. Explain exactly why the fallback clause is necessary for a full-domain choice function. Give a three-alternative example and list the choice from every nonempty menu.

Part II — verified scale-up. Encode finite menus, the search order, the threshold set, and the fallback choice. Ask Codex to generate either a Lean proof or an exhaustive checker establishing equality between sequential choice and $\succ^*$-maximization. Scale from three alternatives to all menus of a user-chosen finite set, and include a deliberately broken implementation without the fallback to exhibit its failure of full-domain nonemptiness.

Problem R003 [Applied]. Choice overload and attraction through menu-dependent attention.

Suggested weekly placement: PS06.

An outside option o is always considered. In a menu with n active products, each active product is independently considered with baseline probability $q_n = 1/n^2$; the consumer chooses the highest-u considered item, or o if none is considered. Utilities are strict. When a target b and its dominated decoy b^- are both offered, the consideration probability of b rises by $\delta = 1/3$, while all other baseline probabilities remain unchanged.

Part I — by hand. Compare $A_2 = \{a, c, o\}$ with $A_3 = \{a, c, d, o\}$ and calculate the probability of choosing o. Next let $u(b) > u(c) > u(b^-) > u(o)$ and compare $A = \{b, c, o\}$ with $B = \{b, b^-, c, o\}$. Calculate the probability of choosing b in both menus. Prove that the resulting attraction inequality cannot be generated by any menu-independent random-utility model.

Part II — verified scale-up. Build an exact enumerator over consideration sets and a Monte Carlo simulation. Confirm the hand calculations, then vary n, δ, and the ranking. Ask Codex to produce automated checks for probability adding-up, the parameter domain, regularity violations, and convergence of the simulation to the exact probabilities. Report which changes generate overload, attraction, both, or neither.

Problem R004 [Proof]. Maximal choice under incompleteness and a GARP failure.

Suggested weekly placement: PS04.

Let $X = \{a, b, c\}$. Define an incomplete preorder R by reflexivity together with bPa; alternative c is incomparable with both a and b. From each nonempty menu A, let $C(A)$ contain all R-maximal elements.

Part I — by hand. Compute $C(A)$ for every nonempty menu. Define direct revealed preference by xR^Dy when some menu chooses x while making y feasible, and define xP^Dy when that menu rejects y. Show that the transitive closure $(R^D)^*$, together with P^D, violates the standard GARP test; identify an explicit witness. Explain why this does not contradict maximality under an incomplete preorder.

Part II — verified scale-up. Represent finite preorders and maximal-choice correspondences in Lean or Python. Verify reflexivity and transitivity of the primitive relation, compute all maximal sets, and mechanically exhibit the revealed-preference path. Then enumerate all three-alternative preorders and classify which induced maximal-choice data satisfy GARP. Distinguish a failure of the data test from a failure of the primitive preorder axioms.

Production and Competitive Markets

Problem R005 [Proof]. Aggregating heterogeneous Cobb–Douglas firms.

Suggested weekly placement: PS08.

Firm j has technology $q_j = A_j \ell_j^{a_j} k_j^{b_j}$ with $A_j > 0$, $a_j, b_j > 0$, and $a_j + b_j < 1$. Its net-output set is $Y_j = \{(-\ell, -k, q) : q \leq A_j \ell^{a_j} k^{b_j},\ \ell, k \geq 0\}$. Output price is $p > 0$ and input prices are $w_\ell, w_k > 0$.

Part I — by hand. Derive the unique profit-maximizing output, both input demands, and profit. State where decreasing returns are used. Prove that the aggregate technology is the Minkowski sum $Y = Y_1 + \cdots + Y_J$ and that its profit function is the sum of the firm profit functions. State the corresponding Hotelling result both at differentiable prices and in subdifferential form, using derivatives $(q_j, -\ell_j, -k_j)$ with respect to (p, w_ℓ, w_k).

Part II — verified scale-up. Generate a heterogeneous population of firms and compute aggregate supply and factor demands. Verify numerically that direct maximization over firm-level allocations agrees with the sum of closed-form firm solutions. Use automatic or symbolic differentiation to check Hotelling's lemma, and test how the aggregate elasticity changes with the distribution of returns-to-scale parameters.

Problem R006 [Applied]. Selection versus capacity destruction in partial equilibrium.

Suggested weekly placement: PS09.

This is the distributional extension of problem 10.A5, the restored Western 2019 final examination question. Use its consumer preferences, unrestricted numeraire budget and producer cost. Here allow any consumer mass $\eta > 0$. Retain its distinction between actual removal and replacement of the producer population.

Part I: by hand. Complete problem 10.A5 first. Reproduce its uniform benchmark, both producer selection experiments, consumption variance and fixed initial wealth comparison as verification targets for the extension below. This reference replaces the previously duplicated uniform prompt; the complete five part question is now in the main chapter bank.

Part II: verified extension. Replace the uniform type distributions by beta distributions and allow an arbitrary survival cutoff. Define each probability law, the actual population measures and their masses before integrating. Derive the market clearing equation, solve it numerically over a parameter grid, and separate the effect of changing the conditional productivity distribution from changing producer mass. Include monotonicity checks and compare numerical integration with the uniform closed form. Hold initial wealth fixed in welfare comparisons and distinguish per consumer moments from population totals.

Equilibrium, Trade, and Computation

Problem R007 [Applied]. CES structural gravity and multilateral resistance.

Suggested weekly placement: PS11.

This is a companion to the full restored gravity exercise 16.A1, not a second copy of its derivation. For comparison with the earlier reserve notation, set $a_i = \beta_i^{1-\sigma}$ and $\tau_{ij} = t_{ij}$, with $\sigma > 1$. In this conditional resistance exercise, nominal incomes and shares are given; it does not independently solve the fixed physical supply economy.

Part I: by hand. Complete parts (a) to (c) of 16.A1. Translate the resulting equations into the a_i notation and use them as verification targets. Keep the distinction between origin revenue, destination expenditure and iceberg shipments.

Part II: verified extension. Begin with the same country set and income shares as your 16.A1 calculation. Vary only asymmetry in the finite positive trade cost matrix, retaining domestic cost one and all trade factors at least one. Solve both normalized resistance systems. Verify every

row and column expenditure sum, positivity and reciprocal index scaling. Compare a normalized iteration with a root solver and report residuals, not only step changes. For synthetic fixed effect recovery, check design rank and include bilateral cost variation that is not absorbed by origin and destination effects.

Problem R008 [Proof]. Why a Banach argument fails even when existence and uniqueness hold.

Suggested weekly placement: PS10.

Use the map g and rectangle $K = [1/2, 2]^2$ from 16.A1(d). The full corrected hand diagnostic is now in the main chapter bank; this reserve entry supplies its numerical stability extension.

Part I: by hand. Complete the fixed point diagnostic in 16.A1(d). Retain its exact fixed point and cycle as verification targets. For the damped rule below, derive the local eigenvalues at the fixed point and distinguish local stability from a contraction bound over the whole rectangle.

Part II — verified scale-up. Use interval arithmetic or Lean-supported inequalities to certify invariance of K and the derivative calculation. Compare plain iteration, the damped rule $P^{n+1} = (1 - \lambda)P^n + \lambda g(P^n)$ for $\lambda = 1/2$, and a root solver from multiple starting values. Ask Codex to create tests that distinguish existence, uniqueness, local stability, and global contraction; the report must not infer one property from another.

Problem R009 [Proof]. A regular two-good economy with three equilibria.

Suggested weekly placement: PS10.

Normalize prices to $(q, 1)$. Consumer 1 chooses $a \geq 0, b > 0$, has $u_1(a, b) = a - \frac{1}{8}b^{-8}$, and owns $(2, r)$. Consumer 2 chooses $c > 0, d \geq 0$, has $u_2(c, d) = -\frac{1}{8}c^{-8} + d$, and owns $(r, 2)$, where $r = 2^{8/9} - 2^{1/9}$.

Part I — by hand. Derive both consumers' interior demands and prove that $a(q), d(q) > 0$ for every $q > 0$, so boundary demand cannot create an omitted equilibrium. Derive the good-2 excess-demand function. Show that $q \in \{1/2, 1, 2\}$ are equilibria and that there are no other positive normalized equilibria. Verify that each equilibrium is regular and locally isolated. Compute $z_2'(1)$ and compare it with the gross-substitutes requirement $z_2'(1) \geq 0$. Explain why regularity does not imply uniqueness.

Part II — verified scale-up. Put $t = q^{1/9}$ and use a computer algebra system to certify the positive-root count without relying only on a plot. Check all equilibrium residuals and derivatives at high precision. Choose one scalar endowment perturbation, initialize continuation at each baseline equilibrium, and trace each local branch only until regularity fails or the declared small

parameter interval ends. Connect the local numerical result to the implicit-function theorem. Optionally formalize the exact three roots and their nonzero derivatives in Lean.

Problem R010 [Applied]. A complete-markets Radner equilibrium by hand.

Suggested weekly placement: PS11.

There are two consumers, two states A, B, and two goods in each state. Both consumers have equal state probabilities and state utility $\log c_1 + \log c_2$. Each owns $\omega > 0$ units of good 2 in each state. Consumer 1's good-1 endowment is $(\omega/2, 3\omega/2)$ across states, and consumer 2's is reversed. Good 1 is the state numeraire and r_s is the state-s price of good 2. Arrow security s pays one unit of good 1 in state s. Consumer i holds θ_{is}, securities are in zero net supply, and the date-0 asset budget is $Q_A\theta_{iA} + Q_B\theta_{iB} = 0$.

Part I — by hand. Normalize the state-B security price to $Q_B = 1$ and write $q = Q_A$. Derive individual security demands as functions of q, r_A, r_B. Show that commodity clearing under equal spot prices requires $r_A = r_B = 1$ and that the equilibrium security-price ratio is one. Calculate portfolios and state consumption, and verify both commodity and security clearing. Explain the role of strict concavity and price normalization in the uniqueness statement.

Part II — verified scale-up. Generalize the computation to S states and a full set of Arrow securities. Build the sequential budget matrix, solve the equilibrium numerically, and verify state-by-state feasibility and zero net security supply. Ask Codex or Lean to check the linear identity mapping sequential budgets into the Arrow–Debreu present-value budget, clearly separating the algebraic equivalence from equilibrium existence.

Weekly Problem Sets: Fall 2026

The end-of-chapter questions are the source of record. The weekly sets below are question-only release views for ECON 803; chapter case studies are assigned separately. Required problems are the board-scale core, while extensions supply a proof or formal/computational bridge. Solutions are not included in the public book.

PS01: Utility maximization and demand

Meeting September 11, 2026; due September 18, 2026.

Required 1.1, 1.2, 1.4

Extension 1.3, 1.6

PS02: Duality and welfare

Meeting September 18, 2026; due September 25, 2026.

Required 2.1, 2.3, 2.5

Extension 2.2, 2.4

PS03: Revealed preference and recoverability

Meeting September 25, 2026; due October 2, 2026.

Required 3.1, 3.3, 3.6

Extension 3.2, 3.5

Reserve optional R001

PS04: Abstract choice and order foundations

Meeting October 2, 2026; due October 9, 2026.

Required 4.1, 4.3, A.1

Extension A.2

Reserve optional R002, R004

PS05: Expected utility and risk

Meeting October 9, 2026; due October 16, 2026.

Required 5.1, 5.2

Extension 5.3

PS06: Stochastic choice and attention

Meeting October 16, 2026; due October 23, 2026.

Required 6.1, 6.3, 7.1

Extension 7.2

Reserve optional R003

PS07: Aggregation

Meeting October 23, 2026; due October 30, 2026.

Required 8.1, 8.2

Extension None

PS08: Production and cost

Meeting October 30, 2026; due November 6, 2026.

Required 9.1, 9.3

Extension 9.2

Reserve optional R005

Advanced exam preparation: 9.A5, 9.A2. Further practice: 9.A1, 9.A3, 9.A4, 9.A6. Attempt the recommended pair by hand after the foundations; the remaining problems form a revision bank. These are additional preparation, not extra weekly requirements unless assigned in class. Suggested times exclude tool extensions.

PS09: Competitive markets and exchange

Meeting November 6, 2026; due November 13, 2026.

Required 10.1, 10.2, 11.1

Extension 10.3

Reserve optional R006

Advanced exam preparation: 10.A5, 10.A6. Further practice: 10.A1, 10.A2, 10.A3, 10.A4, 10.A7, 10.A8. Attempt the recommended pair by hand after the foundations; the remaining problems form a revision bank. These are additional preparation, not extra weekly requirements unless assigned in class. Suggested times exclude tool extensions.

PS10: Welfare and positive equilibrium

Meeting November 13, 2026; due November 20, 2026.

Required 11.2, 12.1, 12.2

Extension 11.3, 12.3

Reserve optional R008, R009

Advanced exam preparation: 11.A1, 11.A7. Further practice: 11.A2, 11.A3, 11.A4. Attempt the recommended pair by hand after the foundations; the remaining problems form a revision bank. These are additional preparation, not extra weekly requirements unless assigned in class. Suggested times exclude tool extensions.

PS11: Empirical equilibrium and asset markets

Meeting November 20, 2026; due November 27, 2026.

Required 13.1, 14.1, 14.2

Extension 13.2

Reserve optional R007, R010

Advanced exam preparation: 11.A5, 11.A6. Further practice: 12.A1. Attempt the recommended pair by hand after the foundations; the remaining problems form a revision bank. These are additional preparation, not extra weekly requirements unless assigned in class. Suggested times exclude tool extensions.

PS12: Matching

Meeting November 27, 2026; due December 4, 2026.

Required 15.1, 15.2, A.3

Extension None

PS13: Computable equilibrium

Meeting December 4, 2026; due December 7, 2026.

Required 16.1, 16.2

Extension None

Advanced exam preparation: 16.A2, 16.A3. Further practice: 16.A1. Attempt the recommended pair by hand after the foundations; the remaining problems form a revision bank. These are additional preparation, not extra weekly requirements unless assigned in class. Suggested times exclude tool extensions.

PS13 is formative because classes end on December 7. Order-theory exercises are referenced where useful but remain in the mathematical appendix.

Appendix A

Order Theory

This mathematical appendix collects the order-theoretic concepts used throughout the book. The orientation is economic: xRy means that x is ranked at least as high as y. Unless stated otherwise, X is a nonempty set.

A.1 Binary Relations

Definition A.1. A binary relation on X is a subset $R \subseteq X \times X$. We write xRy in place of $(x, y) \in R$. The inverse relation R^{-1} is defined by $xR^{-1}y \iff yRx$. The identity relation is $\Delta_X = \{(x, x) : x \in X\}$.

A.1.1 Properties

A relation R on X is:

Reflexive if xRx for every $x \in X$.

Complete (or total) if, for every $x, y \in X$, either xRy or yRx (or both). Completeness implies reflexivity by taking $x = y$.

Symmetric if xRy implies yRx.

Asymmetric if xRy implies $\neg(yRx)$. Every asymmetric relation is irreflexive.

Antisymmetric if xRy and yRx imply $x = y$. Antisymmetry does not mean the negation of symmetry.

Transitive if xRy and yRz imply xRz.

Acyclic if there do not exist $n \geq 1$ and $x_0, \ldots, x_n \in X$ such that $x_0 = x_n$ and $x_{k-1} R x_k$ for every $k = 1, \ldots, n$. Acyclicity is normally imposed on a strict relation.

Example A.2. The equality relation on X is reflexive, symmetric, transitive, and antisymmetric, but it is incomplete whenever X contains two distinct elements.

A.1.2 Important Classes of Relations

Definition A.3. An equivalence relation is reflexive, symmetric, and transitive. A preorder is reflexive and transitive. A partial order is reflexive, transitive, and antisymmetric. A total preorder is a complete preorder. A linear order (or total order) is a complete partial order. A set equipped with a partial order is a partially ordered set, or poset.

In consumer theory, a rational weak preference relation is modeled as a total preorder. Antisymmetry is generally inappropriate for weak preference because distinct alternatives may be indifferent.

Example A.4. The usual relation $\geq$ is a linear order on $\mathbb{R}$. The componentwise relation on $\mathbb{R}^L$, defined by $xRy \iff x_\ell \geq y_\ell$ for every ℓ, is a partial order but is not complete when $L \geq 2$. For example, $(1, 2)$ and $(2, 1)$ are incomparable.

The lexicographic relation on $\mathbb{R}^2$ is defined by $x R_{\text{lex}} y$ if $x_1 > y_1$ or if $x_1 = y_1$ and $x_2 \geq y_2$. It is a linear order. Although complete and transitive, it has no continuous real-valued representation on $\mathbb{R}^2$ with its usual topology.

A.2 Symmetric and Asymmetric Parts

Definition A.5. The symmetric part I_R and asymmetric part P_R of a relation R are defined by

$$x I_R y \iff xRy \text{ and } yRx, \qquad x P_R y \iff xRy \text{ and } \neg(yRx).$$

Proposition A.6. *If R is a preorder, then I_R is an equivalence relation and P_R is asymmetric and transitive.*

Proof. Reflexivity and symmetry of I_R follow immediately, and its transitivity follows from transitivity of R. Asymmetry of P_R is built into its definition. If $x P_R y$ and $y P_R z$, then transitivity gives xRz. If zRx, then zRy, contradicting $y P_R z$. Thus $x P_R z$. □

A.3 Equivalence Classes and Quotient Orders

Definition A.7. Let E be an equivalence relation. The equivalence class of x is $[x]_E = \{y \in X : yEx\}$. The quotient set is $X/E = \{[x]_E : x \in X\}$.

Example A.8. If $X = \{a, b, c\}$ and the only distinct equivalent elements are a and b, then $X/E = \{\{a, b\}, \{c\}\}$. For a preference preorder, the equivalence classes of I_R are indifference classes.

Proposition A.9. *Let R be a preorder and let I_R be its symmetric part. On X/I_R, define*

$$[x]_{I_R} \,\bar{R}\, [y]_{I_R} \iff xRy.$$

Then $\bar{R}$ is well defined and is a partial order.

Proof. Suppose xI_Rx',yI_Ry', and xRy. Then $x'Rx$,xRy, and yRy'; transitivity therefore gives $x'Ry'$, which proves that the definition is independent of representatives. Reflexivity and transitivity are inherited from R. If both $[x]\bar{R}[y]$ and $[y]\bar{R}[x]$, then xI_Ry, so $[x] = [y]$. Thus $\bar{R}$ is antisymmetric. □

A.4 Maximal and Greatest Elements

Definition A.10. For a nonempty subset $S \subseteq X$, the set of**maximal elements** is

$$\text{Max}(S, R) = \{x \in S : \nexists y \in S \text{ such that } yP_Rx\}.$$

The set of**greatest elements** is

$$\text{Great}(S, R) = \{x \in S : xRy \text{ for every } y \in S\}.$$

A greatest element dominates every feasible alternative; a maximal element is merely undominated. Every greatest element is maximal. If R is complete on S, then the two notions coincide. Under a partial order there is at most one greatest element, while there may be many maximal elements. Under a preorder, several indifferent alternatives may all be greatest.

Example A.11. Let $S = \{a, b, c\}$ and define $R = \Delta_S \cup \{(a, b)\}$. Thus aP_Rb and every other pair of distinct alternatives is incomparable. Then a and c are maximal, but there is no greatest element because a and c are incomparable.

Proposition A.12. *Every nonempty finite subset of a set equipped with a total preorder has at least one greatest element and therefore at least one maximal element.*

Proof. Proceed by induction on the cardinality of S. The claim is immediate for a singleton. Suppose g is greatest in a finite set S, and add $a \notin S$. Completeness gives either aRg or gRa. In the first case, transitivity and gRy for every $y \in S$ imply that a is greatest in $S \cup \{a\}$. In the second case, g remains greatest. □

Finiteness matters. For example, $(\mathbb{R}, \geq)$ is linearly ordered but has no greatest element.

A.5 Transitive Closure

Definition A.13. For relations $Q, R \subseteq X \times X$, composition is oriented by $x(R \circ Q)z \iff \exists y \in X : \ xQy$ and yRz. Let $R^1 = R$, and recursively define $R^{n+1} = R^n \circ R$. The transitive closure of R is

$$R^+ = \bigcup_{n \geq 1} R^n.$$

The reflexive-transitive closure is $R^* = \Delta_X \cup R^+$, where Δ_X is the identity relation defined above.

Proposition A.14. *The relation R^+ is the smallest transitive relation containing R.*

Proof. Concatenating an R-path of length m with one of length n yields a path of length $m + n$, so R^+ is transitive. Any transitive relation containing R must contain every R^n and hence their union. □

A.6 Order Extensions

Definition A.15. Let R be a preorder on X. We call R^{ext} an extension of R if $R \subseteq R^{\text{ext}}$ and $P_R \subseteq P_{R^{\text{ext}}}$. Thus every original weak and strict comparison is preserved.

Theorem A.16. *(Szpilrajn extension theorem) Every partial order on a set has a linear-order extension. See[Szpilrajn, 1930].*

Proof. Proof sketch. The standard proof applies Zorn's lemma to the set of partial orders extending the given order, ordered by inclusion; this use of Zorn invokes the axiom of choice. The union of every chain is again a partial order, so a maximal extension exists. The key extension lemma is that if x and y are incomparable, adjoining xRy and taking the transitive closure

preserves antisymmetry: a new nontrivial cycle would contain the added edge and would imply an original path from y back to x, contradicting incomparability. Thus two incomparable elements could be consistently compared, contradicting maximality. Hence the maximal extension is complete and therefore linear. □

Corollary A.17. *Every preorder has a total-preorder extension.*

Proof. Let I_R be the symmetric part of the preorder R. The induced relation $\bar{R}$ on X/I_R is a partial order by the earlier proposition. Next, let $\bar{R}^{\text{ext}}$ be a linear-order extension of $\bar{R}$, and define on X

$$xR^{\text{ext}}y \iff [x]_{I_R}\, \bar{R}^{\text{ext}}\, [y]_{I_R}.$$

Then R^{ext} is a total preorder. It contains R, and antisymmetry of $\bar{R}^{\text{ext}}$ ensures that every comparison in P_R remains strict. □

A.7 Checklist for Economic Applications

- Use a total preorder for rational weak preference; use its asymmetric part for strict preference and its symmetric part for indifference.
- Do not call a strict relation a partial order: strict relations are irreflexive, whereas partial orders are reflexive.
- Distinguish maximal from greatest alternatives whenever the relation may be incomplete.
- Use quotient notation X/I_R, not set-difference notation, when collapsing indifference classes.
- State finiteness, compactness, continuity, or another attainment condition whenever existence of a maximal or greatest alternative is claimed.

A.8 Problems

Part I: By hand

Problem A.1 [Core]. Classifying relations.

Assume $|X| \geq 2$. For each relation below, determine whether it is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, transitive, and complete: equality on X; strict set inclusion on 2^X; weak set inclusion on 2^X; and the componentwise order on $\mathbb{R}^2$.

Give a counterexample for every failed property.

Part II: Finite preference completion and multiple utilities — chapter use case and verified scale-up

Problem A.2 [Proof]. Quotienting a preorder.

Let R be a preorder on X and define xI_Ry if and only if xRy and yRx.

(a) Prove that I_R is an equivalence relation.

(b) Define $[x]\,\bar{R}\,[y]$ if and only if xRy. Prove that this definition is independent of representatives.

(c) Prove that $\bar{R}$ is a partial order on X/I_R.

Problem A.3 [Proof]. Maximal, greatest, and finite existence.

Let R be a preorder on a nonempty finite set S.

(a) Prove that S has an R-maximal element.

(b) Give an example with several maximal elements and no greatest element.

(c) Prove that if R is complete, maximal and greatest elements coincide.

Chapter use case

Seed: Problems A.1, A.2, A.3. Fixed: all unlisted primitives and assumptions. Scaled: the number $|X|$ of alternatives in a finite preorder, beginning with the hand example. Tool: Codex + Lean. Verification: Verify the preorder, quotient construction, and each proposed linear extension. Szpilrajn extension is currently verified; quotient pullback, intersection of all extensions, and the finite multiple-utility corollary remain explicit formal targets.

Problem A.4 [Proof]. Order extensions and multiple representation.

Let R be a preorder on a finite set X.

(a) Pass to the quotient partial order on X/I_R and extend it to a linear order.

(b) Pull the extension back to a complete preorder on X.

(c) Show that R is the intersection of its complete preorder extensions and deduce a finite multiple-utility representation.

Appendix B

Fixed Points and Equilibrium Existence

The main chapter uses Brouwer for a function. This appendix supplies its compact budget and truncation details, then explains what changes when a consumer has several optimal bundles. The fixed point theorems are classical results used here. They and the analytic economic existence proofs are not claimed as locally Lean verified.

B.1 Compact budgets and demand at boundary prices

Retain finitely many consumers, $\omega_i \gg 0$, $\omega = \sum_i \omega_i$, $B = [0, 2\omega]$ and $\Delta = \{p \geq 0 : \sum_\ell p_\ell = 1\}$. Define $C_i(p) = \{x \in B : p \cdot x \leq p \cdot \omega_i\}$ and $D_i^B(p) = \arg\max_{x \in C_i(p)} u_i(x)$, using a continuous representation of the stated preferences. These are artificial compact budget problems. Ordinary unrestricted demand need not exist at a free good and is not silently extended to the closed simplex.

Lemma B.1. *Compact budget continuity. The correspondence C_i is nonempty, compact valued and continuous on Δ. Under continuous preferences, D_i^B is nonempty, compact valued and upper hemicontinuous. Convex preferences make its values convex; strict convexity makes it a single valued continuous function.*

For the budget claim, zero is feasible and strictly affordable because $p \cdot \omega_i \geq \min_\ell \omega_{i\ell} > 0$. Closed graph and the common compact box give upper hemicontinuity. To see lower hemicontinuity, take $x \in C_i(p)$ and any sequence $p^n \to p$. If $p \cdot x < p \cdot \omega_i$, then x remains feasible

eventually. Otherwise $p \cdot x = p \cdot \omega_i > 0$. The bundles $x^n = \alpha_n x$, with $\alpha_n = \min\{1, (p^n \cdot \omega_i)/(p^n \cdot x)\}$ for large n, are feasible and converge to x. The maximum theorem gives the demand conclusions; convex upper contour sets give convex demand values, and strict convexity rules out two distinct maximizers.

Upper hemicontinuity here means that optimal bundles cannot acquire a new limiting value: if prices converge and a corresponding sequence of chosen bundles converges, the limit is optimal at the limiting prices. Compactness of the common box makes this closed graph description equivalent to upper hemicontinuity. It is not a license to select one optimum arbitrarily and assume that the selection is continuous.

B.2 Why the artificial cap disappears

Lemma B.2. *Removing truncation. Suppose $p \gg 0$, $x_i \in D_i^B(p)$ and $\sum_i x_i \leq \omega$. Under continuous, convex and strongly monotone preferences, x_i also solves the unrestricted budget problem. Its budget binds.*

Every $x_i \leq \omega$ lies strictly below the artificial ceiling. Any slack budget would allow a small positive increase of a commodity inside B, contradicting strong monotonicity. Now suppose an unrestricted affordable y is strictly preferred to x_i. By continuity, $y' = \alpha y$ is still strictly preferred for some $\alpha < 1$ sufficiently close to one. Since wealth is positive, $p \cdot y' < p \cdot \omega_i$. For small $t > 0$, $v = (1-t)x_i + ty'$ lies inside B and has strict budget slack. Convex preferences give $v \succeq_i x_i$. A sufficiently small increase of one coordinate preserves both constraints and makes the bundle strictly preferred to v, hence to x_i. This contradicts capped optimality. Under strict convexity a small mixture with y is already strictly preferred, giving the shorter argument in the main chapter.

B.3 Kakutani and nonunique demand

Theorem B.3. *Kakutani fixed point theorem. Let $K \subseteq \mathbb{R}^m$ be nonempty, compact and convex. If $\Phi : K \rightrightarrows K$ is upper hemicontinuous and has nonempty compact convex values, then some $v^* \in K$ satisfies $v^* \in \Phi(v^*)$. Brouwer is the single valued continuous special case.*

[Kakutani, 1941]

Proposition B.4. *Walrasian existence with convex preferences. Consider finitely many consumers with consumption sets $\mathbb{R}^L_+$, individual endowments $\omega_i \gg 0$, and complete, transitive,*

continuous, convex and strongly monotone preferences. Then a Walrasian equilibrium with $p^* \gg 0$ *exists. Demand need not be unique. This includes the stricter preference assumptions of the main chapter's Brouwer theorem as a special case.*

Use the compact domain $K = \Delta \times B^I$. For a proposed allocation x, let $z(x) = \sum_i x_i - \omega$ and let the auxiliary price response be $P(x) = \arg\max_{r \in \Delta} r \cdot z(x)$. Its values are the price weights maximizing the value of the proposed imbalance, not a market trading rule. Define

$$\Phi(p, x) = P(x) \times \prod_{i=1}^{I} D_i^B(p).$$

The budget lemma gives nonempty compact convex demand values and upper hemicontinuity. The auxiliary price response has the same properties by the maximum theorem and linearity in prices. Their finite product is therefore an upper hemicontinuous correspondence from the nonempty compact convex domain into itself. Kakutani gives a fixed point with each household choosing optimally in its capped budget and the auxiliary price maximizing the imbalance's value.

$$p \cdot z(x) \leq 0, \qquad p \cdot z(x) = \max_{r \in \Delta} r \cdot z(x) = \max_{\ell} z_\ell(x).$$

Thus every excess demand coordinate is nonpositive. Aggregate feasibility puts every individual choice below $\omega < 2\omega$. If a price were zero, its good could be increased within the cap and would strictly improve utility, so all prices are positive. Every budget now binds. Walras' law and nonpositive excess demands give exact clearing. The cap removal lemma proves unrestricted household optimality, completing the economic existence proof.

Strict convexity was needed only to replace demand correspondences by continuous functions in the main proof. The appendix removes that restriction, not the positive endowment or strong desirability assumptions. Leontief edges, zero price equilibria, general consumption sets, production and quasi equilibria require the corresponding hypotheses of a broader theorem. They are not established merely by naming Kakutani.

[Arrow and Debreu, 1954, Debreu, 1959, Mas-Colell et al., 1995]

Appendix C

Technical Theorem Records

A starred theorem in the main text is a reader-level statement. This appendix records its exact environment, full equivalences, proof boundary, and formal-verification status. The star never means that omitted assumptions may be guessed and never means that a theorem is Lean verified.

C.1 WGARP and Coherent CMU

Result ID: CT-RP-WGARP-CMU. Let $L \geq 2$, $X = \mathbb{R}^L_+$, and $\mathcal{O} = \{(p^t, x^t)\}_{t\in\mathbb{T}}$ be finite, with $p^t \in \mathbb{R}^L_{++}$, $x^t \in X \setminus \{0\}$, and observed wealth $w^t = p^t \cdot x^t$. Equip $C(X)$ with the topology of uniform convergence on compact subsets. A compact coalition structure is a compact collection Ω of nonempty compact subsets of $C(X)$, under the induced Hausdorff topology.

Theorem C.1. *Full WGARP--CMU characterization. Under the preceding displayed environment, the following six statements are equivalent.*

1. The data are rationalized by an asymmetric, strictly increasing, continuous preference function $r : X \times X \to \mathbb{R}$.
2. The data are rationalized by a coherent, strictly increasing, continuous CMU function.
3. The data are rationalized by a skew-symmetric, coherent, continuous CMU function whose coalitions consist of strictly increasing concave utilities.
4. The data satisfy WGARP.
5. For every $s, t \in \mathbb{T}$ there exist $R^{s,t} \in \mathbb{R}$ and $\lambda^s_{s,t}, \lambda^t_{s,t} > 0$ such that

$$R^{s,t} = -R^{t,s}, \qquad \lambda^s_{s,t} = \lambda^s_{t,s}, \qquad R^{s,t} \geq \lambda^s_{s,t} p^s \cdot (x^s - x^t).$$

6. There are antisymmetric numbers $V^{s,t} = -V^{t,s}$ such that

$$\begin{aligned} p^s \cdot (x^s - x^t) \geq 0 &\Longrightarrow V^{s,t} \geq 0, \\ p^s \cdot (x^s - x^t) > 0 &\Longrightarrow V^{s,t} > 0 \end{aligned} \qquad (s, t \in \mathbb{T}).$$

Proof record. The easy implications from a rationalizing preference function to WGARP and between the two numerical certificate systems are finite arguments. The reverse representation direction constructs symmetric pairwise Afriat utilities and assembles them into compact coherent coalitions. That research-level construction, including compactness and skew-symmetry, is proof-delegated to the exact finite-data theorem of[Aguiar et al., 2026]; this book does not relabel the delegation as an independent proof.

Boundary note. Coherence means pairwise intersection and need not imply a common utility in every coalition. A general coherent CMU may be incomplete or intransitive. The representation rationalizes the observed budgets but does not guarantee nonempty demand at all counterfactual budgets.

C.2 WARP and Strict Coherent CMU

Result ID: CT-RP-WARP-CMU. A preference function is strictly star-concave when, for every $x \neq y$ and $\alpha \in (0, 1)$,

$$r(\alpha x + (1 - \alpha)y, x) > \alpha r(x, x) + (1 - \alpha) r(y, x).$$

This condition concerns the first argument along line segments ending at the second argument; it is not global concavity of the two-argument function.

Theorem C.2. *Full WARP--CMU characterization. In the same finite budget environment, the following are equivalent: (i) WARP; (ii) rationalization by an asymmetric, strictly star-concave, continuous preference function; (iii) rationalization by a coherent, strictly star-concave, continuous CMU; and (iv) rationalization by a skew-symmetric, coherent, continuous CMU whose coalitions consist of strictly increasing strictly concave utilities.*

Proof record. The proof uses strict pairwise utility rationalizations and is separately delegated to the WARP theorem and online appendix of[Aguiar et al., 2026]. It is not obtained by merely replacing the acronym WGARP in the preceding theorem.

C.3 Formalization Record and Boundaries

CT-RP-CMU-SINGLETON. Textbook proof complete; local Lean target. The exact singleton reduction is elementary and introduces no compactness machinery beyond the singleton case.

CT-RP-CMU-CLD. Lean core verified through the pinned WGARP project and the compiled declaration `Econ803.Consumer.compensated_law_of_demand`. The certificate covers the asymmetric, strictly increasing preference-function result stated in the chapter.

CT-RP-WGARP-CMU and CT-RP-WARP-CMU. Cited theorems, not yet Lean complete. A complete formalization requires the pairwise Afriat construction, compact-open and Hausdorff topology, coherence, and the strict-rationalization dependencies.

The model does not by itself recover transitivity, global convexity, classical UMP--EMP duality, Slutsky symmetry, scalar-utility welfare comparisons, or global existence of counterfactual demand. Those conclusions remain owned by the benchmark chapters and require their displayed assumptions.

Appendix D

Glossary of Symbols

This glossary records the book's persistent notation. Each blue symbol links back to its first substantive use in the text. There, the blue first use links here. A symbol described as scoped may be reused only after the new local meaning is stated explicitly.

D.1 Sets and indices

L Number of commodities; $L \in \mathbb{N}_{>0}$. Scope: Book-wide.

ℓ, k Commodity-coordinate indices in $\{1, \dots, L\}$. Scope: Book-wide.

$\mathbb{R}^n_+$ Vectors in $\mathbb{R}^n$ whose coordinates are all nonnegative. Scope: Book-wide.

$\mathbb{R}^n_{++}$ Vectors in $\mathbb{R}^n$ whose coordinates are all strictly positive; equivalently $x \gg 0$. Scope: Book-wide.

X Ambient feasible set; in consumer chapters, the consumption set, normally $\mathbb{R}^L_+$. Scope: Scoped.

$i = 1, \dots, I$ Consumer index and number of consumers. Scope: Book-wide.

$j = 1, \dots, J$ Firm index and number of firms. Scope: Book-wide.

D.2 Operators

$a \cdot b$ Euclidean inner product. Scope: Book-wide.

$\|a\|_2$ Euclidean norm of a vector a. Scope: Book-wide.

$\mathbf{1}_n$ All-ones column vector in $\mathbb{R}^n$; the dimension is shown whenever it is not immediate. Scope: Book-wide.

I_n $n \times n$ identity matrix; the subscript records its dimension. Scope: Book-wide.

$a^\top$ Transpose of a vector or matrix; primes never denote transpose. Scope: Book-wide.

a' Alternative observation or object of the same type as a. Scope: Book-wide.

$A \setminus B$ Elements of A that do not belong to B. Scope: Book-wide.

$D_z f$ Jacobian of a vector-valued map f with respect to z. Scope: Book-wide.

$\nabla_z f$ Column gradient of a scalar-valued function f with respect to z. Scope: Book-wide.

$\leq, \geq, \gg$ Vector inequalities are componentwise; $x \gg 0$ means every coordinate is strictly positive. Scope: Book-wide.

2^X Power set of X. Scope: Book-wide.

$\arg\max, \arg\min$ Sets of maximizers and minimizers; a singleton selection is identified explicitly. Scope: Book-wide.

$\Pr, \mathbb{P}, \mathbb{E}$ Probability and expectation under the probability space declared in the surrounding model. Scope: Book-wide.

$\mathbb{I}\{\cdot\}$ Indicator of the event inside braces. Scope: Book-wide.

$\log$ Natural logarithm; information quantities are therefore measured in nats. Scope: Book-wide.

D.3 Optimization

$\mathcal{L}$ Lagrangian of the optimization problem currently under study. Scope: Scoped.

D.4 Choice and preference

$\succeq, \succ, \sim$ Weak preference and its asymmetric and symmetric parts. Scope: Book-wide.

$c : \mathcal{A} \rightrightarrows X$ Nonempty-valued choice correspondence on a family of feasible menus. Scope: Book-wide.

$\succeq^C$ Direct weak revealed choice: $x \succeq^C y$ when some menu contains both and selects x. Scope: Abstract-choice chapter.

$\succ^C$ Direct strict revealed choice: some menu selects x and rejects available y. Scope: Abstract-choice chapter.

$\succeq^{C*}$ Reflexive-transitive closure of direct weak revealed choice $\succeq^C$. Scope: Abstract-choice chapter.

D.5 Revealed preference

$\mathbb{T} = \{1, \ldots, T\}$ Finite set of observation indices. Scope: Revealed-preference chapters.

$\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$ Dataset of price-choice observations indexed by $\mathbb{T}$. Scope: Book-wide.

$w^t = p^t \cdot x^t$ Observed expenditure, equal to wealth under budget exhaustion. Scope: Revealed-preference chapters.

$\succeq^{R,D}$ Direct weak revealed preference generated by affordability. Scope: Budget-demand chapters.

$\succ^{R,D}$ Strict direct revealed preference generated by strict affordability. Scope: Budget-demand chapters.

$\succeq^R$ Reflexive-transitive closure of budget-based $\succeq^{R,D}$. Scope: Budget-demand chapters.

(U^t, λ^t) Utility-level and positive marginal-utility-of-wealth certificates satisfying the Afriat inequalities. Scope: Afriat theorem.

$\succeq_\alpha^{R,D}, \succ_\alpha^{R,D}, \succeq_\alpha^R$ Efficiency-adjusted direct weak, direct strict, and revealed-closure relations at expenditure fraction α. Scope: CCEI section.

$\mathcal{S}_\mathcal{O}(p, w)$ Bundles consistent with GARP when appended to dataset $\mathcal{O}$ at a new budget. Scope: Consumer chapter.

$\mathrm{CCEI}(\mathcal{O})$ Largest common expenditure fraction for which the adjusted dataset satisfies GARP. Scope: Consumer chapter.

$r : X \times X \to \mathbb{R}$ Two-argument numerical comparison rule used to define maximal demand. Scope: CMU section.

$x_r(p, w)$ Bundles in $B(p, w)$ that weakly defeat every feasible alternative under preference function r. Scope: CMU section.

Ω Compact collection of nonempty compact coalitions of continuous utility functions. Scope: CMU section.

U A nonempty compact set of continuous utility functions whose members evaluate a comparison unanimously. Scope: CMU section.

$r_\Omega(x, y)$ Coalitional multi-utility preference function $\max_{U\in\Omega} \min_{u\in U}[u(x) - u(y)]$. Scope: CMU section.

D.6 Consumer theory

x Consumption bundle; arguments such as $x(p, w)$ distinguish demand from a generic bundle. Scope: Book-wide default.

p Commodity-price column vector; normally $p \in \mathbb{R}^L_{++}$. Scope: Book-wide default.

w Scalar wealth or income available to a consumer. Scope: Book-wide default.

a_ℓ Strictly positive taste weights normalized to sum to one; equal expenditure shares only in the Cobb-Douglas case. Scope: Workhorse demand models.

$b_\ell(p, w), \boldsymbol{b}$ Expenditure share $p_\ell x_\ell(p, w)/w$ and its column vector; independent of wealth in the homothetic models considered here. Scope: Workhorse demand models.

ρ CES power parameter with $\rho < 1$ and Cobb-Douglas limit at zero; some production parameterizations use a different sign convention. Scope: CES consumer model.

σ Constant substitution elasticity $1/(1 - \rho) > 0$; distinguished from Marshallian own-price elasticity. Scope: CES consumer model.

γ Nonnegative subsistence bundle; discretionary wealth is $w - p \cdot \gamma$. Scope: Stone-Geary model.

$\mathcal{P}(p)$ Unit expenditure $e(p, 1)$ under the specified degree-one utility normalization; subscript TL denotes the translog index. Scope: Workhorse dual models.

Γ Symmetric matrix $(\gamma_{\ell k})$ of log-price quadratic coefficients, with zero row sums; distinct from the Stone-Geary vector γ. Scope: Translog expenditure model.

$B(p, w)$ Bundles affordable at prices p and wealth w: $\{x \in X : p \cdot x \leq w\}$. Scope: Book-wide.

$x(p, w)$ Marshallian demand correspondence; treated as a function only when uniqueness is assumed. Scope: Book-wide.

$h(p, \bar{u})$ Expenditure-minimizing demand correspondence at prices p and target utility $\bar{u}$. Scope: Book-wide.

u Direct utility; in uncertainty chapters it is the Bernoulli utility of a prize. Scope: Book-wide default.

$v(p, w)$ Maximum attainable utility at prices p and wealth w. Scope: Book-wide.

$e(p, \bar{u})$ Minimum expenditure required to attain utility $\bar{u}$ at prices p. Scope: Book-wide.

$\mu_p(x)$ Expenditure at reference prices p required to attain the utility of bundle x: $e(p, u(x))$. Scope: Welfare-analysis sections.

$(\mathcal{G}, d)$ Compact metric space of differentiated-product characteristics and its economically specified distance. Scope: Many-product section.

b_i Consumer i's bliss or ideal point in product-characteristic space. Scope: Ideal-point model.

$C_j(p)$ Types who choose product j at price vector p; a generalized Voronoi cell under distance-based utility. Scope: Ideal-point model.

$\mathcal{M}_+(\mathcal{G})$ Finite nonnegative Radon measures on product space; includes atomic finite shopping baskets. Scope: Continuum-product model.

$\langle p, x \rangle$ Expenditure pairing $\int_{\mathcal{G}} p(g)x(dg)$ between a continuous price schedule and a basket measure. Scope: Continuum-product model.

$B_r(g)$ Closed radius-r ball around product point g in the declared product metric. Scope: Many-product section.

$\mathcal{E}^H(A; \varphi_B)$ Compensated percentage response of demand mass in region A to the continuous log-price shock profile φ_B concentrated near region B. Scope: Continuum-product model.

$S(p, w)$ $D_p x(p, w) + D_w x(p, w) x(p, w)^\top$. Scope: Book-wide.

P Strictly positive commodity-price domain $\mathbb{R}^L_{++}$. Scope: Consumer chapters.

W Strictly positive wealth domain $\mathbb{R}_{++}$. Scope: Consumer chapters.

D.7 Behavioral demand

$\widehat{p}$ Price vector used in the consumer's perceived first-order condition. Scope: Sparse-max chapter.

x^{SM} Sparse-max demand satisfying perceived-price tangency and the true budget. Scope: Sparse-max chapter.

$a = (a_1, \ldots, a_L)$ Coordinatewise attention weights in $[0,1]^L$. Scope: Sparse-max chapter.

$\mathcal{C}$ Cost of attention or information acquisition. Scope: Attention chapter.

$\mathcal{A}$ Finite set of actions available to the rationally inattentive decision maker. Scope: Rational-inattention section.

μ_0 Full-support prior distribution over the finite state space $\mathcal{S}$. Scope: Rational-inattention section.

$q(a \mid s)$ Conditional probability of action a in state s; equivalently, the direct information-and-decision channel. Scope: Rational-inattention section.

$\mathcal{I}(q)$ Shannon mutual information between the state and the action induced by channel q. Scope: Rational-inattention section.

D.8 Choice under uncertainty

Z Prize or consequence set; finite unless the infinite-outcome section explicitly broadens it. Scope: Uncertainty chapter.

μ, ν, η Probability mass functions or probability measures over prizes. Scope: Uncertainty chapter.

$\Delta(Z)$ Set of lotteries over Z; for general spaces the measurable structure is stated explicitly. Scope: Uncertainty chapter.

δ_z Lottery assigning probability one to prize z. Scope: Uncertainty chapter.

$U(\mu)$ Expected Bernoulli utility of lottery μ. Scope: Uncertainty chapter.

F_μ Cumulative distribution function of lottery μ on an ordered prize space. Scope: Uncertainty chapter.

$\succeq^{\mathrm{FOSD}}$ First-order stochastic-dominance relation, oriented toward the dominating distribution. Scope: Uncertainty chapter.

$\mathrm{CE}(\mu; u)$ Sure prize with the same utility as lottery μ. Scope: Uncertainty chapter.

$\mathrm{RP}(\mu; u)$ Expected prize minus certainty equivalent. Scope: Uncertainty chapter.

$\mathrm{PP}(x, \varepsilon; u)$ Probability adjustment making the specified binary risk indifferent to a sure prize. Scope: Uncertainty chapter.

ϖ_s Strictly positive state probabilities summing to one. Scope: State-contingent chapter.

r Coefficient of relative risk aversion in the power-utility family. Scope: Risk-aversion section.

$\mathcal{S}$ Finite state space used in the surrounding model, including sequential-trade applications. Scope: Scoped.

D.9 Probability

Σ Borel sigma-algebra on the prize space. Scope: Infinite-outcome section.

$\Rightarrow$ Weak convergence of probability measures. Scope: Infinite-outcome section.

D.10 Aggregation

$x^A(p, \mathbf{w})$ Sum of individual Marshallian demands. Scope: Aggregate-demand chapter.

$\mathbf{w}$ Vector of individual wealth levels. Scope: Aggregate-demand chapter.

W Sum of individual wealth levels. Scope: Aggregate-demand chapter.

$a_i(p), b(p)$ Consumer-specific intercept and common wealth coefficient in Gorman indirect utility. Scope: Aggregate-demand chapter.

$\mathcal{W}$ Social welfare function or explicitly stated welfare aggregate. Scope: Scoped.

D.11 Discrete choice

$B_X(p, w)$ Affordable members of the finite bundle catalogue at prices p and wealth w. Scope: Bundle-to-discrete-choice bridge.

$\delta_{\widetilde{p}}(x)$ Quasilinear subutility net of bundle expenditure in units of the outside good: $v(x) - \widetilde{p} \cdot x$. Scope: Bundle-to-discrete-choice bridge.

U^*_{ij} Latent utility of chooser i from alternative j. Scope: Discrete-choice chapter.

P_{ij} Probability that chooser i selects alternative j from the stated menu. Scope: Discrete-choice chapter.

τ Strictly positive Type-I extreme-value scale parameter. Scope: Discrete-choice chapter.

$(\Omega, \mathcal{F}, \mathbb{P})$ Probability space supporting random utility. Scope: Random-utility section.

$\boldsymbol{U}$ Measurable utility vector indexed by alternatives. Scope: Random-utility section.

$P = \{P_B\}_{B \in \mathcal{B}}$ Family of choice-probability distributions, one for each nonempty menu B. Scope: Stochastic-choice section.

$\mathcal{R}_Y$ Finite set of strict linear orders on the alternative universe Y. Scope: Random-utility section.

$\mathcal{V}(v)$ Expected maximum random utility as a function of the mean-utility vector $v \in \mathbb{R}^A$. Scope: Consumer-surplus section.

D.12 Firm theory

y Net-output vector, with positive coordinates denoting outputs and negative coordinates inputs. Scope: Book-wide default.

Y Technologically feasible net-output plans. Scope: Firm chapter.

f Single-output production function. Scope: Firm chapter.

$\pi_Y(p)$ Support function of production set Y. Scope: Firm chapter.

$\mathcal{Y}_Y(p)$ Profit-maximizing production plans. Scope: Firm chapter.

$\boldsymbol{z}$ Nonnegative vector of physical inputs in a single-output technology. Scope: Firm chapter.

$\boldsymbol{w}$ Strictly positive vector of input prices. Scope: Firm chapter.

p_q Scalar price of output q. Scope: Single-output model.

$C(\boldsymbol{w}, q)$ Minimum input expenditure required to produce at least q. Scope: Firm chapter.

$\mathcal{Z}(\boldsymbol{w}, q)$ Cost-minimizing input correspondence. Scope: Firm chapter.

D.13 Equilibrium

ω Endowment vector; subscripts identify agents and superscripts identify observations or states when declared. Scope: Book-wide default.

θ_{ij} Consumer i's ownership share of firm j. Scope: Competitive-equilibrium chapter.

$z(p)$ Aggregate demand minus aggregate resources at prices p. Scope: Equilibrium chapters.

D.14 Sequential trade

R Matrix of state-contingent security payoffs. Scope: Sequential-trade chapter.

q Date-0 security-price vector. Scope: Sequential-trade chapter.

ψ Strictly positive state-price vector satisfying $q = R^\top \psi$ when markets are complete. Scope: Sequential-trade chapter.

D.15 Matching

M, W Two finite sides of the one-to-one matching market. Scope: Matching chapter.

φ Involutive matching map, with an unmatched agent mapped to themselves. Scope: Matching chapter.

D.16 Order theory

R Generic binary relation on a nonempty ground set X. Scope: Scoped.

R^{-1} Inverse relation: $xR^{-1}y$ iff yRx. Scope: Appendix A.

I_R Symmetric part $R \cap R^{-1}$. Scope: Book-wide.

P_R Asymmetric part of R. Scope: Appendix A.

$[x]_E$ Equivalence class of x under E. Scope: Appendix A.

X/E Set of equivalence classes of X under an equivalence relation E. Scope: Book-wide.

$\bar{R}$ Partial order induced by a preorder on its quotient by I_R. Scope: Appendix A.

$\mathrm{Max}(S, R)$ Set of R-maximal elements of feasible subset S. Scope: Appendix A.

$\mathrm{Great}(S, R)$ Set of R-greatest elements of feasible subset S. Scope: Appendix A.

Δ_X Diagonal or identity relation on X. Scope: Appendix A.

R^n n-fold relational composition, with composition orientation stated in the text. Scope: Appendix A.

R^+ Least transitive relation containing R. Scope: Appendix A.

R^* $\Delta_X \cup R^+$. Scope: Appendix A.

R^{ext} Complete-preorder extension obtained from a linear extension of the quotient order. Scope: Appendix A.

References

Jason Abaluck and Abi Adams-Prassl. What do consumers consider before they choose? identification from asymmetric demand responses. *The Quarterly Journal of Economics*, 136(3): 1611–1663, 2021. doi: 10.1093/qje/qjab008.

S. N. Afriat. The construction of utility functions from expenditure data. *International Economic Review*, 8(1):67–77, 1967. doi: 10.2307/2525382.

Victor H. Aguiar and Nail Kashaev. Stochastic revealed preferences with measurement error. *The Review of Economic Studies*, 88(4):2042–2093, July 2021. doi: 10.1093/restud/rdaa067.

Victor H. Aguiar, Nail Kashaev, and Roy Allen. Prices, profits, proxies, and production. *Journal of Econometrics*, 235(2):666–693, 2023. doi: 10.1016/j.jeconom.2022.06.007. URL `https://arxiv.org/abs/1810.04697`.

Victor H. Aguiar, Per Hjertstrand, Roberto Serrano, and Ozgur Evren. A rationalization of the weak axiom of revealed preference. arXiv:1906.00296, version 2, 2026.

David Ahn, Syngjoo Choi, Douglas Gale, and Shachar Kariv. Estimating ambiguity aversion in a portfolio choice experiment. *Quantitative Economics*, 5(2):195–223, July 2014. doi: 10.3982/QE243.

Amazon. Amazon stores. Company information page, 2026. URL `https://www.aboutamazon.com/what-we-do/amazon-store`. Accessed 19 September 2026.

Simon P. Anderson, André de Palma, and Jacques-François Thisse. Demand for differentiated products, discrete choice models, and the characteristics approach. *The Review of Economic Studies*, 56(1):21–35, 1989. doi: 10.2307/2297747.

Kenneth J. Arrow and Gerard Debreu. Existence of an equilibrium for a competitive economy. *Econometrica*, 22(3):265–290, 1954. doi: 10.2307/1907353.

Steven Berry and Ariel Pakes. The pure characteristics demand model. *International Economic Review*, 48(4):1193–1225, 2007. doi: 10.1111/j.1468-2354.2007.00459.x.

Steven Berry, James Levinsohn, and Ariel Pakes. Automobile prices in market equilibrium. *Econometrica*, 63(4):841–890, 1995. doi: 10.2307/2171802.

Truman F. Bewley. Existence of equilibria in economies with infinitely many commodities. *Journal of Economic Theory*, 4(3):514–540, 1972. doi: 10.1016/0022-0531(72)90136-6.

Christian Broda and David E. Weinstein. Product creation and destruction: Evidence and price implications. *American Economic Review*, 100(3):691–723, 2010. doi: 10.1257/aer.100.3.691.

Donald J. Brown and Rosa L. Matzkin. Testable restrictions on the equilibrium manifold. *Econometrica*, 64(6):1249–1262, 1996. doi: 10.2307/2171830.

Erik Brynjolfsson, Yu Jeffrey Hu, and Michael D. Smith. Consumer surplus in the digital economy: Estimating the value of increased product variety at online booksellers. *Management Science*, 49(11):1580–1596, 2003. doi: 10.1287/mnsc.49.11.1580.20580.

Erik Brynjolfsson, Yu Jeffrey Hu, and Duncan Simester. Goodbye pareto principle, hello long tail: The effect of search costs on the concentration of product sales. *Management Science*, 57 (8):1373–1386, 2011. doi: 10.1287/mnsc.1110.1371.

Pierre-André Chiappori, Robert J. McCann, and Lars P. Nesheim. Hedonic price equilibria, stable matching, and optimal transport: Equivalence, topology, and uniqueness. *Economic Theory*, 42(2):317–354, 2010. doi: 10.1007/s00199-009-0455-z.

Syngjoo Choi, Raymond Fisman, Douglas Gale, and Shachar Kariv. Consistency and heterogeneity of individual behavior under uncertainty. *American Economic Review*, 97(5):1921–1938, 2007. doi: 10.1257/aer.97.5.1921.

Laurits R. Christensen, Dale W. Jorgenson, and Lawrence J. Lau. Transcendental logarithmic utility functions. *American Economic Review*, 65(3):367–383, 1975. URL `https://www.jstor.org/stable/1804840`.

Rahul Deb, Yuichi Kitamura, John K. H. Quah, and Jörg Stoye. Revealed price preference: Theory and empirical analysis. *The Review of Economic Studies*, 90(2):707–743, 2023. doi: 10.1093/restud/rdac041.

Gerard Debreu. *Theory of Value: An Axiomatic Analysis of Economic Equilibrium.* Yale University Press, New Haven, CT, 1959.

Gerard Debreu. Excess demand functions. *Journal of Mathematical Economics*, 1(1):15–21, 1974.

Gerard Debreu and Herbert Scarf. A limit theorem on the core of an economy. *International Economic Review*, 4(3):235–246, 1963. doi: 10.2307/2525306.

W. Erwin Diewert. A generalization of the symmetric translog functional form. Discussion Paper 23-02, Vancouver School of Economics, University of British Columbia, June 2023. Draft dated 14 June 2023.

Dennis Epple, Brett Gordon, and Holger Sieg. A new approach to estimating the production function for housing. *American Economic Review*, 100(3):905–924, 2010. doi: 10.1257/aer.100.3.905.

Robert C. Feenstra and James A. Levinsohn. Estimating markups and market conduct with multidimensional product attributes. *The Review of Economic Studies*, 62(1):19–52, 1995. doi: 10.2307/2297840.

David Gale and Lloyd S. Shapley. College admissions and the stability of marriage. *American Mathematical Monthly*, 69(1):9–15, 1962. doi: 10.2307/2312726.

Oded Galor. From stagnation to growth: Unified growth theory. In Philippe Aghion and Steven N. Durlauf, editors, *Handbook of Economic Growth*, volume 1A, pages 171–293. Elsevier, 2005. doi: 10.1016/S1574-0684(05)01004-X.

Oded Galor and David N. Weil. Population, technology, and growth: From malthusian stagnation to the demographic transition and beyond. *American Economic Review*, 90(4):806–828, 2000. doi: 10.1257/aer.90.4.806.

Amit Gandhi, Salvador Navarro, and David Rivers. On the identification of gross output production functions. *Journal of Political Economy*, 128(8):2973–3016, 2020. doi: 10.1086/707736. URL `https://publish.uwo.ca/~drivers2/research/Production_Final_Full.pdf`.

Amit Gandhi, Zhentong Lu, and Xiaoxia Shi. Estimating demand for differentiated products with zeroes in market share data. *Quantitative Economics*, 14(2):381–418, 2023. doi: 10.3982/QE1593.

John Geanakoplos. Nash and walras equilibrium via brouwer. *Economic Theory*, 21:585–603, 2003. doi: 10.1007/s001990000076.

W. M. Gorman. Community preference fields. *Econometrica*, 21(1):63–80, 1953. doi: 10.2307/1906943.

Faruk Gul and Wolfgang Pesendorfer. Random expected utility. *Econometrica*, 74(1):121–146, 2006.

Charles A. Holt and Susan K. Laury. Risk aversion and incentive effects. *American Economic Review*, 92(5):1644–1655, 2002. doi: 10.1257/000282802762024700.

Harold Hotelling. Stability in competition. *The Economic Journal*, 39(153):41–57, 1929. doi: 10.2307/2224214.

Harold Hotelling. Edgeworth's taxation paradox and the nature of demand and supply functions. *Journal of Political Economy*, 40(5):577–616, 1932.

Shizuo Kakutani. A generalization of brouwer's fixed point theorem. *Duke Mathematical Journal*, 8(3):457–459, 1941. doi: 10.1215/S0012-7094-41-00838-4.

Kelvin J. Lancaster. A new approach to consumer theory. *Journal of Political Economy*, 74(2): 132–157, 1966. doi: 10.1086/259131.

Nicholas Li. An engel curve for variety. *The Review of Economics and Statistics*, 103(1):72–87, 2021. doi: 10.1162/rest_a_00879.

Rolf R. Mantel. On the characterization of aggregate excess demand. *Journal of Economic Theory*, 7(3):348–353, 1974.

Andreu Mas-Colell. A model of equilibrium with differentiated commodities. *Journal of Mathematical Economics*, 2(2):263–295, 1975. doi: 10.1016/0304-4068(75)90028-2.

Andreu Mas-Colell, Michael D. Whinston, and Jerry R. Green. *Microeconomic Theory*. Oxford University Press, New York, 1995.

Filip Matějka and Alisdair McKay. Rational inattention to discrete choices: A new foundation for the multinomial logit model. *American Economic Review*, 105(1):272–298, 2015. doi: 10.1257/aer.20130047.

Daniel McFadden. Econometric models of probabilistic choice. In Charles F. Manski and Daniel McFadden, editors, *Structural Analysis of Discrete Data with Econometric Applications*, pages 198–272. MIT Press, Cambridge, MA, 1981.

Brent Neiman and Joseph Vavra. The rise of niche consumption. *American Economic Journal: Macroeconomics*, 15(3):224–264, 2023. doi: 10.1257/mac.20210263.

Joris Pinkse, Margaret E. Slade, and Craig Brett. Spatial price competition: A semiparametric approach. *Econometrica*, 70(3):1111–1153, 2002. doi: 10.1111/1468-0262.00320.

Matthew Polisson and John K.-H. Quah. Revealed preference tests under risk and uncertainty. Technical Report 13/24, University of Leicester, December 2013a. URL `https://www.le.ac.uk/economics/research/RePEc/lec/leecon/dp13-24.pdf`.

Matthew Polisson and John K.-H. Quah. Revealed preference in a discrete consumption space. *American Economic Journal: Microeconomics*, 5(1):28–34, 2013b. doi: 10.1257/mic.5.1.28.

Matthew Polisson, John K.-H. Quah, and Ludovic Renou. Revealed preferences over risk and uncertainty. *American Economic Review*, 110(6):1782–1820, 2020. doi: 10.1257/aer.20180210.

John K.-H. Quah. Survey lecture on revealed preference analysis. Lecture slides, Queensland University of Technology, November 2015. URL `https://www.johnquah.com/uploads/7/5/6/7/75676677/survey-qutweb.pdf`. Presented 16 November 2015.

Roy Radner. Existence of equilibrium of plans, prices, and price expectations in a sequence of markets. *Econometrica*, 40(2):289–303, 1972. doi: 10.2307/1909407.

Lloyd Shapley and Herbert Scarf. On cores and indivisibility. *Journal of Mathematical Economics*, 1(1):23–37, 1974.

Ronald W. Shephard. *Cost and Production Functions*. Princeton University Press, Princeton, NJ, 1953.

Christopher A. Sims. Implications of rational inattention. *Journal of Monetary Economics*, 50 (3):665–690, 2003. doi: 10.1016/S0304-3932(03)00029-1.

Hugo Sonnenschein. Market excess demand functions. *Econometrica*, 40(3):549–563, 1972.

Edward Szpilrajn. Sur l'extension de l'ordre partiel. *Fundamenta Mathematicae*, 16:386–389, 1930.

Simone Tonin. Existence of equilibrium prices: A pedagogical proof, 2018. URL `https://arxiv.org/abs/1808.03129`. Version 2, 23 September 2018.

Hal R. Varian. The nonparametric approach to demand analysis. *Econometrica*, 50(4):945–973, 1982. doi: 10.2307/1912771.